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INTERSTATE POWER & LIGHT COMPANY

PSD Air Construction Permit Application (80% Complete)

Morgan Valley Energy Center

PROJECT NO. 185510

REVISION 0

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Contents

Tables

List of Abbreviations

Abbreviation	Term/Phrase/Name
%	percent
°F	degrees Fahrenheit
AERMOD	AMS/EPA Regulatory Model
AVO	audio/visual/olfactory
BACT	Best Available Control Technology
CAA	Clean Air Act
CAAA	Clean Air Act Amendments
CAIR	Clean Air Interstate Rule
CAQT	critical air quality threshold
CEMS	continuous emission monitor system
CF_4	perfluoromethane
CFR	Code of Federal Regulations
CH_4	methane
CI	compression ignition
CO	carbon monoxide
CO_2	carbon dioxide
CO_2e	carbon dioxide equivalent
DLN	dry low NO_x
EOR	enhanced oil recovery
EPA	U.S. Environmental Protection Agency
ESP	electrostatic precipitator
FGR	flue gas recirculation
ft/s	feet per second
g/hp-hr	gram per horsepower hour
g/kW-hr	gram per kilowatt hour
g/m^2	grams per square meter
GCP	good combustion practices
GEP	Good Engineering Practice
GHG	greenhouse gas
GWP	global warming potential

Abbreviation	Term/Phrase/Name
H_2O	water
H_2SO_4	sulfuric acid
HAP	Hazardous Air Pollutant
HHV	higher heating value
hp	horsepower
IAC	Iowa Administrative Code
ICE	internal combustion engine
IEC	International Electrotechnical Commission
Iowa DNR	Iowa Department of Natural Resources
IPL	Interstate Power and Light Company
kg/GJ	kilograms per gigajoule
kPa	kilopascal
kV	kilovolt
kW	kilowatt
LAER	Lowest Achievable Emission Rate
lb/hr	pounds per hour
lb/lb-mol	pound per pound-mole
lb/MMBtu	pounds per million British thermal units
lb/MW-hr	pound per megawatt hour
lb/VMT	pounds per vehicle mile traveled
lb/yr	pounds per year
LNB	low-NO_x burner
MACT	Maximum Achievable Control Technology
MECL	minimum emissions compliance load
mg/L	milligrams per liter
MMBtu/hr	million British thermal units per hour
MW	megawatt
N_2O	nitrous oxide
NAAQS	National Ambient Air Quality Standards
NAICS	North American Industrial Classification System
NESHAP	National Emission Standards for Hazardous Air Pollutants
ng/J	nanogram per Joule
NMHC	non-methane hydrocarbon

Abbreviation	Term/Phrase/Name
NO_2	nitrogen dioxide
NO_x	nitrogen oxides
NSPS	New Source Performance Standards
NSR	New Source Review
O_2	oxygen
PEMS	parametric emission monitoring system
PM	particulate matter
PM_{10}	particulate matter less than 10 microns in diameter
$PM_{2.5}$	particulate matter less than 2.5 microns in diameter
ppb	parts per billion
ppbvd	parts per billion dry volume
ppm	parts per million
PSD	Prevention of Significant Deterioration
psia	pounds per square inch
RACT	Reasonable Available Control Technology
RBLC	RACT/BACT/LAER Clearinghouse
RICE	Reciprocating Internal Combustion Engines
RMP	Risk Management Plan
SCR	selective catalytic reduction
SER	significant emission rate
SF_6	sulfur hexafluoride
SIC	Standard Industrial Classification
SNCR	selective non-catalytic reduction
SO_2	sulfur dioxide
SO_3	sulfur trioxide
TCEQ	Texas Commission on Environmental Quality
tpy	tons per year
ULSD	ultra-low sulfur diesel
VMT	vehicle miles traveled
VOC	volatile organic compounds

1.0 Introduction

Interstate Power and Light Company (IPL) is submitting this Prevention of Significant Deterioration (PSD) air construction permit application for the proposed construction of a greenfield combustion turbine generation facility (Morgan Valley Energy Center), in Linn County, Iowa (Project). This Project is intended to help meet projected electrical demand for IPL's customers. Each turbine will have a maximum heat input of 2,270 million British thermal units per hour (MMBtu/hr) (Higher Heating Value, HHV) when firing natural gas, based on ambient conditions of -40 degrees Fahrenheit (°F) and 75 percent humidity. In addition to the turbines, four natural gas hot water boilers, an emergency diesel generator, an emergency diesel fire pump, two fuel oil belly tanks, comfort building heaters, haul road traffic fugitives, natural gas piping components, and circuit breakers will be part of this Project.

This permit application is divided into the following sections:

- Part 1 – Introduction
- Part 2 – Project Description
- Part 3 – Emissions Estimates (This section provides estimates of emissions associated with the Project.)
- Part 4 – Regulatory Review (This section identifies applicable State and Federal air quality regulations.)
- Part 5 – Best Available Control Technology (BACT) Analysis
- Part 6 – Air Dispersion Modeling (This section provides model descriptions and data requirements for the air quality impact assessment as well as interpretation, analysis, and comparison of the modeling results with applicable air quality regulations.)
- Part 7 – Additional Impact Analysis (This section addresses other potential air quality-related impacts [i.e., growth, soil, vegetation, and visibility].)

The construction permit application forms required by the Iowa Department of Natural Resources (Iowa DNR) are included as part of this application.

1.1 Project Equipment

The Project will consist of three simple-cycle natural gas-fired combustion turbines, with each turbine having a maximum heat input of 2,270 MMBtu/hr, HHV, when firing natural gas, based on ambient conditions of -40 degrees Fahrenheit (°F) and 75 percent humidity. In addition to the combustion turbines, the following equipment will be included as part of the Project:

- Four natural gas-fired hot water boilers
- One emergency diesel generator
- One emergency diesel fire pump
- Two fuel oil belly tanks
- Haul road traffic fugitives
- Natural gas piping components (fugitives)
- Sulfur hexafluoride (SF_6) containing circuit breakers
- Comfort building heaters

1.2 Project Emissions

Potential emissions from the Project are shown in Table 1-1 which includes start-up and shutdown emissions for the combustion turbines and auxiliary equipment emissions. A full description of equipment associated with the Project is provided in Part 2.0 of this application.

Based on the projected potential emissions, the Project is a Prevention of Significant Deterioration (PSD) major source because potential emissions of one or more regulated New Source Review (NSR) pollutants exceed the PSD major source threshold of 250 tons per year for a source category not listed in 40 CFR 52.21(b)(1). Accordingly, Table 1-1 includes a comparison of Project emissions increases to the applicable PSD significant emission rates (SERs) to identify pollutants subject to PSD review and to demonstrate which emission increases are considered significant under PSD regulations.

The Project will be an area source of Hazardous Air Pollutants (HAPs) (i.e. less than 25 tons per year of total HAPs and less than 10 tons per year of any single HAP).

Table 1-1: Project Potential Emissions and PSD Significant Emission Rates

Pollutant[a]	Project Potential Emissions[b] (tons per year)	PSD Significant Emission Rates[c] (tons per year)
NO_x	**319**	40
CO	**1,152**	100
SO_2	18.0	40
VOC	**105**	40
PM	**61.6**	25
PM_{10}[d]	**61.6**	15
$PM_{2.5}$[d]	**61.6**	10
CO_2e	**1,516,388**	75,000[e]
H_2SO_4 mist	2.7	7
Lead	5.2×10^{-5}	0.6

(a) NO_x = nitrogen oxides; CO = carbon monoxide; SO_2 = sulfur dioxide; VOC = volatile organic compounds; PM = particulate matter; PM_{10} = particulate matter 10 microns or less in diameter; $PM_{2.5}$ = particulate matter 2.5 microns or less in diameter; CO_2e = carbon dioxide equivalents; H_2SO_4 mist = sulfuric acid mist
(b) Numbers in ***bold*** indicate potential emissions greater than PSD significant emission rate (SER)
(c) 40 CFR 52.21(b)(23)(i)
(d) Filterable plus condensable
(e) 40 CFR 52.21(b)(49)(iv)(a)

1.3 BACT

A "top-down" BACT analysis was performed for each of the pollutants in Table 1-1 that was above its corresponding PSD significant emission rate: nitrogen oxides (NO_x), carbon monoxide (CO), particulate matter (PM)/ particulate matter of 10 microns in diameter or smaller (PM_{10})/particulate matter of 2.5 microns in diameter or smaller ($PM_{2.5}$), volatile organic compounds (VOC), and greenhouse gases (CO_2e).

Although Table 1-1 shows sulfur dioxide (SO_2) and sulfuric acid (H_2SO_4) mist emissions below their respective PSD significance thresholds, IPL intends to preserve optionality for its customers. Therefore, IPL is including SO_2 and H_2SO_4 mist in the pollutants subject to BACT review.

State-of-the-art pollution control equipment has been selected as BACT for the Project. Emissions of NO_x from the combustion turbines will be controlled by dry low-NO_x (DLN) burners. Emissions of CO and VOC will be controlled by good combustion practices. Use of low sulfur fuel (natural gas) and good combustion practices will control emissions of SO_2, H_2SO_4 mist, PM/PM_{10}/$PM_{2.5}$. Greenhouse gas emissions will be controlled with the use of natural gas fuel, monitoring and control of excess air, and efficient turbine design. Table 5-1 in Section 5.0 displays the BACT results for the combustion turbines, while Table 5-2 outlines the BACT results for auxiliary equipment.

1.4 Air Quality Analysis

The existing air quality in the Linn County area is designated as attainment or unclassifiable in regard to the NAAQS for all criteria pollutants. An air dispersion modeling analysis was performed for the pollutants subject to PSD to assess potential ambient air quality impacts associated with the Project. The modeling was performed in accordance with approved Iowa DNR and U.S. Environmental Protection Agency (EPA) modeling guidance.

The modeling analysis (included in Part 6.0 of this application) demonstrates that operation of the Project will not cause or contribute to a violation of the NAAQS or PSD increments, as applicable.

1.5 Additional Impacts Analysis

The potential impacts of the proposed Project on visibility, soils, vegetation, and growth are discussed in Part 7.0 of this application. As indicated by the analysis, the addition of the Project will not have a significant impact on visibility, soils, growth, or vegetation in the surrounding area.

2.0 Project Description

The Project will be located at a greenfield site in Linn County, Iowa. The Project location is shown in Figure A-1 (Appendix A). Linn County is currently designated as an attainment/unclassified area for all criteria pollutants in 40 CFR Part 81.

2.1 Combustion Turbines and Emission Controls

The three simple-cycle turbines will be Siemens SGT6-5000F, F-class combustion turbines, each with a maximum heat input of 2,270 MMBtu/hr, each, HHV (which occurs on a negative (-) 40.0 °F ambient temperature and 75 percent relative humidity day). The combustion turbines are proposed to be permitted to operate for 3,500 hours per year, each, including start-up and shutdown events.

To control emissions of NO_x, the combustion turbines will be equipped with DLN burners, which will achieve a NO_x emission rate of 5 parts per million (ppm) at 15% O_2 for all loads down to minimum emission compliance load (MECL). According to vendor performance data, the DLN combustion system is capable of maintaining a NO_x emission rate of 5 ppm at 15% O_2 across the full operating envelope, including ambient temperatures ranging from -40°F to 105°F and relative humidity levels between 30 and 75 percent. This demonstrates that even under extreme low-temperature or high-humidity conditions, the DLN system remains effective. To minimize emissions of CO, VOC, SO_2, H_2SO_4 mist and $PM/PM_{10}/PM_{2.5}$, the combustion turbines will be controlled by using low sulfur fuel (natural gas) and good combustion practices. This will achieve a CO emission rate of 4 ppm at 15% O_2 for 100 percent load down to 71 percent and 9 ppm at 15% O_2 for loads less than 71 percent. Greenhouse gas emissions will be controlled with the use of natural gas fuel, monitoring and control of excess air, and efficient turbine design.

2.2 Hot Water Boilers

Four natural gas-fired hot water boilers (6 MMBtu/hr, each) will be constructed to warm the natural gas prior to combustion in the Project's combustion turbines. The hot water boilers will be permitted to operate up to 8,760 hours per year per heater, and each heater will be designed with low-NO_x burners.

2.3 Emergency Diesel Generator

One emergency diesel generator will be built to support the facility in case of a power interruption. The emergency diesel generator will each have a maximum power output of 2,000 kilowatts (kW) and will be fired solely by ultra-low sulfur diesel (ULSD). IPL proposes to operate the emergency diesel generators for up to 100 hours annually for testing and maintenance purposes but has evaluated emissions based on 500 hours per year, in case a true emergency does occur. The basis for the potential to emit calculation for emergency equipment was outlined in a September 6, 1995, memorandum from the EPA Director of Air Quality Planning and Standards.

2.4 Emergency Diesel Fire Pump

An emergency diesel fire pump will be built to support the Project in case of a fire. The emergency diesel fire pump will have a maximum power output of 422 horsepower (hp) and will be fired solely by ULSD. IPL proposes to operate the emergency diesel fire pump for up to 100 hours annually for testing and maintenance purposes but has evaluated emissions based on 500 hours per year in case of operation in an emergency. The basis for the potential to emit calculation for emergency equipment was outlined in a September 6, 1995, memorandum from the EPA Director of Air Quality Planning and Standards.

2.5 Haul Road Traffic Fugitives

Miscellaneous supplies associated with facility operations will be transported to and from the site via trucks. Up to 60 trucks per year are expected for delivery of materials to the site or removal of materials off-site. Deliveries and pick-ups associated with facility include materials such as fuel oil for emergency equipment. To mitigate onsite road emissions from these deliveries, IPL will pave the primary facility roads.

Natural gas required to be fired in the turbines, hot water boilers, and comfort building heaters will be delivered to the site via pipeline and not by truck delivery.

2.6 Natural Gas Fugitives (Piping Components)

The proposed Project will include natural gas piping components from the natural gas line that will enter the Project site to provide gas for the combustion turbines, hot water boilers, and comfort building heaters. These natural gas piping components are potential sources of methane and VOC emissions due to fugitive emissions from valves, open ended lines and flanges.

2.7 Sulfur Hexafluoride (SF_6) Containing Equipment

The Project will include circuit breakers that utilize SF_6 gas for electrical insulation. SF_6 is a potent greenhouse gas and the circuit breaker emissions are included in the greenhouse gas emissions for the facility. IPL proposes to install the following circuit breaker equipment to support this Project:

- Three 20-kilovolt (kV) circuit breakers
- Three 345-kV circuit breakers

2.8 Insignificant Activities

The following insignificant activities will be installed as part of the Project:

2.8.1 Diesel Belly Tanks

The Project will include one 1,260-gallon diesel belly tank and one 465-gallon diesel belly tank. These tanks will store fuel oil for the emergency diesel generator and the emergency diesel fire pump, respectively.

2.8.2 Comfort Building Heaters

To provide comfort heating for the equipment and administrative buildings, IPL proposes to install 40 natural gas-fired comfort building heaters as part of the Project. Since final design specifications are still pending, a 20 percent safety margin has been applied to the total heat input. As a result, the combined heat input from all installed heaters will not exceed 80.8 MMBtu/hr, with each individual heater rated below 10 MMBtu/hr. Emission calculations are based on this projected capacity.

3.0 Emissions Estimates

Emissions of air contaminants will result from the combustion of natural gas in the combustion turbines. There will also be emissions of air contaminants generated from the following equipment: four natural gas hot water boilers, an emergency diesel generator, an emergency diesel fire pump, haul road traffic fugitives, circuit breakers, natural gas piping components, comfort building heaters, and two diesel belly tanks.

Process flow diagram figures for the combustion turbine process and additional equipment are located in Appendix A. Each emission point's control device descriptions, corresponding controlled emission rates, and procedures for estimating emissions are discussed in detail in the sections below. Tables summarizing the emissions estimates are included in Appendix B.

3.1 Combustion Turbines

The following sections summarize the combustion turbines hours of operation, emissions estimates for various operating loads, and start-up/shutdown operation.

3.1.1 Combustion Turbine Steady-State Operation Emissions

Emissions from the combustion turbines are dependent on ambient temperature conditions and the turbine's operating load, which can vary from MECL, approximately 50 percent, to 100 percent. To account for representative seasonal climatic variations, potential emissions from the proposed combustion turbines were analyzed at the MECL, 75, 70, and 100 percent load for ambient temperatures ranging from negative (-) 40.0 °F to 105.0°F. The projected emissions were based on data provided by the combustion turbine manufacturer, 40 CFR Part 98 – Subpart C, and AP-42 emission factors. SO_2 emissions were based on a natural gas sulfur content of 0.5 grains/100 standard cubic feet. Detailed calculations of the combustion turbines' emissions are provided in Appendix B of this application.

Based on the above assumptions, the maximum expected hourly emission rates for normal operation (excluding start-up and shutdown) for the combustion turbines are shown in Table 3-1.

Table 3-1: Maximum Expected Hourly Emission Rates per Combustion Turbine

Pollutant	100% Load	75% Load	70% Load	MECL
		pounds per hour		
NO_x	42.6	35.0	33.6	27.7
CO[a]	20.8	17.0	36.8	30.4
$PM/PM_{10}/PM_{2.5}$	11.0	8.4	8.2	7.2
SO_2	3.3	2.7	2.6	2.2
VOC	3.0	2.4	2.3	1.9
H_2SO_4 mist	0.5	0.4	0.4	0.3
Lead	0.0	0.0	0.0	0.0
CO_2e	278,237	228,568	219,561	181,912

(a) CO emissions will meet 4 ppm at 15% O_2 for 100% load down to 71% load and 9 ppm at 15% O_2 for loads less than 71%.

3.1.2 Combustion Turbines Start-up and Shutdown Emissions

The combustion turbines' emissions are based on expected 365 start-up and shutdown events per year, per turbine. Potential start-up and shutdown emissions per event for NOx, CO, VOC and particulate were provided by Siemens. Emissions of SO_2, H_2SO_4 mist and CO_2e are calculated using ratios derived from full-load operating conditions. One start-up/shutdown event is equivalent to one start-up plus one shutdown.

Potential start-up and shutdown emissions are shown in Table 3-2. Detailed calculations of the potential start-up and shutdown emissions are provided in Appendix B.

Table 3-2: Potential Natural Gas Turbine Start-up and Shutdown Emissions Per Turbine

Pollutant	Start-up Emissions	Shutdown Emissions	Start-up and Shutdown Emissions[a]
	lb/start	lb/shutdown	tons per year
NO_x	76.8	48.5	22.9
CO	1,060.6	645.6	311.4
$PM/PM_{10}/PM_{2.5}$	6.2	3.9	1.8
SO_2	1.9	1.1	0.5
VOC	91.5	55.9	26.9
H_2SO_4 mist	0.3	0.2	0.1
Lead	0.0	0.0	0.0
CO_2e	159,522	89,963	45,531

(a) Emissions are based on 365 start-up and shutdown events per turbine.

3.1.3 Combustion Turbines Total Annual Emissions

The following assumptions were considered in the determination of total annual emissions calculations for this Project:

- Combustion turbine operation of 3,500 hours of operation per year per turbine
- 365 start-up/shutdowns events per year per turbine
- Worst-case emissions based on operation at full load or 365 start-up/shutdown events with the remainder of the hours for the year at full load for each turbine.

3.2 HAP Emissions

The Project will be an area source of Hazardous Air Pollutants (HAPs) (i.e. less than 25 tons per year of total HAPs and less than 10 tons per year of any single HAP). HAP emission calculations are included in Appendix B. Formaldehyde emission factors for the three Project Combustion Turbines are based on vendor-supplied emission data. Supporting details are provided in the turbine performance sheet included in Appendix B. The vendor data reflects results from factory acceptance testing and performance guarantees, which were conducted under representative operating conditions. These values are considered more accurate than generalized AP-42 factors because they account for the specific turbine model, fuel type, and expected load profile of the project. Vendor performance data is included in Appendix B.

3.3 Hot Water Boiler Emissions

Four 6-MMBtu/hr natural gas-fired hot water boilers will be installed at the facility to heat the natural gas prior to being combusted in the combustion turbine. As a worst-case estimate, it is assumed that annual operations will be 8,760 hours per year for each heater. Emissions for the hot water boilers were estimated based on AP-42 Section 1.4 emission factors. Greenhouse gas emissions were estimated based on the emission factors in 40 CFR Part 98. Detailed calculations are provided in Appendix B.

3.4 Emergency Diesel Generator Emissions

One 2,000-kW diesel generator will be installed for emergency power use at the facility. The generator will be fired with ULSD. Emissions for the emergency diesel generator were estimated assuming 500 hours per year of operation in a true emergency. Emissions for this unit were estimated based on New Source Performance Standards (NSPS) Subpart IIII emission limits and AP-42 Section 3.4 emission factors. Greenhouse gas emissions were estimated based on 40 CFR Part 98 emission factors. Detailed calculations of the emergency diesel generator emissions are provided in Appendix B.

3.5 Emergency Diesel Fire Pump

One 422-hp emergency diesel fire pump will be installed at the facility. The fire pump will be fired with ULSD. Emissions for the emergency diesel fire pump were estimated assuming 500 hours of operation should a true emergency occur. Emissions for this unit were estimated based on NSPS Subpart IIII emission limits and AP-42 Section 3.3 emission factors. Greenhouse gas emissions were estimated based on 40 CFR Part 98 emission factors. Detailed calculations of diesel fire pump emissions are provided in Appendix B.

3.6 Haul Road Traffic Fugitives Calculation Method

Emissions from haul roads due to traffic were estimated using Equation 1 from AP-42, Section 13.2.1.3 for the paved roads, size-specific emission calculation equation below:

$$E = k * (sL)^{0.91} * (W)^{1.02}$$

Where:

- E = pounds per vehicle miles traveled (lb/VMT)
- sL = silt loading grams per square meter (g/m^2)
- W = mean vehicle weight (tons)
- k = constant (AP-42 Table 13.2.1-1)

The mean vehicle weight is calculated by averaging the loaded and unloaded vehicle weights. A silt loading value of 8.2 g/m^2 was selected for all paved roads to provide a conservative emissions estimate and is consistent with the mean silt loading for paved roads at quarry facilities presented in AP-42 Table 13.2.1-3.

For paved roads, vehicle miles traveled (VMT) is calculated as follows:

$$VMT = length\ of\ path\ haul\ road\ vehicle\ travels * maximum\ trips\ (hourly\ or\ annual)$$

Whether a vehicle travels the haul road twice (back and forth) or once (when traveling in a loop) was accounted for when calculating the miles traveled for each haul road route. Detailed calculations of haul road emissions are provided in Appendix B.

3.7 Natural Gas Fugitives Calculations Method

Fugitive emissions will come from small leaks in equipment connections throughout the natural gas pipeline on IPL's property to support the Project sources at Morgan Valley Energy Center. The estimated number of flanges, open ended lines, and valves were determined from engineering plans for the facility. The emissions were then estimated using the 1995 Protocol for Equipment Leak Emission Estimates- EPA-453/R-95-017. The emissions for natural gas VOC fugitive emissions were calculated using the minimum methane content of natural gas. Further, to determine natural gas CO_2e fugitive emissions the maximum methane content was used. Detailed calculations of Project natural gas fugitives located at Morgan Valley Energy Center are provided in Appendix B.

3.8 SF_6-Containing Equipment Calculation Method

Annual potential-to-emit emissions of SF_6 from the circuit breakers were based on maximum leakage rate of 0.5 percent per year, the amount of SF_6 in each size of circuit breaker, and SF_6's global warming potential (GWP). Detailed calculations are provided in Appendix B of this application.

3.9 Insignificant Activities

Insignificant emissions were included in the emissions calculations to be incorporated into the overall site-wide emissions calculations. Emissions from Insignificant activities can be found in Appendix B.

3.9.1 Diesel Belly Tanks Calculation Method

The Project will include a 1,260-gallon diesel belly tank to support the emergency generator and a 465-gallon diesel belly tank for the emergency fire pump. Emissions from loading and breathing losses were estimated for the belly tank using the EPA TANKS emission software. A detailed report of the fuel belly tanks emissions is provided in Appendix B.

3.9.2 Comfort Building Heaters Calculation Method

The Project will include approximately 40 natural gas fired comfort building heaters. Since final design specifications are still pending, a 20 percent safety margin has been applied to the total heat input. As a result, the combined heat input from all installed heaters will not exceed 80.8 MMBtu/hr, with each individual heater rated below 10 MMBtu/hr. Emissions for the comfort building heaters were estimated based on AP-42 emission factors. Greenhouse gas emissions were estimated based on the emission factors in 40 CFR Part 98. Detailed calculations are provided in Appendix B.

4.0 Regulatory Review

The Project is subject to various Federal and State air regulations. Part 4.0 contains a discussion of applicable Federal and Iowa Administrative Code (IAC) provisions. Where applicable, reference to general limitations is provided when there is no specific requirement that applies to an emission source.

In certain instances, there may be multiple applicable regulatory requirements that identify differing levels of emission limitations. For instance, where a BACT emission limitation is established for a specific pollutant and a NSPS regulation is also applicable, the BACT limitation may be more stringent than an applicable NSPS emission limitation for the same pollutant. In these situations, it is understood that compliance with the most restrictive requirement would demonstrate compliance with other less stringent requirements.

4.1 PSD Regulations

PSD review applies to a physical change of a major stationary source located in an area designated as attainment or unclassified that would result in a significant emissions increase of a regulated New Source Review (NSR) pollutant and a significant net emissions increase of that pollutant pursuant to 40 CFR 52.21. PSD review consists of the following:

- A BACT analysis
- An air quality analysis
- An analysis of additional impacts on visibility, soils, vegetation, and growth

Three criteria were evaluated to determine PSD applicability to the Project (EPA, 1990):

- Whether the Project is sufficiently large (in terms of its emissions) to be a "major stationary source" or "major modification"
- Whether the source is in an area designated as "attainment" or "unclassified"
- Whether the Project would result in a "significant emissions increase" or a "significant net emissions increase" of a "regulated NSR pollutant" as defined by 40 CFR 52.21

Regulated NSR pollutants in Iowa include NO_x, SO_2, CO, PM, PM_{10}, $PM_{2.5}$, VOC, CO_2e, H_2SO_4 mist, fluorides, and lead. The definition of a "major stationary source" is given in 40 CFR 52.21. The Project is not one of the "listed" 28 source categories specified in the PSD regulations as it consists of simple-cycle combustion turbines. Therefore, the applicable major source threshold is 250 tons per year of any regulated NSR pollutant. The facility will be considered a major stationary source because the potential emissions of one or more regulated NSR pollutants exceed 250 tons per year. Thus, the Project meets the first criterion for PSD applicability.

The Project is to be located in an attainment/unclassified area for all criteria pollutants; thus, it meets the second criterion for PSD applicability.

The maximum potential emissions from the Project are listed in Table 1-1, which include start-up and shutdown emissions from the combustion turbines. The Project would result in a "significant emission increase" for the following regulated NSR pollutants: NO_x, $PM/PM_{10}/PM_{2.5}$, VOC, CO, and CO_2e. Thus, the Project meets the third and final criterion for PSD applicability.

The PSD regulations in 40 CFR 52.21 require the following issues be addressed:

- Determination of BACT on a case-by-case basis, taking into account costs as well as energy, environmental, and economic impacts;
- Demonstration that the increase in emissions would not cause or contribute to an exceedance of the NAAQS or PSD increments;
- Analysis of the impairment, if any, to visibility, soils, vegetation, and growth.

Section 5.0 contains the BACT analyses for the regulated NSR pollutants. Section 6.0, which details the air dispersion modeling analysis, and Section 7.0, which contains the additional impacts analysis, will be included in the final air permit submittal.

4.2 New Source Performance Standards

Per 40 CFR Part 60, (adopted by reference in IAC 567-23.1(2)), the Project is subject to NSPS. Relevant NSPS standards are listed below, and if applicable, a description of how IPL plans to meet the standards.

4.2.1 Subpart KKKKa

40 CFR Part 60 Subpart KKKKa, Standards of Performance for Stationary Combustion Turbines, is applicable to stationary combustion turbines that commenced construction, modification, or reconstruction after December 13, 2024 and have a base-load rated heat input greater than or equal to 10.7 gigajoules per hour (10 MMBtu/hr), based on the higher heating value of the fuel. This rule provides emission standards for NO_x and SO_2.

The Project combustion turbines are designed and expected to operate at an annual utilization rate less than or equal to 45 percent with a design efficiency less than 38 percent, as defined in Table 1 to 40 CFR Part 60, Subpart KKKKa.

Nitrogen Oxides (NO_x) Emission Standards

As specified in 40 CFR 60.4320a(b)(3) and Table 1 to Subpart KKKKa, the Project's combustion turbines are subject to a NO_x limit of 9 ppm at 15 percent oxygen.

Part-Load and Special Operating Condition Standards

During any hour in which the turbine satisfies one or more of the special-condition criteria in Table 1—including ambient temperatures below 0 °F, operation at less than 70 percent of base-load rating, turbine tuning, byproduct-fuel firing, or other listed conditions—the applicable input-based NO_x emissions standard for turbines greater than 300 MMBtu/hr is 96 ppm at 15 percent O_2.

Consistent with 40 CFR § 60.4320a(b)(1), if the turbine operates at less than 70 percent of its base-load rating at any point during an operating hour, the part-load NO_x emission standard applies for the entire operating hour (96 ppm at 15 percent O_2).

Sulfur Dioxide (SO_2) Emission Standards

Per 40 CFR 60.4330a(a), the Project's combustion turbines are subject to a SO_2 limit of either 110 nanograms per Joule (ng/J) (0.90 pounds per megawatt-hour) gross energy output or 26 ng/J (0.060 pounds per million British thermal unit) heat input.

Compliance Demonstration

In accordance with 40 CFR 60.4333a of NSPS Subpart KKKKa, IPL proposes to demonstrate compliance with the applicable NO_x emission limits using a continuous emissions monitoring system as described in 40 CFR 60.4345a(c). As required by 40 CFR 60.4333a(b), IPL will conduct an initial NO_x performance (stack) test in accordance with 40 CFR 60.8 and the applicable test methods in 40 CFR 60.4400a or 60.4405a; use of a NO_x CEMS replaces only the requirement for subsequent performance testing and does not eliminate the required initial test.

For SO_2 compliance, IPL will conduct an initial performance test according to 40 CFR 60.8 and maintain on-site records documenting that the sulfur content of all fuels combusted meets the requirements of 40 CFR 60.4370a, as specified in 40 CFR 60.4333a (d)(3).

4.2.2 Subpart TTTTa

NSPS 40 CFR Part 60, Subpart TTTTa, Standards of Performance for Greenhouse Gas Emissions for Modified Coal-fired Steam Electric Generating Units and New Construction and Reconstruction Stationary Combustion Turbine Electric Generating Units, regulates CO_2 emissions from electric generating units under Clean Air Act 111b. The standards apply to any electric generating unit, including a steam generating unit, integrated gasification combined cycle unit, or stationary combustion turbine that commences construction or reconstruction after May 23, 2023, that has a fossil-fuel heat input capacity greater than 250 MMBtu/hr of fossil fuel and serves a generator capable of selling greater than 25 MW of electricity to a utility power distribution system.

The combustion turbines will be subject to NSPS Subpart TTTTa, if the regulation remains in effect (is not stayed or remanded by the courts). The rule establishes CO_2 emission limits for natural gas-fired combustion turbines, with compliance options based on operational load classification.

Emissions and modeling for this Project are based on 3,500 hours of annual operation, which is expected to align with the intermediate-load classification under the regulation. IPL will determine the appropriate compliance pathway during permitting, and will provide additional data if Subpart TTTTa remains in force at the time of permit issuance.

Should NSPS Subpart TTTTa be revised or repealed as is currently proposed, NSPS 40 CFR Part 60, Subpart TTTT, Standards of Performance for Greenhouse Gas Emissions for Electric Utility Generating Units could be applicable to the turbines, depending on EPA's regulatory guidance or rulemaking.

4.2.3 Subpart IIII

NSPS 40 CFR Part 60, Subpart IIII applies to stationary compression ignition (CI) internal combustion engines (ICE) and the manufacturers or owners and operators of these engines as follows:

1. Manufacturers of stationary CI ICE with a displacement of less than 30 liters per cylinder where the model year is 2007 or later for non-fire pump engines and the model year listed or later model years for fire pump engines (2008 or 2011)
2. Owners and operators of stationary CI ICE that commenced construction after July 11, 2005, where the CI ICE are manufactured after April 1, 2006 (non-fire pump engines), or manufactured as a National Fire Protection Association fire pump engine after July 1, 2006

For purposes of this application, Subpart IIII is applicable to the emergency diesel generator. The engines will meet the definition of “emergency stationary internal combustion engine” under this subpart as follows:

- There is no time limit on the use of emergency stationary ICE in emergency situations.
- The engine may be operated for a maximum of 100 hours per calendar year for testing and maintenance, except as indicated below.
- 50 hours of the 100 hours per calendar year allocated may be used for non-emergency situations.

Emergency Generator

Based on the size (horsepower) and use (emergency) and assuming IPL purchases certified model year 2007 or later CI ICE with a displacement that will be less than 10 liters per cylinder, the emergency generators will be certified in accordance with the limits in 40 CFR 60.4202(a)(2), which refer to the limits in 40 CFR 1039, appendix I. As the emergency generator will be greater than 560 kW and manufactured after 2006, Table 2 to Appendix I – Tier 2 Emissions Standards, indicates the following applicable emission standards:

- 6.4 g/kW-hr (4.8 g/hp-hr) for NMHC plus NO_x
- 3.5 g/kW-hr (2.6 g/hp-hr) for CO
- 0.20 g/kW-hr (0.15 g/hp-hr) for PM

The emergency generator will also be subject to the exhaust opacity limits in 40 CFR 1039.105, with single-cylinder engines, constant speed engine, and engines certified to a PM emission standard or family emission limits of 0.07 g/KW-hr or lower being exempt from these limits:

- 20 percent during the acceleration mode
- 15 percent during the lugging mode
- 50 percent during the peaks in either the acceleration or lugging modes

Emergency Fire Pump

Based on the size (horsepower) and use (emergency) and IPL purchasing certified model year 2010 or later CI ICE with a displacement that is less than 30 liters per cylinder, the emergency fire pump will be certified in accordance with the limits in 40 CFR 60.4202(d), which refer to the limits in Table 4 to Subpart IIII of 40 CFR 60. As the emergency fire pump will be 422 hp and manufactured after 2009, the following applicable emission standards are applicable:

- 4.0 g/kW-hr (3.0g/hp-hr) for NMHC plus NO_x
- 3.5 g/kW-hr (2.6 g/hp-hr) for CO
- 0.20 g/kW-hr (0.15 g/hp-hr) for PM

Compliance with this subpart will be shown by purchasing engines certified to meet the applicable emission standards for the model year and maximum engine power depending on the date of purchase. IPL will install an emergency diesel engine that is certified to meet the applicable emission standards based on the date that the unit will be installed.

Pursuant to 40 CFR 60.4207(b), owners and operators of CI ICE subject to Subpart IIII with a displacement of less than 30 liters per cylinder that use diesel fuel must purchase diesel fuel that meets the requirements of 40 CFR 1090.305 for non-road diesel fuel. As stated in 40 CFR 1090.305, non-road diesel fuel must be limited to 15 parts ppm maximum sulfur content. The cetane index is limited to a minimum of 40 and the maximum aromatic content is limited to 35 volume percent.

IPL will be subject to the applicable requirements of this rule for the emergency generator and emergency fire pump. Although emissions were calculated using 500 hours per year for permitting purposes, IPL will limit actual operation of the emergency generator and fire pump to 100 hours for testing and maintenance to maintain its classification as an emergency stationary internal combustion engine under 40 CFR Part 60, Subpart IIII.

4.2.4 Subpart Dc – Not Applicable

Steam generating units that commenced construction, modification, or reconstruction after June 9, 1989, and have a heat input capacity of less than 29 MW (100 MMBtu/hr) but greater than or equal to 2.9 MW (10 MMBtu/hr) are subject to 40 CFR Part 60, Subpart Dc. Since the hot water boilers and comfort building heaters each have a heat input capacity below 10 MMBtu/hr, the units do not meet the applicability criteria and are not subject to this rule.

4.2.5 Subparts GG and KKKK – Not Applicable

Stationary combustion turbines that commenced construction, modification, or reconstruction after December 13, 2024, and that have a base load heat input capacity of 10 MMBtu/hr or greater, are subject to NSPS 40 CFR Part 60, Subpart KKKKa and are exempt from the requirements of Subpart GG and KKKK. Section 4.2.1, above, covers Subpart KKKKa.

4.2.6 Subpart Kb – Not Applicable

40 CFR Part 60, Subpart Kb applies to each storage vessel with a capacity greater than or equal to 75 cubic meters used to store volatile organic liquids for which construction, reconstruction, or modification is commenced after July 23, 1984. The fuel belly tanks will each have a capacity less than 75 cubic meters; therefore, it they will not be subject to Subpart Kb.

4.3 National Emission Standards for Hazardous Air Pollutants and Maximum Available Control Technology

National Emission Standards for Hazardous Air Pollutants (NESHAP) are contained in 40 CFR Part 63 (adopted by reference in IAC 567-23.1(4)). NESHAP are emissions standards set by the EPA for specific source categories. The NESHAP require the maximum degree of emission reduction of certain HAP emissions that the EPA determines to be achievable, which is known as the maximum achievable control technology (MACT) standards.

The following MACT standards are relevant to the Project.

4.3.1 Subpart ZZZZ

The Reciprocating Internal Combustion Engines (RICE) MACT (40 Part 63, Subpart ZZZZ) is applicable to stationary RICE located at major or area sources of HAP emissions.

The emergency generator and emergency fire pump will be new sources located at an area source, as defined under 40 CFR 63.6590(c)(1). Therefore, the emergency generator and emergency fire pump will comply with Subpart ZZZZ by meeting the requirements of 40 CFR Part 60 Subpart IIII pursuant to 40 CFR

63.6590(c)(1). There are no additional requirements, including notification, under Subpart ZZZZ that apply to the proposed engines.

4.3.2 Subpart JJJJJJ

40 CFR Part 63, Subpart JJJJJJ applies to industrial, commercial, or institutional boilers located at an area source of HAPs. Because the site is an area source of HAPs, this regulation is applicable to the hot water boilers and comfort building heaters.

The hot water boilers and comfort building heaters meet the criteria for applicability under Subpart JJJJJJ, they are not subject because they will be natural gas fired as specified in 40 CFR 63.11195(e).

4.3.3 Subpart YYYY – Not Applicable

EPA promulgated MACT standards for new stationary combustion turbines on March 5, 2004. These standards apply to stationary combustion turbines for which construction commenced after January 14, 2003, but only if they are located at facilities that are major sources of HAPs. Because the site will be an area source of HAPs, the proposed combustion turbines are not subject to the requirements of Subpart YYYY.

4.3.4 Subpart DDDDD – Not Applicable

40 CFR Part 63, Subpart DDDDD applies to industrial, commercial, or institutional boilers and process heaters located at a major source of HAPs. Because the site is not major for HAPs, this regulation is not applicable to the hot water boilers or comfort building heaters.

4.4 Iowa Administrative Code

This section describes the regulations which apply to the Project, according to the IAC.

4.4.1 567 IAC 21.1(3) – Emissions Inventory

The facility must maintain records of fuel use, air contaminants emitted, and emission rates in regard to both actual and allowable air contaminant emissions. IPL will comply with this rule.

4.4.2 567 IAC 21.4(455B) – Circumvention of Rules

The facility shall not build, erect, install, or use any article, machine, equipment, or other contrivance which, without resulting in a reduction in the amount of air contaminants released to the atmosphere, reduces or conceals an emission which would otherwise constitute a violation of the IAC. IPL will comply with this rule.

4.4.3 567 IAC 22.1(1) – Permits Required

New or modified equipment is required to obtain a construction permit. IPL is submitting an application prior to installing or modifying any equipment at the facility.

4.4.4 567 IAC 23.1(2) – New Source Performance Standards

The NSPS pursuant 40 CFR Part 60 are adopted into the state regulations. IPL will comply with all the applicable requirements of 40 CFR Part 60.

4.4.5 567 IAC 23.1(4) – Emission Standards for Hazardous Air Pollutants for Source Categories

The federal emission standards for HAPs for source categories pursuant to 40 CFR Part 63 are adopted into the state regulations. IPL will comply with all the applicable requirements of 40 CFR Part 63.

4.4.6 567 IAC 23.1(6) – Calculation of Emission Limitations Based Upon Stack Height

This rule sets limits for the maximum stack height credit to be used in ambient air quality modeling for the purpose of setting an emission limitation and calculating the air quality impact of a source. The rule does not limit the actual physical stack height for any source. IPL will comply with this rule and the stacks will be below the good engineering practice (GEP) stack heights.

4.4.7 567 IAC 23.2(455B) – Opening Burning

The rule states that no person shall allow, cause, or permit open burning of combustible materials (except for disaster rubbish, trees, and tree trimming, as described in 567 IAC 23.2(3), sections "a" and "b"). IPL will comply with this rule.

4.4.8 567 IAC 23.3(2) "a" – General Emission Rates

The general emission rate for particulate matter is 0.1 grains per standard cubic foot (gr/dscf) of exhaust gas. The Project will be subject to this regulation.

4.4.9 567 IAC 23.3(2) "b" (3) – Combustion for Indirect Heating

For a new unit of less than 150 MMBtu/hr of heat input, the maximum allowable emissions from such new unit shall be 0.6 lb/MMBtu of particulates. The proposed hot water boilers and comfort building heaters are subject to this regulation and will be well below this limit at 0.0075 lb/MMBtu of particulate.

4.4.10 567 IAC 23.3(2) "c" – Fugitive Dust

No person shall cause, suffer, or allow any material to be handled, processed, transported, or stored; a building or its appurtenances to be constructed, altered, repaired, or demolished; or a road to be used without taking reasonable precautions to prevent particulate matter from becoming airborne and to prevent visible fugitive dust at the property line. This Project is not expected to produce significant fugitive dust during operation and will comply with this regulation.

4.4.11 567 IAC 23.3(2) "d" – Visible Emissions

The state standard for visible emissions is 40 percent opacity. The Project equipment will meet this limit.

4.4.12 567 IAC 23.3(3) "b" – Sulfur Dioxide Emissions

The combustion of number 1 or number 2 fuel oil may not exceed a sulfur content of 0.5 percent by weight. The state standard for processes capable of emitting sulfur dioxide imposes a limit of 2.5 lb/MMBtu SO_2 from a liquid fuel-burning unit. The emergency diesel generator and emergency fire pump will meet this limit as the fuel will have a sulfur content of 15 ppm.

4.4.13 567 IAC 23.3(3) "e" – Sulfur Dioxide Emissions

The state standard for processes capable of emitting sulfur dioxide imposes a limit of 500 ppmv on SO_2 emissions from any process other than sulfuric acid manufacture. The Project equipment will meet this limit.

4.4.14 567 IAC 24.128(455B) – Acid Rain Program

The State of Iowa has adopted the acid rain program by reference. IPL will comply with all of the applicable requirements of Title IV of the Clean Air Act, as explained in Section 4.7.

4.5 Linn County Code of Ordinances

This section describes the regulations which apply to the Project, according to the Linn County Code of Ordinances.

4.5.1 Chapter 10, Article III, Section 10-58 – Permits for New or Existing Stationary Sources

No person shall construct, install, reconstruct, or alter any equipment or control equipment without obtaining a permit. IPL is applying for a permit.

4.5.2 Chapter 10, Article III, Section 10-60 – Visible Emissions

The Linn County standard for visible emissions is 20 percent opacity. The Project equipment will meet this limit.

4.5.3 Chapter 10, Article III, Section 10-61 – Emissions from Fuel Burning Equipment

Emissions of particulate matter allowed to be emitted from a combustion source, per Section 10-61, is based off of the heat input of the source. Based on the heat input of the combustion turbines, hot water boilers, emergency generator, emergency fire pump, and building comfort heaters, lb/MMBtu limits for particulate matter emissions were determined (using the equation below) and compared to their actual maximum PM emissions.

For equipment greater than 10 MMBtu/hr and less than 4,000 MMBtu/hr, the following equation shall be used.

$$Y = 1.02\ X^{-0.231}$$

Where,

Y = allowable rate of emission in lb/MMBtu
X = maximum equipment capacity rate in MMBtu/hr

Table 4-1 displays the limits compared to their actual emission rate. All emission units meet the standards listed and are fueled by only pipeline quality natural gas or diesel.

Table 4-1: PM Limits Compared to PM Emissions for Linn County Code of Ordinance Chapter 10, Article III, Section 10-61

Emission Unit	Size (MMBtu/hr)[a]	PM Limit (lb/MMBtu)[a, b]	Actual PM Rate (lb/MMBtu)
Combustion turbines	2,270	0.17	0.0048
Hot water boilers	6.0	0.6	0.0075
Emergency generator	21.84	0.5	0.13
Emergency fire pump	2.95	0.6	0.13
Building heaters	Various: all < 10.0	0.6	0.0075

(a) MMBtu/hr = million British thermal units per hour, lb/MMBtu = pound per million British thermal units
(b) Fuel burning equipment with heat inputs less than 10 MMBtu, 0.6 lb/MMBtu PM limit applies.

4.5.4 Chapter 10, Article III, Section 10-63 – Open Burning

The rule states that no person shall allow, cause, or permit open burning of combustible materials (except for disaster rubbish, trees, and tree trimming, as described in sections "d" and "h"). IPL will comply with this regulation.

4.5.5 Chapter 10, Article III, Section 10-65 – Sulfur Compounds

The combustion of number 1 or number 2 fuel oil may not exceed a sulfur content of 0.5 percent by weight. The state standard for processes capable of emitting sulfur dioxide imposes a limit of 1.5 pounds of sulfur dioxide per 1.0 MMBtu of heat input from a liquid fuel-burning unit. IPL will be in compliance with this regulation by limiting the sulfur content of liquid fuels and complying with the applicable sulfur dioxide emission limits.

4.5.6 Chapter 10, Article III, Section 10-66 – Fugitive Dust

No person shall cause, suffer, or allow any material to be handled, processed, transported, or stored; a building or its appurtenances to be constructed, altered, repaired, or demolished; or a road to be used without taking reasonable precautions to prevent particulate matter from becoming airborne and to prevent visible fugitive dust at the property line. This Project is not expected to produce significant fugitive dust during operation and will comply with this regulation.

4.6 Mandatory Reporting of Greenhouse Gases – 40 CFR Part 98

40 CFR Part 98 requires facilities that emit 25,000 metric tons or more per year of greenhouse gases to submit annual reports to EPA. The Project will exceed the reporting threshold. Therefore, IPL will report GHG emissions as required.

4.7 40 CFR Part 68 – Chemical Accident Prevention – Not Applicable

40 CFR Part 68, Accidental Release Prevention Provisions, under Clean Air Act (CAA) Section 112(r), Prevention of Accidental Releases, establishes a general duty for owners and operators of stationary sources who produce, process, handle, or store any of a number of regulated substances, to prevent and mitigate accidental releases of these substances by preparing detailed risk assessments and implementing a number of safety procedures through the preparation of a risk management plan (RMP).

The specific requirements of the RMP for affected facilities are established In 40 CFR Part 68, Accidental Release Prevention Provisions. These regulations require the owner or operator of an affected source to prepare and implement an RMP to detect and prevent or minimize accidental releases of regulated substances, and to provide a prompt emergency response to any such release to protect human health and the environment. Affected facilities are those stationary sources that store, use, or handle any of the 140 listed hazardous chemicals or flammable/explosive substances in amounts greater than the listed threshold quantities. The proposed Project does not plan to have any hazardous substances on-site that would require an RMP, therefore, this regulation does not apply.

4.8 40 CFR Part 72 – Acid Rain Program

Title IV of the Clean Air Act imposes stringent requirements on electrical utilities and is enforced through the administration of the Title IV Acid Rain Permit Program, which is designed to achieve reductions in emissions of SO_2 and NO_x. The centerpiece of the Title IV program is the establishment of an SO_2 emissions allowance and trading program. The Project will be subject to the requirements of 40 CFR 72. IPL will submit the necessary Acid Rain Permit application forms to the Iowa DNR in accordance with this rule.

5.0 Best Available Control Technology Analysis

Federal PSD regulations at 40 CFR 52.21 requires the application of BACT for each regulated NSR pollutant which will result in a significant net emissions increase as a result of the Project. As indicated in Part 1, the Project will result in significant net emission increases of NO_x, CO, PM_{10}, $PM_{2.5}$, VOC, and greenhouse gases (CO_2e). These pollutants will be subject to PSD review, as such, a BACT analysis was performed for each of these regulated NSR pollutants in accordance with 40 CFR Part 52.21.

Although sulfur dioxide (SO_2) and sulfuric acid (H_2SO_4) mist emissions are below their respective PSD significance thresholds, IPL intends to preserve optionality for its customers. Therefore, IPL is including SO_2 and H_2SO_4 mist in the pollutants subject to BACT review.

The Project will consist of three F-class simple-cycle combustion turbines. The combustion turbines will be designed to utilize natural gas. In addition to the combustion turbines, emergency diesel generator, fuel oil belly tanks, haul roads, natural gas piping components, circuit breakers, and comfort building heaters will be included as part of the Project. This Part describes the BACT analysis for all new equipment proposed for the Project

The BACT analysis was performed using the "top-down" approach, which is described in this Part. A summary of the BACT emission limits and the associated control technologies for the simple-cycle combustion turbine are shown in Table 5-1. BACT emission limits and associated control technologies for the auxiliary equipment are listed in Table 5-2.

Table 5-1: Summary of BACT Results: Simple-Cycle Turbines

Pollutant	BACT Control	BACT Emission Rate[a,b,c]	Compliance Method and Averaging Period[a]
NO_x	Dry low-NO_x burners	5 ppm[d]	24-hour operating rolling CEMS
CO	Good combustion practices	4 ppm (100% to 71% load)[d] 9 ppm (less than 71% load)[d]	24-hour operating rolling CEMS
$PM/PM_{10}/PM_{2.5}$	Combustion controls and low ash fuels	11 lb/hr	Average of 3-run stack test
SO_2	Combustion controls and low sulfur fuels	3.3 lb/hr	Utilization of Natural Gas
H_2SO_4 Mist	Combustion controls and low sulfur fuels	0.5 lb/hr	Utilization of Natural Gas
VOC	Good combustion practices	1 ppm[d]	Average of 3-run stack test
Greenhouse gases	Use of natural gas as a fuel, monitoring and control of excess air, and efficient turbine design	120 lb CO_2/MMBtu	Utilization of Natural Gas

(a) ppm = parts per million; lb/hr = pounds per hour; lb/MMBtu = pound per million British thermal unit; CEMS = continuous emission monitoring system
(b) Concentration at 15 % O_2 while operating at MECL and greater under steady state conditions, unless otherwise noted
(c) Limit valid for loads of 100% load down to MECL, unless otherwise noted.
(d) Excludes start-up and shutdown. Annual limits represent BACT for start-up and shutdown for these pollutants.

Table 5-2: Summary of BACT Results: Auxiliary Equipment

Equipment	Pollutant	Control[a]	BACT Emission Rate[a]
Hot Water Boilers (each)	NO_x	LNB/GCP/clean fuels	0.05 lb/MMBtu
	CO	GCP/clean fuels	0.082 lb/MMBtu
	$PM/PM_{10}/PM_{2.5}$	GCP/clean fuels	0.0075 lb/MMBtu
	SO_2	GCP/clean fuels	pipeline quality natural gas
	H_2SO_4 Mist	GCP/clean fuels	pipeline quality natural gas
	VOC	GCP/clean fuels	0.0054 lb/MMBtu
	Greenhouse gases (CO_2e)	GCP/clean fuels	117.1 lb/MMBtu
Emergency Diesel Generator	NO_x	Purchase NSPS-certified generators, GCP/clean fuels	6.4 g/kW-hr
	CO		3.5 g/kW-hr
	$PM/PM_{10}/PM_{2.5}$		0.2 g/kW-hr
	SO_2		ULSD
	H_2SO_4 Mist		ULSD
	VOC		7.05×10^{-4} lb/hp-hr
	Greenhouse gases (CO_2e)		163.6 lb/MMBtu
Emergency Fire Pump	NO_x	Purchase NSPS-certified generators, GCP/clean fuels	4.0 g/kW-hr
	CO		3.5 g/kW-hr
	$PM/PM_{10}/PM_{2.5}$		0.2 g/kW-hr
	SO_2		ULSD
	H_2SO_4 Mist		ULSD
	VOC		2.51×10^{-3} lb/hp-hr
	Greenhouse gases (CO_2e)		163.6 lb/MMBtu
Haul Roads	$PM/PM_{10}/PM_{2.5}$	Paved roads and Fugitive Dust Control Plan	
Natural Gas Piping Components	GHG, VOC	A/V/O Detection System	
Circuit Breakers	GHG	Leak monitoring	<0.5% loss rate
Diesel Belly Tanks	VOC	--	7.68×10^{-4} tpy
Comfort Building Heaters	NO_x	GCP/clean fuels	0.098 lb/MMBtu
	CO	GCP/clean fuels	0.082 lb/MMBtu
	$PM/PM_{10}/PM_{2.5}$	GCP/clean fuels	0.0075 lb/MMBtu
	SO_2	GCP/clean fuels	pipeline quality natural gas
	H_2SO_4 Mist	GCP/clean fuels	pipeline quality natural gas
	VOC	GCP/clean fuels	0.0054 lb/MMBtu
	Greenhouse gases (CO_2e)	GCP/clean fuels	117.1 lb/MMBtu

(a) LNB = low-NOx burners; GCP = good combustion practices; lb/MMBtu = pound per million British thermal units; tpy = tons per year; g/kW-hr = gram per kilowatt-hour

BACT is an emission limitation based on the maximum degree of reduction which the Iowa DNR determines is achievable, on a case-by-case basis, taking into account energy, environmental, and economic impacts and other costs.

The Iowa DNR has directed by policy that the BACT be determined using a “top-down” process. The “top-down” process was outlined in a December 1, 1987, memorandum from the EPA Assistant Administrator for Air and Radiation.

While there is no legal requirement to perform the BACT analysis utilizing a specific criteria or process, the Iowa DNR follows the EPA-developed guidance that establishes a five-step “top-down” BACT process/methodology (EPA, 1990).

For purposes of this PSD application, IPL has prepared this BACT analysis consistent with EPA’s top-down approach, which consists of the following steps:

- Step 1 – Identify all potential control technologies
- Step 2 – Determine technical feasibility (of potential technologies)
- Step 3 – Rank control technologies by control effectiveness
- Step 4 – Evaluate most effective controls and document results
- Step 5 – Select BACT

Each of these steps is discussed in further detail below.

<u>Step 1 – Identify all potential control technologies</u>. The first step in a “top-down” analysis is to identify, for all applicable emission units, all “available” control options. Available control options are defined as those air pollution control technologies or techniques that have a practical potential for application to the emissions unit and the regulated pollutant under evaluation and have been demonstrated in practice. Air pollution control technologies and techniques include the application of production processes or available methods, systems, and techniques, including innovative fuel combustion techniques and add-on controls.

<u>Step 2 – Determine technical feasibility (of potential options)</u>. In the second step, the technical feasibility of the control options identified in Step 1 is evaluated with respect to source-specific factors. A demonstration of technical infeasibility should be documented and should show, based on physical, chemical, and engineering principles, that technical difficulties would preclude the successful use of the control option on the emissions unit under review. Technically infeasible control options are then eliminated from further consideration in the BACT analysis.

<u>Step 3 – Rank control technologies by control effectiveness</u>. All remaining control alternatives not eliminated in Step 2 are ranked and then listed in order of overall control effectiveness for the pollutant under review, with the most effective control alternative at the top. A list should be prepared for each pollutant and for each emissions unit (or grouping of similar units) subject to a BACT analysis.

<u>Step 4 – Evaluate most effective controls and document results</u>. After the identification of available and technically feasible control technology options, the energy, environmental, and economic impacts are taken into account, in this Step. For each control option an objective evaluation of each impact is presented. Both beneficial and adverse impacts should be discussed and, where possible, quantified. If IPL accepts the top alternative in the listing as BACT, IPL will proceed to consider whether impacts of unregulated air pollutants or impacts in other media would justify selection of an alternative control option. If there are no outstanding

issues regarding collateral environmental impacts, the analysis ends, and the results proposed as BACT. If the top candidate is shown to be inappropriate, due to energy, environmental, or economic impacts, the rationale for this finding is documented and the next level of control is analyzed.

<u>Step 5 – Select BACT</u>. The final BACT determination is presented in this Step.

<u>Greenhouse Gas BACT Process</u>

Based on EPA Greenhouse Gas Guidance (EPA, 2011), the Greenhouse BACT process is similar to the five Steps summarized above. Steps 1 and 2 identify potential control strategies and then eliminate technologically infeasible options. Step 3 ranks the remaining technically feasible control technologies. Step 4 evaluates the most effective control technologies from an environmental, energy, and economic perspective. And finally, Step 5 selects the most appropriate BACT.

The BACT analysis for the Project is also based on the following concepts:

- Emission limits are defined on a “case-by-case” analysis that considers site specific factors
- Emission limits must be “achievable” on a long-term, day in and day out, basis
- The technology must be available and feasible for a specific project
- BACT does not redefine the facility as proposed (including fuels)

There is no prescriptive approach to performing a case-by-case control technology and emission limit analysis. PSD permitting authorities determine emission limits on a case-by-case basis. These case-by-case determinations must consider source-specific and site-specific characteristics. This is not a “cookie-cutter” approach and there is no single right answer to determining the appropriate emission limits for a specific source or for a specific pollutant.

The Iowa DNR is not required to set any emission limit at the most stringent level that has been demonstrated by a facility using similar emissions control technology. Similarly, an emission limit does not need to be set at the most stringent emission limit found in another permit. Rather, the Iowa DNR has the authority and is required to evaluate and determine the correct emissions limits and control technologies for a project based on project-specific factors, including location. The case-by-case process does not require that each subsequent determination identify emission limitations that are equal to or more stringent than the previous determination.

Further, in establishing the emission limits, the BACT must confirm that emission limits are achievable by the specific facility that is subject to the emission limits: (1) over the life of the facility; and (2) during all operating conditions, not just ideal conditions. The use of a safety factor or margin is well-established in the air permitting context to appropriately account for the uncertainty and operational variability that will occur over the life of a facility. This safety factor must be sufficient to allow permit holders to comply on a continuous basis. Emission limits should not be based on the lowest emissions rate or highest control efficiency ever documented by a similar facility for a short-term period. The emission limits must account for a full range of operating conditions and the inherent variability of complex fuel combustion and air pollution control systems.

To be considered in the permitting process, a control technology must be commercially available (i.e., it must be offered for sale on a commercial scale through commercial channels). Permittees are not required to explore research and development projects to determine whether a specific technology is suitable. In addition, to be considered feasible technology for purposes of inclusion in an analysis, a particular technology must have been previously demonstrated, on a long-term basis, at commercial scale. In fact, even 2-3 years of operating history on a commercial scale has been determined to be insufficient to demonstrate that a particular technology is feasible.

The air permit process cannot redefine the source. IPL has defined the "proposed facility," including the goals, objectives, purpose and basic design. Requiring alteration as to the type of power generating unit and/or range of fuels to be used would redefine the source.

Fuels can be an inherent part of a project design. In such cases, the air permitting process cannot be used to require a fuel other than the fuels proposed by IPL. As Congress explained, "the Administrator may consider the use of clean fuels to meet BACT requirements <u>if a permit applicant proposes</u> to meet such requirements by using clean fuel. <u>In no case is the Administrator compelled to require the mandatory use of clean fuels by a permit applicant</u>." (emphasis added). S. Rep. No. 101-228 at 338 (1989).

The first step in the "top-down" BACT process is the identification of potentially available control technologies. One of the ways to identify available control technologies is to review previous BACT determinations for similar sources. EPA's RACT/BACT/LAER Clearinghouse (RBLC) database was reviewed to identify recent BACT determinations for similar projects. This database is maintained on EPA's Technology Transfer Network website at www.epa.gov/catc. Advanced queries of the database were conducted to identify control technology determinations for sources similar to the proposed simple-cycle combustion turbine and applicable auxiliary equipment. The queries are summarized in Table 5-3, below. The results of the RBLC query can be found in Appendix C.

Table 5-3: RBLC Query Information

Equipment	Process Type Lookup Code	Look-up Dates
Simple-Cycle Combustion Turbines	15.110 – Simple-Cycle Turbines > 25 MW, Natural Gas	January 2015 to February 2026
Hot water boilers	13.310 – Boilers < 100 MMBtu/hr, Natural Gas	January 2015 to February 2026
Emergency Diesel Generator	17.110 – Internal Combustion Engines >500 hp, Fuel Oil	January 2015 to February 2026
Emergency Diesel Fire Pump	17.210 – Internal Combustion Engines <500 hp, Fuel Oil	January 2015 to February 2026
Haul Roads	99.140 – Paved Roads	January 2015 to February 2026
Natural Gas Piping Components	99.999 – Other Miscellaneous Sources	January 2015 to February 2026
Circuit Breakers	99.999 – Other Miscellaneous Sources	January 2015 to February 2026
Diesel Belly Tanks	42.005 – Petroleum Liquid Stored in a Fixed Roof Tank	January 2015 to February 2026
Comfort Building Heaters	13.310 – Boilers < 100 MMBtu/hr, Natural Gas	January 2015 to February 2026

To identify previous control technology determinations for comparable sources, queries were run using the "standard search" in which the RBLC database was searched using the following parameters:

- Draft Determinations and RBLC Permits issued during or after the dates presented in Table 5-3
- Process type lookup codes presented in Table 5-3

After the queries were run, combustion turbines that were not similar (e.g., digester gas-fired, fuel oil-fired, cogeneration units, combined-cycle turbines, boilers, etc.) were eliminated from the search. Information on turbine emissions was sorted from the remaining combustion turbine listing. A discussion of control options identified in the RBLC database is included in each subsection.

In some cases, the RBLC listings are not clearly categorized and cover both simple- and combined-cycle installations. Also, it should be noted that all RBLC listings in California represent Lowest Achievable Emission Rate (LAER); although they are often listed as BACT, BACT and LAER are essentially the same in California. LAER is a much more stringent requirement than BACT and involves application of control technology regardless of cost. This is not the case for the proposed Project, which is subject only to BACT.

5.1 BACT for Nitrogen Oxides – Simple-Cycle Turbines

The following sections outline the top-down steps for NO_x emissions from the combustion turbines.

5.1.1 Step 1. Identify All Potential Control Strategies

NO_x is primarily formed in combustion processes in two ways:

1. The combination of elemental nitrogen with oxygen in the combustion air within the high temperature environment of the combustor (thermal NO_x)
2. The reactions of nitrogen with hydrocarbon radicals from the fuel (prompt NO_x)
3. The oxidation of nitrogen contained in the fuel (fuel NO_x)

Natural gas contains negligible amounts of fuel-bound nitrogen. Therefore, it is assumed that essentially all NO_x emissions from the combustion turbine will originate as thermal NO_x. The rate of formation of thermal NO_x is a function of residence time and free oxygen and is exponential with peak flame temperature.

The new combustion turbines will be subject to NO_x limits per NSPS Subpart KKKKa and thus the BACT determination and resulting emission limits must be at least as stringent as the NSPS. Part 4 identifies the applicable Subpart KKKKa limits for the combustion turbines.

Control of NO_x emissions from combustion turbines is generally aimed at either the prevention of NO_x formation or the capture and oxidation of post-combustion NO_x. Since the rate of formation of thermal NO_x is a function of residence time and free oxygen, and is exponential with peak flame temperature, “front-end” control techniques are aimed at controlling one or more of these variables. These controls include the XONON™ catalytic combustors and low-NOx burners (LNB). The XONON™ system uses a catalyst to keep the system temperatures lower while LNB offer a staged combustion process, resulting in a lower peak flame temperature. Water injection reduces the combustion temperature, thereby reducing the formation of NO_x.

Other control methods utilize add-on control equipment to remove NO_x from the exhaust gas stream after its formation. The most common control technique involves the injection of ammonia into the gas stream to reduce the NO_x to molecular nitrogen and water with the use of a catalyst. Ammonia or urea can either be injected into the system without the use of a catalyst (selective non-catalytic reduction [SNCR]) or with the use of a catalyst (SCR). Finally, EM_x™ (formerly $SCONO_x$™), a multi-pollutant control technology, relies upon a catalyst similar to SCR to reduce NO_x emissions but does so without injecting ammonia into the exhaust gas stream.

The output from the RBLC search provided in Appendix C shows that a variety of emission limits and control technologies have been applied to combustion turbines for natural gas combustion. The most stringent limits found during a review of EPA's database were for facilities located in ozone non-attainment areas. These facilities were required to meet such low emission limits since they were subject to LAER requirements.

Typical BACT determinations for simple-cycle units that are located in attainment areas were in the 2 to 96 ppm range using LNB, SCR, or a combination of these technologies. The lower emission rates listed utilize SCR.

5.1.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling NO_x emissions are evaluated for technical feasibility in the following sections.

5.1.2.1 XONON™ System

The XONON™ system controls NO_x emissions by preventing their formation. The key to the XONON™ system is the utilization of a chemical process versus a flame to combust fuel, thus limiting temperature and NO_x formation. The XONON™ system is an integral part of the combustor. The fuel and air that are supplied to the combustor are thoroughly mixed before entering the catalyst. The catalyst is responsible for combusting the fuel to release its energy. Due to the low catalyst operating temperatures, the nitrogen molecules are not involved in the reaction chemistry; they pass through the catalyst unchanged, thereby eliminating NO_x formation. The XONON™ system does have the same high outlet temperature, and some NO_x is formed in the post-combustion process.

Currently, the XONON™ system has not had wide-scale application. It has been demonstrated on a 1.5-MW unit in California, with the unit operating in a base load capacity (24 hours a day, 7 days a week). Tests are underway to apply this technology to other types and sizes of turbines; however, testing data is currently unavailable. As the combustion turbine is expected to experience repeated start-ups and shutdowns, it is unclear how the changing load conditions would affect the XONON™ system.

In September 2006, the XONON™ Combustion System (referred to as the XONON™ Cool Combustion® technology) was sold to Kawasaki Heavy Industries, Ltd. (KHI). Kawasaki is currently making the technology available to the gas turbine generators in the 1- to 1.4-MW range (KHI, 2008). The technology is not commercially available for larger F class turbines. As this is a large simple-cycle peaking project, and the XONON™ system has yet to demonstrate applicability for such units, **the XONON™ system has been deemed technically infeasible for this Project.**

5.1.2.2 EM_x™ System (formerly $SCONO_x$™)

The EM_x™ system (formerly $SCONO_x$™) uses a single catalyst to remove NO_x emissions from combustion exhaust gas by oxidizing nitric oxide to nitrogen dioxide (NO_2) and then absorbing the NO_2 onto a catalytic surface using a potassium carbonate absorber coating. The potassium carbonate coating reacts with NO_2 to form potassium nitrites and nitrates, which are deposited onto the catalyst surface. The optimal temperature window for operation of the EM_x™ catalyst ranges from 300°F to 700°F.

The exhaust gases of the Project's simple-cycle turbine will be around 1,000°F, which is greater than the optimum operating temperature range of the catalyst.

Commercial experience with the SCONOx™ control system is extremely limited. The NO_x reduction system was commercially demonstrated first at the 32-MW (GE LM2500 turbine) Sunlaw Federal Cogeneration Facility located in Vernon, California in 1998. NOx emissions from the process were <2 ppm during 100-percent operation, and <1 ppm for 90-percent operation. Other installations of the technology include a 15-MW (Solar Titan 130 turbines) installation at the University of California, San Diego, and a 45-MW (Alstom GTX100 turbine) installation at the City of Redding Municipal Electric Plant. A number of smaller installations are also operating: two 5-MW installations at the Wyeth BioPharma cogeneration facility in Andover, Massachusetts, and at the Montefiore Medical Center in New York, New York. Actual NO_x emissions from these smaller installations are typically below 1.5 ppm, with substantial periods below 1.0 ppm. Alstom Power, one of the EM_X™ licensees, engineered and installed the technology on one of their GTX100 (43-MW class) gas turbines.

The application of this technology has not advanced and there are no examples of application on larger F class turbines. **As a result, this technology is considered to be technically infeasible for the proposed Project**

5.1.2.3 Selective Non-Catalytic Reduction (SNCR)

SNCR is a post-combustion NO_x control technology in which a reagent (ammonia or urea) is injected into the exhaust gases to react chemically with NO_x, forming nitrogen and water. SNCR is "selective" because the reagent reacts primarily with NOx.

The success of this process in reducing NO_x emissions is highly dependent on the ability to uniformly mix the reagent into the flue gas at a zone in the exhaust stream at which the flue gas temperature is within a narrow range, typically from 1,700°F to 2,000°F. To achieve the necessary mixing and reaction, the residence time of the flue gas within this temperature window should be at least 0.5 to 1.0 seconds. The consequences of operating outside the optimum temperature range are severe. Outside the upper end of the temperature range, the reagent will be converted to NO_x. Below the lower end of the temperature range, the reagent will not react with the NO_x and the ammonia slip concentrations (ammonia discharge from the stack) will be very high. The exhaust gases of the Project's simple-cycle turbine will be around 1,000°F, outside the normal operating temperature range of a SNCR.

SNCR is a proven and reliable technology. SNCR was first applied commercially in 1974 and has been installed on numerous applications worldwide. Applications include utility boilers and a broad range of industrial applications including wood-fired boilers, coal-fired boilers, co-generation boilers, pulp and paper boilers, steel industry furnaces, refinery process units, process heaters, cement kilns, municipal waste combustors, glass-melting furnaces, hazardous waste incinerators, and other combustion-type sources. Urea-based SNCR has been applied commercially to sources ranging in size from a 60-MMBtu/hr (gross heat input) paper mill sludge incinerator to a 640-MW pulverized coal-fueled, wall-fired electric utility boiler [Institute of Clean Air Companies (ICAC), 2006]. SNCR has not been applied to any combustion turbines according to the RBLC database. Because of the comparatively low exhaust temperatures, fuel and energy requirements, environmental implications and economic considerations; **SNCR is considered to be technically infeasible for the combustion turbines under consideration for this Project.**

5.1.2.4 Selective Catalytic Reduction (SCR)

SCR is a post-combustion technology that employs ammonia in the presence of a catalyst to convert NO_x to nitrogen and water. The function of the catalyst is to lower the activation energy of the NO_x decomposition reaction. Technical factors related to this technology include the catalyst reactor design, optimum operating temperature, sulfur content of the fuel, de-activation due to aging, ammonia slip emissions, and the design of the ammonia injection system.

SCR represents state-of-the-art post-combustion NO_X control technology for simple-cycle combustion turbines. SCR technology has been permitted as LAER and BACT for simple-cycle turbines at 2 to 5 ppm NO_x. Conventional SCR uses a metal honeycomb or “foil” catalyst support structure and requires a heat recovery steam generator to drop flue gas temperatures to less than 600°F.

Because of the high exhaust temperature of a simple-cycle turbine, a conventional SCR system is not technically feasible. Instead, a high temperature “zeolite”-based SCR system has been introduced for use on certain simple-cycle turbines. Zeolite is a sodium alumina silicate ceramic material with a design operating temperature of approximately 800 to 1,000°F. Only a few natural gas-fired installations were identified that use these high-temperature systems. Two have had major problems such as catastrophic catalyst failures; the third has not yet acquired a long enough history to sufficiently evaluate its operational effectiveness. Although vendors reported that they have catalysts that they believe will operate under high temperature conditions, they did not identify many combustion turbines successfully operating with SCR under simple-cycle conditions.

Another option to utilizing a high temperature SCR is to dilute the exhaust air with ambient air to reduce the temperature prior to the conventional SCR catalyst to reduce the temperature to allow for a medium or traditional catalyst. This requires a lot of extra duct work to allow time for the exhaust to be lowered to the appropriate temperature for a vanadium catalyst.

For simple-cycle turbines, each turbine exhausts directly to its own stack, making a single, combined SCR reactor technically impractical. A shared system would require extensive ducting and mixing to equalize flow and temperature and would complicate ammonia injection control when turbines operate at different loads. It would also introduce a single point of failure, where catalyst maintenance or malfunction could impact all units simultaneously. Therefore, single SCR technology per turbine is feasible but one large combined SCR is not technically feasible.

SCR is considered to be technically feasible for simple-cycle combustion turbines.

5.1.2.5 Low-NOx Burners

LNB are currently available from most turbine manufacturers. This technology reduces combustion temperatures, thereby reducing NO_x formation. In a conventional combustor, the air and fuel are introduced at an approximately stoichiometric ratio and air/fuel mixing occurs at the flame-front where diffusion of fuel and air reaches the combustible limit. A lean premixed combustor design premixes the fuel and air prior to combustion. Premixing results in a homogenous air/fuel mixture, which minimizes localized fuel-rich pockets that produce elevated combustion temperatures and higher NO_x emissions. A lean air-to-fuel ratio approaching the lean flammability limit is maintained, and the excess air serves as a heat sink to lower combustion temperatures, which lowers NO_x formation. A pilot flame is used to maintain combustion stability in this fuel-lean environment.

LNB represent a general class of designs that reduce NO_x compared to conventional diffusion combustors. Dry low NO_x (DLN) burners, referred to by Siemens as "Ultra DLN", are a specific advanced subset of this technology that achieves single-digit NO_x. For large-frame turbines such as the Siemens 5000F, DLN combustors represent the commercially available baseline. Therefore, DLN is identified as the technically feasible combustion technology baseline, and conventional LNB are not considered separately.

Other commercially available burner technologies, such as conventional DLN combustors, were not selected as the NO_x baseline because the Siemens 5000F simple-cycle combustion turbine is equipped with "Ultra DLN" burners (Siemens terminology) that achieve a guaranteed emission rate of 5 ppm_v NO_x at 15% O_2. This rate is achieved by DLN alone, without SCR, and represents the lowest LNB emission rate for large-frame F-class turbines.

Since the installed combustors already reflect the best available combustion design for minimizing NO_x without post-combustion controls, they establish the appropriate baseline for this Project. DLN burners come standard on the combustion turbine.

LNB (DLN burners) are available, demonstrated, and technically feasible for units in either simple-cycle or combined-cycle configurations. The LNB combustion technology alone can achieve NO_x emission rates as low as 5 to 9 ppm (corrected to 15% O_2 dry conditions) when firing natural gas in some combustion turbine models. **Low-NO_x burners are currently available for these turbines and are a technically feasible control option for this Project for natural gas combustion.**

5.1.2.6 Water Injection

Water or steam injection is utilized to control emissions of NO_x in turbines while combusting fuel oil. LNB on turbines is typically utilized on dual fuel turbines when combusting natural gas and water injection is utilized to control emissions while combusting fuel oil. **Because the turbines will not combust fuel oil, water injection is not considered feasible.**

5.1.2.7 Flue Gas Recirculation

With flue gas recirculation, that is seen in some boiler technologies, a portion of the flue gas is recirculated back into the combustion zone to lower flame temperature and reduce thermal NO_x formation. Flue gas recirculation is widely used in industrial boilers but is impractical for combustion turbines. Turbine designs require precise airflow and pressure ratios, and introducing recirculated exhaust gas can disrupt compressor performance and turbine efficiency.

Dry low-NO_x systems already achieve low flame temperatures through lean premixing.

Flue gas recirculation is considered to be technically infeasible for the combustion turbines under consideration for this Project.

5.1.2.8 Overfire Air

Overfire air is a staged combustion approach where a portion of the combustion air is introduced above the primary flame zone to complete combustion at lower temperatures. Overfire air is effective in large coal-fired boilers but is not technically compatible with turbine combustors, which rely on tightly integrated combustion chamber designs.

Dry low-NO_x systems already employ staged combustion principles internally. External overfire air injection is not feasible due to turbine geometry and would duplicate existing design features.

Overfire air is considered technically infeasible for the combustion turbines under consideration for this Project.

5.1.2.9 Reburning

Reburning is when a secondary fuel (often natural gas or biomass) is injected downstream of the primary flame to create a reducing environment that converts NO_x to molecular nitrogen in boilers. Reburning has been applied in utility boilers but is not practical for combustion turbines, where fuel injection and flame stability are tightly controlled.

Dry low-NO_x combustors already minimize NO_x formation at the source. Reburning would add complexity without significant incremental reduction, and could negatively affect turbine performance.

Reburning is considered to be technically infeasible for the combustion turbines under consideration for this Project.

5.1.2.10 Summary of the Technically Feasible Control Options

Technically feasible NO_x control options for the simple-cycle combustion turbines are summarized in Table 5-4. The expected performance has been determined considering the performance of existing systems, vendor guarantees, permitted emission limits, and the design requirements for the combustion turbine.

Table 5-4: Summary of Technical Feasibility of NO_x Control Technologies for Simple-Cycle Combustion Turbines

Control System		Technical Feasibility	Comments
Combustion controls	Low-NO_x burners	Feasible	Standard on combustion turbines for natural gas operation.
	Water injection	Not feasible	Water Injection is typically only utilized for fuel oil combustion.
	Flue gas recirculation	Not feasible	Technically impractical
	Overfire air	Not feasible	Technically impractical
	Reburning	Not feasible	Technically impractical
Post combustion controls	XONON™	Not feasible	Not a demonstrated technology for large combustion turbines.
	EMx™	Not feasible	Not a demonstrated technology for large combustion turbines.
	Selective non-catalytic reduction	Not feasible	Exhaust temperature is too low.
	Selective catalytic reduction	Feasible	SCR is considered technically feasible for simple-cycle turbines.

5.1.3 Step 3. Rank the Technically Feasible Control Technologies

Add-on controls may be used for natural gas combustion in the turbine. The combustion turbines under consideration come with low-NO_x burners as part of their standard package; therefore, low-NO_x burners are used as the baseline for the proposed combustion turbine.

The technically feasible NO_x control technologies for the combustion turbines are ranked by control effectiveness in Table 5-5, with low-NO_x burners serving as the baseline against which add-on controls are evaluated.

Table 5-5: Ranking of Technically Feasible NOx Control Technologies for Simple-Cycle Combustion Turbines

Control System	Reduction (%)	Controlled Emission Level (ppm)[b]
Selective Catalytic Reduction	60 to 90[a]	2
Low-NO_x Burners	N/A (Baseline)	5

(a) SCR efficiency varies with load and exhaust temperature: 60 to 90% reflects realistic simple-cycle performance.

(b) 2 ppm at 15% O_2 is considered the floor for SCR control on any combustion turbine.

5.1.4 Step 4. Evaluate the Most Effective Controls

Recent BACT determinations have indicated a level of 2 to 25 ppm for NO_x emissions from simple-cycle units that are fired with natural gas (Appendix C). The combustion turbines will have LNB advanced technology that allows emissions to be controlled to 5 ppm NO_x.

The energy, environmental, and economic considerations for the feasible control options are discussed below.

5.1.4.1 Selective Catalytic Reduction

Energy Impacts

An SCR system results in a loss of energy due to the pressure drop across the SCR catalyst. To compensate for the energy loss in the SCR system, additional natural gas combustion is required to maintain the net energy output, which also results in additional air pollutant emissions.

Environmental Impacts

SCR systems consist of an ammonia injection system and a catalytic reactor. Urea can be decomposed in an external reactor to form ammonia for use in a SCR. Unreacted ammonia may escape through to the exhaust gas. This is commonly called “ammonia slip.” It is estimated that ammonia slip from an SCR on a unit this size could be 10 ppm and may be considered to be an environmental impact. The ammonia that is released may also react with other pollutants in the exhaust stream to create fine particulates in the form of ammonium salts. In addition, the storing of the ammonia on-site is another environmental and safety concern. SCR catalysts must also be replaced on a routine basis. In some cases, these catalysts may be classified as hazardous waste. This typically requires either returning the material to the manufacturer for recycling and reuse or disposal in designated landfills.

Economic Impacts

The costs associated with an SCR system for the new 5000F combustion turbines operating in simple-cycle mode are shown in Appendix D. It is anticipated that the catalyst will need to be replaced once every five years; this replacement cost is annualized and included in the annual cost of the SCR system, not in the capital equipment cost. Because the turbines will operate in simple-cycle mode with intermittent dispatch, the reduced annual operating hours may show catalyst aging and extend effective life, which supports the five-year replacement interval assumption. The overall total capital investment of installing an SCR system is approximately $28,116,000. On an annual basis, the SCR system would cost approximately $3,998,350, which results in a cost per ton of NO_x removed of $89,389 while removing only 44.7 tons of NO_x per year, when operating at steady state. Therefore, any control of NO_x by add-on controls would result in costs that are not considered economically feasible.

5.1.4.2 Low-NO_x Burners

Energy Impacts

Low-NO_x burners are usually accompanied by an efficiency penalty (typically 2 to 3 percent) and an increase in power output (typically 5 to 6 percent). The increase in power output results from the increase in mass flow required to maintain turbine inlet temperature at manufacturer's specifications. Because there is a power increase, no energy impacts are associated with low-NO_x burners.

Environmental Impacts

The low-NO_x burner system may increase CO and VOC emissions on a lb/hr basis; however, the potential increase in CO and VOC emissions does not outweigh the advantages of decreased NO_x emissions to reduce health effects.

Economic Impacts

The turbine manufacturer currently installs low-NO_x burners as standard equipment on natural gas-fired combustion turbines. Since the low-NO_x burners are considered standard equipment on the turbine, there is no annualized cost of the control.

5.1.5 Step 5. Proposed NO_x BACT Determination

The BACT recommended for control of NO_x emissions from the simple-cycle combustion turbines is low-NOx burners. These controls will meet NO_x emission limits, excluding start-up/shutdown, of 5 ppm at 15% O_2 during steady state conditions for 100 percent load down to MECL.

Compliance will be determined via CEMS on a 24-hour average basis.

Simple-cycle BACT rates shown in the RBLC for advanced class combustion turbines range from 5 ppm (for F-class turbines without add-on controls) to 25 ppm. Table 5-6 shows RBLC entries for combustion turbines that have a lower emission rate than this Project.

Table 5-6: RBLC Summary

Facility	Controls	Emission Limit	Units	Type
Shell Chem Appalachia/Petrochemicals		2	ppm @ 15% O_2	LAER
Liquefaction Plant	SCR, DLN combustors, and good combustion practices	2	ppm @ 15% O_2	BACT
Cricket Valley Energy Center	dry low NOx burners in combination with selective catalytic reduction	2	ppmvd @ 15% O_2	LAER
Puente Power		2.5	ppmvd	other case-by-case (LAER – CA)
Commonwealth LNG Facility	Good combustion practices and use clean fuel. Dry low-NOx and selective catalytic reduction.	2.5	ppmvd	BACT
Lone Star Power Station	Dry-low NOx (DLN) combustors and an aqueous ammonia-based SCR system	2.5	ppmvd	LAER
Bayonnne Energy Center	Selective Catalytic Reduction, water injection, use of natural gas a low NOx emitting fuel	2.5	ppmvd@15% O_2	LAER
Lake Charles LNG Export Terminal	LNB + SCR	3.1	ppmvd @15% O_2	BACT

Further review of the sources listed in Table 5-6, five of the sources were subject to LAER. Sources subject to LAER would not be representative of this Project because our sources are subject to BACT. The Liquefaction Plant facility included much smaller turbines than the Project (1,113 MMBtu/hr vs. 2,270 MMBtu/hr) and the turbines are not electrical generating units. Puente Power was not F-class combustion turbines (not the same make/model) and was also never constructed and thus has not been demonstrated in practice. Commonwealth LNG Facility and Lake Charles LNG Export Terminal are both gas processing facilities and are therefore not applicable to this Project are they are a different type of facility.

BACT was selected to be 5 ppm at 15% O_2 which is the lowest emission rate selected as BACT for simple-cycle turbines. This determination is consistent with recent permitting actions, including Omaha Public Power District (OPPD) Cass County Station, which installed the same make and model of turbine (Siemens 5000F) and was permitted with a 5 ppm at 15% O_2 NO_x BACT limit using low-NO_x burners. To meet this BACT rate, the Project's F-class turbines will also utilize low-NO_x burners.

5.2 BACT for Carbon Monoxide – Simple-Cycle Combustion Turbines

The following sections outline the top-down BACT determination steps for CO emissions from the proposed combustion turbines.

5.2.1 Step 1. Identify Potential Control Strategies

CO is a product resulting from incomplete combustion. Incomplete combustion occurs when there is not enough oxygen to allow the fuel to react completely to produce CO_2 and water. CO emissions from combustion turbines are a function of oxygen availability (excess air), flame temperature, residence time at flame temperature, combustion zone design, and turbulence. Control of CO is typically accomplished by providing adequate fuel residence time and a high temperature in the combustion zone to complete combustion. These control factors, however, also tend to result in increased emissions of NO_x. Conversely, a lower NO_x emission rate achieved through flame temperature control (by water injection or dry lean pre-mix) can result in higher levels of CO emissions. A compromise is usually established where the flame temperature reduction is set to achieve the lowest NO_x emission rate possible while keeping CO emissions to an acceptable level.

Post-combustion control involves the use of catalytic oxidation while front-end control involves controlling the combustion process to suppress CO formation.

A survey of the RBLC database (Appendix C) indicated that almost all CO BACT determinations are based on good combustion practices. The control technologies identified in Section 5.2.2 for reducing CO emissions are based on recent BACT determinations for F-class turbines and also technologies that have historically been applied to combustion turbines.

5.2.2 Step 2. Identify Technically Feasible Control Technologies

The technical feasibility for applying several CO control technologies to the proposed combustion turbines are discussed in this section.

5.2.2.1 EM_x™ System

The EM_x™ system was described in the BACT analysis for NO_x. The EM_x™ system simultaneously oxidizes CO to CO_2, NO to NO_2, and then absorbs NO_2 onto the surface of a catalyst using a potassium carbonate absorber coating. VOCs are also removed by the catalyst system. The system does not use ammonia and operates most effectively at temperatures ranging from 300°F to 700°F. Operation of EM_x™ requires natural gas, water, steam, electricity, and ambient air. Steam and reformed natural gas are used periodically to regenerate the catalyst bed and are an integral part of the process.

Regeneration of the catalyst is accomplished by passing a dilute hydrogen reducing gas across the surface of the catalyst in the absence of oxygen. Hydrogen in the gas reacts with nitrites and nitrates to form water and nitrogen. CO_2 in the gas reacts with potassium nitrite and nitrates to form potassium carbonate, which is the absorbing surface coating on the catalyst. Regeneration gas is produced by reacting natural gas with a carrier gas (such as steam) over a steam-reforming catalyst.

The demonstrated application for EM_x™ is currently limited to combined-cycle combustion turbines under approximately 50 MW in size. The EM_x™ system has not been demonstrated on any type of combustion source other than a combustion turbine. The application of this technology has not advanced and there are no examples of application on larger F-class turbines. **As a result, this technology is considered to be technically infeasible for the proposed Project**

Although the EMx™ system is capable of oxidizing CO to CO_2, it has not been demonstrated on large simple-cycle turbines and is therefore considered technically infeasible for this project. The system's effective operating temperature range (300 to 700°F) is far below the exhaust temperature of simple-cycle turbines, which typically exceeds 900 to 1,100°F. Cooling the exhaust to the EMx™ operating window would require large heat exchangers or quench systems, resulting in excessive backpressure, efficiency losses, condensation/corrosion risks, and significant capital and operating costs. In addition, EMx™ regeneration requires steam-reformed natural gas and hydrogen handling infrastructure, which is incompatible with the fast-start, cycling operation of simple-cycle turbines. For these reasons, EMx™ is not technically or economically feasible for CO control on the proposed units, and conventional oxidation catalysts remain the demonstrated and commercially viable technology for CO reduction in simple-cycle turbine applications.

5.2.2.2 Oxidation Catalyst

Oxidation catalysts are a post-combustion technology which does not rely on the introduction of additional chemicals, such as ammonia with SCR, for a reaction to occur. The oxidation of CO to CO_2 utilizes excess air present in the turbine exhaust; the activation energy required for the reaction to proceed is lowered in the presence of a catalyst. Products of combustion are introduced into a catalytic bed, with the optimum temperature range for these systems being between 700°F and 1,100°F. At higher temperatures, catalyst sintering may occur, potentially causing permanent damage to the catalyst. The addition of a catalyst bed onto the turbine exhaust will create a pressure drop, resulting in back pressure to the turbine. This has the effect of reducing the efficiency of the turbine and the power generating capabilities.

A combined oxidation catalyst system is not feasible for simple-cycle turbines due to the need for precise temperature control and uniform exhaust distribution across the catalyst bed. Mixing exhaust streams from multiple turbines operating independently would create maldistribution, reduce CO/VOC removal efficiency, and complicate compliance monitoring. In addition, a shared catalyst would increase outage risk by tying all units to a single module. The recommended configuration is to equip each turbine with its own oxidation catalyst, with opportunities to share non-critical support systems such as access platforms, monitoring equipment, and spare catalyst elements. A combined oxidation catalyst system is considered to be technically infeasible for the combustion turbines under consideration for this Project.

The use of an oxidation catalyst is considered to be a technically feasible method of controlling CO emissions from the proposed simple-cycle combustion turbines.

5.2.2.3 Combustion Control

CO emissions are generated from the incomplete combustion of carbon in the fuel. Optimization of the combustion chamber designs and operation practices that improve the oxidation process and minimize incomplete combustion is the primary mechanism available for lowering CO emissions. This process is often referred to as combustion controls. The combustion system design in modern combustion turbines provides all of the factors required to facilitate complete combustion, including continuous mixing of air and fuel in the proper proportions, extended residence time, and consistent temperatures in the combustion chamber. As a result, CO emissions from combustion turbines are inherently low. Good combustion practices are typically employed to ensure that the combustion turbines are operated as designed.

Good combustion practices are considered to be a technically feasible method of controlling CO emissions from the proposed simple-cycle combustion turbines.

5.2.2.4 Summary of the Technically Feasible Control Options

The technical feasibility of the CO control options for the proposed simple-cycle combustion turbines is summarized in Table 5-7. Except for the EM_X^{TM} technology, all others discussed are considered to be technically feasible.

Table 5-7: Summary of Technically Feasible CO Control Technologies for Simple-Cycle Combustion Turbines

Control System		Technical Feasibility	Comments
Combustion controls		Feasible	Demonstrated technology, modern combustion turbines are equipped with it.
Post combustion controls	Oxidation catalyst	Feasible	Demonstrated technology for F-class combustion turbines.
	EM_x^{TM}	Not feasible	Not a demonstrated technology for F-class combustion turbines.

5.2.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible CO control technologies for the combustion turbine are ranked by control effectiveness in Table 5-8.

Table 5-8: Ranking of Technically Feasible CO Control Technologies for Simple-Cycle Combustion Turbines

Control System	Reduction (%)	Controlled Emission Level (ppm)[a]
Oxidation catalyst	40 to 90[b]	2
Combustion control	Not applicable (Baseline)	4[c]

(a) ppm = parts per million; 2 ppm is the floor for control for the F-class turbine.

(b) The range reflects both ideal design performance and realistic operating variability.

(c) Valid for 100 percent load down to 75 percent load.

5.2.4 Step 4. Evaluation of Economic, Environmental, and Energy Impacts of Feasible Technologies

The next step is to review each of the technically feasible control options for environmental, energy, and economic impacts.

5.2.4.1 Combustion Control

No energy, environmental, or economic impacts are associated with combustion controls.

5.2.4.2 Oxidation Catalyst

Energy Impacts

With an oxidation catalyst, the output of the turbines would be reduced by the additional backpressure. To replace this lost energy, additional power generation will be required, which will require consumption of additional natural gas.

Environmental Impacts

The installation of an oxidation catalyst would theoretically reduce potential CO emissions by 50 to 80% (down to 2 ppm) for each combustion turbine. The oxidation catalyst oxidizes CO to CO_2 which is released to the atmosphere. CO_2 is a greenhouse gas and is a regulated pollutant. Increasing CO_2 emissions could have a negative impact on the atmosphere. However, the oxidation catalyst will also reduce the amount of methane (CH_4) (also a greenhouse gas). Considering both greenhouse gases, the net effect is an overall decrease in global warming potential since the global warming potential of CH_4 is 28 times larger than CO_2.

As with all controls that utilize catalysts for removal of pollutants, the catalyst must be disposed of after it is spent. The catalyst may be considered hazardous waste and require special treatment or disposal; even if it is not hazardous, it adds to landfills.

Economic Impacts

The capital and annual costs associated with an oxidation catalyst for the combustion turbine operating in simple-cycle mode are shown in Appendix D. The total capital investment of installing an oxidation catalyst on the simple-cycle turbine is approximately $9,876,000. On an annual basis, the oxidation catalyst would cost $1,521,700 which results in a cost per ton of CO and VOC removed of $52,770 while removing only 28.3 tons of CO emissions per year for natural gas operation based on worst-case steady state operations. Therefore, any control of CO by add-on controls would result in costs that would not be economical.

5.2.5 Step 5. Proposed CO BACT Determination

The use of good combustion practices is BACT for the turbines. The CO BACT emissions limit is 4 ppm at 15 % O_2 during steady state conditions for loads from 100 percent down to 71 percent and 9 ppm at 15 % O_2 for loads less than 71 percent. Compliance will be determined via CEMS.

Simple-cycle BACT rates shown in the RBLC for advanced class combustion turbines range from 9 ppm at 15 % O_2 (for F-class turbines without add-on controls) to 25 ppm at 15 % O_2. Table 5-9 shows power generation facilities that have a lower emission rate than this Project.

Table 5-9: RBLC Summary

Facility	Controls	Emission Limit	Units	Type
Commonwealth LNG Facility	Good combustion practices and use of clean fuel.	1.7	ppmvd	BACT
Shell Chem Appalachia/Petrochemicals Complex	No controls listed in RBLC	2	ppmdv @ 15% O_2	BACT
Cricket Valley Energy Center	good combustion practice and oxidation catalyst	2	ppmvd @ 15% O_2	BACT

Further review of the sources listed in Table 5-9 found that these facilities are not representative of the Project. Commonwealth LNG Facility and Lake Charles LNG Export Terminal are both gas processing facilities with much smaller turbines which are not electrical generating units and are therefore not applicable to this Project. Cricket Valley Energy Center is a combined- cycle plant and is not representative of this Project. BACT for this Project was selected to be 4 ppm at 15 % O_2 (for all loads down to 71 percent), which is the lowest emission rate selected as BACT for simple-cycle turbines. This determination is consistent with recent permitting actions, including Omaha Public Power District (OPPD) Cass County Station, which installed the same make and model of turbine (Siemens 5000F) and was permitted with a 4 ppm at 15 % O_2 CO BACT limit using good combustion practices. To meet this BACT rate, the Project's F-class turbines will also utilize good combustion practices.

5.3 BACT for Particulate Matter – Simple-Cycle Combustion Turbines

The following sections outline the top-down steps for particulate matter emissions from combustion turbines.

5.3.1 Step 1. Identify Potential Control Strategies

Particulate ($PM/PM_{10}/PM_{2.5}$) emissions from natural gas combustion sources consist of inert contaminants in natural gas, of sulfates from fuel sulfur or mercaptans used as odorants, of dust drawn in from the ambient air, and particles of carbon and hydrocarbons resulting from incomplete combustion. Therefore, units firing fuels with low ash content and high combustion efficiency exhibit correspondingly low particulate emissions.

From the review of previous BACT determinations (Appendix C), it is evident that all PM BACT determinations are based on good combustion practices and the use of natural gas or low-sulfur fuel. Add-on PM control technologies were not selected as BACT for any of the determinations.

5.3.2 Step 2. Identify Technically Feasible Control Technologies

Add-on particulate control devices such as electrostatic precipitators (ESPs), baghouses, and fabric filters are not technically feasible for application to simple-cycle gas turbines. These technologies are designed for solid-fuel combustion sources—such as coal-fired boilers or industrial furnaces—where flue gas contains significant quantities of entrained ash and particulate matter. In contrast, gas turbines fired on natural gas produce extremely low levels of PM, primarily composed of condensable organic compounds and trace metal oxides.

The high-temperature, high-velocity exhaust stream from a gas turbine presents additional challenges for post-combustion PM control. Installing ESPs or baghouses would require substantial modifications to the turbine exhaust system, including temperature reduction, flow conditioning, and structural retrofits—none of which are standard or proven for gas turbine applications. These systems are not commercially demonstrated on gas-fired turbines and are absent from the RBLC database for simple-cycle units, confirming their lack of technical viability.

In the absence of add-on controls, the most effective and widely demonstrated PM control method for combustion turbines is the use of clean-burning, low ash fuels such as natural gas, combined with combustion controls. This was confirmed by a survey of the RBLC database (Appendix C) which showed no add-on $PM/PM_{10}/PM_{2.5}$ control technologies for simple-cycle combustion turbine units. Proper combustion control and the firing of fuels with negligible or zero ash content (such as natural gas) is the predominant control method listed.

Therefore, based on current industry practice and technical limitations, add-on PM controls such as ESPs and baghouses are not considered technically feasible.

5.3.3 Step 3. Rank the Technically Feasible Control Technologies

Good combustion practices and the use of clean fuels such as natural gas are the only feasible PM control technologies for the proposed combustion turbines. These technologies are compatible control strategies and, considered together, they present the best control option for PM from the proposed units. A ranking is therefore not required to establish the top technology

5.3.4 Step 4. Evaluate the Most Effective Control Technologies

No energy, environmental, or economic impacts are associated with combustion controls; the use of low ash fuel is not an add-on control device.

5.3.5 Step 5. Proposed PM/PM_{10}/$PM_{2.5}$ BACT Determination

The use of low ash fuels and good combustion practices are proposed as BACT for PM/PM_{10}/$PM_{2.5}$ emissions from the proposed simple-cycle combustion turbines. The PM/PM_{10}/$PM_{2.5}$ emission limit are proposed as BACT is 11 lb/hr during steady state conditions for 100% load down to MECL. Compliance will be determined via the average of three runs stack test.

This limit includes front and back half PM/PM_{10}/$PM_{2.5}$ emissions. Add-on controls are not feasible for combustion turbines, thus the BACT emission rates are based on vendor data for this turbine.

5.4 BACT for Sulfur Dioxide – Simple-Cycle Combustion Turbines

The following sections outline the top-down steps for SO_2 emissions from combustion turbines.

5.4.1 Step 1. Identify Potential Control Strategies

The majority of the fuel sulfur combusted in the combustion turbine leaves the turbine as SO_2 or is converted to other forms of sulfur such as H_2SO_4 mist or as ammonium sulfate. There are relatively few BACT entries in the RBLC for SO_2 for simple-cycle combustion turbines. Among the available entries, no add-on control technologies are identified for SO_2 control from combustion turbines. The most common controls listed are use of clean natural gas or low-sulfur fuels.

5.4.2 Step 2. Identify Technically Feasible Control Technologies

There are no add-on controls available for SO_2 from combustion turbines. In the absence of add-on controls, the most effective control method demonstrated for combustion turbines is the use of low sulfur fuel, such as natural gas, and combustion controls. Proper combustion control and the firing of fuels with very low sulfur content (such as natural gas) is the only control method available.

5.4.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible SO_2 control technologies for the combustion turbines are ranked by control effectiveness in Table 5-10.

Table 5-10: Ranking of Technically Feasible SO_2 Control Technologies for Simple-Cycle Combustion Turbines

Control Technology	Reduction (%)	Controlled Emission Level (lb/hr)
Low Sulfur Fuel and Combustion Control	Not applicable (baseline)	3.3

5.4.4 Step 4. Evaluate the Most Effective Control Technologies

No energy, environmental, or economic impacts are associated with combustion controls; the use of low sulfur fuel is not an add-on control device.

5.4.5 Step 5. Proposed SO_2 BACT Determination

The use of pipeline quality natural gas and good combustion control represents BACT for SO_2 control in the proposed simple-cycle combustion turbines. Utilizing pipeline quality natural gas, the turbines will be able to achieve 3.3 lb/hr of SO_2, including during periods of start-up and shutdown.

Compliance will be shown by keeping records of fuel contracts of pipeline quality natural gas.

The SO_2 emissions depend on the sulfur content of the fuel that is combusted in the turbine and the duct burner. The results in the RBLC vary, due to size of combustion turbine and the sulfur content of the fuel.

5.5 BACT for Sulfuric Acid Mist – Simple-Cycle Combustion Turbines

The following sections outline the top-down steps for H_2SO_4 mist emissions from combustion turbines.

5.5.1 Step 1. Identify Potential Control Strategies

The majority of sulfur present in the fuel combusted by a simple-cycle combustion turbine is emitted as sulfur dioxide (SO_2). During the combustion process, a small fraction of the fuel sulfur may be oxidized to sulfur trioxide (SO_3). As the exhaust gas cools following discharge from the turbine and through atmospheric dilution, SO_3 may combine with water vapor present in the exhaust to form sulfuric acid (H_2SO_4) vapor. Under appropriate temperature and humidity conditions, H_2SO_4 vapor may further condense to form sulfuric acid mist.

For simple-cycle combustion turbines firing natural gas, SO_2 oxidation to SO_3 is expected to be limited, as no post-combustion catalytic control devices (e.g., SCR or oxidation catalyst) are present that would promote additional SO_2 oxidation. Accordingly, the primary factors influencing H_2SO_4 mist formation are fuel sulfur content and combustion conditions. The combustion turbine will fire pipeline natural gas with a sulfur content up to 0.5 grains per standard cubic foot on a 12-month rolling average.

There are relatively few BACT entries in the RBLC for H_2SO_4 mist for simple-cycle combustion turbines. Among the available entries, no add-on control technologies are identified for H_2SO_4 mist control from combustion turbines. The most common controls listed are use of clean natural gas or low-sulfur fuels.

5.5.2 Step 2. Identify Technically Feasible Control Technologies

As with SO_2, there are no add-on controls available for H_2SO_4 mist from combustion turbines. In the absence of add-on controls, the most effective control method demonstrated for combustion turbines is the use of low sulfur fuel, such as natural gas and combustion controls. Proper combustion control and the firing of fuels with very low sulfur content is the only known control method available. This was confirmed by a survey of the RBLC database (Appendix D).

5.5.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible H_2SO_4 mist control technologies for the combustion turbines are ranked by control effectiveness in Table 5-11.

Table 5-11: Ranking of Technically Feasible H_2SO_4 Control Technologies for Simple-Cycle Combustion Turbines

Control Technology	Reduction (%)	Controlled Emission Level (lb/hr)
Low sulfur fuel and combustion control	N/A (baseline)	0.5 lb/hr

5.5.4 Step 4. Evaluate the Most Effective Control Technologies

There are no energy, environmental, or economic impacts associated with combustion controls; the use of low sulfur fuel and combustion control is not an add-on control device.

5.5.5 Step 5. Proposed H_2SO_4 Mist BACT Determination

The use of low sulfur fuel and good combustion control represents BACT for H_2SO_4 mist control in the proposed simple-cycle combustion turbine. These operational controls will limit H_2SO_4 mist emissions, including startup and shutdown emissions to 0.5 lb/hr.

Compliance will be determined by keeping records of contracts for pipeline quality natural gas.

H_2SO_4 results in the RBLC for other turbines vary based on turbine make/model size and sulfur content of the fuel. The H_2SO_4 BACT rates represent the lowest emission rates for this make/model turbine with pipeline quality natural gas

5.6 BACT for Volatile Organic Compounds – Simple-Cycle Combustion Turbines

The following sections outline the top-down steps for VOC emissions from combustion turbines.

5.6.1 Step 1. Identify Potential Control Strategies

Like CO, VOC is a product resulting from incomplete combustion. VOC emissions occur when a portion of the natural gas fuel remains unburned or is only partially burned during the combustion process. With natural gas, some organics are unreacted trace constituents of the gas, while others may be products of the heavier hydrocarbon constituents. Partially burned hydrocarbons result from poor air-to-fuel mixing prior to, or during, combustion or incorrect air-to-fuel ratios in the combustion turbine.

The technologies identified for reducing VOC emissions from simple-cycle combustion turbines are the same as identified for CO control: the multi-pollutant control system, an oxidation catalyst (also referred to as a CO catalyst), and combustion controls. The standard technology for reducing VOC emissions is to maintain "good combustion" through proper control and monitoring of the combustion process through monitoring of the air-to-fuel ratio. A survey of the RBLC database (Appendix C) indicates that combustion controls is the most prevalent BACT control. There are a few entries that also installed oxidation catalyst system as LAER or BACT.

5.6.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling VOC emissions are evaluated for technical feasibility in the following sections.

5.6.2.1 EMx™ System

The EMx™ system was described in the BACT analysis for NO_x (Section 5.1.2.2). It is also applicable for controlling VOC and can reduce emissions by up to 20 percent. The system does not use ammonia and operates most effectively at temperatures ranging from 300°F to 700°F. Operation of EM_x™ requires natural gas, water, steam, electricity, and ambient air. Steam and reformed natural gas are used periodically to regenerate the catalyst bed and are an integral part of the process. Because EM_x™ does not use ammonia. there are no ammonia emissions from this technology.

Regeneration of the catalyst is accomplished by passing a dilute hydrogen reducing gas across the surface of the catalyst in the absence of oxygen. Hydrogen in the gas reacts with the nitrites and nitrates to form water and nitrogen. CO_2 in the gas reacts with the potassium nitrite and nitrates to form potassium carbonate, which is the absorbing surface coating on the catalyst. The regeneration gas is produced by reacting natural gas with a carrier gas (such as steam) over a steam-reforming catalyst.

The demonstrated application for EM_x™ is currently limited to combined-cycle combustion turbines under approximately 50 (MW) in size. The EM_x™ system has not been demonstrated on any type of combustion source other than a combustion turbine. There are technical differences between the proposed combustion turbine versus those few sources where this technology has been demonstrated in practice. These significant technical differences preclude a determination that the EM_x™ system has been demonstrated to function efficiently on sources that are similar to the proposed furnaces and boilers (Environmental Resource Management, 2014).

Therefore, the EM_x™ system is considered a technically infeasible method of controlling VOC emissions from the proposed simple-cycle combustion turbines.

Although the EMx™ system is capable of oxidizing VOCs, it has not been demonstrated on large simple-cycle turbines and is therefore considered technically infeasible for this project. The system's effective operating temperature range (300 to 700°F) is far below the exhaust temperature of simple-cycle turbines, which typically exceeds 900 to 1,100°F. Cooling the exhaust to the EMx™ operating window would require large heat exchangers or quench systems, resulting in excessive backpressure, efficiency losses, condensation/corrosion risks, and significant capital and operating costs. In addition, EMx™ regeneration requires steam-reformed natural gas and hydrogen handling infrastructure, which is incompatible with the fast-start, cycling operation of simple-cycle turbines. For these reasons, EMx™ is not technically or economically feasible for VOC control on the proposed units, and conventional oxidation catalysts remain the demonstrated and commercially viable technology for VOC reduction in simple-cycle turbine applications.

5.6.2.2 Oxidation Catalyst

As discussed in Section 5.2.2.2, oxidation catalysts are a post-combustion technology that do not rely on the introduction of additional chemicals, such as ammonia or urea with SCR, for a reaction to occur. The catalyst beds that reduce CO also promote the oxidation of VOC, thereby reducing the VOC emissions out the stack.

Such systems typically achieve a maximum of 30 percent removal of VOC, as opposed to the much higher efficiencies achieved for CO reduction.

A combined oxidation catalyst system is not feasible for simple-cycle turbines due to the need for precise temperature control and uniform exhaust distribution across the catalyst bed. Mixing exhaust streams from multiple turbines operating independently would create maldistribution, reduce CO/VOC removal efficiency, and complicate compliance monitoring. In addition, a shared catalyst would increase outage risk by tying all units to a single module. The recommended configuration is to equip each turbine with its own oxidation catalyst, with opportunities to share non-critical support systems such as access platforms, monitoring equipment, and spare catalyst elements. A combined oxidation catalyst system is considered to be technically infeasible for the combustion turbines under consideration for this Project.

The use of an oxidation catalyst for VOC control is considered to be technically feasible for the simple-cycle combustion turbines.

5.6.2.3 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas to confirm that there is enough oxygen present for complete combustion (controlling the air-to-fuel ratio).

Good combustion practices are a technically feasible method of controlling VOC emissions from the proposed combustion turbines.

5.6.2.4 Summary of the Technically Feasible Control Options

The technical feasibility of the VOC control options for the proposed combustion turbine is summarized in Table 5-12.

Table 5-12: Summary of Technically Feasible VOC Control Technologies for Simple-Cycle Combustion Turbines

Control System		Technical Feasibility	Comments
Combustion controls		Feasible	Standard on the proposed combustion turbine. Not an add-on control
Post combustion controls	Oxidation catalyst	Feasible	Produces CO_2 emissions. Demonstrated technology for combined-cycle combustion turbines.
	EMx™	Not feasible	Not a demonstrated technology for large combustion turbines.

5.6.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible CO control technologies for the combustion turbine are ranked by control effectiveness in Table 5-13.

Table 5-13: Ranking of Technically Feasible VOC Control Technologies for Simple-Cycle Combustion Turbines

Control System	Reduction (%)	Controlled Emission Level (ppm)[a]
Oxidation catalyst	40 to 90[b]	0.6-0.7
Combustion control	Not applicable (Baseline)	1[c]

(a) ppm = parts per million

(b) The range reflects both ideal design performance and realistic operating variability.

(c) Valid for 100 percent load down to MECL.

5.6.4 Step 4. Evaluate the Most Effective Control Technologies

The next step is to review each of the technically feasible control options for environmental, energy, and economic impacts.

5.6.4.1 Combustion Control

No energy, environmental, or economic impacts are associated with combustion controls.

5.6.4.2 Oxidation Catalyst

Energy Impacts

With an oxidation catalyst, the output of the turbines would be reduced by the additional backpressure. To replace this lost energy, additional power generation will be required, which will require consumption of additional natural gas.

Environmental Impacts

The installation of an oxidation catalyst would theoretically reduce potential CO emissions by 50 to 80% for each combustion turbine. The oxidation catalyst oxidizes CO to CO_2 which is released to the atmosphere. CO_2 is a greenhouse gas and is a regulated pollutant. Increasing CO_2 emissions could have a negative impact on the atmosphere. However, the oxidation catalyst will also reduce the amount of methane (CH_4) (also a greenhouse gas). Considering both greenhouse gases, the net effect is an overall decrease in global warming potential since the global warming potential of CH_4 is 28 times larger than CO_2.

As with all controls that utilize catalysts for removal of pollutants, the catalyst must be disposed of after it is spent. The catalyst may be considered hazardous waste and require special treatment or disposal; even if it is not hazardous, it adds to landfills.

Economic Impacts

The capital and annual costs associated with an oxidation catalyst for the combustion turbine operating in simple-cycle mode are shown in Appendix D. The total capital investment of installing an oxidation catalyst on the simple-cycle turbine is approximately $9,876,000. On an annual basis, the oxidation catalyst would cost $1,521,700 which results in a cost per ton of CO and VOC removed of $52,770 while removing only 0.5 tons of VOC emissions per year for both natural gas operation and fuel oil operation, based on worst-case normal operation emissions. Therefore, any control of VOC by add-on controls would result in costs that would not be economical.

5.6.5 Step 5. Proposed VOC BACT Determination

The use of good combustion practices is BACT for the turbines. The CO BACT emissions limit is 1 ppm at 15 % O_2 during steady state conditions for 100 percent load down to MECL. Compliance will be determined via the average of three run stack tests.

One facility in the RBLC showed an emission rate of 0.7 ppmvd at 15 % O_2. The Cricket Valley facility turbines are combined-cycle and it was subject to LAER so it is not comparable to this Project.

5.7 BACT for Greenhouse Gases – Simple-Cycle Combustion Turbines

The following sections outline the top-down steps for greenhouse gas (GHG) emissions from combustion turbines.

5.7.1 Step 1. Identify All Potential Control Strategies

For the proposed simple-cycle combustion turbine, the CO_2e emissions are due to CO_2, CH_4, and nitrous oxide (N_2O) emissions. The GWP of CH_4 and N_2O emissions are normalized to the warming potential of carbon dioxide (as CO_2e) by multiplying the CH_4 emissions by 28 and the N_2O emissions by 265. Despite the higher warming potentials of CH_4 and N_2O compared to CO_2, it is expected that CO_2 emissions will still account for over 99 percent of the CO_2e GWP for these units, based on published emission factors for natural gas-fired turbines.

There are two broad strategies for reducing CO_2 emissions from stationary combustion processes such as combustion turbines. The first is to minimize the production of CO_2 through the use of low-carbon fuels and through aggressive energy-efficient design. The use of gaseous fuels, such as natural gas, reduces the production of CO_2 during the combustion process relative to burning solid fuels (e.g., coal or coke) and liquid fuels (e.g., distillate or residual oils). Additionally, a highly efficient operation requires less fuel for process heat, which directly impacts the amount of CO_2 produced. Establishing an aggressive basis for energy recovery and facility efficiency will reduce CO_2 production and the costs to recover it.

The second strategy for CO_2 emission reduction is carbon capture and sequestration. The inherent design of the combustion turbines produces a dilute CO_2 stream for potential capture.

The CO_2 emissions from combustion turbines can theoretically be controlled through pre-combustion methods or through post-combustion methods. In the pre-combustion approach, oxygen instead of air is used to combust the fuel and a concentrated CO_2 exhaust gas is generated. This approach significantly reduces the capital and energy cost of removing CO_2 from conventional combustion processes using air as an oxygen source, but it incurs significant capital and energy costs associated with separating oxygen from the air.

Post-combustion methods are applied to conventional combustion techniques using air and carbon-containing fuels in order to isolate CO_2 from the combustion exhaust gases. Because the air used for combustion contains nearly 80 percent nitrogen, the CO_2 concentration in the exhaust gases is only 5 to 20 percent depending on the amount of excess air and the carbon content of the fuel. RBLC results are shown in Appendix C.

5.7.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling GHG emissions are evaluated for technical feasibility in the following sections.

5.7.2.1 Fuels

Fuel has a significant impact on GHG formation as discussed below.

5.7.2.1.1 Low-Carbon Fuels

Numerous fuels are available for use. As Table 5-14 shows, combustion of natural gas yields 40 to 50 percent less CO_2 than does combustion of coal and petroleum coke and approximately 30 percent less CO_2 than does combustion of residual oil. Accordingly, the use of natural gas as the primary fuel and fuel oil as the backup fuel in the proposed combustion turbines is considered to be the most effective CO_2 control technique. Utilizing low carbon fuels (natural gas and fuel oil) is technically feasible for the combustion turbines and is an inherent part of the Project's design.

Table 5-14: CO_2 Emission Factors

Fuel	kilograms CO_2 per MMBtu
Petroleum coke	113.67
Coal (anthracite)	103.69
Distillate fuel oil No. 2	73.96
Natural gas	53.06

Source: Title 40 CFR 98: Table C-1 to Subpart C of Part 98 – Default CO_2 Emission Factors and Types of Fuel

5.7.2.1.2 Combustion of Biogenic Sources

The proposed combustion turbine has not been designed to accommodate fibrous biomass, such as woody biomass, which is the most likely biomass available in sufficient quantities for the unit from the surrounding area. For both regulatory and technical feasibility issues, **biogenic sources are not a feasible option since they are not part of the original design**.

5.7.2.2 Energy Efficiency

The evaluation of energy efficiency, continuous excess air monitoring and control and the selection of efficient turbine design, are discussed below.

5.7.2.2.1 Continuous Excess Air Monitoring and Control

Excessive amounts of combustion air in turbines results in energy-inefficient operation because more fuel combustion is required in order to heat the excess air to combustion temperatures. This inefficiency can be alleviated using state-of-the-art instrumentation for monitoring and controlling the excess air levels in the combustion process, which reduces the heat input by minimizing the amount of combustion air needed for safe and efficient combustion. Additionally, lowering excess air levels, while maintaining good combustion, reduces not only CO_2 emissions but also NO_x emissions. The combustion turbine will be equipped with oxygen monitors.

5.7.2.2.2 Selection of Efficient Turbine Design

Energy efficiency reduces CO_2 emissions by maximizing the operation of the combustion turbines, thereby reducing the amount of fuel burned per megawatt-hour produced.

Combustion control optimization and energy efficient equipment is a main control strategy for emissions of greenhouse gases. The combustion turbines that are under consideration for this Project is highly efficient. Energy efficiency is technically and economically feasible. Potential options that may increase efficiency include the following:

- Airfoil-shaped compressor rotor blades designed to increase compressor efficiency
- Improved thermal barrier coating application on some turbine airfoils
- 13 stage high efficiency compressor design with modulating inlet guide vanes and inter-stage air extraction for cooling and sealing air
- Inlet air filtration system utilizing high efficiency media filters to remove combustion air contaminants
- On and off-line compressor water wash capability to remove deposits and other contaminants from compressor blades to maintain and improve compressor efficiency
- LNB for improved performance, enhanced operability, and lower emissions
- Extended turndown for increased spinning reserve capability and lower fuel costs
- Low turndown ratios for reduced CO emissions at part loads
- Outlet temperature correction control systems enhance part load performance and emissions
- Advanced hot gas path components with three-dimensional airfoil shapes, improved materials, improved sealing, more effective cooling to achieve increased turbine efficiency
- Higher firing temperatures to increase turbine performance and overall turbine efficiency
- Blade tip clearance reduction

5.7.2.3 Add-On Control Devices

Another method of GHG control is an add-on control device.

5.7.2.3.1 Oxidation Catalyst

N_2O emissions are reduced by passing the combustion gases over a catalyst, converting N_2O to nitrogen plus oxygen. Similarly, VOC emissions, such as CH_4, may be converted from CH_4 to CO_2 plus water. For the same reasons given above in the discussion for CO BACT controls, oxidation catalysts are **technically feasible for the control of GHG emissions from the proposed simple-cycle combustion turbines.**

5.7.2.3.2 Thermal Oxidation

Several types of thermal oxidation technology are available. All these technologies oxidize CH_4 to CO_2 and water, by raising the temperature of the treated gas stream to approximately 1,600°F for approximately one to two seconds. Given sufficient mixing, this residence time and temperature is capable of achieving at least a 98 percent reduction in CH_4 emissions for these processes.

Secondary pollutants, however, are produced by thermal oxidation, including NO_x and CO from the combustion of natural gas used to heat the process stream. Thermal oxidation technologies also may employ some form of heat recovery, either recuperative or regenerative, to reduce economic, environmental and energy costs. In the case of a combustion turbine, it is expected that approximately 5.0 lb/hr of CH_4 while firing natural gas will be produced at full load (with an exhaust flow rate of approximately 2,600,000 standard cubic feet per minute while firing natural gas at full load). The exhaust gas stream is thus both high volume and very dilute in CH_4, so it would need to be concentrated to the point that the CH_4 would be capable of combustion. Also, additional CO_2 would be produced due to the need for combusting natural gas to heat the CH_4 to the oxidation point. This would reduce the overall effectiveness in reducing CO_2e emissions due to CH_4 because additional CO_2 would be produced as a result of combusting the CH_4. **Therefore, thermal oxidation is technically infeasible for the control of GHG emissions from the proposed simple-cycle combustion turbines.**

5.7.2.4 Carbon Capture and Sequestration

Carbon capture and sequestration is a general term which is used for approaches that capture and separate CO_2 from an exhaust stream, and then store it in a place which will keep it from the atmosphere for a long time. The two general categories of CO_2 capture are: pre-combustion CO_2 capture and post-combustion CO_2 capture.

5.7.2.4.1 Pre-Combustion CO_2 Capture

Pre-combustion CO_2 capture is used in gasification plants, where the CO_2 is captured from the syngas prior to combustion in the turbine, where it is relatively concentrated in the gas stream. This facility is not a gasification plant; therefore, **pre-combustion capture is technically infeasible for the control of CO_2 emissions from the proposed simple-cycle combustion turbines**.

5.7.2.4.2 Post-Combustion CO_2 Capture

Post-combustion CO_2 capture is used for units such as pulverized coal plants. In these units, the flue gas concentration of CO_2 runs between 10-15 percent by volume and is released at atmospheric pressure. This results in a high actual volume of gas to be treated. Trace impurities in the airflow tend to reduce the effectiveness of the CO_2-adsorbing process and compressing the captured CO_2 from atmospheric pressure to pipeline pressure represents a large parasitic load. The currently available process is costly and energy intensive, so research is being done on ways to increase the solvent capture efficiency and reduce the cost. These approaches include investigating the use of alternative solvents, solid sorbents or membranes. Of these potentially more efficient approaches, most are currently at laboratory/bench scale, so are not technically feasible. Pilot scale processes are starting to be placed in service, such as a 48 MW slipstream project at Brindisi, Italy, started in March 2011, which is limited to capturing less than 10,000 tons of CO_2 per year. A larger 235-MW slipstream project for the 1,300 MW Mountaineer Power Plant near New Haven, West Virginia was built with technology that used chilled ammonia to trap CO_2. The pilot project removed up to 300,000 metric tons of CO_2; however, the project was abandoned due to diminishing Federal and State support for clean coal technology. No commercially available post-combustion CO_2 capture systems are known to have been installed at large power plants other than pilot-scale demonstration projects. Therefore, **post-combustion capture is technically infeasible for the control of CO_2 emissions from the proposed simple-cycle combustion turbines**.

5.7.2.5 CO_2 Sequestration

CO_2 sequestration involves transporting CO_2 to a suitable geologic location where it can be injected as a supercritical fluid into deep, underground rock formations for permanent storage. Identifying a suitable site within an economically viable distance from the Project site will require site-specific quantitative risk assessment. Four trapping methods are known: mineral trapping, physical adsorption, hydrodynamic trapping, and solubility trapping.

5.7.2.5.1 Mineral Trapping

The mineral trapping method traps CO_2 by undergoing a chemical reaction with various minerals, resulting in the formation of a carbonate mineral. This process can be rapid or very slow, depending on the chemistry of the rock and water at the site. Mineral trapping is expected to result in the most stable, permanent form of geological CO_2 sequestration. Experiments have shown that basalt formations can rapidly transform injected CO_2 into carbonate minerals, beginning precipitation in a few months' time and completing conversion within 100 years or less, depending on depth of injection. Sandstone formations low in carbonates may also be suitable candidates, depending on the mineral contents of the formations. These methods have been demonstrated only on a laboratory scale; therefore, mineral trapping is **not technically feasible for the proposed simple-cycle combustion turbines**.

5.7.2.5.2 Physical Adsorption

The physical adsorption process traps CO_2 molecules are trapped in micropore wall surfaces of coal organic matter or organic rich shales. The hydrostatic pressure in the formation controls the adsorption process. The injection of CO_2 can also result in driving off CH_4 for collection by other wells, helping the economics. Some coal beds in the US are being tested for CO_2 storage/CH_4 recovery, but this is currently at a pilot phase. Defining the depths and lateral distribution of coal strata that might be suitable for this approach has not been done, due to the significant depths required for CO_2 sequestration. Significant research and exploration efforts would be required to determine whether such coal beds occur at the required depths beneath central Iowa. Use of coal beds in Iowa would require much further study to locate a suitable site for sequestration, and since the results of pilot phase testing of this technique are not known, these factors combined render the use of coal beds **not technically feasible for the proposed simple-cycle combustion turbines.**

5.7.2.5.3 Hydrodynamic Trapping

With hydrodynamic trapping, the pore space of a salt-water aquifer takes the injected CO_2 in a geologic setting where the aquifer is capped by an impermeable rock layer to trap the CO_2 well below the near-surface environment. For storage purposes, the aquifer should be saline enough to be non-potable, and deep enough (over 2,700 feet) to confirm that the pressure is sufficient to keep the compressed CO_2 in a supercritical liquid phase. Since the sedimentary bedrock strata in the site vicinity are over 10,000 feet thick, the possibility exists that geologically suitable strata exist somewhere within these layered rock formations. However, in the absence of oil and gas exploratory test holes, the locations, depths, and character of such strata are not known, and would have to be discovered and defined by extensive exploratory drilling and testing. As the state of Iowa does not currently have primacy for the Class VI regulations (governing injection wells), EPA rules that require a minimum of 10,000 milligrams per liter (mg/L) total dissolved solids to qualify as saline enough to be suitable for injection will probably apply. Discovering locations which exceed 10,000 mg/L would require significant exploration and test wells to characterize the site and determine the aquifer suitability. At these depths, defining suitable geology would be rendered costly and problematic. Multiple oil and gas fields exist in the region, but a serious limitation to feasibility in an existing oil or gas field is the great likelihood of significant numbers of "penetrations" (old, either documented or undocumented wells and test holes that may or may not be adequately plugged and abandoned). Also, the additional surface infrastructure that would be needed to inject CO_2 would be massive, problematic, and likely infeasible. Pilot-scale projects injecting CO_2 into saline aquifers are underway in Illinois and Texas at depths of over 6,000 feet and these are the closest known sites that have been initially characterized for potential long-term sequestration, but the studies are in their early stages. Therefore, hydrodynamic trapping is **technically infeasible for the control of CO_2 emissions from the proposed simple-cycle combustion turbines** at this time.

5.7.2.5.4 Solubility Trapping

With solubility trapping, the CO_2 dissolves in the water or forms carbonic acid, becoming slightly heavier and, theoretically, sinking to the bottom of the aquifer. Solubility trapping also occurs during CO_2 flooding for enhanced oil recovery (EOR). In this case, the CO_2 dissolves into the oil, and is trapped by the immobile, non-recoverable oil. CO_2 flooding has been used for years for EOR, resulting in some existing injection infrastructure at oil fields (using both solubility trapping and hydrodynamic trapping), although the sequestration effects were not originally monitored, and the volumes injected for such operations are minuscule. However, oil fields have stored crude oil and natural gas for millions of years, and the geologic conditions that trap oil and gas are also suitable for CO_2 storage. If the CO_2 is used for EOR, the cost of transporting it to the oil field may be partially offset. Since the sedimentary bedrock strata in the site vicinity are over 10,000 feet thick, the possibility exists that oil and gas fields involving geologically suitable strata exist somewhere within these layered rock formations within the region. However, defining suitable geologic conditions in an existing oil or gas field, including the locations, depths, and character of such strata would have to be defined by extensive exploratory drilling and testing. Multiple oil and gas fields exist in Texas, however, as was the case with hydrodynamic trapping, there is a likelihood of undocumented penetrations. Also, additional surface infrastructure that would be needed to inject CO_2.would be massive, problematic, and likely infeasible. Therefore, solubility trapping is **technically infeasible for the control of CO_2 emissions from the proposed simple-cycle combustion turbines** at this time.

5.7.2.5.5 Summary of CO_2 Sequestration

To summarize, existing CO_2 capture technologies have not been applied at large simple-cycle power plants, as the economic costs are prohibitive, and while more efficient approaches are being investigated, none have currently been developed past the pilot-stage. Further, the CO_2 capture processes require a constant slipstream to operate effectively, and these peaking combustion turbines will not have a constant pure CO_2 stream while load-following.

A published cost estimate for a 235-MW slipstream pilot project in West Virginia is $668 million, so scaling that linearly to a size capable of handling this Project would be approximately $2.5 billion. Potential carbon sequestration sites may exist in Iowa but the technologies to use them are mostly still in the pilot-scale phase of development, and IPL would need to do much more investigation in order to discover where the sites are, if any, and characterize them enough to demonstrate the long-term viability of the locations. Defining suitable geologic conditions in an existing oil or gas field, including the locations, depths, and character of suitable strata, and defining penetrations (potentially leaky wells and test holes, some of which are likely to exist but are undocumented) into the geological traps comprising existing oil and gas fields, would have to be defined by extensive exploratory drilling and testing. One of the closest known sites for sequestration is the Nevada Biorefinery located in Nevada, Iowa, approximately 100 miles from the plant. The cost to construct a pipeline as determined from a similar project (Iowa Power & Light Ottumwa – Iowa Department of Natural Resources project 11-219) to this Project's site would be approximately $1.4 million/mile of pipeline, or about $140 million. The capital cost estimated for this comparable project was nearly $2.1 billion for capture equipment and pipeline construction alone prior to any costs for gas compression, additional injection and monitoring wells necessary to handle the volume of CO_2 produced, pipeline right-of-way, operation and maintenance costs, etc. As can be seen from the above discussion, the qualitative cost estimate of capture and sequestration is quite high, the technological effectiveness for the capture equipment for a unit of this size has not been demonstrated in practice yet, and there is uncertainty as to whether locations capable of storing the large amounts of CO_2 that would be produced per year exist within a closer radius of the plant. These considerations are sufficient to eliminate this option without requiring a more detailed site-specific technological or economic analysis.

5.7.2.6 Summary of Technically Feasible Control Technologies

The technical feasibility of the greenhouse gas control options for the proposed combustion turbine is summarized in Table 5-15.

Table 5-15: Summary of Technically Feasible Greenhouse Gas Control Technologies for Combustion Turbines

Control System		Technical Feasibility	Comments
Fuel Selection	Low Carbon Fuels	Feasible	Natural gas has been selected as the primary fuel for this Project
	Combustion of Biogenic Sources	Not Feasible	--
Energy Efficiency	Continuous Excess Air Monitoring and Control	Feasible	Standard for the turbines under consideration
	Efficient Turbine Design	Feasible	Standard for the turbines under consideration
Post Combustion Controls	Oxidation Catalyst	Feasible	Will reduce CH_4 emissions but create CO_2
	Thermal Oxidation	Not Feasible	--
Carbon Capture	Pre-combustion CO_2 capture	Not Feasible	--
	Post-combustion CO_2 capture	Not Feasible	--
Carbon Sequestration	Mineral Trapping	Not Feasible	--
	Physical Adsorption	Not Feasible	--
	Hydrodynamic Trapping	Not Feasible	--
	Solubility Trapping	Not Feasible	--

5.7.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible control technologies are low-carbon fuel (natural gas), monitoring and control of excess air, efficient turbine design, and oxidation catalysts. The use of low-carbon fuels and aggressive energy-efficient design to reduce CO_2 emissions is inherent in the design of the proposed combustion turbine under consideration and is considered the baseline condition. Table 5-16 presents the ranking of the greenhouse gas technologies deemed feasible for the Project. While these technologies are “ranked” in order of their presentation, they are more appropriately considered as a suite of measures that would be implemented to allow the Project to generate and consume power in the most efficient manner and thereby achieve BACT for greenhouse gases.

Table 5-16: Greenhouse Gas Control Technology Ranking for the Combustion Turbines

Technology	Ranking	Applied to Project
Simple-Cycle Combustion Turbines (employing efficient, state-of-the-art design)	1	Yes
Clean Fuel – Natural Gas, Ultra-low sulfur fuel oil	2	Yes
Oxidation Catalyst	3	No
Operational Design – Control of Excess Air	4	Yes
Carbon Capture and Sequestration (CCS)[a]	5	No

(a) CCS has been identified as an infeasible potential control strategy; however, it is being further evaluated for environmental, energy, and economic impacts.

5.7.4 Step 4. Evaluate the Most Effective Control Technologies

The next step is to review each of the technically feasible control options for environmental, energy, and economic impacts.

5.7.4.1 Environmental, Energy, and Economic Feasibility of Control Options

Because IPL is proposing to utilize all of the feasible technologies for reducing greenhouse gases from the proposed combustion turbine besides catalytic oxidation, only detailed analysis for the oxidation catalyst is provided to compare the available control technologies' relative environmental, energy and economic impacts.

5.7.4.1.1 Oxidation Catalyst

An oxidation catalyst works to reduce CH_4 emissions according the following equation:

$$CH_4 + 2O_2 = CO_2 + 2H_2O$$

Substituting in the molecular weights of CH_4 (16.043 pound per pound mole [lb/lb-mol]) and CO_2 (44.0096 lb/lb-mol), the removal of 1 pound of CH_4 results in the release of 2.7 pounds of CO_2. However, CH_4 has a GWP of 28, whereas the GWP of CO_2 is 1. Substituting in the GWPs, the removal of 1 pound of CH_4 results in a net reduction of 22.3 lb CO_2 as CO_2e.

It is also important to note the increase in CO_2e emissions from the oxidation of CO to CO_2 in accordance with the following reaction:

$$2CO + O_2 = 2CO_2$$

CO_2 will be emitted at a rate of approximately 1.5 pounds per pound of CO. Therefore, it is expected that there will still be a net decrease in CO_2e, even with the additional CO_2 that is produced from the oxidation catalyst with the oxidation of CO and CH_4.

Additional negative environmental, energy, or economic impacts from the use of an oxidation catalyst are discussed in Step 4 of the combustion turbine CO BACT (Section 5.2.4).

5.7.4.1.2 Carbon Capture and Sequestration

Carbon capture and sequestration (CCS) has been identified as an infeasible potential control strategy however, it is being further evaluated for environmental, energy, and economic impacts. For purposes of this analysis, it is assumed that CCS would capture and store 90 percent of the emitted CO_2, which is consistent with preliminary engineering studies for carbon capture on pure CO_2-streams at coal-fired power plants.

Economic Impacts of CCS

According to the Report of the Interagency Task Force on Carbon Capture and Storage, the incremental costs of new natural gas-fired combined-cycle combustion turbines with CCS relative to new conventional natural gas-fired combined cycle combustion turbines, are approximately $114 per metric ton ($103 per short ton) of CO_2 avoided. Note, the $103 per short ton of CO_2 is in 2010 dollars (and would therefore be higher in if converted to today's dollars) and also does not include the cost of a pipeline for transportation and storage/sequestration costs. The $103 per ton of CO_2 cost (in 2010 dollars), which does not include the cost of pipeline transport and subsequent storage/sequestration, is well above the range of cost effectiveness values considered to be reasonable and acceptable in BACT determinations for control of GHG emissions. If the cost of the pipeline were to be included in the overall cost, the estimated cost at this time for constructing a CO_2 pipeline is approximately $1.4 million per mile. In review of the applications for Class VI wells, the closest location (which is not yet through the permitting process) would be located in Illinois, likely in Henry County. This is almost 80 miles away and would make the pipeline cost almost $112 million, which, especially when added to the cost of the carbon capture system would make the application of CCS cost prohibitive.

The economic impacts of CCS on the Project would be enormous and would result in the Project not being considered. The Interagency Task Force report (2010) stated that the CCS cost would increase the cost of a natural gas-fired combined-cycle combustion turbine plant by 80 percent. A similar economic impact would occur with the application of CCS to the simple-cycle combustion turbines proposed.

Energy Impacts of CCS

The adverse energy impacts of CCS as applied to the combustion turbines are significant. Based on the Interagency Task Force report (2010), the energy impacts are due to the significant amount of electric energy needed to compress the CO_2, and steam needed to regenerate the CO_2 capture system reagent.

Environmental Impacts of CCS

The adverse environmental impacts of CCS as applied to the Project are a result of the significant amount of electric and steam energy required to compress the captured CO_2 and solvent regeneration.

Due to the significant adverse energy, economic, and environmental impacts and the comparatively insubstantial environmental benefits that would result from the application of CCS, it is eliminated from consideration as BACT for the Project. Additionally, the technical feasibility of CO_2 injection into a nearby saline formation or EOR field is unknown and would require extensive testing to confirm the feasibility of these storage options.

No adverse impacts were found for the remaining control methods.

5.7.5 Step 5. Proposed Greenhouse Gas BACT Determination

BACT for greenhouse gas emissions from the combustion turbines is proposed to be the use of natural gas as a fuel, monitoring and control of excess air, and efficient turbine design. These controls will meet CO_2

emission limits of 120 lb/MMBtu while combusting natural gas. Compliance will be shown by keeping records of natural gas combusted in the turbines.

5.8 BACT for Start-up and Shutdown Emissions – Simple-Cycle Combustion Turbines

The following sections outline the top-down BACT steps for start-up and shutdown emissions from the combustion turbine.

5.8.1 Step 1. Identify Potential Control Strategies

Criteria pollutants will be emitted during start-up and shutdown of the combustion turbine. Start-up emissions are generally higher for CO, NO_x, and VOC than for normal operation. A start-up is defined when fuel is first introduced until it reaches MECL and a shutdown is defined when a shutdown is initiated until the unit stops operating. Detailed calculations of the potential start-up and shutdown emissions are provided in 3.1.2.

IPL expects up to 365 start-up/shutdowns per turbine per year, as discussed in Section 3.1.2.

5.8.2 Step 2. Identify Technically Feasible Control Technologies

Controls that may be used during normal operation are not available to control start-up and shutdown emissions. In addition, the manufacturer will only guarantee emissions down to MECL, indicating that this is where stability in these emissions is reached. To minimize emissions, however, start-up and shutdown shall be limited to 35 minutes for start-up and 20 minutes for shutdown.

Therefore, no technically feasible control technologies for start-up and shutdown emissions from the combustion turbine have been identified.

5.8.3 Step 3. Rank the Technically Feasible Control Technologies

Since no technically feasible control technologies for start-up and shutdown emissions have been identified, ranking of such control technologies is not applicable.

5.8.4 Step 4. Evaluate the Most Effective Control Technologies

Since no technically feasible control options for start-up and shutdown emissions have been identified, evaluation of environmental, energy or economic impacts of such control technologies is not applicable.

5.8.5 Step 5. Proposed Start-up and Shutdown BACT

BACT for start-up and shutdown emissions is limiting the time that the units are in start-up or shutdown. The turbines will therefore meet an annual tons per year limit that includes start-up and shutdown. The proposed BACT for start-up and shutdown emissions for NO_x, CO and VOC are shown in Table 5-17.

Table 5-17: Combustion Turbine Start-up and Shutdown BACT Limits, Each Turbine

Pollutant	Start-up/Shutdown BACT Annual Emissions Per Turbine[a]
	tons per year
NO_x	90.4
CO	359.6
VOC	31.6

(a) BACT emissions are based on total annual emissions, including start-up and shutdown emissions, per turbine

5.9 BACT Analysis for Hot Water Boilers

The four natural gas-fired hot water boilers are rated at 6.0 MMBtu/hr, each, and are proposed to operate 8,760 hours per year of natural gas. The RBLC has limited information on BACT conclusions for the hot water boilers (Appendix C). The RBLC tables also show high variability for emission rates for each pollutant. For all pollutants, very few entries with add-on controls were listed because hot water boilers are so small.

5.9.1 BACT for Nitrogen Oxides – Hot Water Boilers

The following sections outline the top-down steps for NO_x emissions from the hot water boilers.

5.9.1.1 Step 1. Identify Potential Control Strategies

One entry in the RBLC listed SCR as an add-on NO_x control technique available for units of this size, with an emission rate of 0.006 lb/MMBtu, however, this RBLC facility was subject to LAER. Ultra-low NO_x burners, LNB, along with combustion controls, are listed as BACT in the RBLC for the hot water boilers. NO_x emissions listed in the RBLC range from 0.011 to 0.1 lb/MMBtu for similar sized hot water boilers utilizing low-NO_x burners and combustion controls in attainment areas.

The hot water boilers installed for the Project will be equipped with low-NO_x burners, with an emission rate of 0.050 lb/MMBtu. This value is consistent with other BACT units with low-NO_x burners listed in the RBLC.

5.9.1.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling NO_x emissions are evaluated for technical feasibility in the following sections.

5.9.1.2.1 SCR

None of the units in attainment areas listed SCR as BACT. However, it is possible for an SCR to be installed on a unit of this size.

As a result, an SCR system is considered technically feasible for the hot water boilers.

5.9.1.2.2 Low-NO_x Burners

Low-NO_x burners are currently available from most natural gas hot water boiler manufacturers. This technology seeks to reduce combustion temperatures, thereby reducing NO_x. In a conventional combustor, the air and fuel are introduced at an approximately stoichiometric ratio, and air/fuel mixing occurs at the flame front where diffusion of fuel and air reaches the combustible limit. A lean premixed combustor design premixes the fuel and air prior to combustion. Premixing results in a homogenous air/fuel mixture, which minimizes localized fuel-rich pockets that produce elevated combustion temperatures and higher NO_x emissions. A lean air-to-fuel ratio approaching the lean flammability limit is maintained, and the excess air serves as a heat sink to lower combustion temperatures, which lowers NO_x formation. A pilot flame is used to maintain combustion stability in this fuel-lean environment.

Low-NO_x burners are available on the hot water boilers and are considered both baseline and technically feasible.

5.9.1.2.3 Ultra-Low-NO_x Burners

Ultra-low-NO_x burners (ULNB) are not available on these small 6.0 MMBtu/hr hot water boilers. The units in the RBLC that installed ULNB are larger at 10 MMBtu/hr. **As such, ULNB is not considered technically feasible for hot water boilers.**

5.9.1.2.4 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas to confirm that there is enough oxygen present for complete combustion.

As a result, combustion control is considered baseline for the hot water boilers and is technically feasible.

5.9.1.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible NO_x control technologies for the 6.0 MMBtu/hr hot water boilers are ranked by control effectiveness in Table 5-18.

Table 5-18: Ranking of NO_x Control Technologies for the Hot water boilers

Control Technology	Reduction (%)	Controlled Emission Level (lb/MMBtu)
SCR	95	0.005
Low-NOx burners	50	0.05
Combustion control	N/A (baseline)	0.1

5.9.1.4 Step 4. Evaluate the Most Effective Control Technologies

Each technically feasible control technology was evaluated for energy, environmental, and economic impacts. These impacts are discussed below for each control technology.

5.9.1.4.1 SCR

Energy and Environmental Impacts

Energy and environmental impacts for an SCR system are discussed in Section 5.1.4.1.

Economic Impacts

The capital costs and annualized costs associated with an SCR system for each hot water boiler was evaluated as detailed in Appendix D. The estimated total capital investment of installing an SCR system on the hot water boiler is approximately $87,790 with total annualized costs of $43,600. Based on an annual NO_x reduction of only 1.1 tons of NO_x per year this results in a cost-effectiveness value of approximately $38,140 per ton of NO_x removed. Therefore, any control of NO_x by add-on controls would result in costs that would not be economical.

SCR is not proposed as BACT for the hot water boilers because it is not economically feasible.

5.9.1.4.2 Low-NO_x Burners and Combustion Control

Because the low-NO_x burners come standard on most hot water boilers and combustion control is accomplished through operation of the hot water boilers, there are no incremental energy, environmental, or economic impacts associated with these controls.

5.9.1.5 Step 5. Proposed NO_x Hot Water Boilers BACT Determination

Low-NO_x burners and combustion control were selected as BACT for the hot water boilers; add-on controls are not practical on this small unit since the economic impacts are high. The low-NO_x burners can achieve an emission rate of 0.050 lb/MMBtu during steady state operation.

5.9.2 BACT for Carbon Monoxide – Hot Water Boilers

The following sections outline the top-down steps for CO emissions from hot water boilers.

5.9.2.1 Step 1. Identify Potential Control Strategies

The RBLC does not list add-on controls for hot water boilers in the BACT determinations for control of CO emissions from hot water boilers this size. As with the combustion turbines, good combustion control will help control emissions of CO from the hot water boilers. CO emissions listed in the RBLC range from 0.037 to 0.084 lb/MMBtu for similar sized hot water boilers utilizing combustion controls and clean fuels.

5.9.2.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling CO emissions are evaluated for technical feasibility in the following sections.

5.9.2.2.1 Oxidation Catalyst

The oxidation catalyst system is an add-on control that converts CO and VOC to CO_2 by use of an oxidation catalyst. Section 5.2.2.2 describes the oxidation catalyst system for gas-fired units.

An oxidation catalyst system is considered technically feasible for the hot water boilers.

5.9.2.2.2 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas to confirm that there is enough oxygen present for complete combustion.

Good combustion practices are a technically feasible method of controlling CO emissions from the hot water boilers.

5.9.2.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible CO control technologies for the 6.0 MMBtu/hr hot water boilers are ranked by control effectiveness in Table 5-19.

Table 5-19: Ranking of CO Control Technologies for the Hot Water Boilers

Control Technology	Reduction (%)	Controlled Emission Level (lb/MMBtu)
Oxidation Catalyst	50	0.041
Combustion Control	N/A (baseline)	0.082

5.9.2.4 Step 4. Evaluate the Most Effective Control Technologies

Each technically feasible control technology was evaluated for energy, environmental, and economic impacts. These impacts are discussed below for each control technology.

5.9.2.4.1 Oxidation Catalyst

Energy and Environmental Impacts

Energy and environmental impacts of an oxidation catalyst are discussed in Section 5.2.4.2.

Economic Impacts

The control cost analysis for an oxidation catalyst system for the hot water boiler is displayed in Appendix D. An oxidation catalyst system for this size unit would require a total capital investment of $23,560. On an annual basis the oxidation catalyst would cost $41,420 which results in a cost per ton of CO and VOC removed of $36,370 while removing only 1.1 tons of CO emissions per year.

The cost is considered economically infeasible; therefore, an oxidation catalyst for control of CO emissions from the hot water boilers is not considered BACT.

5.9.2.4.2 Combustion Control

Combustion control is accomplished through operation of the hot water boilers, there are no incremental energy, environmental, or economic impacts associated with these controls.

5.9.2.5 Step 5. Proposed Carbon Monoxide Hot Water Boilers BACT Determination

Since add-on controls are not feasible on such a small gas-fired unit, combustion control was selected as BACT for CO from the hot water boilers at an emission rate of 0.082 lb/MMBtu.

BACT for CO emissions from the hot water boilers is good combustion practices.

5.9.3 BACT for Particulate Matter – Hot Water Boilers

The following sections outline the top-down steps for $PM/PM_{10}/PM_{2.5}$ emissions from the hot water boilers.

5.9.3.1 Step 1. Identify Potential Control Strategies

The RBLC does not list any control strategies other than good combustion practices and low ash fuel (natural gas). No add-on controls were identified for significant removal of these pollutants from the hot water boiler exhaust.

5.9.3.2 Step 2. Identify Technically Feasible Control Technologies

The only technically feasible control option is combustion control for $PM/PM_{10}/PM_{2.5}$.

5.9.3.3 Step 3. Rank the Technically Feasible Control Technologies

The only technically feasible control option is combustion control for $PM/PM_{10}/PM_{2.5}$.

5.9.3.4 Step 4 and 5. Evaluate the Most Effective Control Technologies and Proposed Particulate Matter Hot Water Boilers BACT Determination

Since add-on controls are not feasible on such a small gas-fired unit, combustion control was selected as BACT for $PM/PM_{10}/PM_{2.5}$ from the hot water boilers at an emission rate of 0.0075 lb/MMBtu.

5.9.4 BACT for Sulfur Dioxide – Hot Water Boilers (Steps 1-5)

There are no add-on control technologies for controlling SO_2 emissions from hot water boilers. As with the combustion turbines, using low sulfur fuel and controlling combustion is the only technologically feasible control option.

BACT is use of lower sulfur fuel (pipeline quality natural gas) and good combustion practices.

5.9.5 BACT for Sulfuric Acid Mist – Hot Water Boilers (Steps 1-5)

There are no add-on control technologies for controlling H_2SO_4 emissions from hot water boilers. As with the combustion turbines, using low sulfur fuel and controlling combustion is the only technologically feasible control option.

BACT is the use of lower sulfur fuel (pipeline quality natural gas) and good combustion practices.

5.9.6 BACT for Volatile Organic Compounds – Hot Water Boilers

The following sections outline the top-down steps for VOC emissions from hot water boilers.

5.9.6.1 Step 1. Identify Potential Control Strategies

The RBLC does not list add-on controls for hot water boilers in the BACT determinations for control of VOC emissions. As with the combustion turbines, good combustion control will help control emissions of VOC from the hot water boilers.

5.9.6.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling VOC emissions are evaluated for technical feasibility in the following sections.

5.9.6.2.1 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas to confirm that there is enough oxygen present for complete combustion.

Good combustion practices are a technically feasible method of controlling VOC emissions from the hot water boilers.

5.9.6.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible VOC control technologies for the 6.0 MMBtu/hr hot water boilers is ranked by control effectiveness in Table 5-20.

Table 5-20: Ranking of VOC Control Technologies for the Hot Water Boilers

Control Technology	Reduction (%)	Controlled Emission Level (lb/MMBtu)
Oxidation Catalyst	50	0.0027
Combustion Control	N/A (baseline)	0.0054

5.9.6.4 Step 4. Evaluate the Most Effective Control Technologies

Each technically feasible control technology was evaluated for energy, environmental, and economic impacts. These impacts are discussed below for each control technology.

5.9.6.4.1 Oxidation Catalyst

Energy and Environmental Impacts

Energy and environmental impacts of an oxidation catalyst are discussed in Section 5.6.4.2.

Economic Impacts

The control cost analysis for an oxidation catalyst system for the hot water boiler is displayed in Appendix D. An oxidation catalyst system for this size unit would require a total capital investment of $23,560. On an annual basis the oxidation catalyst would cost $41,420 which results in a cost per ton of CO and VOC removed of $36,370 while removing only 0.07 tons of VOC emissions per year.

The cost is considered economically infeasible; therefore, an oxidation catalyst for control of VOC emissions from the hot water boilers is not considered BACT.

5.9.6.4.2 Combustion Control

Combustion control is accomplished through operation of the hot water boilers, there are no incremental energy, environmental, or economic impacts associated with these controls.

5.9.6.5 Step 5. Proposed Volatile Organic Compounds Hot Water Boilers BACT Determination

Since add-on controls are not feasible on such a small gas-fired unit, combustion control was selected as BACT for VOC from the hot water boilers at an emission rate of 0.0054 lb/MMBtu.

BACT for VOC emissions from the hot water boilers is good combustion practices.

5.9.7 BACT for Greenhouse Gases – Hot Water Boilers (Steps 1-5)

The hot water boilers as proposed will be fired exclusively on natural gas and used to pre-heat natural gas fuel prior to combustion in the turbines. The units are each rated at approximately 6.0 MMBtu/hr and will be permitted to be fired a total of 8,760 hours per year each. GHG emissions from this unit are based on the emission rates provided in 40 CFR Part 98, 117.1 lb CO_2e/MMBtu. These GHG emissions are also *de minimis*, when compared to the turbine GHG emissions or the facility total GHG emissions. The basic GHG BACT reasoning presented for the turbines essentially applies to these hot water boilers as well. IPL proposes that GHG BACT for these units will be the following:

- Use of clean fuels (exclusive use of natural gas)

- Requiring IPL to maintain the unit according to the manufacturer's specifications, and to operate the unit in the most efficient manner possible, i.e. good combustion practices
- Tune the units every two years according to the manufacturer's specifications
- Record the annual hours of operation and annual fuel use and report the GHG emissions annually. The GHG emissions from this unit may be included in the facility-wide annual GHG limit.

5.10 BACT Analysis for Emergency Diesel Generator

One 2,000-kW emergency diesel generator will be installed for the Project. The emergency diesel generator will be limited to 500 hours per year of non-emergency use (100 of those hours are for testing and maintenance purposes) and will utilize ultra-low sulfur transportation grade distillate fuel oil, with a sulfur content of no more than 0.0015 percent by weight. The emergency diesel generator will comply with the applicable NSPS requirements. The RBLC has limited information on BACT conclusions for small engines such as the emergency diesel generator (Appendix C). The RBLC tables also show high variability for emission rates for each pollutant. For all pollutants, no add-on controls were listed in the RBLC for engines of comparable size.

BACT can be no less stringent than the NSPS Subpart IIII limits, which are discussed in Section 4.2.3.

5.10.1 BACT for Nitrogen Oxides – Emergency Diesel Generator

The following sections outline the top-down steps for NO_x emissions from the emergency diesel generator.

5.10.1.1 Step 1. Identify Potential Control Strategies

The emergency diesel generator will only operate 100 hours per year for testing and maintenance as described. In a true emergency, the diesel generator could operate up to 500 hours. Post-combustion control technologies (such as SCR) would be technically feasible for application to these diesel-fired emergency engines but are generally not required for emergency units. For typical testing and maintenance, the generator would only operate up to an hour at a time, and the SCR would barely be up to temperature to control emissions. Further, there are issues with starting up the emergency generator quickly with an SCR and the emergency generator must be able to start up quickly in a true emergency. With an SCR, the generator would not be able to operate as quickly as needed to assist with the emergency on the site. Additionally, the fuel oil that is combusted would quickly poison and/or foul an SCR catalyst in a short amount of operating time. Therefore, SCR is considered not technically feasible for the emergency generator.

5.10.1.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling NO_x emissions are evaluated for technical feasibility in the following sections.

5.10.1.2.1 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas to confirm that there is enough oxygen present for complete combustion.

As a result, combustion control is considered baseline for the emergency diesel generator and is technically feasible.

5.10.1.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible NO_x control technologies for the emergency diesel generator are ranked by control effectiveness in Table 5-21.

Table 5-21: Ranking of NO_x Control Technologies for the Emergency Diesel Generator

Control Technology	Reduction (%)	Controlled Emission Level (g/kW-hr)
Combustion Control	Not applicable (baseline)	6.4

5.10.1.4 Step 4. Evaluate the Most Effective Control Technologies

Each technically feasible control technology was evaluated for energy, environmental, and economic impacts. These impacts are discussed below for each control technology.

5.10.1.4.1 Combustion Control

Combustion control is accomplished through operational control of the engines; therefore, there are no energy, environmental, or economic impacts associated with this control.

5.10.1.5 Step 5. Proposed NO_x Emergency Diesel Generator BACT Determination

Combustion control was selected as BACT for NO_x for the emergency diesel generator; add-on controls are not practical on a unit this size, with limited operation. The emergency diesel generator will be able to achieve an emission rate of 6.4 g/kW-hr of NO_x emissions on an on-going basis. This emission rate is for emergency use units from NSPS IIII, which restricts testing and maintenance to 100 hours per year. Therefore, 6.4 g/kW-hr is the BACT emission limitation for NO_x emissions from the emergency diesel generator. Compliance will be demonstrated by purchasing a certified generator.

5.10.2 BACT for Carbon Monoxide – Emergency Diesel Generator

The following sections outline the top-down steps for CO emissions from the emergency diesel generator.

5.10.2.1 Step 1. Identify Potential Control Strategies

For an engine that only operates 100 hours per year for testing and maintenance as described, an oxidation catalyst is not considered feasible. Similarly to the SCR for an emergency engine, the unit will typically only be started up for testing and maintenance for a short amount of time (typically up to one hour). The oxidation catalyst would barely be up to operating temperature before the engine would be shutdown for testing. Additionally, there have been issues noted with starting up quickly in a true emergency with add-on controls. The fuel oil that is combusted would quickly poison and/or foul the oxidation catalyst in a short amount of operating time. Therefore, for these reasons, the only feasible control is combustion control.

5.10.2.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling CO emissions are evaluated for technical feasibility in the following sections.

5.10.2.2.1 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas, helping maintain adequate oxygen levels for complete combustion.

Good combustion practices are a technically feasible method of controlling CO emissions from the emergency diesel generator and therefore are BACT for the emergency diesel generator.

5.10.2.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible CO control technologies for the emergency diesel generator are ranked by control effectiveness in Table 5-22.

Table 5-22: Ranking of CO Control Technologies for the Emergency Diesel Generator

Control Technology	Reduction (%)	Controlled Emission Level (g/kW-hr)
Combustion Control	Not applicable (baseline)	3.5

5.10.2.4 Step 4. Evaluate the Most Effective Control Technologies

Each technically feasible control technology was evaluated for energy, environmental, and economic impacts. These impacts are discussed below for each control technology.

5.10.2.4.1 Combustion Control

Combustion control is accomplished through operational control of the engines; therefore, there are no energy, environmental, or economic impacts associated with this control.

5.10.2.5 Step 5. Proposed CO Emergency Diesel Generator BACT Determination

Combustion control was selected as BACT for CO for the emergency diesel generator; add-on controls are not practical on a unit this size, with limited operation, and the economic impacts are high. The emergency diesel generator will be able to achieve 3.5 g/kW-hr of CO emissions on an on-going basis. This emission rate is for emergency use units from NSPS IIII, which also restricts the engine to only 100 hours per year for testing and maintenance. **Therefore, 3.5 g/kW-hr is the BACT emission limitation for CO emissions from the emergency diesel generator.** Compliance will be demonstrated by purchasing a certified generator.

5.10.3 BACT for Particulate Matter – Emergency Diesel Generator

The following sections outline the top-down steps for $PM/PM_{10}/PM_{2.5}$ emissions from the emergency diesel generator.

5.10.3.1 Step 1. Identify Potential Control Strategies

The RBLC does not list any control strategies other than good combustion practices and low ash fuel (ultra-low sulfur diesel) for the emergency diesel generator. No add-on controls were identified for significant removal of these pollutants from the engine's exhaust.

5.10.3.2 Step 2. Identify Technically Feasible Control Technologies

The only technically feasible control option is combustion control for $PM/PM_{10}/PM_{2.5}$.

5.10.3.2.1 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas, helping maintain adequate oxygen levels for complete combustion.

Good combustion practices are a technically feasible method of controlling PM/PM_{10}/$PM_{2.5}$ emissions from the emergency diesel generator and therefore are BACT for the generator.

5.10.3.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible PM/PM_{10}/$PM_{2.5}$ control technologies for the emergency diesel generator are ranked by control effectiveness in Table 5-23.

Table 5-23: Ranking of PM/PM_{10}/$PM_{2.5}$ Control Technologies for the Emergency Diesel Generator

Control Technology	Reduction (%)	Controlled Emission Level (g/kW-hr)
Combustion Control	Not applicable (baseline)	0.2

5.10.3.4 Steps 4. Evaluate the Most Effective Control Technologies

Since no add-on controls were identified, combustion control with low ash fuel was selected as BACT for PM/PM_{10}/$PM_{2.5}$ at an emission rate of 0.2 g/kW-hr for the emergency diesel generator.

5.10.3.5 Step 5. PM/PM_{10}/$PM_{2.5}$ BACT Emission Limitation

Combustion control was selected as BACT for PM/PM_{10}/$PM_{2.5}$ for the emergency diesel generator; add-on controls are not practical on a unit this size and with limited operation. The emergency diesel generator will be able to achieve 0.2 g/kW-hr of PM/PM_{10}/$PM_{2.5}$ emissions on an on-going basis. This emission rate is for emergency use units from NSPS IIII, which also restricts to no more than 100 hours for testing and maintenance. **Therefore, 0.2 g/kW-hr is the BACT emission limitation for PM/PM_{10}/$PM_{2.5}$ emissions from the emergency diesel generator.** Compliance will be demonstrated by purchasing a certified generator.

5.10.4 BACT for Sulfur Dioxide – Emergency Diesel Generator (Steps 1-5)

There are no add-on control technologies for controlling SO_2 emissions from an emergency diesel generator. As with the combustion turbines, using low sulfur fuel and controlling combustion is the only technologically feasible control option.

BACT is use of lower sulfur fuel (ULSD) and good combustion practices.

5.10.5 BACT for Sulfuric Acid Mist – Emergency Diesel Generator (Steps 1-5)

There are no add-on control technologies for controlling H_2SO_4 emissions from the emergency diesel generator. As with the combustion turbines, using ultra-low sulfur diesel and controlling combustion is the only technologically feasible control option.

BACT is use of lower sulfur fuel (ULSD) and good combustion practices.

5.10.6 BACT for Volatile Organic Compounds – Emergency Diesel Generator

The following sections outline the top-down steps for VOC emissions from the emergency diesel generator.

5.10.6.1 Step 1. Identify Potential Control Strategies

For an engine that only operates up to 100 hours per year for testing and maintenance, add-on controls are not considered feasible. Section 5.10.2.1 for CO emissions describes the reasons that oxidation catalyst is not considered technically feasible for the emergency generator.

5.10.6.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling VOC emissions are evaluated for technical feasibility in the following sections.

5.10.6.2.1 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas, helping maintain adequate oxygen levels for complete combustion.

Good combustion practices are a technically feasible method of controlling VOC emissions from the emergency diesel generator and therefore are BACT for the generator.

5.10.6.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible VOC control technologies for the emergency diesel generator are ranked by control effectiveness in Table 5-24.

Table 5-24: Ranking of VOC Control Technologies for the Emergency Diesel Generator

Control Technology	Reduction (%)	Controlled Emission Level (lb/hp-hr)
Combustion Control	N/A (baseline)	7.05×10^{-4}

5.10.6.4 Step 4. Evaluate the Most Effective Control Technologies

Each technically feasible control technology was evaluated for energy, environmental, and economic impacts. These impacts are discussed below for each control technology.

5.10.6.4.1 Combustion Control

Combustion control is accomplished through operational control of the engines; therefore, there are no energy, environmental, or economic impacts associated with this control.

5.10.6.5 Step 5. Proposed Volatile Organic Compounds Diesel Emergency Generator BACT Determination

Combustion control was selected as BACT for VOC for the emergency diesel generator; add-on controls are not practical on these small units with limited operation and the economic impacts are high. The emergency diesel generator will be able to achieve 7.05×10^{-4} lb/hp-hr during steady state operations. Therefore, the BACT emission limitation for VOC emissions from the emergency diesel generator is 7.05×10^{-4} lb/hp-hr. Compliance will be demonstrated by purchasing a certified generator.

5.10.7 BACT for Greenhouse Gases – Emergency Diesel Generator (Steps 1-5)

The emergency diesel generator is proposed to be used for no more than 500 hours per year in a true emergency and would be limited to 100 hours per year for testing and maintenance. The design of the engine is dictated by the manufacturer, not by the end-user. As such, the Project is limited to commercially available options, which include the engine meeting EPA certified emergency engine requirements.

Consistent with its rationale for the BACT determination for greenhouse gas emissions from the combustion turbine, BACT for the emergency diesel generator involves selection of the most efficient stationary emergency diesel generator that can meet the facility's needs. Total greenhouse gas emissions from the emergency diesel generator are estimated at 893 tons CO_2e per year. These greenhouse gas emissions are also *de minimis* when compared to the turbine greenhouse gas emissions.

A certified emergency engine is the most fuel-efficient option for these purposes. Further, because emissions of greenhouse gases are directly correlated to operation of the unit, BACT requires that the engine shall only be operated for maintenance, readiness testing, and during emergencies and other periods authorized by the permitting agency and/or the permit.

Because operation of the emergency diesel generator will be limited by permit conditions for reliability- and maintenance-related activities and IPL will be required to keep records of the operation of the emergency diesel generators and its fuel usage. Therefore, IPL believes no additional conditions are required to enforce this greenhouse gas BACT determination.

5.11 BACT Analysis for Emergency Fire Pump

One 422-hp emergency diesel fire pump will be installed for the Project. The emergency diesel fire pump will be limited to 500 hours per year of non-emergency use (100 of those hours are for testing and maintenance purposes) and will utilize ultra-low sulfur transportation grade distillate fuel oil, with a sulfur content of no more than 0.0015 percent by weight. The emergency diesel fire pump will comply with the applicable NSPS requirements. The RBLC has limited information on BACT conclusions for small engines such as the emergency diesel fire pump (Appendix C). The RBLC tables also show high variability for emission rates for each pollutant. For all pollutants, no add-on controls were listed because the add-on controls were determined to not be economically feasible due to engine size.

BACT can be no less stringent than the NSPS Subpart IIII limits, which are discussed in Section 4.2.3.

5.11.1 BACT for Nitrogen Oxides – Emergency Fire Pump

The following sections outline the top-down steps for NO_x emissions from the emergency diesel fire pump.

5.11.1.1 Step 1. Identify Potential Control Strategies

The emergency fire pump will only operate 100 hours per year for testing and maintenance as described. In a true emergency, the diesel fire pump could operate up to 500 hours. Post-combustion control technologies (such as SCR) would be technically feasible for application to the diesel-fired emergency fire pump but is generally not required for emergency units. For typical testing and maintenance, the fire pump would only operate up to an hour at a time, and the SCR would barely be up to temperature to control emissions. Further, there are issues with starting up the emergency fire pump quickly with an SCR and the emergency fire pump must be able to start up quickly in a true emergency. With an SCR, the fire pump would not be able to operate as quickly as needed to assist with the emergency on the site. Additionally, the fuel oil that is combusted would quickly poison and/or foul an SCR catalyst in a short amount of operating time. Therefore, SCR is considered not technically feasible for the emergency fire pump.

5.11.1.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling NO_x emissions are evaluated for technical feasibility in the following sections.

5.11.1.2.1 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas to confirm that there is enough oxygen present for complete combustion.

As a result, combustion control is considered baseline for the emergency diesel fire pump and is technically feasible.

5.11.1.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible NO_x control technologies for the emergency diesel fire pump are ranked by control effectiveness in Table 5-25.

Table 5-25: Ranking of NO_x Control Technologies for the Emergency Fire Pump

Control Technology	Reduction (%)	Controlled Emission Level (g/kW-hr)
Combustion Control	Not applicable (baseline)	4.0

5.11.1.4 Step 4. Evaluate the Most Effective Control Technologies

Each technically feasible control technology was evaluated for energy, environmental, and economic impacts. These impacts are discussed below for each control technology.

5.11.1.4.1 Combustion Control

Combustion control is accomplished through operational control of the fire pump; therefore, there are no energy, environmental, or economic impacts associated with this control.

5.11.1.5 Step 5. Proposed NO_x Emergency Diesel Fire Pump BACT Determination

Combustion control was selected as BACT for NO_x for the emergency diesel fire pump; add-on controls are not practical on a unit this size, with limited operation. The emergency diesel fire pump will be able to achieve 4.0 g/kW-hr of NO_x emissions on an on-going basis. This emission rate is for emergency use units from NSPS IIII, which restricts testing and maintenance to 100 hours per year. Therefore, 4.0 g/kW-hr is the BACT emission limitation for NO_x emissions from the emergency diesel fire pump. Compliance will be demonstrated by purchasing a certified engine.

5.11.2 BACT for Carbon Monoxide – Emergency Fire Pump

The following sections outline the top-down steps for CO emissions from the emergency diesel fire pump.

5.11.2.1 Step 1. Identify Potential Control Strategies

For an engine that only operates 100 hours per year for testing and maintenance as described, an oxidation catalyst is not considered feasible. Similarly to the SCR for an emergency engine, the unit will typically only be started up for testing and maintenance for a short amount of time (typically up to one hour). The oxidation catalyst would barely be up to operating temperature before the engine would be shutdown for testing. Additionally, there have been issues noted with starting up quickly in a true emergency with add-on controls. The fuel oil that is combusted would quickly poison and/or foul the oxidation catalyst in a short amount of operating time. Therefore, for these reasons, the only feasible control is combustion control.

5.11.2.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling CO emissions are evaluated for technical feasibility in the following sections.

5.11.2.2.1 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas to confirm that there is enough oxygen present for complete combustion.

As a result, combustion control is considered baseline for the emergency diesel fire pump and is technically feasible.

5.11.2.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible CO control technologies for the emergency diesel fire pump are ranked by control effectiveness in Table 5-26.

Table 5-26: Ranking of CO Control Technologies for the Emergency Fire Pump

Control Technology	Reduction (%)	Controlled Emission Level (g/kW-hr)
Combustion Control	Not applicable (baseline)	3.5

5.11.2.4 Step 4. Evaluate the Most Effective Control Technologies

Each technically feasible control technology was evaluated for energy, environmental, and economic impacts. These impacts are discussed below for each control technology.

5.11.2.4.1 Combustion Control

Combustion control is accomplished through operational control of the engine; therefore, there are no energy, environmental, or economic impacts associated with this control.

5.11.2.5 Step 5. Proposed CO Emergency Fire Pump BACT Determination

Combustion control was selected as BACT for CO for the emergency diesel fire pump; add-on controls are not practical on this small unit with limited operation and economic impacts are high. The emergency diesel fire pump will be able to achieve 3.5 g/kW-hr of CO emissions on an on-going basis. This emission rate is for emergency use units from NSPS IIII, which also restricts the engine to only 100 hours per year for testing and maintenance. Therefore, 3.5 g/kW-hr is the BACT emission limitation for CO emissions from the emergency diesel fire pump. Compliance will be demonstrated by purchasing a certified engine.

5.11.3 BACT for Particulate Matter – Emergency Fire Pump

The following sections outline the top-down steps for $PM/PM_{10}/PM_{2.5}$ emissions from the emergency diesel fire pump.

5.11.3.1 Step 1. Identify Potential Control Strategies

The RBLC does not list any control strategies other than good combustion practices and low ash fuel (natural gas) for the emergency diesel fire pump. Vendors have stated there is no precedent for a particulate filter on the emergency diesel fire pump; therefore, a diesel particulate filter is considered experimental control technology not viable for the diesel fire pump.

No add-on controls were identified for significant removal of these pollutants from the engine's exhaust.

5.11.3.2 Step 2. Identify Technically Feasible Control Technologies

The only technically feasible control option is combustion control for $PM/PM_{10}/PM_{2.5}$.

5.11.3.2.1 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas, helping maintain adequate oxygen levels for complete combustion.

Good combustion practices are a technically feasible method of controlling $PM/PM_{10}/PM_{2.5}$ emissions from the emergency fire pump and therefore are BACT for the fire pump.

5.11.3.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible $PM/PM_{10}/PM_{2.5}$ control technologies for the emergency diesel fire pump are ranked by control effectiveness in Table 5-27.

Table 5-27: Ranking of $PM/PM_{10}/PM_{2.5}$ Control Technologies for the Emergency Fire Pump

Control Technology	Reduction (%)	Controlled Emission Level (g/kW-hr)
Combustion Control	Not applicable (baseline)	0.2

5.11.3.4 Step 4. Evaluate the Most Effective Control Technologies

Since no add-on controls were identified, combustion control with low ash fuel was selected as BACT for $PM/PM_{10}/PM_{2.5}$ at an emission rate of 0.2 g/kW-hr for the emergency diesel fire pump.

5.11.3.5 Step 5. $PM/PM_{10}/PM_{2.5}$ BACT Emission Limitation

Combustion control was selected as BACT for $PM/PM_{10}/PM_{2.5}$ for the emergency fire pump; add-on controls are not practical on a unit this size and with limited operation. The emergency fire pump will be able to achieve 0.2 g/kW-hr of $PM/PM_{10}/PM_{2.5}$ emissions on an on-going basis. This emission rate is for emergency use units from NSPS IIII, which also restricts to no more than 100 hours for testing and maintenance. **Therefore, 0.2 g/kW-hr is the BACT emission limitation for $PM/PM_{10}/PM_{2.5}$ emissions from the emergency fire pump.** Compliance will be demonstrated by purchasing a certified engine.

5.11.4 BACT for Sulfur Dioxide – Emergency Fire Pump (Steps 1-5)

There are no add-on control technologies for controlling SO_2 emissions from an emergency diesel fire pump. As with the combustion turbines, using low sulfur fuel and controlling combustion is the only technologically feasible control option.

BACT is use of lower sulfur fuel (ULSD) and good combustion practices.

5.11.5 BACT for Sulfuric Acid Mist – Emergency Fire Pump (Steps 1-5)

There are no add-on control technologies for controlling H_2SO_4 emissions from the emergency diesel fire pump. As with the combustion turbine, using low sulfur fuel and controlling combustion is the only technologically feasible control option.

BACT is use of lower sulfur fuel (ULSD) and good combustion practices.

5.11.6 BACT for Volatile Organic Compounds – Emergency Fire Pump

The following sections outline the top-down steps for VOC emissions from the emergency diesel fire pump.

5.11.6.1 Step 1. Identify Potential Control Strategies

For an engine that only operates up to 100 hours per year for testing and maintenance, add-on controls are not considered feasible. Section 5.11.2.1 for CO emissions describes the reasons that oxidation catalyst is not considered technically feasible for the emergency diesel fire pump.

5.11.6.2 Step 2. Identify Technically Feasible Control Technologies

The primary methods for controlling VOC emissions are evaluated for technical feasibility in the following sections.

5.11.6.2.1 Combustion Control

"Good combustion practices" include operational and design elements to control the amount and distribution of excess air in the flue gas to confirm that there is enough oxygen present for complete combustion.

Good combustion practices are a technically feasible method of controlling VOC emissions from the emergency diesel fire pump and therefore are BACT for the fire pump.

5.11.6.3 Step 3. Rank the Technically Feasible Control Technologies

The technically feasible VOC control technologies for the emergency diesel fire pump are ranked by control effectiveness in Table 5-28.

Table 5-28: Ranking of VOC Control Technologies for the Emergency Fire Pump

Control Technology	Reduction (%)	Controlled Emission Level (lb/hp-hr)
Combustion Control	N/A (baseline)	0.0025

5.11.6.4 Step 4. Evaluate the Most Effective Control Technologies

Each technically feasible control technology was evaluated for energy, environmental, and economic impacts. These impacts are discussed below for each control technology.

5.11.6.4.1 Combustion Control

Combustion control is accomplished through operational control of the engines; therefore, there are no energy, environmental, or economic impacts associated with this control.

5.11.6.5 Step 5. Proposed Volatile Organic Compounds Emergency Fire Pump BACT Determination

Combustion control was selected as BACT for VOC for the emergency diesel fire pump; add-on controls are not practical on these small units with limited operation and economic impacts are high. The emergency diesel fire pump will be able to achieve 0.0025 lb/hp-hr during steady state operations. Therefore, the BACT emission limitation for VOC emissions from the emergency fire pump is 0.0025 lb/hp-hr. Compliance will be demonstrated by purchasing a certified engine.

5.11.7 BACT for Greenhouse Gases – Emergency Fire Pump (Steps 1-5)

The emergency fire pump is proposed to be used for no more than 500 hours per year in a true emergency, but will typically only be testing/maintained for operations for up to 100 hours per year. The design of the engine is dictated by the manufacturer, not by the end-user. As such, the Project is limited to commercially available options.

Consistent with its rationale for the BACT determination for greenhouse gas emissions from the combustion turbines, BACT for the emergency fire pump involves selection of the most efficient stationary emergency diesel fire pump that can meet the facility's needs. Total greenhouse gas emissions from the emergency diesel fire pump are estimated at 120 tons CO_2e per year. These greenhouse gas emissions are also *de minimis* when compared to the combustion turbine greenhouse gas emissions.

Further, because emissions of greenhouse gases are directly correlated to operation of the unit, BACT requires that the fire pump shall only be operated for maintenance, readiness testing, and during emergencies and other periods authorized by the permitting agency and/or the permit.

Because operation of the emergency diesel fire pump will be limited by permit conditions for reliability-and maintenance related activities and IPL will be required to keep records of the operation of the emergency diesel fire pump and its fuel usage. Therefore, IPL believes no additional conditions are required to enforce this greenhouse gas BACT determination.

5.12 BACT for Particulate Matter ($PM/PM_{10}/PM_{2.5}$) – Haul Road Fugitives

Haul roads will be located onsite and delivery truck traffic will travel on paved roads. Emissions of particulate matter will be filterable only and speciated into PM, PM_{10}, and $PM_{2.5}$. However, control technologies will control all sizes of particulate.

5.12.1 Step 1. Identify Potential Control Strategies

In a review of the RBLC, the following control technologies for particulate emissions from roads were identified:

1. Chemical dust suppression and surfactant application,
2. Watering, sweeping and vacuuming,
3. Paving, and
4. Traffic and speed restrictions

5.12.2 Step 2. Identify Technically Feasible Control Technologies

All of the options listed, except chemical dust suppression and surfactant application, are potentially applicable control technologies considered technically feasible for the Project. Chemical dust suppression and surfactant application are generally used for unpaved surface and are considered infeasible for this Project as the facility roads will be paved.

5.12.3 Step 3. Rank the Technically Feasible Control Technologies

The third step in the BACT analysis is to rank the remaining control technologies in order of control effectiveness. Table 5-29 provides a listing of $PM/PM_{10}/PM_{2.5}$ control technologies by effectiveness.

Table 5-29: Efficiency Ranking of Particulate Control Technologies for Haul Roads

Control Technology	Approximate Control Efficiency (percent)
Water flushing followed by sweeping of paved roads	up to 96
Water flushing of paved roads	up to 69
Vacuum sweeping of paved roads	up to 58
Paving	--
Speed/traffic restrictions	--

Source: EPA Control of Open Fugitive Dust Sources

5.12.4 Step 4. Evaluate Most Effective Control Technologies

The fourth step in the BACT analysis is to evaluate the most effective control technology based on energy, environmental, and economic impacts.

5.12.4.1 Water Flushing and/or Vacuum Sweeping of Paved Roads

Energy and Environmental Impacts

Besides the additional use of fuel and water, there are not many energy impacts associated with water flushing paved roads. However, utilizing an additional truck on the haul roads to water the roads and/or vacuum sweep will create more particulate in the air, since the emissions are based on the number of vehicles travelling on the haul roads.

Economic Impacts

The control cost analysis for the water truck is displayed in Appendix D. Per research of current market prices of water trucks, a new water truck can cost upwards of $100,000. Using the conservative estimate of $100,000 as the initial capital cost, owning and operating a water truck costs approximately $11,700 annually, with a cost per ton of $PM/PM_{10}/PM_{2.5}$ value of $339,818, while controlling 0.035 tons per year of PM. Therefore, for this Project, use of a water truck is not considered economically feasible. Further, vacuum sweeping trucks are at least or even more expensive than a water truck, thus vacuum sweeping on the haul roads is also considered not economically feasible.

5.12.4.2 Fugitive Dust Control Plan

Based on a review of the RBLC, the implementation of a Fugitive Dust Control Plan is considered a control method accepted as BACT for particulate emissions from roads at similar facilities. No specific BACT emission limits associated with the previously mentioned control methods were obtained from the RBLC.

5.12.5 Step 5. Select BACT

IPL proposes to develop, maintain, and implement a Fugitive Dust Control Plan as BACT for the paved roads. The plan will establish best management practices to minimize fugitive dust emissions, measures such as, but not limited to:

- Maintaining regular schedules for cleaning paved roads
- Conducting visibility emission observations to identify dust sources
- Taking immediate corrective action when fugitive dust is observed, such as applying water or chemical suppressants or reducing/halting truck traffic to mitigate emissions.

5.13 BACT for Greenhouse Gases (GHG) – Natural Gas Fugitives

The Project will include natural gas piping components from the natural gas line that will enter the Project site to provide gas for the combustion turbine, hot water boilers, and comfort building heaters. These natural gas piping components are potential sources of methane emissions due to emissions from valves, flanges, and open ended lines.

Methane is not a VOC but is regulated as a GHG with a GWP of 28 when expressed as CO_2e.

5.13.1 Step 1. Identify Potential Control Strategies

Greenhouse gas emissions (methane) and VOCs may leak out of certain components within the pipeline system, anywhere there is a connection, valve or flange. Per a review of the RBLC database (Appendix C), the following technologies were identified as potential control options for these piping fugitives:

- Physical inspection: an audio/visual/olfactory (AVO) leak detection program
- Good operating processes
- Certified low-leaking valves

5.13.2 Step 2. Identify Technically Feasible Control Technologies

A leak detection program based on AVO monitoring is determined to be partially feasible. Audio and visual inspection of the piping components will be possible for the piping that is above ground. Additionally, good operating practices and certified low-leaking valves are also feasible for natural gas fugitive emissions. **Therefore, the audio and visual inspections, good operating practices, and certified low-leaking valves listed in Step 1 are technically feasible for natural gas.**

5.13.3 Step 3. Rank the Technically Feasible Control Technologies

Leak detection programs are used to inspect fugitive components to identify leaks. Leaks identified by the inspections are then repaired within a specified time period, thus reducing the emissions.

The top-ranked control strategy is an audio and visual monitoring program that utilizes physical inspections. Based on available data, piping components are generally assigned control efficiencies ranging from 30 to 97 percent for valves, relief valves, and sampling connections (Texas Council on Environmental Quality [TCEQ], 2018).

Certified low-leaking valves are a remaining control technology with 80 percent control of CO_2e.

Good operating processes are considered baseline for the purposes of this BACT analysis. Table 5-30 summarizes the control efficiencies for the various control technology options.

Table 5-30: GHG and VOC Technology Rankings for Natural Gas Fugitives

Rank	Control Technology	Percent Control (%)
1	Audio and visual detection	97
2	Certified low-leaking valves	80
3	Good operating process	Not applicable (baseline)

Source: TCEQ, 2018

5.13.4 Step 4. Evaluate Most Effective Control Technologies

The next step is to review each of the technically feasible control options for environmental, energy, and economic impacts.

5.13.4.1 Audio and Visual Detection (Fugitive Monitoring Plan)

A fugitive monitoring plan could be developed using audio and visual detection to identify potential leaks. The implementation of a plan would involve development costs and ongoing operational expenses - including personnel required to conduct inspections – these costs are considered economically feasible for the purposes of this analysis.

In addition to economic considerations, the plan may result in minor energy and environmental impacts, such as increased energy use for monitoring equipment or incidental emissions during inspections. However, these impacts are not considered technically infeasible.

5.13.4.2 Low-Leaking Valves

Low-leaking valves may be available for use in this project; however, the final design of the piping components has not yet been determined. If such valves are compatible with the selected design, they will be incorporated into the system. While the installation of low-leaking valves would result in additional costs, these are not considered economically infeasible for the purposes of this BACT analysis.

Furthermore, the use of low-leaking valves may introduce minor energy impacts due to pressure differentials. At this time, these impacts are not expected to be significant.

5.13.5 Step 5. Select BACT

Fugitive emissions will come from small leaks in equipment connections from the natural gas pipeline on IPL's property to support the Project sources at Morgan Valley Energy Center. Any GHG emissions from the piping components will be fugitive emissions. Fugitive emissions are, by their nature, very difficult to monitor directly, as they are not emitted from a discrete emission point. Since the uncontrolled CO_2e emissions from the natural gas piping represent less than 0.05 percent of the total sitewide CO_2e emissions, any emission control techniques applied to the piping fugitives will provide minimal additional CO_2e emission reductions over the baseline. Therefore, IPL proposes to create a monitoring plan (based on audio and visual detection) for the fugitive piping components under IPLs control to reduce GHG emissions. The proposed work practice is consistent with the BACT determinations identified above.

Based on the top-down analysis for natural gas, an audio and visual leak detection program developed by IPL is BACT for natural gas components.

5.14 BACT for Greenhouse Gases (GHG) – SF_6-Containing Circuit Breakers

SF_6 is a very potent GHG with a GWP of 23,500, which means that it is 23,500 times more potent as a GHG than CO_2. SF_6 is a gaseous dielectric used in circuit breakers. The Project is expected to have three 20-kV circuit breakers and three 345-kV circuit breakers that will all contain small amounts of SF_6. Leakage is expected to be minimal and is expected to occur only as a result of circuit interruption and at extremely low temperatures.

Emissions of SF_6 from the circuit breakers are shown in Appendix B. Annual potential to emit emissions of SF_6 from the circuit breakers were based on maximum leakage rate of 0.5 percent per year, the amount of SF_6 and in each size of circuit breaker, and the GWP. Project potential emissions of CO_2e leakage from all proposed circuit breakers combined are estimated to be 75.4 tons per year.

The following sections outline the top-down steps for GHG emissions from the circuit breakers.

5.14.1 Step 1 and Step 2. Identify Potential Control Strategies and Eliminate Technologically Infeasible Options

The first steps in a top-down BACT analysis are to determine the potential control strategies and then determine if the control strategy is technically feasible for the Project. There are no add-on control technologies for SF_6; only inherent controls are available. The following control strategies have been identified and considered in determining BACT for SF_6 emissions from circuit breakers:

1. Use state-of-the-art SF_6 technology with leak detection systems to limit fugitive emissions.

The use of state-of-the-art gas-filled circuit breakers using SF_6 with leak detection to limit fugitive emissions is the proposed control option. Modern circuit breakers are designed as a totally enclosed-pressure system with far lower potential for SF_6 emissions than older circuit breakers. The current International Electrotechnical Commission (IEC) standards are that new equipment be built to low leakage limits (less than 0.5 percent per year) (Blackman, et al., 2019). The effectiveness of these leak-tight closed systems is further enhanced by equipping them with an alarm that provides a warning when SF_6 has leaked from the breaker. **Therefore, this type of technology is available to limit emissions, is feasible for use, and is the baseline established for this BACT analysis.**

2. Substitution of another, non-greenhouse-gas substance for SF_6 such as the use of a different dielectric oil or compressed air (air-blast) circuit breaker as the dielectric material in the breakers.

Historically, circuit breakers used dielectric oil or compressed air (air-blast) as insulating and arc-quenching media prior to the widespread adoption of SF_6. These technologies were common in high-voltage applications, SF6 has become the industry standard due to its superior dielectric strength, compact equipment design, and reliable arc interruption capabilities.

Substituting oil or air-blast breakers for SF_6-based equipment is not technically feasible for this Project for several reasons:

- Oil breakers are no longer manufactured for new installations and are only available as used equipment, which does not meet modern reliability standards.
- Air-blast breakers, though historically used at 345 kV, are now obsolete and not commercially available for new projects. They require large footprints, complex pneumatic systems, and produce significant noise.
- For 20-kV applications, air-blast breakers are not offered by vendors and are incompatible with current design standards.

Based on vendor input and current market availability, neither oil nor air-blast circuit breakers are viable alternatives for the 20-kV or 345-kV breakers proposed in this Project. **Therefore, these technologies are not considered available or feasible control options for reducing SF_6-related greenhouse gas emissions.**

3. Use an emerging technology to replace SF_6 with a material that has similar dielectric and arc-quenching properties, but without the drawbacks of oil and air-blast breakers.

The availability of emerging technology alternatives to SF_6 was researched. According to the most recent report released by the EPA SF_6 Partnership, there is no clear alternative to SF_6 (EPA, 2015). Research and development efforts have been focused on finding substitutions for SF_6 that have comparable insulating and arc quenching properties in high-voltage applications (U.S. Climate Change Technology Program, 2003). Most studies have concluded "there is no replacement gas immediately available to use as an SF_6 substitute" for high-voltage applications (Siemens Industry, Inc., 2013). **Therefore, the alternative to use an emerging technology to replace SF_6 is not an available control technology.**

Table 5-31 displays the control options and feasibility for SF_6.

Table 5-31: Summary of Potential GHG Control Technologies

GHG Technology	Evaluation Status
State-of-the-art SF_6 technology with leak detection systems	Considered and applied
Oil/air-blast circuit breakers	Considered (Not Feasible)
Use of emerging technology to replace SF_6	Considered (Not Feasible)

5.14.2 Step 3. Rank the Technically Feasible Control Technologies

Table 5-32 presents the ranked technically feasible control options.

Table 5-32: GHG Technology Rankings for Circuit Breaker Equipment Leaks

Control Technology	Emission Rate (short tons CO_2e/year)	Emissions Reduction (short tons CO_2e/year)
State-of-the-art SF_6 technology with leak detection systems	75.4	N/A

5.14.3 Step 4. Evaluate the Most Effective Control Technologies

The next step is to review each of the technically feasible control options for environmental, energy, and economic impacts.

5.14.3.1 Environmental, Energy, and Economic Feasibility of Control Options

Purchasing leak detection systems for the circuit breakers will come with a cost: however, the costs are considered economically feasible for this Project.

5.14.4 Step 5. GHG BACT Emission Limitation

The proposed BACT for the circuit breakers consists of the following:

- State-of-the-art enclosed-pressure SF_6 circuit breakers with a guaranteed loss rate of 0.5 percent by weight or less by year; and
- Low-pressure detection system with alarm system

A review of the RBLC for circuit breakers containing SF_6 (most of them combined-cycle plants) have a similar or the same BACT determination. As shown in Appendix C, a leak detection rate of 0.5 percent from enclosed pressured design with leak detection alarms is BACT.

5.14.5 Compliance with GHG BACT for Circuit Breakers

Any SF_6 emissions from the circuit breakers will be fugitive emissions. Fugitive emissions are, by their nature, very difficult to monitor directly, as they are not emitted from a discrete emission point. Therefore, IPL proposes the following compliance demonstrations, recordkeeping and monitoring requirements:

- Follow manufacturer recommendations for maintenance and repair of the affected breakers, with recovery and recycling of SF_6 removed during maintenance procedures.
- Install a low-pressure detection system with an alarm system on each SF_6 circuit breaker to measure pressure changes.
- Create alarms based on the pressure readings in the breakers, so that leaks can be detected before a substantial portion of SF_6 is lost.
- Upon a detectable pressure drop that is 10 percent of the original pressure (accounting for ambient air conditions), perform maintenance on a breaker to fix seals within 20 days of the detection of the pressure drop.
- Keep a log of all detected leaks and maintenance procedures potentially affecting SF_6 emissions from circuit breakers that are part of this Project.
- For a period of at least 5 years, track and maintain records of annual SF_6 leakage amounts due to breakers that are part of this Project. The leakage amounts will be assumed equal to the inventory of SF_6 replaced in the breakers each calendar year.

These proposed work practices are consistent with the BACT determinations identified above.

5.15 BACT for Diesel Belly Tank

The following sections outline the top-down BACT steps for emissions of VOC from the diesel belly tank.

5.15.1 STEP 1. Identify Potential Control Strategies

The Project will include a 1,260-gallon diesel belly tank to support the emergency generator and a 465-gallon diesel belly tank for the emergency fire pump. These are factory-integrated, equipment-mounted tanks located beneath the respective units. Diesel fuel has a very low vapor pressure, resulting in minimal VOC emissions during storage and transfer. Emission control technologies designed for high vapor pressure liquids, such as floating roofs, are not affected or appropriate for diesel storage.

See Appendix C for RBLC query results. Fixed-roof tanks identified in the RBLC are generally associated with larger, standalone above ground storage tanks, rather than small, equipment-mounted diesel belly tanks. While submerged fill is listed in the RBLC, it is typically applied to larger tanks with higher throughput and emission potential.

5.15.2 STEP 2. Identify Technically Feasible Control Technologies

Use of diesel belly tanks integrated with the equipment package, fixed roof tanks and submerged fill tanks are discussed below for technical feasibility.

5.15.2.1 Fixed Roof Tank

Installing a fixed roof tank would require a standalone bulk storage tank and a custom fuel transfer system, increasing capital costs and operational complexity. A fixed roof tank is not standard practice for diesel storage associated with emergency equipment of this size. Therefore, the fixed roof tank is not considered technically feasible because this would also change the design of the facility by removing the belly tank and instead installing a larger stand-alone fixed roof tank. A fixed roof is not available for the small integrated belly tanks proposed for this project to support the emergency equipment.

5.15.2.2 Submerged Fill Tank

Installing submerged fill would require a standalone tank and a custom fuel transfer system, increasing capital costs and operational complexity. Given the low emission potential (7.68×10^{-4} tons/year), the environmental benefit does not justify the added cost. Submerged fill is also not standard practice for diesel belly tanks of this size. Therefore, submerged fill is not technically feasible for this project.

5.15.3 STEP 3. Rank the Technically Feasible Control Technologies

There are no feasible controls for the belly tanks.

5.15.4 STEP 4. Evaluate the Most Effective Control Technologies

The next step is to review each of the technically feasible control options for environmental, energy, and economic impacts. There are no technically feasible controls to evaluate in this step.

5.15.5 STEP 5. Select BACT

The proposed BACT for the diesel belly tanks is the use of diesel belly tanks integrated into the equipment package. This approach is consistent with industry standards and regulatory precedent for low-volatile fuels. Given the extremely low potential VOC emissions (7.68×10^{-4} tons/year), diesel belly tanks represent the most reasonable and cost-effective control technology.

5.16 BACT for Building Comfort Heaters

The Project will include approximately 40 natural gas fired comfort building heaters. Each heater is expected to be less than 10.0 MMBtu/hr in size. The RBLC has limited information on BACT conclusions for natural gas heaters (Appendix C). The RBLC tables also show high variability for emission rates for each pollutant. For all pollutants, very few entries with add-on controls were listed because the building comfort heaters are so small.

5.16.1 BACT for Nitrogen Oxides – Building Comfort Heaters

The following sections outline the top-down steps for NO_x emissions from the building comfort heaters.

5.16.1.1 Steps 1-5. Identify Control Strategies and Select BACT

Low-NO_x burners, along with combustion controls, are listed as BACT in the RBLC for natural gas heaters. Because these heaters are so small, low-NO_x burners and add-on controls, such as SCR are not available for these small heaters. Combustion control was selected as BACT for NO_x emissions from the comfort building heaters at an emission rate of 0.098 lb/MMBtu.

5.16.2 BACT for Carbon Monoxide –Comfort Building Heaters

The following sections outline the top-down steps for CO emissions from comfort building heaters.

5.16.2.1 Steps 1-5. Identify Control Strategies and Select BACT

Combustion controls are the only controls listed as BACT in the RBLC for natural gas heaters CO emissions. Because these heaters are so small, add-on controls, such as oxidation catalysts, are not available for these small heaters. Combustion control was selected as BACT for CO from the comfort building heaters at an emission rate of 0.082 lb/MMBtu.

5.16.3 BACT for Volatile Organic Compounds –Comfort Building Heaters

The following sections outline the top-down steps for VOC emissions from comfort building heaters.

5.16.3.1 Steps 1-5. Identify Control Strategies and Select BACT

Combustion controls are the only controls listed as BACT in the RBLC for natural gas heaters VOC emissions. Because these heaters are so small, add-on controls, such as oxidation catalysts, are not available for these small heaters. Combustion control was selected as BACT for VOC from the comfort building heaters at an emission rate of 0.0054 lb/MMBtu.

5.16.4 BACT for Sulfur Dioxide – Comfort Building Heaters (Steps 1-5)

There are no add-on control technologies for controlling SO_2 emissions from the comfort building heaters. As with the combustion turbines, using low sulfur fuel and controlling combustion is the only technologically feasible control option.

BACT is use of lower sulfur fuel (pipeline quality natural gas) and good combustion practices.

5.16.5 BACT for Sulfuric Acid Mist – Comfort Building Heaters (Steps 1-5)

There are no add-on control technologies for controlling H_2SO_4 mist emissions from the comfort building heaters. As with the combustion turbines, using low sulfur fuel and controlling combustion is the only technologically feasible control option.

BACT is the use of lower sulfur fuel (pipeline quality natural gas) and good combustion practices.

5.16.6 BACT for Particulate Matter – Comfort Building Heaters

The following sections outline the top-down steps for PM/PM_{10}/$PM_{2.5}$ emissions from comfort building heaters.

5.16.6.1 Steps 1-5. Identify Control Strategies and Select BACT

Combustion controls and the use of natural gas as a fuel are the only controls listed as BACT in the RBLC for natural gas heater PM emissions. Because these heaters are so small, add-on controls are not available for these small heaters. Combustion control was selected as BACT for PM from the comfort building heaters at an emission rate of 0.0075 lb/MMBtu.

5.16.7 BACT for Greenhouse Gases – Comfort Building Heaters (Steps 1-5)

The natural gas comfort building heaters as proposed will be fired exclusively on natural gas and used to heat the buildings at Morgan Valley Energy Center. The units will be permitted to be fired a total of 8,760 hours per year each. GHG emissions from these units are based on the emission rates provided in 40 CFR Part 98, 117.1 lb CO_2e/MMBtu. These GHG emissions are also *de minimis*, when compared to the combustion turbine GHG emissions or the facility total GHG emissions. The basic GHG BACT reasoning presented for the combustion turbines essentially applies to these heaters as well. GHG BACT for the comfort building heaters is the use of natural gas (exclusive use of natural gas).

6.0 Air Dispersion Modeling

Since the Project is subject to PSD review, an air dispersion modeling analysis is required for each regulated NSR pollutant that exceeds its PSD significance level. According to the emission calculations for this Project, NO_x, CO, PM, PM_{10}, $PM_{2.5}$, VOC, and CO_2e are subject to PSD review; as a result, an air quality analysis was performed for NO_x, CO, PM_{10}, and $PM_{2.5}$ using the EPA-approved American Meteorological Society (AMS)/EPA Regulatory Model (AERMOD). Consistent with Iowa DNR and EPA guidance, AERMOD modeling of PM, VOC, and CO_2e were not conducted, since there are no modeling thresholds for these pollutants.

Although SO_2 and H_2SO_4 mist are included in the Project's PSD review for BACT purposes, their projected emissions do not exceed the federal PSD significance levels. Therefore, SO_2 does not trigger PSD NAAQS or increment modeling requirements, and there is no modeling threshold for H_2SO_4 mist.

However, the Iowa DNR has established additional modeling requirements for non-PSD pollutants in its *Air Dispersion Modeling Guidelines for Non-PSD, Pre-Construction Permit Applications* (Version Date 7/28/2025). Under this state-specific framework, non-PSD modeling is required when emissions exceed the corresponding Iowa significant emission rate. According to the SO_2 emission calculations for this Project and as determined by Iowa DNR Form MD, SO_2 is greater than the significant emission rate. Therefore, SO_2 non-PSD air dispersion modeling will be performed for the Project.

A summary of the models, the modeling techniques, and modeling results for the Project will be discussed in this section for the final air permit submittal.

7.0 Additional Impacts Analysis

The additional impacts analysis requirement under PSD includes the ambient air quality impact analysis, soils and vegetation impacts, visibility impairment, and growth analysis for the Project. This section will be submitted with the final air permit submittal.

8.0 References

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Appendix A – Figures

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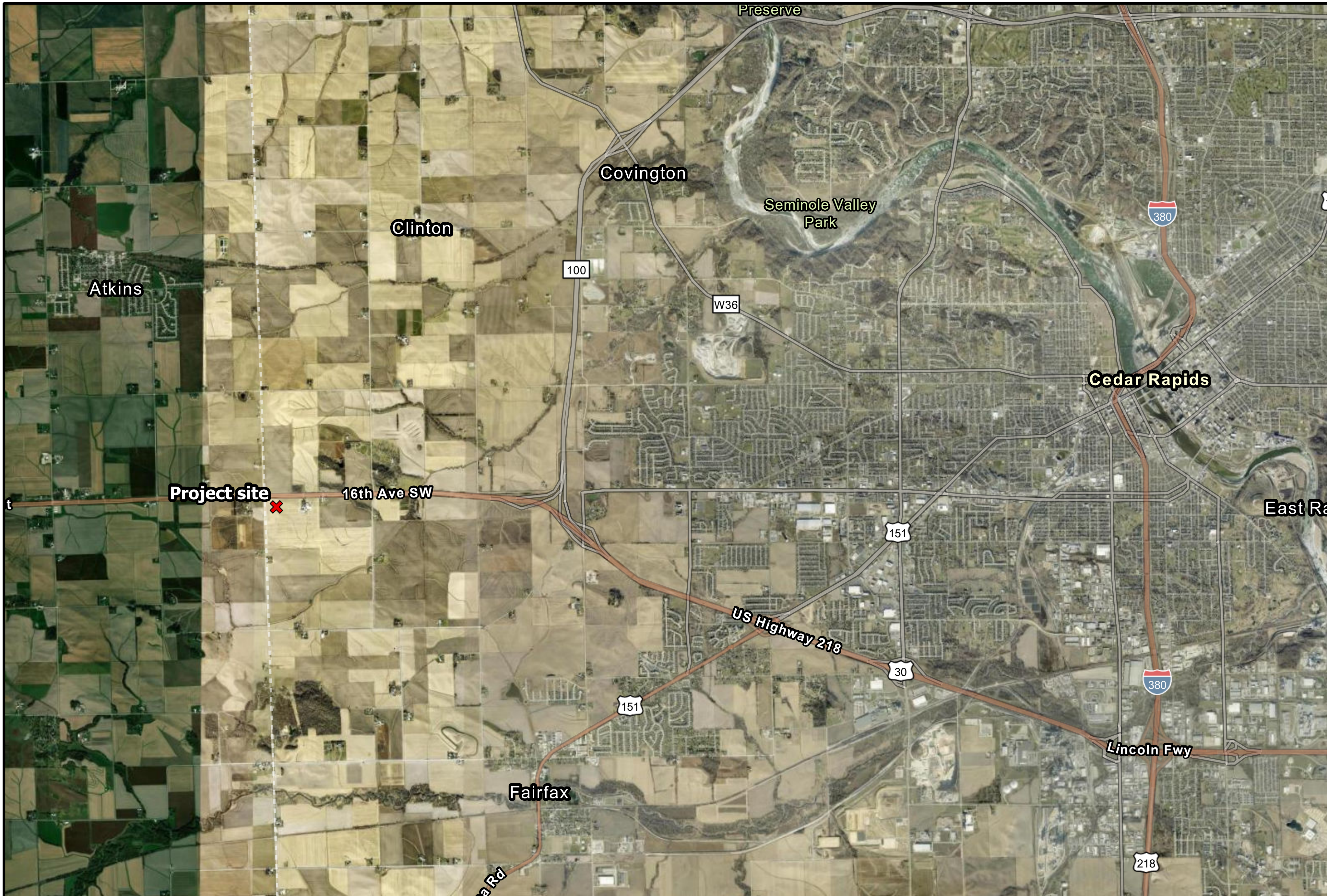

LEGEND

Project Site

REFERENCE

N

5,000 2,500 0 5,000

Feet

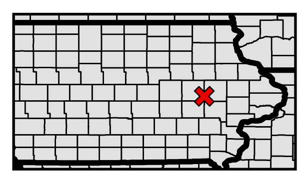

Figure A-1
Project Site
Morgan Valley Energy Center
Interstate Power & Light Company

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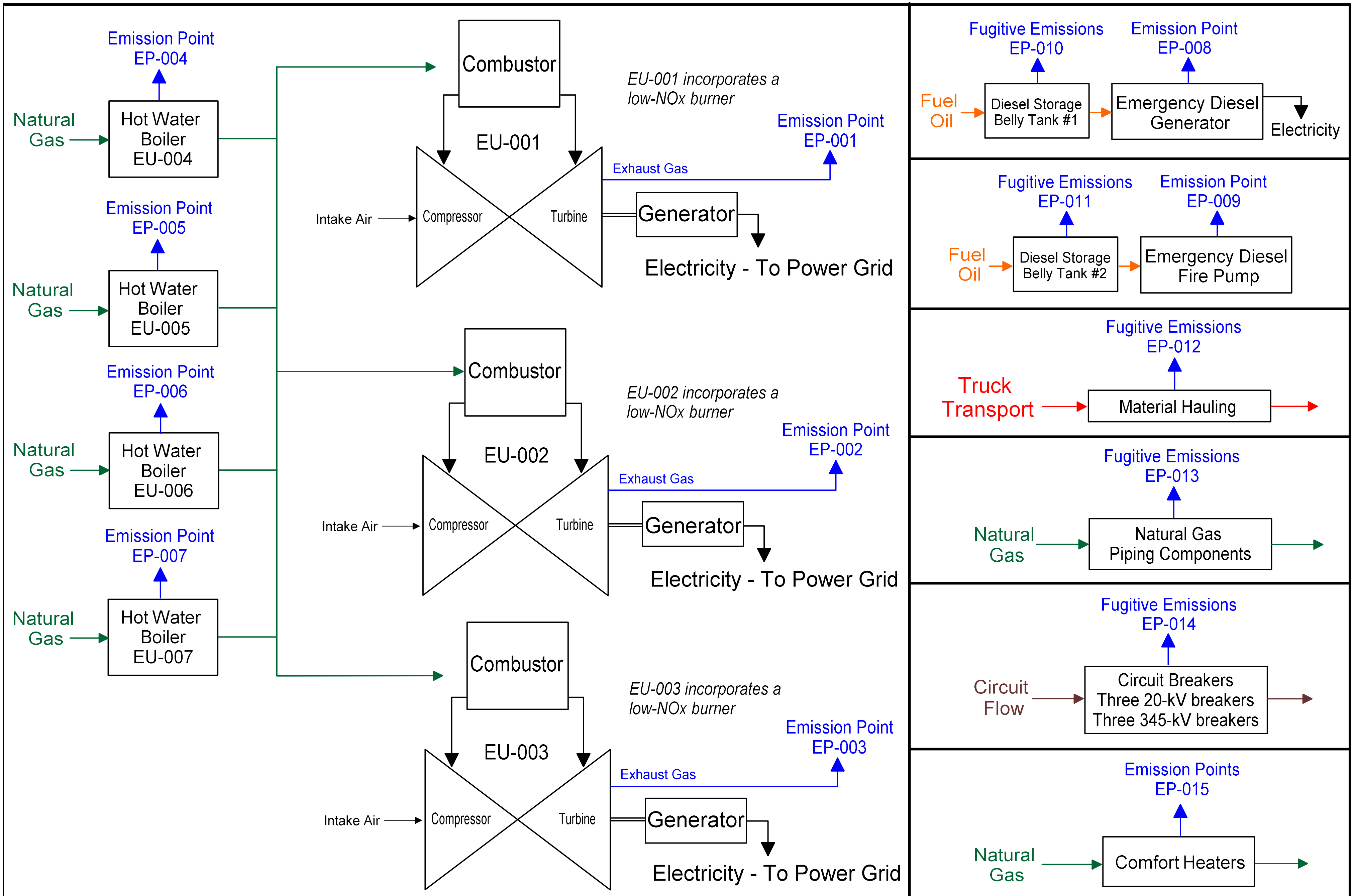

Fuel Oil
Truck Transport
Natural Gas
Circuit Flow
Emissions
Processes

BURNS McDONNELL

Figure A-2
Process Flow Diagram
Morgan Valley Energy Center
Interstate Power & Light Company

Appendix B – Calculations

Interstate Power and Light Company - Morgan Valley Energy Center
Project PTE

Pollutant	Combustion Turbines[a] (tpy)	Hot Water Boilers (tpy)	Emergency Generator (tpy)	Emergency Fire Pump (tpy)	Storage Tanks (tpy)	Haul Road Fugitives (tpy)	Piping Fugitives (tpy)	SF_6 Equipment (tpy)	Building Comfort Heaters (tpy)	Potential Emissions[b] (tpy)	PSD Major Source Thresholds (tpy)
NO_X	271	5.2	7.1	0.7	--	--	--	--	34.7	**319**	40
CO	1,109	8.7	3.9	0.6	--	--	--	--	29.2	**1,152**	100
PM	57.9	0.8	0.2	0.03	--	0.05	--	--	2.6	**61.6**	25
PM_{10}	57.9	0.8	0.2	0.03	--	0.009	--	--	2.6	**61.6**	15
$PM_{2.5}$	57.9	0.8	0.2	0.03	--	0.002	--	--	2.6	**61.6**	10
SO_2	17.5	0.06	8.1E-03	0.22	--	--	--	--	0.2	18.0	40
VOC	95.0	0.6	0.5	0.27	7.68E-04	--	6.6	--	1.9	**105**	40
H_2SO_4	2.7	9.5E-03	1.2E-03	9.9E-08	--	--	--	--	0.03	2.7	7
Lead	0.0	5.15E-05	--	--	--	--	--	--	--	5.2E-05	0.6
CO_2e (sum)	1,460,743	12,309	893	120	--	--	783	75	41,464	**1,516,388**	75,000

(a) Represents worse-case emissions scenario

(b) Values in **bold** exceed

Assumptions				
Unit	**Parameter**	**Units**		
Simple-Cycle Combustion Turbines				
Turbine	3	Combustion Turbines		
	3,500	Total hours per year, including SUSD, each		
	3,173	Hours of natural gas steady state per year, each		
	365	Number of natural gas startup/shutdowns per year, each		
	327	Hours of natural gas startup/shutdown per year, each		
Auxiliary Equipment			**Size**	**Units (each)**
Hot Water Boilers	8,760	Hours per year per unit	6	MMBtu/hr
	4	Hot water boilers		
Emergency Generator	500	Hours per year	2000	kW
	1	Emergency Generator		
Emergency Fire Pump	500	Hours per year	422	hp
	1	Emergency Fire Pump		
Natural gas heating value	1,020	Btu/cf		
Fuel Oil Heating Value	137,000	Btu/gal		

Interstate Power and Light Company - Morgan Valley Energy Center
Combustion Turbine Annual Emissions Analysis

Pollutant	Maximum steady state hourly emission rate (lb/hr)	NG Start-up/ Shutdown Emissions per turbine (tpy)
NO_X	42.6	22.9
CO	36.8	311.4
$PM/PM_{10}/PM_{2.5}$	11.0	1.8
SO_2	3.3	0.5
VOC	3.0	26.9
H_2SO_4	0.5	0.1
Lead	0.0	0.0
CO_2	277,964	45,486
CH_4	5.0	0.8
N_2O	0.5	0.1
CO_2e (sum)	278,237	45,531

Pollutant	Scenario 1	Scenario 2	Worst-Case Emissions
	tpy		
NO_X	271	224	**271**
CO	1,109	193	**1,109**
$PM/PM_{10}/PM_{2.5}$	57.9	57.8	**57.9**
SO_2	17.5	17.5	**17.5**
VOC	95.0	15.8	**95.0**
H_2SO_4	2.7	2.7	**2.7**
Lead	0.0	0.0	**0.0**
CO_2	1,459,311	1,459,311	**1,459,311**
CH_4	26.3	26.3	**26.3**
N_2O	2.6	2.6	**2.6**
CO_2e (sum)	1,460,743	1,460,743	**1,460,743**

Scenario 1

Turbines	3	
NG	3,173	hours per year per turbine
NG SUSD	327	hours per year per turbine

NG= Natural Gas
SUSD= Startup Shutdown

Total Emissions (all turbines combined)

Pollutant	NG	NG SSSD	SUM
	tpy		
NO_X	202.7	68.6	**271.3**
CO	175.1	934.1	**1,109.3**
$PM/PM_{10}/PM_{2.5}$	52.3	5.5	**57.9**
SO_2	15.9	1.6	**17.5**
VOC	14.3	80.7	**95.0**
H_2SO_4	2.4	0.3	**2.7**
Lead	0.0	0.0	**0.0**
CO_2	1,322,851.5	136,459.5	**1,459,311.0**
CH_4	23.8	2.5	**26.3**
N_2O	2.4	0.2	**2.6**
CO_2e (sum)	1,324,149.5	136,593.4	**1,460,742.9**

Scenario 2

Turbines	3	
NG	3,500	hours per year per turbine
NG SUSD	0	hours per year per turbine

Total Emissions (all turbines combined)

Pollutant	NG	NG SSSD	SUM
	tpy		
NO_X	223.7	0	**223.7**
CO	193.2	0	**193.2**
$PM/PM_{10}/PM_{2.5}$	57.8	0	**57.8**
SO_2	17.5	0	**17.5**
VOC	15.8	0	**15.8**
H_2SO_4	2.7	0	**2.7**
Lead	0.0	0	**0.0**
CO_2	1,459,311.0	0	**1,459,311.0**
CH_4	26.3	0	**26.3**
N_2O	2.6	0	**2.6**
CO_2e (sum)	1,460,742.9	0	**1,460,742.9**

Interstate Power and Light Company - Morgan Valley Energy Center
Combustion Turbine

	Natural Gas - Per Turbine				
	100	75	70	Minimum Emissions Compliance Load	
Pollutants	lb/hr	lb/hr	lb/hr	lb/hr	Source
NO_X	42.6	35.0	33.6	27.7	Vendor Data
CO	20.8	17.0	36.8	30.4	Vendor Data
$PM/PM_{10}/PM_{2.5}$	11.0	8.4	8.2	7.2	Vendor Data
SO_2[a]	3.3	2.7	2.6	2.2	AP-42
VOC	3.0	2.4	2.3	1.9	Vendor Data
H_2SO_4	0.5	0.4	0.4	0.3	Conversion
Lead	0.0E+00	0.00E+00	0.00E+00	0.00E+00	AP-42
CO_2	277,964	228,344	219,346	181,734	Vendor Data
CH_4	5.00	4.1	3.9	3.27	Federal Register[b]
N_2O	0.5	0.4	0.4	0.3	Federal Register[c]
CO_2e (sum)	278,237	228,568	219,561	181,912	

a) SO_2 emissions for the proposed turbine were calculated using a natural gas sulfur content of 0.5 grains/100 scf.
b) Federal Register - Subpart C of Part 98 (1.00×10^{-3} kg/MMbtu)
c) Federal Register - Subpart C of Part 98 (1.00×10^{-4} kg/MMbtu)

Diameter (ft)	21.3	21.3	21.3	21.3
Temperature (°F)	991.0	1,038.0	1,038.0	1,049.0
Velocity (ACFM)	2,766,748	2,432,123	2,369,260	2,061,974
Velocity (ft/s)	129.0	113.4	110.5	96.1
MMBtu/hr	2,270	1,865	1,791	1,484

Natural Gas Startup/Shutdown Emissions, Per Turbine

Pollutant	Start-up Emissions (lb/start)[b]	Shutdown Emissions (lb/shutdown)[c]	Number of Starts Per Turbine	Start-up/ Shutdown Emissions (tpy)
NOx[a]	76.8	48.5	365	22.9
CO[a]	1,060.6	645.6	365	311.4
$PM/PM_{10}/PM_{2.5}$[a]	6.2	3.9	365	1.8
SO_2	1.9	1.1	365	0.5
VOC[a]	91.5	55.9	365	26.9
H_2SO_4	0.3	0.2	365	0.1
Lead	0.0	0.0	365	0.0
CO_2	159,366	89,875	365	45,486
CH_4	2.9	1.6	365	0.8
N_2O	0.3	0.2	365	0.1
CO_2e	159,522	89,963	365	45,531

(a) Start-up emissions based on vendor data for turbine without controls
(b) Start-up emissions: 34.4 minutes
(c) Shutdown emissions: 19.4 minutes

Sulfuric Acid Mist

	Conversion Percent	
Assume 10% of SO2 is converted to SO3	10	SO2 + 1/2 O2 = SO3
Assume 100% of SO3 is converted to H2SO4	100	SO3 + H2O = H2SO4

Name	lb/hr SO_2	lb/hr SO_2 converted to SO_3	lb/hr SO_3 created	lb/hr H_2SO_4 created
100	3.3	0.33	0.42	0.51
75	2.7	0.27	0.34	0.42
70	2.6	0.26	0.33	0.40
50	2.2	0.22	0.27	0.33

Molecular Weights	
SO_2	64.1
SO_3	80.1
H_2SO_4	98.1

Vendor Data

SIEMENS energy

US5153 - Alliant Energy - Morgan Valley Energy Center (Iowa)
US5153-Alliant Energy-SGT6-5000F SS Emissions on NG (5 ppmvd GT NOX)-SC Stack-R3 Option Estimated Performance and Emissions - 5 ppm NOX (Simple Cycle Stack)
October 30, 2025
Gas Turbine in Simple Cycle / Ultra Low NOX Combustor
SGen6-1000A (113/55) Generator; PF = 0.85

	CASE 1	CASE 2	CASE 5	CASE 8	CASE 11	CASE 12	CASE 13	CASE 3	CASE 6	CASE 9	CASE 14	CASE 15	CASE 16	CASE 4	CASE 7	CASE 10
FUEL TYPE	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas	Natural Gas
LOAD LEVEL	100	100	100	100	100	100	100	75	75	75	70	70	70	50	50	50
NET FUEL HEATING VALUE, Btu/lb_m (LHV)	20,194	20,194	20,194	20,194	20,194	20,194	20,194	20,194	20,194	20,194	20,194	20,194	20,194	20,194	20,194	20,194
GROSS FUEL HEATING VALUE, Btu/lb_m (HHV)	22,366	22,366	22,366	22,366	22,366	22,366	22,366	22,366	22,366	22,366	22,366	22,366	22,366	22,366	22,366	22,366
AMBIENT DRY BULB TEMPERATURE, °F	89.6	89.6	50.6	-6.5	105.0	105.0	-40.0	89.6	50.6	-6.5	89.6	50.6	-6.5	89.6	50.6	-6.5
AMBIENT WET BULB TEMPERATURE, °F	75.0	75.0	43.3	-7.2	77.6	77.6	-40.1	75.0	43.3	-7.2	75.0	43.3	-7.2	75.0	43.3	-7.2
AMBIENT RELATIVE HUMIDITY, %	51.8	51.8	56.0	75.0	30.0	30.0	75.0	51.8	56.0	75.0	51.8	56.0	75.0	51.8	56.0	75.0
AMBIENT PRESSURE, psia	14.233	14.233	14.233	14.233	14.233	14.233	14.233	14.233	14.233	14.233	14.233	14.233	14.233	14.233	14.233	14.233
COMPRESSOR INLET TEMPERATURE, °F	89.6	76.5	50.6	0.0	105.0	80.3	0.0	89.6	50.6	13.5	89.6	50.6	13.5	89.6	50.6	20.0
INLET PRESSURE LOSS, in H_2O (Total)	3.78	3.98	4.00	3.72	3.45	3.91	4.05	2.46	2.63	2.73	2.34	2.49	2.59	1.78	1.90	2.05
EXHAUST PRESSURE LOSS, in H_2O (Total)	3.23	3.46	3.57	3.54	2.90	3.39	3.54	2.19	2.43	2.51	2.08	2.30	2.38	1.58	1.75	1.83
EXHAUST PRESSURE LOSS, in H_2O (Static)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
INLET AIR COND. (HEATER/COOLER)	NO	Evap. Cooler	NO	NO	NO	Evap. Cooler	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
EVAPORATIVE COOLER STATUS / EFFECTIVENESS, %	0	90	0	0	0	90	0	0	0	0	0	0	0	0	0	0
EVAPORATIVE COOLER REQ WATER FLOW, lb_m/hr	0	13,016	0	0	0	24,207	0	0	0	0	0	0	0	0	0	0
FUEL FLOW, lb_m/hr	91,783	96,532	99,595	101,288	85,710	95,290	101,493	75,019	80,892	83,375	72,146	77,677	80,090	59,239	63,601	66,356
HEAT INPUT, MMBtu/hr (LHV)	1,853	1,949	2,011	2,045	1,731	1,924	2,050	1,515	1,634	1,684	1,457	1,569	1,617	1,196	1,284	1,340
HEAT INPUT, MMBtu/hr (HHV)	2,053	2,159	2,228	2,265	1,917	2,131	2,270	1,678	1,809	1,865	1,614	1,737	1,791	1,325	1,423	1,484
GAS TURBINE PERFORMANCE																
GROSS POWER OUTPUT, kW	206,886	221,497	234,286	238,455	187,834	217,526	229,368	155,164	175,715	178,838	144,819	164,000	166,916	103,443	117,143	119,577
EFFICIENCY (LHV), %	38.09	38.77	39.75	39.78	37.03	38.57	38.19	34.95	36.70	36.24	33.92	35.67	35.21	29.50	31.12	30.45
EFFICIENCY (HHV), %	34.39	35.01	35.89	35.92	33.43	34.83	34.48	31.55	33.14	32.72	30.62	32.21	31.79	26.64	28.10	27.49
HEAT RATE (LHV), Btu/kWh	8,959	8,801	8,585	8,578	9,215	8,846	8,936	9,764	9,297	9,415	10,060	9,565	9,690	11,565	10,964	11,206
HEAT RATE (HHV), Btu/kWh	9,922	9,747	9,508	9,500	10,206	9,798	9,897	10,814	10,296	10,427	11,142	10,593	10,732	12,809	12,143	12,412
EXHAUST GAS																
EXHAUST FLOW, lb_m/hr	4,303,506	4,470,431	4,595,141	4,629,611	4,053,152	4,417,203	4,632,041	3,475,422	3,726,162	3,836,459	3,386,259	3,626,400	3,734,680	2,949,639	3,160,740	3,262,785
EXHAUST FLOW, ACFM	2,899,237	2,994,144	3,011,410	2,960,386	2,766,748	2,968,866	2,960,287	2,432,123	2,518,926	2,531,792	2,369,260	2,450,998	2,464,137	2,061,974	2,134,303	2,166,381 2,605,686
EXHAUST TEMPERATURE, °F	1,056	1,045	1,025	992	1,078	1,049	991	1,115	1,071	1,038	1,115	1,071	1,038	1,115	1,071	1,049
OXYGEN, Vol. %	12.64	12.47	12.84	12.88	12.73	12.44	12.88	12.55	12.83	12.93	12.66	12.94	13.04	13.12	13.43	13.47
CARBON DIOXIDE, Vol. %	3.68	3.71	3.75	3.80	3.65	3.71	3.80	3.72	3.76	3.77	3.67	3.70	3.72	3.45	3.47	3.52
WATER, Vol. %	9.30	9.81	7.71	7.18	9.07	10.01	7.13	9.37	7.72	7.13	9.28	7.62	7.04	8.88	7.19	6.66
NITROGEN, Vol. %	73.51	73.14	74.81	75.25	73.67	72.98	75.30	73.48	74.80	75.27	73.52	74.84	75.31	73.67	75.01	75.45
ARGON, Vol. %	0.87	0.87	0.89	0.90	0.88	0.87	0.90	0.87	0.89	0.90	0.87	0.89	0.90	0.88	0.89	0.90
MOLECULAR WEIGHT	28.28	28.23	28.46	28.52	28.30	28.20	28.53	28.27	28.46	28.52	28.28	28.46	28.53	28.30	28.49	28.55
STACK EMISSIONS (Based on USEPA Test Methods):																
NO_X, ppmvd @ 15% O_2	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
NO_X, lb_m/hr as NO_2	38.6	40.6	41.8	42.5	36.1	40.1	42.6	31.5	33.9	35.0	30.3	32.6	33.6	24.8	26.6	27.7
CO, ppmvd @ 15% O_2	4	4	4	4	4	4	4	4	4	4	9	9	9	9	9	9
CO, lb_m/hr	18.8	19.8	20.4	20.7	17.6	19.5	20.8	15.4	16.5	17.0	33.2	35.7	36.8	27.2	29.1	30.4
VOC, ppmvd @ 15% O_2 as CH_4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
VOC, lb_m/hr as CH_4	2.7	2.8	2.9	3.0	2.5	2.8	3.0	2.2	2.4	2.4	2.1	2.3	2.3	1.7	1.9	1.9
Formaldehyde, ppbvd @ 15% O_2	91	91	91	91	91	91	91	91	91	91	91	91	91	91	91	91
Formaldehyde, lbm/hr	0.459	0.482	0.497	0.505	0.428	0.476	0.506	0.375	0.403	0.416	0.360	0.387	0.399	0.295	0.316	0.329
Particulate Matter, lb_m/hr	9.3	9.6	10.0	11.0	8.8	9.5	11.0	7.5	8.1	8.4	7.3	7.9	8.2	6.4	6.9	7.2
CO_2, lb_m/hr	251,370	264,376	272,765	277,404	234,739	260,976	277,964	205,459	221,543	228,344	197,591	212,737	219,346	162,242	174,188	181,734

NOTES:

- All data is ESTIMATED, NOT guaranteed and is for ONE unit.
- Performance based on 5 ppmvd engine NO_X.
- Gross power output is at the generator terminals, minus excitation losses. It does not include SGT-PAC™ auxiliary load losses.
- IGV schedule may be adjusted during commissioning. Part load performance will be adjusted accordingly.
- Efficiency and Heat Rate without sensible heat are being considered in all cases.
- The anti-icing system may be in operation at cold ambient in order to maintain emission compliance.
- With anti-icing system in operation, GT output and efficiency will be lower than what is shown.
- Natural Gas fuel composition, by moles, is: 87.07% CH_4, 9.217% C_2H_6, 0.659% C_3H_8, 0.038% iC_4H_{10}, 0.049% nC_4H_{10}, 0.007% iC_5H_{12}, 0.008% nC_5H_{12}, 0.005% nC_6H_{14}, 2.076% N_2, 0.81% CO_2, 0.05% H_2, 0.011% He, and assumes 0.5 grains S/100 SCF.
- Average fuel gas temperature is 15°C (59°F). Sensible Heat of the fuel is not included in the calculated Heat Input values.
- Gas fuel must be in compliance with the Siemens Energy Gas Fuel Specification.
- Emissions exclude ambient air contributions and assume steady-state conditions.
- VOC consist of total hydrocarbons excluding methane and ethane and are expressed in terms of methane (CH_4).
- Particulates are per US EPA Method 5 and 202 (front and back half).
- CO_2 mass flow rate (lb_m/hr) calculated per USEPA 40CFR75 Appendix G, Equation G-4, based on standard fuel F_C.
- Please be advised that the information contained in this transmittal has been prepared and is being transmitted per customer request specifically for information purposes only.
  Data included in any permit application or Environmental Impact Statement are strictly the customer's responsibility. Siemens Energy is available to review permit application data upon request.

SIEMENS energy

SGT6-5000F in Simple Cycle Operation on Natural Gas

Total Estimated Startup & Shutdown Emissions and Fuel Use

Vendor Data

Mode	Ramp Rate (MW/min)	Time (min)	Total Pounds Per Event				
			NO_X	CO	VOC	PM	Fuel Use
Startup Emissions (GT Ignition to 100% Load, Steady-State)	13.4	34.4	76.8	1,061	91.5	6.2	45,757
Shutdown Emissions (100% Load to Fuel Cut Off)	13.4	19.4	48.5	646	55.9	3.9	23,913

General Notes

1.) All data is ESTIMATED, NOT guaranteed and is for ONE unit.

2.) Emissions are at the exhaust stack outlet and exclude ambient air contributions.

3.) Emissions are based on new and clean conditions.

4.) Natural Gas Fuel must be in compliance with the Siemens Energy Fuel Specification.

5.) NO_X as NO_2.

6.) VOC consist of total hydrocarbons excluding methane and ethane and are expressed in terms of methane (CH_4).

7.) Particulate (PM) emissions are per USEPA Methods 5/202.

8.) Fuel use is based on a fuel heating value of approximately 22,690 Btu/lb_m (HHV).

9.) Please be advised that the information contained in this transmittal has been prepared and is being transmitted per customer request specifically for information purposes only. Data included in any permit application or Environmental Impact Statement is strictly the customer's responsibility. Siemens Energy is available to review permit application data upon request.

Startup / Shutdown Notes

1.) Estimated data are based on the ramp rates noted above, time from GT ignition to synchronization in ~5-minutes, and will be higher for longer times and/or slower ramp rates.

2.) Shutdown data is based on the ramp rates noted with no holds at FSNL (Full Speed No Load / 0%) prior to fuel cut off.

3.) All startup prerequisites have been met, including any required equipment to be in operation.

4.) Operator actions do not extend start-up or shutdown.

5.) It is assumed that there is no restriction from the interconnected utility for loading the GT within the SU times considered and the generator can be synchronized to the grid in ≤ 30 seconds.

6.) CEMS may calculate emissions differently.

Interstate Power and Light Company - Morgan Valley Energy Center
Auxiliary Combustion Sources Emissions Calculations

Hot Water Boilers

Size (each)	6.00	MMBtu/hr
HHV	1,020	Btu/cf
Operation	8,760	hours/year
Quantity	4	heaters

Hot Water Boiler Stack Parameters

Height (ft)	Temp. (F)	Velocity (ft/sec)	Diameter (ft)	ACFM	Stack Discharge Type	Fuel
30.00	350.00	14.00	2.33	3,592	Vertical	Natural Gas

	Emission Factors			Emissions - one heater		Emissions - four heaters	
Pollutant	lb/MMcf	lb/MMBtu	Source	lb/hr	tpy	lb/hr	tpy
NO_X	50	0.05	AP-42[a]	0.29	1.3	1.18	5.2
CO	84	0.082	AP-42[a]	0.49	2.2	1.98	8.7
$PM/PM_{10}/PM_{2.5}$	7.6	0.0075	AP-42[a]	0.045	0.2	0.179	0.8
SO_2	0.60	0.0006	AP-42[a]	3.5E-03	0.02	0.01	0.06
VOC	5.5	0.0054	AP-42[a]	0.03	0.1	0.13	0.6
H_2SO_4 Mist	--	--	Mass Balance	5.4E-04	2.4E-03	2.2E-03	9.5E-03
CO_2	--	117.0	Federal Register[b]	702	3,074	2807	12,297
CH_4	--	0.0022	Federal Register[b]	0.013	0.06	0.053	0.23
N_2O	--	0.00022	Federal Register[b]	1.3E-03	0.006	5.3E-03	0.023
CO_2e	--	117.1	Federal Register[b]	703	3,077	2810	12,309

(a) AP-42 Section 1.4 (7/98)
(b) Federal Register - Subpart C of Part 98

Emergency Diesel Generator

Size	2,000	KW
	2,680	hp
	159.40	gal/hr
	21.84	MMBtu/hr
Operation	500	hours/year
Sulfur Content	0.0015	%
Fuel oil heating value	137,000	Btu/gal
Quantity	1	generator

Emergency Diesel Generator Stack Parameters

Height (ft)	Temp. (F)	Velocity (ft/sec)	Diameter (ft)	ACFM	Stack Discharge Type	Fuel
15.6	987	327.39	1.12	19,209	Vertical	Diesel

	Emission Factors				Emissions	
Pollutant	g/kw-hr	lb/hp-hr	lb/MMBtu	Source	lb/hr	tpy
NO_X	6.4	--	--	NSPS[a]	28.2	7.1
CO	3.5	--	--	NSPS[a]	15.4	3.9
$PM/PM_{10}/PM_{2.5}$	0.2	--	--	NSPS[a]	0.9	0.2
SO_2	--	1.21E-05	--	AP-42[b]	0.03	8.1E-03
VOC	--	7.05E-04	--	AP-42[b]	1.9	0.5
H_2SO_4 Mist	--	--	--	Mass Balance	5.0E-03	1.2E-03
CO_2	--	--	163.1	Federal Register[c]	3,561	890
CH_4	--	--	0.0066	Federal Register[c]	0.1	0.04
N_2O	--	--	0.0013	Federal Register[c]	0.03	7.2E-03
CO_2e	--	--	163.6	Federal Register[c]	3,572	893

(a) NSPS Limits - 40 CFR Part 60, Subpart IIII, (40 CFR 60.4202(a)(2) and 40 CFR Part 1039, Appendix I)
(b) AP-42 Section 3.4 (04/25)
(c) Federal Register - Subpart C of Part 98

Interstate Power and Light Company - Morgan Valley Energy Center
Auxiliary Combustion Sources Emissions Calculations

Emergency Diesel Fire Pump

Size	315	KW
	422	hp
	21.50	gal/hr
	2.95	MMBtu/hr
Operation	500	hours/year
Sulfur Content	0.0015	%
Fuel oil heating value	137,000	Btu/gal
Quantity	1	pump

Emergency Diesel Fire Pump Stack Parameters

Height (ft)	Temp. (F)	Velocity (ft/sec)	Diameter (ft)	ACFM	Stack Discharge Type	Fuel
22	895	229.18	0.50	2,700	Vertical	Diesel

	Emission Factors				Emissions	
Pollutant	**g/kw-hr**	**lb/hp-hr**	**lb/MMBtu**	**Source**	**lb/hr**	**tpy**
NO_X	4.0	--	--	NSPS[a]	2.8	0.7
CO	3.5	--	--	NSPS[a]	2.4	0.6
$PM/PM_{10}/PM_{2.5}$	0.2	--	--	NSPS[a]	1.39E-01	0.03
SO_2	--	0.0021	--	AP-42[b]	0.87	0.22
VOC	--	0.0025	--	AP-42[b]	1.1	0.27
H_2SO_4 Mist	--	--	--	Mass Balance	1.3E-01	9.9E-08
CO_2	--	--	163.1	Federal Register[c]	480.3	120.1
CH_4	--	--	0.0066	Federal Register[c]	1.9E-02	4.9E-03
N_2O	--	--	0.0013	Federal Register[c]	3.9E-03	9.7E-04
CO_2e	--	--	163.6	Federal Register[c]	482	120

(a) NSPS Limits - 40 CFR Part 60, Subpart IIII, (40 CFR 60.4205(c) and Table 4 to 40 CFR Part 60 Subpart IIII)
(b) AP-42 Section 3.3 (04/25)
(c) Federal Register - Subpart C of Part 98

Sulfuric Acid Mist

	Conversion Percent	
Assume 10% of SO2 is converted to SO3	10	SO2 + 1/2 O2 = SO3
Assume 100% of SO3 is converted to H2SO4	100	SO3 + H2O = H2SO4

Name	lb/hr SO_2	lb/hr SO_2 converted to SO_3	lb/hr SO_3 created	lb/hr H_2SO_4 created	tons / year H_2SO_4
Natural Gas Heater	0.004	3.5E-04	4.4E-04	5.4E-04	2.4E-03
Emergency Diesel Generator	0.03	3.3E-03	4.1E-03	5.0E-03	1.2E-03
Booster Pump	0.87	8.7E-02	1.1E-01	1.3E-01	9.9E-08

Molecular Weights	
SO_2	64.1
SO_3	80.1
H_2SO_4	98.1

CO_2 Equivalent Ratios

Greenhouse Gas			CO_2 Equivalent Ratio[a]
Carbon Dioxide	124-38-9	CO_2	1
Methane	74-82-8	CH_4	28
Nitrous Oxide	10024-97-2	N_2O	265

(a) 40 CFR 98: Table A-1 to Subpart A of Part 98

Interstate Power and Light Company - Morgan Valley Energy Center
Greenhouse Gas Emissions from SF_6 in Circuit Breakers

Inputs	
Number of 20 kV breakers	3
Quantity of SF_6 in each 20 kV breaker (lbs)	23
Number of 345 kV breakers	3
Quantity of SF_6 in each 345 kV breaker (lbs)	405
Global Warming Potential of SF_6 (100yr)[a]	23,500

(a) 40 CFR 98: Table A-1 to Subpart A of Part 98

Fugitive Emissions of SF6 due to leakage

	Number of Units	Quantity of SF_6 per Breaker (lbs)	Emissions of SF_6 Per Breaker[a] (lbs/yr)	Total SF_6 Emissions (lbs/yr)	Global Warming Potential	Total CO_2e Emissions (tons/yr)
20 kV Breaker	3	23	0.115	0.35	23,500	4.1
345 kV Breaker	3	405	2.025	6.08	23,500	71.4
			Total	**6.08**		**75.4**

(a) Based on a maximum SF_6 leakage rate of 0.5% per year

Paved Haul Road Emissions

$E = k * (sL)^{0.91} * (W)^{1.02}$ Equation 1 from AP 42 Section 13.2.1.3.

where E is the particulate emission factor having the units matching k

Parameter	Value	Description of parameter
sL	8.2	Quarry Facility Mean Silt Loading Value for Paved Roads, g/m^2 (AP-42 Table 13.2.1-3)
W	see below	Mean vehicle weight [(loaded truck weight + unloaded truck weight)/2], tons
VMT	see below	Vehicle miles traveled (length traveled round trip)
VMT/hr	see below	Vehicle miles traveled per hour = VMT*maximum trips per hour
VMT/yr	see below	Vehicle miles traveled per year = VMT*maximum trips per year

	k (lb/VMT)
PM2.5	0.00054
PM10	0.0022
PM30 (TSP)	0.011

Notes: Constant k, lb/VMT is from AP 42 Table 13.2.1-1

	Vehicle Type	Paved	Max # Trips/hour	Max # Trips/yr[a]	VMT - Length (round trip)		Truck Weight[b]		Factor "E" lb PM/VMT	Factor "E" lb PM10/VMT	Factor "E" lb PM2.5/VMT	Traveled VMT/hr	Traveled VMT/yr	Emissions		Emissions		Emissions		# segments	Control (%)
					meters	(miles)	Loaded tons	Unloaded tons						Uncontrolled lb PM/hr	Uncontrolled PM tpy	Uncontrolled lb PM10/hr	Uncontrolled PM10 tpy	Uncontrolled lb PM2.5/hr	Uncontrolled PM2.5 tpy		
Miscellaneous Deliveries paved (single-trip: loop)	generic haul truck	yes	1.00	60	1,124	0.70	40	15	2.19	0.44	0.11	0.70	41.91	1.53	0.05	0.31	0.009	0.075	0.002	94	0%

(a) The maximum trucks per year are expected for delivery or removal;

(b) Based on generic truck weight of the trucks that will be traveling onsite

Interstate Power and Light Company - Morgan Valley Energy Center
Model Haul Road Inputs

	Feet	Meters
vehicle height (VH)	12.0	3.66
Road width	20.0	6.10

Volume Sources	**Feet**	**Meters**
Top of Plume Height=1.7 x VH	20.4	6.22
Volume Source Release Height = 0.5 x Top of Plume height	10.2	3.11
Width of Plume = Road width + 6 m	39.69	12.10
Initial Sigma Z = Top of Plume / 2.15 *[Init. Vertical Dimension]*	9.49	2.89
Initial Sigma Y =Width of Plume / 2.15 *[Init. Horizontal Dimension]*	18.46	5.63
Volume Source Spacing = W = sigma yo*2.15	39.69	12.10

Source:
U.S. Environmental Protection Agency. Haul Road Workgroup Final Report Submission to EPA-OAQPS. Memo from Tyler Fox. March 2, 2012.

Natural Gas

				VOC[b]			Methane	CO_2e[c,d]		
Equipment Type	**Service**	**Quantity**	**Factor[a] (kg/hr/source)**	**Maximum emissions (lb/hr)**	**Maximum theoretical emissions (tpy)**	**Potential to emit (tpy)**	**Maximum emissions (lb/hr)**	**Maximum emissions (lb/hr)**	**Maximum theoretical emissions (tpy)**	**Potential to emit (tpy)**
Flanges	Natural Gas	980	3.90E-04	0.19	0.85	0.85	0.82	22.9	100.3	100.3
Open Ended Lines	Natural Gas	75	2.00E-03	0.08	0.33	0.33	0.32	9.0	39.4	39.4
Valves	Natural Gas	545	4.50E-03	1.2	5.4	5.4	5.2	147.0	643.7	643.7
			Total	**1.5**	**6.6**	**6.6**	**6.4**	**178.8**	**783**	**783**

(a) 1995 Protocol for Equipment Leak Emission Estimates- EPA-453/R-95-017
(b) Since methane is not a VOC, the maximum VOC is calculated at the minimum methane content.
77.07% minimum wt% methane
(c) Since methane is GHG, the maximum CO_2e is calculated at the maximum methane content.
97.07% maximum wt% methane
(d) Methane Global Warming Potential (40 CFR 98) was applied 28

Interstate Power and Light Company - Morgan Valley Energy Center
Storage Tank

Tank #	Material Stored	Size	VOC Emissions	
		Gallons	lb/year	Tons/year
Diesel Generator Belly Tank	#2 Fuel Oil	1,260	1.325	6.63E-04
Diesel Fire Pump Belly Tank	#2 Fuel Oil	465	0.211	1.06E-04
			TOTAL: (tpy VOC)	**7.68E-04**

TANKS 5.1d Inputs

Description	Diesel Generator Belly Tank	Units	Diesel Fire Pump Belly Tank	Units
Tank Type	Horizontal Fixed Roof Tank		Horizontal Fixed Roof Tank	
Tank ID	Tank_IPL		Tank_IPL	
Location (meteorological data)	Waterloo, Iowa		Waterloo, Iowa	
Tank Shape	Cylinder		Cylinder	
Shell Length*	26.00	ft	5.00	ft
Shell Diameter*	5.00	ft	5.00	ft
Maximum Liquid Height (ft)	4.00	ft	4.00	ft
Minimum Liquid Height (ft)	2.00	ft	2.00	ft
Tank Bottom Type	Flat		Flat	
Liquid Heel Type	No Heel		No Heel	
Vapor Space Pressure at Normal Operating Conditions (psig)	0.00	psig	0.00	psig
Insulated or Underground	Not Insulated		Not Insulated	
Liquid Bulk Temperature Calculation Method	AP-42		AP-42	
Shell Color/Shade	White		White	
Shell Condition	good		good	
Roof Color/Shade	White		White	
Roof Condition	good		good	
Roof Type	Flat		Flat	
Roof Height	n/a		n/a	
Slope (Cone Roof)	n/a		n/a	
Vacuum Settings (psig)	-0.03	psig	-0.03	psig
Pressure Settings (psig)	0.03	psig	0.03	psig
Tank Equipped with Control Device	No control device		No control device	
Input Type	Annual Values		Annual Values	
Chemical Category of Liquid	AP-42 Petroleum Liquids (Table 7.1-2)		AP-42 Petroleum Liquids (Table 7.1-2)	
Sum of Increases in Liquid Level Method	AP-42 Calculation		AP-42 Calculation	
Working Loss Turnover Factor Method	AP-42 Calculation		AP-42 Calculation	
Chemical Name	No.2 Fuel Oil (Diesel)		No.2 Fuel Oil (Diesel)	
Size	159.40	gal/yr	21.50	gal/yr
Operation	500	hr/yr	500	hr/yr
Annual Throughput	79,700	gal/yr	10,750	gal/yr
Working Loss	1.020	lb/yr	0.153	lb/yr
Standing Loss	0.305	lb/yr	0.059	lb/yr
Total losses	1.325	lb/yr	0.211	lb/yr
Total Emissions	**6.63E-04**	**tpy**	**1.06E-04**	**tpy**

*The minimum shell length and diameter allowed in TANKS 5.1 is 5 feet.

Interstate Power and Light Company - Morgan Valley Energy Center
Building Comfort Heaters Emissions Calculations

Preliminary Estimated Building Heating Design		
Estimated Size MMBtu/hr	Quantity	Total MMBtu/hr
0.2	1.0	0.2
0.4	30.0	12
0.46	2.0	0.92
7.75	7.0	54.25

Collective Size[a]	80.8	MMBtu/hr
HHV	1,020	Btu/cf
Operation	8,760	hours/year

(a) Total heat input includes a 20% safety factor as final building heating design is not finalized.

Pollutant	Emission Factors lb/MMcf	Emission Factors lb/MMBtu	Source	Total Emissions[a] lb/hr	Total Emissions[a] tpy
NO_X	100	0.098	AP-42[b]	7.9	34.7
CO	84	0.082	AP-42[b]	6.7	29.2
$PM/PM_{10}/PM_{2.5}$	7.6	0.0075	AP-42[b]	0.6	2.6
SO_2	0.6	0.0006	AP-42[b]	0.05	0.2
VOC	5.5	0.0054	AP-42[b]	0.4	1.9
H_2SO_4 Mist	--	--	Mass Balance	7.28E-03	0.03
CO_2	--	117.0	Federal Register[c]	9,457	41,421
CH_4	--	0.0022	Federal Register[c]	0.2	0.8
N_2O	--	0.00022	Federal Register[c]	0.02	0.08
CO_2e	--	117.1	Federal Register[c]	9,467	41,464

(a) Total emissions are based on the collective size, which is the the sum of all 40 comfort heat units
(b) AP-42 Section 1.4 (7/98)
(c) Federal Register - Subpart C of Part 98

Sulfuric Acid Mist

	Conversion Percent	
Assume 10% of SO2 is converted to SO3	10	SO2 + 1/2 O2 = SO3
Assume 100% of SO3 is converted to H2SO4	100	SO3 + H2O = H2SO4

Name	lb/hr SO2	lb/hr SO2 converted to SO3	lb/hr SO3 created	lb/hr H2SO4 created	tons / year H2SO4
Building Comfort Heaters	0.048	4.8E-03	5.9E-03	7.3E-03	0.03

Molecular Weights	
SO2	64.1
SO3	80.1
H2SO4	98.1

CO_2 Equivalent Ratios

Greenhouse Gas			CO_2 Equivalent Ratio[a]
Carbon Dioxide	124-38-9	CO_2	1
Methane	74-82-8	CH_4	28
Nitrous Oxide	10024-97-2	N_2O	265

(a) 40 CFR 98: Table A-1 to Subpart A of Part 98

Interstate Power and Light Company - Morgan Valley Energy Center
Building Comfort Heaters Emissions Calculations

		Size	Stack Height	Temperature	Exit Velocity	Stack Diameter	Exhaust Flow	NO_x		CO		$PM/PM_{10}/PM_{2.5}$		SO_2		VOC		H_2SO_4		CO_2		CH_4		N_2O		CO_2e		Total HAPs	
Model Source ID	Description	(MMBtu/hr)	(ft)	(°F)	(fps)	(ft)	acfm	lb/hr	tpy	lb/hr	tpy	lb/hr	tpy	lb/hr	tpy	lb/hr	tpy	lb/hr	tpy	lb/hr	tpy	lb/hr	tpy	lb/hr	tpy	lb/hr	tpy	lb/hr	tpy
514_BH1	Building Comfort Heater 1	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH2	Building Comfort Heater 2	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH3	Building Comfort Heater 3	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH4	Building Comfort Heater 4	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH5	Building Comfort Heater 5	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH6	Building Comfort Heater 6	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH7	Building Comfort Heater 7	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH8	Building Comfort Heater 8	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH9	Building Comfort Heater 9	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH10	Building Comfort Heater 10	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH11	Building Comfort Heater 11	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH12	Building Comfort Heater 12	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH13	Building Comfort Heater 13	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH14	Building Comfort Heater 14	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH15	Building Comfort Heater 15	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH16	Building Comfort Heater 16	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH17	Building Comfort Heater 17	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH18	Building Comfort Heater 18	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH19	Building Comfort Heater 19	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH20	Building Comfort Heater 20	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH21	Building Comfort Heater 21	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH22	Building Comfort Heater 22	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH23	Building Comfort Heater 23	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH24	Building Comfort Heater 24	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH25	Building Comfort Heater 25	0.48	20.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH26	Building Comfort Heater 26	0.48	40.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH27	Building Comfort Heater 27	0.48	40.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH28	Building Comfort Heater 28	0.48	40.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH29	Building Comfort Heater 29	0.48	40.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH30	Building Comfort Heater 30	0.48	40.00	-459.67	6.79	0.50	80	0.05	0.21	0.04	0.17	3.6E-03	0.02	2.8E-04	0.001	0.003	0.01	0.00	0.00	56.15	245.93	0.001	0.005	0.0001	0.0005	56.21	246.19	0.001	0.004
514_BH31	Rooftop Air Unit 1	0.55	40.00	-459.67	7.81	0.50	92	0.05	0.24	0.05	0.20	4.1E-03	0.02	3.2E-04	0.001	0.003	0.01	0.00	0.00	64.57	282.82	0.001	0.01	0.0001	0.0005	64.64	283.11	0.001	0.004
514_BH32	Rooftop Air Unit 2	0.55	40.00	-459.67	7.81	0.50	92	0.05	0.24	0.05	0.20	4.1E-03	0.02	3.2E-04	0.001	0.003	0.01	0.00	0.00	64.57	282.82	0.001	0.01	0.0001	0.0005	64.64	283.11	0.001	0.004
514_BH33	Makeup Air Unit 1	9.30	35.00	-459.67	14.62	1.50	1,550	0.91	3.99	0.77	3.35	0.07	0.30	5.5E-03	0.02	0.05	0.22	0.00	0.00	1087.89	4764.95	0.02	0.09	0.002	0.01	1089.01	4769.85	0.02	0.08
514_BH34	Makeup Air Unit 2	9.30	35.00	-459.67	14.62	1.50	1,550	0.91	3.99	0.77	3.35	0.07	0.30	5.5E-03	0.02	0.05	0.22	0.00	0.00	1087.89	4764.95	0.02	0.09	0.002	0.01	1089.01	4769.85	0.02	0.08
514_BH35	Makeup Air Unit 3	9.30	35.00	-459.67	14.62	1.50	1,550	0.91	3.99	0.77	3.35	0.07	0.30	5.5E-03	0.02	0.05	0.22	0.00	0.00	1087.89	4764.95	0.02	0.09	0.002	0.01	1089.01	4769.85	0.02	0.08
514_BH36	Makeup Air Unit 4	9.30	35.00	-459.67	14.62	1.50	1,550	0.91	3.99	0.77	3.35	0.07	0.30	5.5E-03	0.02	0.05	0.22	0.00	0.00	1087.89	4764.95	0.02	0.09	0.002	0.01	1089.01	4769.85	0.02	0.08
514_BH37	Makeup Air Unit 5	9.30	35.00	-459.67	14.62	1.50	1,550	0.91	3.99	0.77	3.35	0.07	0.30	5.5E-03	0.02	0.05	0.22	0.00	0.00	1087.89	4764.95	0.02	0.09	0.002	0.01	1089.01	4769.85	0.02	0.08
514_BH38	Makeup Air Unit 6	9.30	35.00	-459.67	14.62	1.50	1,550	0.91	3.99	0.77	3.35	0.07	0.30	5.5E-03	0.02	0.05	0.22	0.00	0.00	1087.89	4764.95	0.02	0.09	0.002	0.01	1089.01	4769.85	0.02	0.08
514_BH39	Makeup Air Unit 7	9.30	35.00	-459.67	14.62	1.50	1,550	0.91	3.99	0.77	3.35	0.07	0.30	5.5E-03	0.02	0.05	0.22	0.00	0.00	1087.89	4764.95	0.02	0.09	0.002	0.01	1089.01	4769.85	0.02	0.08
514_BH40	Gas Fired Water Heater	0.24	30.00	-459.67	13.58	0.25	40	0.02	0.10	0.02	0.09	1.8E-03	7.8E-03	1.4E-04	6.2E-04	0.00	0.01	0.00	0.00	28.07	122.97	0.001	0.002	0.0001	0.0002	28.10	123.09	0.0004	0.002
Total		**80.8**						**7.9**	**34.7**	**6.7**	**29.2**	**0.6**	**2.6**	**0.05**	**0.2**	**0.4**	**1.9**	**0.01**	**0.03**	**9,457**	**41,421**	**0.2**	**0.8**	**0.02**	**0.08**	**9,467**	**41,464**	**0.1**	**0.7**

Interstate Power and Light Company - Morgan Valley Energy Center
Project HAP Emissions

Hours of Operation

Combustion Turbines (each) =	3,500	hours per year
Hot Water Boilers (each) =	8,760	hours per year
Building Comfort Heaters (each) =	8,760	hours per year
Emergency Diesel Generator =	500	hours per year
Emergency Diesel Fire Pump =	500	hours per year

HAP	Project HAPs Emissions tpy
Formaldehyde	2.7
Toluene	1.6
Hexane	0.8
Xylene	0.8
TOTAL	7.3

Size

	MMBtu/hr	MMCF/hr	Quantity
Combustion Turbines (per turbine) =	2,270.0	2.225	3
Hot Water Boilers (each) =	6.0	0.006	4
Building Heaters (total) =	80.8	0.079	40
Emergency Generator =	21.8	--	1
Emergency Diesel Fire Pump =	2.9	--	1

			Natural Gas - Internal Combustion			Natural Gas - External Combustion					Fuel Oil - Large Diesel Engines			Fuel Oil - Small Diesel Engines			Project HAPs
			Emission Factor[a]	Combustion Turbines		Emission Factor[b]	Hot Water Boilers		Building Heaters		Emission Factor[c]	Emergency Generator		Emission Factor[d]	Emergency Diesel Fire Pump		Total
Chemical	CAS	POM?	lb/MMBtu	lb/hr (each)	tpy (total)	lb/MMcf	lb/hr (each)	tpy (total)	lb/hr (total)	tpy (total)	lb/MMBtu	lb/hr	tpy	lb/MMBtu	lb/hr	tpy	tpy
2-Methylnaphthalene	97-57-6	POM				2.4E-05	1.4E-07	2.5E-06	1.9E-06	8.3E-06							1.1E-05
3-Methylchloranthrene	56-49-5	POM				1.8E-06	1.1E-08	1.9E-07	1.4E-07	6.2E-07							8.1E-07
7,12-Dimethylbenz(a)anthracene	--	POM				1.6E-05	9.4E-08	1.6E-06	1.3E-06	5.6E-06							7.2E-06
Acenaphthene	83-32-9	POM				1.8E-06	1.1E-08	1.9E-07	1.4E-07	6.2E-07	4.7E-06	1.0E-04	2.6E-05	1.4E-06	4.2E-06	1.0E-06	2.7E-05
Acenaphthylene	203-96-8	POM				1.8E-06	1.1E-08	1.9E-07	1.4E-07	6.2E-07	9.2E-06	2.0E-04	5.0E-05	5.1E-06	1.5E-05	3.7E-06	5.5E-05
Acetaldehyde	75-07-0		4.0E-05	9.1E-02	4.8E-01						2.5E-05	5.5E-04	1.4E-04	7.7E-04	2.3E-03	5.6E-04	4.8E-01
Acrolein	107-02-8		6.4E-06	1.5E-02	7.6E-02						7.9E-05	1.7E-03	4.3E-04	9.3E-05	2.7E-04	6.8E-05	7.7E-02
Anthracene	120-12-7	POM				2.4E-06	1.4E-08	2.5E-07	1.9E-07	8.3E-07	1.2E-06	2.7E-05	6.7E-06	1.9E-06	5.5E-06	1.4E-06	9.2E-06
Benz(a)anthracene	56-55-3	POM				1.8E-06	1.1E-08	1.9E-07	1.4E-07	6.2E-07	6.2E-07	1.4E-05	3.4E-06	1.7E-06	4.9E-06	1.2E-06	5.4E-06
Benzene	71-43-2		1.2E-05	2.7E-02	1.4E-01	2.1E-03	1.2E-05	2.2E-04	1.7E-04	7.3E-04	7.8E-04	1.7E-02	4.2E-03	9.3E-04	2.7E-03	6.9E-04	1.5E-01
Benzo(a)pyrene	50-32-8	POM				1.2E-06	7.1E-09	1.2E-07	9.5E-08	4.2E-07	2.6E-07	5.6E-06	1.4E-06	1.9E-07	5.5E-07	1.4E-07	2.1E-06
Benzo(b)fluoranthene	205-99-2	POM				1.8E-06	1.1E-08	1.9E-07	1.4E-07	6.2E-07	1.1E-06	2.4E-05	6.0E-06	9.9E-08	2.9E-07	7.3E-08	6.9E-06
Benzo(g,h,l)perylene	191-24-2	POM				1.2E-06	7.1E-09	1.2E-07	9.5E-08	4.2E-07	5.6E-07	1.2E-05	3.0E-06	4.9E-07	1.4E-06	3.6E-07	3.9E-06
Benzo(k)fluoranthene	205-82-3	POM				1.8E-06	1.1E-08	1.9E-07	1.4E-07	6.2E-07	2.2E-07	4.8E-06	1.2E-06	1.6E-07	4.6E-07	1.1E-07	2.1E-06
1,3-Butadiene	106-99-0		4.3E-07	9.8E-04	5.1E-03									3.9E-05	1.2E-04	2.9E-05	5.2E-03
Chrysene	218-01-9	POM				1.8E-06	1.1E-08	1.9E-07	1.4E-07	6.2E-07	1.5E-06	3.3E-05	8.4E-06	3.5E-07	1.0E-06	2.6E-07	9.4E-06
Dibenzo(a,h)anthracene	53-70-3	POM				1.2E-06	7.1E-09	1.2E-07	9.5E-08	4.2E-07	3.5E-07	7.6E-06	1.9E-06	5.8E-07	1.7E-06	4.3E-07	2.9E-06
Dichlorobenzene	25321-22-6					1.2E-03	7.1E-06	1.2E-04	9.5E-05	4.2E-04							5.4E-04
Ethyl benzene	100-41-4		3.2E-05	7.3E-02	3.8E-01												3.8E-01
Fluoranthene	206-44-0	POM				3.0E-06	1.8E-08	3.1E-07	2.4E-07	1.0E-06	4.0E-06	8.8E-05	2.2E-05	7.6E-06	2.2E-05	5.6E-06	2.9E-05
Fluorene	86-73-7	POM				2.8E-06	1.6E-08	2.9E-07	2.2E-07	9.7E-07	1.3E-05	2.8E-04	7.0E-05	2.9E-05	8.6E-05	2.2E-05	9.3E-05
Formaldehyde	500-00-0		vendor value	0.506	2.7E+00	7.5E-02	4.4E-04	7.7E-03	5.9E-03	2.6E-02	7.9E-05	1.7E-03	4.3E-04	1.2E-03	3.5E-03	8.7E-04	2.7E+00
Hexane	110-54-3					1.8E+00	1.1E-02	1.9E-01	1.4E-01	6.2E-01							8.1E-01
Indeno(1,2,3-cd)pyrene	193-39-5	POM				1.8E-06	1.1E-08	1.9E-07	1.4E-07	6.2E-07	4.1E-07	9.0E-06	2.3E-06	3.8E-07	1.1E-06	2.8E-07	3.3E-06
Naphthalene	91-20-3		1.3E-06	3.0E-03	1.5E-02	6.1E-04	3.6E-06	6.3E-05	4.8E-05	2.1E-04	1.3E-04	2.8E-03	7.1E-04	8.5E-05	2.5E-04	6.2E-05	1.7E-02
PAH	--		2.2E-06	5.0E-03	2.6E-02												2.6E-02
Phenanathrene	85-01-8	POM				1.7E-05	1.0E-07	1.8E-06	1.3E-06	5.9E-06	4.1E-05	8.9E-04	2.2E-04	2.9E-05	8.7E-05	2.2E-05	2.5E-04
Propylene	--										2.8E-03	6.1E-02	1.5E-02	2.6E-03	7.6E-03	1.9E-03	1.7E-02
Proplylene Oxide	75-56-9		2.9E-05	6.6E-02	3.5E-01												3.5E-01
Pyrene	129-00-0	POM				5.0E-06	2.9E-08	5.2E-07	4.0E-07	1.7E-06	3.7E-06	8.1E-05	2.0E-05	4.8E-06	1.4E-05	3.5E-06	2.6E-05
Toluene	108-88-3		1.3E-04	3.0E-01	1.5E+00	3.4E-03	2.0E-05	3.5E-04	2.7E-04	1.2E-03	2.8E-04	6.1E-03	1.5E-03	4.1E-04	1.2E-03	3.0E-04	1.6E+00
Xylene	1330-20-7		6.4E-05	1.5E-01	7.6E-01						1.9E-04	4.2E-03	1.1E-03	2.9E-04	8.4E-04	2.1E-04	7.6E-01
Beryllium	--					1.20E-05	7.1E-08	1.2E-06	9.5E-07	4.2E-06							5.4E-06
Arsenic	--					2.00E-04	1.2E-06	2.1E-05	1.6E-05	6.9E-05							9.0E-05
Cadmium	--					1.10E-03	6.5E-06	1.1E-04	8.7E-05	3.8E-04							5.0E-04
Chromium	--					1.40E-03	8.2E-06	1.4E-04	1.1E-04	4.9E-04							6.3E-04
Cobalt	--					8.40E-05	4.9E-07	8.7E-06	6.7E-06	2.9E-05							3.8E-05
Manganese	--					3.80E-04	2.2E-06	3.9E-05	3.0E-05	1.3E-04							1.7E-04
Mercury	--					2.60E-04	1.5E-06	2.7E-05	2.1E-05	9.0E-05							1.2E-04
Nickel	--					2.10E-03	1.2E-05	2.2E-04	1.7E-04	7.3E-04							9.5E-04
Selenium	--					2.40E-05	1.4E-07	2.5E-06	1.9E-06	8.3E-06							1.1E-05
		TOTAL		1.2	6.4		0.01	0.19	0.15	0.66		0.10	0.024		0.019	0.005	7.3

(a) Emission factors from AP-42 Section 3.1, Updated 4/2000
(b) Emission factors from AP-42 Section 1.4, Updated 7/1998
(c) Emission factors from AP-42 Section 3.4, Updated 04/2025
(d) Emission factors from AP-42 Section 3.3, Updated 04/2025

			Emission Factor[a]	Combustion Turbines		Emission Factor[b]	Dew Point Heater		Building Heaters		Emission Factor[c]	Emergency Generator		Emission Factor[d]	Booster Pump		Total
Chemical	CAS	POM?	lb/MMBtu	lb/hr (each)	tpy (total)	lb/MMcf	lb/hr (each)	tpy (total)	lb/hr (each)	tpy (total)	lb/MMBtu	lb/hr	tpy	lb/MMBtu	lb/hr	tpy	tpy
Lead						5.0E-04	2.9E-06	5.2E-05	4.0E-05	1.7E-04							2.3E-04
Copper						8.50E-04	5.0E-06	8.8E-05	6.7E-05	3.0E-04							3.8E-04
Vanadium						2.30E-03	1.4E-05	2.4E-04	1.8E-04	8.0E-04							1.0E-03
Zinc						2.90E-02	1.7E-04	3.0E-03	2.3E-03	1.0E-02							1.3E-02

(a) Emission factors from AP-42 Section 3.1, Updated 4/2000
(b) Emission factors from AP-42 Section 1.4, Updated 7/1998
(c) Emission factors from AP-42 Section 3.4, Updated 04/2025
(d) Emission factors from AP-42 Section 3.3, Updated 04/2025

Appendix C – RBLC Tables

Table C-1a: RBLC NOx Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
WV-0028	WAVERLY POWER PLANT	PLEASANTS ENERGY LLC	3/13/2018	167.8	MW	Dry LNB	69	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hR	Pipeline quality natural gas & dry-low-NOX burners	86.38	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Pipeline quality natural gas & dry-low-NOX burners	86.38	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Pipeline quality natural gas & dry-low-NOX burners	240	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Pipeline quality natural gas & dry-low-NOX burners	240	LB/HR	BACT
IN-0261	VERMILLION GENERATING STATION	DUKE ENERGY INDIANA, LLC VERMILLION GENERATING STA	2/28/2017	80	MW	GOOD COMBUSTION PRACTICES	250	LB/H	BACT
NY-0106	GREENIDGE STATION	GREENIDGE GENERATION LLC	9/7/2016	107	MW	Advanced low NOx burners, closed-coupled and staged over-fire air, Selective Non-Catalytic Reduction, and Selective Catalytic Reduction.	0.03	LB/MMBTU	LAER
IL-0121	INVENERGY NELSON EXPANSION LLC	INVENERGY	9/27/2016	190	MW	Dry low-NOx combustion technology for natural gas and low-NOx combustion technology and water injection for ULSD.	0.033	LB/MMBTU	BACT
PA-0305	SHELL CHEM APPALACHIA/PETROCHEMICALS	SHELL CHEMICAL APPALACHIA	6/18/2015	0	Three 40.6 MW turbines		2	PPMDV @ 15% O2	LAER
AK-0088	LIQUEFACTION PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	7/7/2022	1113	MMBtu/hr	SCR, DLN combustors, and good combustion practices	2	PPMV @ 15% O2	BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	228	mw	dry low NOx burners in combination with selective catalytic reduction	2	PPMVD @ 15% O2	LAER
CA-1238	PUENTE POWER		10/13/2016	262	MW		2.5	PPMVD	OTHER CASE-BY-
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	575	mm btu/hr	Good combustion practices and use clean fuel. Dry low-NOx and selective catalytic reduction.	2.5	PPMVD	BACT
*TX-0989	LONE STAR POWER STATION	ENTERGY TEXAS INC	3/17/2025	0		Dry-low NOx (DLN) combustors and an aqueous ammonia-based SCR system	2.5	PPMVD	LAER
NJ-0086	BAYONNNE ENERGY CENTER	BAYONNNE ENERGY CENTER LLC	8/26/2016	2143980	MMBTU/YR	Selective Catalytic Reduction, water injection, use of natural gas a low NOx emitting fuel	2.5	PPMVD@15 %O2	LAER
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		LNB + SCR	3.1	PPMVD @15%O2	BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	540	mm btu/hr	DLN and SCR	5	PPMVD	BACT
*TN-0187	TENNESSEE VALLEY AUTHORITY - JOHNSONVILLE COMBUSTION	TENNESSEE VALLEY AUTHORITY	8/31/2022	465.8	MMBtu/hr	dry low-NOx burners selective catalytic reduction	5	PPMVD @ 15% O2	BACT
TX-0788	NECHES STATION	APEX TEXAS POWER LLC	3/24/2016	232	MW	Dry low-NOx burners (DLN), good combustion practices	9	PPM	BACT
VA-0326	DOSWELL ENERGY CENTER	DOSWELL LIMITED PARTNERSHIP DOSWELL ENERGY CENTER	10/4/2016	1961	MMBTU/HR	Low NOx Burners/Combustion Technology	9	PPM	BACT
WV-0026	WAVERLY FACILITY	PLEASANTS ENERGY, LLC	1/23/2017	1571	mmbtu/hr	Dry Low-NOx Combustion System (DLNB), Water Injection	9	PPM	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	927	MM BTU/h	Dry Low NOx Combustor Design, Good Combustion Practices, and Natural Gas Combustion.	9	PPMV	BACT
TX-0819	GAINES COUNTY POWER PLANT	SOUTHWESTERN PUBLIC SERVICE COMPANY	4/28/2017	227.5	MW	Dry Low NOx burners (control), natural gas, good combustion practices, limited operating hours (prevention)	9	PPMV	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	229	MW		9	PPMVD	BACT

Table C-1a: RBLC NOx Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	240	MW Each	dry low-NOx combustors, water injection when using ULSD, good combustion practices, clean fuels	9	PPMVD	BACT
TX-0694	INDECK WHARTON ENERGY CENTER	INDECK WHARTON, L.L.C.	2/2/2015	220	MW	DLN combustors	9	PPMVD	BACT
TX-0826	MUSTANG STATION	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	8/16/2017	162.8	MW	Dry low-NOx burners	9	PPMVD	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	920	MW	Dry low NOx burners	9	PPMVD	BACT
TX-0900	ECTOR COUNTY ENERGY CENTER	ECTOR COUNTY ENERGY CENTER LLC	8/17/2020	0		Equipped with dry-low NOx burners with best management practices and good combustion practices. Minimize the duration of startup and shutdown events to less than 60 minutes per event. Limit MSS by 140 lb/hr maximum allowable emission rate for each turbine.	9	PPMVD	BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		LOW NOX BURNERS AND SCR	9	PPMVD	BACT
TX-0975	FREESTONE PEAKERS PLANT	FPEC, LLC	6/13/2024	221	MW	Dry low-NOx burners (DLNB) and good combustion practices.	9	PPMVD	BACT
TX-0734	CLEAR SPRINGS ENERGY CENTER (CSEC)	NAVASOTA SOUTH PEAKERS OPERATING COMPANY II, LLC.	5/8/2015	183	MW	dry low-NOx (DLN) burners	9	PPMVD @ 15% O2	BACT
TX-0764	NACOGDOCHES POWER ELECTRIC GENERATING PLANT	NACOGDOCHES POWER, LLC	10/14/2015	232	MW	Dry Low NOx burners, good combustion practices, limited operations	9	PPMVD @ 15% O2	BACT
TX-0768	SHAWNEE ENERGY CENTER	SHAWNEE ENERGY CENTER, LLC	10/9/2015	230	MW	Dry Low NOx burners	9	PPMVD @ 15% O2	BACT
TX-0769	VAN ALSTYNE ENERGY CENTER (VAEC)	NAVASOTA NORTH COUNTRY PEAKERS OPERATING COMPANY I	10/27/2015	183	MW	DLN burners	9	PPMVD @ 15% O2	BACT
TX-0777	UNION VALLEY ENERGY CENTER	NAVASOTA SOUTH PEAKERS OPERATING COMPANY I, LLC.	12/9/2015	183	MW	dry low NOX burners	9	PPMVD @ 15% O2	BACT
TX-0794	HILL COUNTY GENERATING FACILITY	BRAZOS ELECTRIC COOPERATIVE	4/7/2016	171	MW	Emission controls consist of dry low-NOx combustors (DLN). DLN combustors use two stages of combustion, transitioning from initial startup with fuel and flame in the primary nozzles only, through a lean lean stage with fuel and flame in the primary and secondary nozzles, to fuel in the secondary stage only, extinguishing the primary flame, and in full operation, premix mode, with fuel to both nozzles, but flame only in the second stage. When natural gas and air are well-mixed before combustion, the flame temperature and resulting NOx emissions are greatly reduced compared to conventional diffusion flame combustion.	9	PPMVD @ 15% O2	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Pipeline quality natural gas & dry-low-NOX burners	9	PPMVD @15%O2	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Pipeline quality natural gas & dry-low-NOX burners	9	PPMVD @15%O2	BACT
TX-0733	ANTELOPE ELK ENERGY CENTER	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	5/12/2015	202	MW	Dry Low NOx burners	9	PPMVD AT 15% O2	BACT
FL-0355	FORT MYERS PLANT	FLORIDA POWER & LIGHT (FPL)	9/10/2015	2262.4	MMBtu/hr gas	DLN and wet injection (for ULSD operation)	9	PPMVD@15 % O2	BACT
FL-0354	LAUDERDALE PLANT	FLORIDA POWER & LIGHT	8/25/2015	2100	MMBtu/hr (approx)	Dry-low-NOx combustion system. Wet injection when firing ULSD.	9	PPMVD@15 %O2	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	386	MMBtu/hr	DLN combustors and Good Combustion Practices	15	PPMV @ 15% O2	BACT

Table C-1a: RBLC NOx Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	1069	mm btu/hr	good combustion practices and dry low nox burners	15	PPMVD	BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	180.2	MW	Each turbine is limited to 15 ppmvd concentration at 15% O2 on a 3-hour average through an upgrade to â€˜HR3â€™ Dry Low-NOx (DLN) burners. 14 ppmvd at 15% O2 is achieved on an annual average and also includes during periods when wet compression (evaporative air cooling) is applied. MSS - Limited to 50 startups and 50 shutdowns per year for each turbine. Startup and shutdown events are each expected to last less than an hour in duration.	15	PPMVD	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	667	MMBTU/H	Dry low NOx burners (DLNB) and good combustion practices.	25	PPM	BACT
MI-0447	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	1/7/2021	667	MMBTU/H	DLNB and good combustion practices.	25	PPM	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	667	MMBTU/H	DLNB and good combustion practices.	25	PPM	BACT
IN-0264	MONTPELIER GENERATING STATION	AES OHIO GENERATION, LLC	1/6/2017	270.9	MMBTU/H	WATER INJECTION	25	PPMV	BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	333	mm btu/hr	Dry Low NOX burners and good combustion practices	25	PPMVD	BACT
LA-0343	SABINE PASS LNG TERMINAL	SABINE PASS LNG LP AND SABINE PASS LIQUEFACTION LL	9/6/2019	0		good combustion practices	96	PPMV	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	0		Minimizing duration of startup/shutdown, using good air pollution control practices and safe operating practices.	0.01	TON/YR	BACT

Table C-1b: RBLC CO Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	667	MMBTU/H	Dry low NOx burners and good combustion practices.	9	LB/H	BACT
MI-0447	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	1/7/2021	667	MMBTU/H	Dry low NOx burners and good combustion practices	9	LB/H	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	667	MMBTU/H	Dry low NOx burners and good combustion practices.	9	LB/H	BACT
VA-0326	DOSWELL ENERGY CENTER	DOSWELL LIMITED PARTNERSHIP DOSWELL ENERGY CENTER	10/4/2016	1961	MMBTU/HR	Pipeline Quality Natural Gas	13.99	LB	BACT
IN-0261	VERMILLION GENERATING STATION	DUKE ENERGY INDIANA, LLC VERMILLION GENERATING STA	2/28/2017	80	MW	GOOD COMBUSTION PRACTICES	525	LB/H	BACT
WV-0028	WAVERLY POWER PLANT	PLEASANTS ENERGY LLC	3/13/2018	167.8	MW	Combustion Controls	33.9	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hR	Good combustion practices & use of pipeline quality natural gas	800.08	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of pipeline quality natural gas	800.08	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of pipeline quality natural gas	2000	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of pipeline quality natural gas	2000	LB/HR	BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	333	mm btu/hr	good combustion practices and fueled by natural gas	0.062	LB/MM BTU	BACT
NY-0106	GREENIDGE STATION	GREENIDGE GENERATION LLC	9/7/2016	107	MW		0.095	LB/MMBTU	BACT
IN-0264	MONTPELIER GENERATING STATION	AES OHIO GENERATION, LLC	1/6/2017	270.9	MMBTU/H	NATURAL GAS AS PRIMARY FUEL; GOOD COMBUSTION PRACTICES	0.2	LB/MMBTU	BACT
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	575	mm btu/hr	Good combustion practices and use of clean fuel.	1.7	PPMVD	BACT
PA-0305	APPALACHIA/PETROCHEMICALS COMPLEX	SHELL CHEMICAL APPALACHIA	6/18/2015	0	Three 40.6 MW turbines		2	PPMDV @ 15% O2	BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	228	mw	good combustion practice and oxidation catalyst	2	PPMVD @ 15% O2	BACT
TX-0694	INDECK WHARTON ENERGY CENTER	INDECK WHARTON, L.L.C.	2/2/2015	220	MW	DLN combustors	4	PPMVD	BACT
FL-0354	LAUDERDALE PLANT	FLORIDA POWER & LIGHT	8/25/2015	2100	MMBtu/hr (approx)	Good combustion minimizes CO formation	4	PPMVD@15 %O2	BACT
AK-0088	LIQUEFACTION PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	7/7/2022	1113	MMBtu/hr	Oxidation Catalyst and good combustion practices	5	PPMV @ 15% O2	BACT
NJ-0086	BAYONNNE ENERGY CENTER	BAYONNNE ENERGY CENTER LLC	8/26/2016	2143980	MMBTU/YR	Add-on control is CO Oxidation Catalyst, and use of natural gas as fuel for pollution prevention	5	PPMVD@15 %O2	CASE-BY-CASE
*TN-0187	JOHNSONVILLE COMBUSTION TURBINE	TENNESSEE VALLEY AUTHORITY	8/31/2022	465.8	MMBtu/hr	oxidation catalyst	5	PPMVD @ 15% O2	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of pipeline quality natural gas	6	PPMVD AT 15% O2	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of pipeline quality natural gas	6	15% OXYGEN	BACT
TX-0788	NECHES STATION	APEX TEXAS POWER LLC	3/24/2016	232	MW	good combustion practices	9	PPM	BACT

Table C-1b: RBLC CO Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
WV-0026	WAVERLY FACILITY	PLEASANTS ENERGY, LLC	1/23/2017	1571	mmbtu/hr	Good Combustion Practices	9	PPM	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	229	MW		9	PPMVD	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	240	MW Each		9	PPMVD	BACT
TX-0819	GAINES COUNTY POWER PLANT	SOUTHWESTERN PUBLIC SERVICE COMPANY	4/28/2017	227.5	MW	Good combustion practices; limited operating hours	9	PPMVD	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	920	MW	Dry low NOx burners	9	PPMVD	BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		Oxidization catalyst, good combustion practices and the use of gaseous fuel	9	PPMVD	BACT
TX-0975	FREESTONE PEAKERS PLANT	FPEC, LLC	6/13/2024	221	MW	Good combustion practices	9	PPMVD	BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	180.2	MW	Each turbine is limited to 9 ppmvd at 15% O2 on a 3-hour average, which meets Tier I BACT. Good combustion practices are used. MSS - Limited to 50 startups and 50 shutdowns per year for each turbine. Startup and shutdown events are each expected to last less than an hour in duration.	9	PPMVD	BACT
TX-0733	ANTELOPE ELK ENERGY CENTER	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	5/12/2015	202	MW	Good combustion practices; limited operating hours	9	PPMVD @ 15% O2	BACT
TX-0734	CLEAR SPRINGS ENERGY CENTER (CSEC)	NAVASOTA SOUTH PEAKERS OPERATING COMPANY II, LLC.	5/8/2015	183	MW	DLN burners and good combustion practices	9	PPMVD @ 15% O2	BACT
TX-0764	NACOGDOCHES POWER ELECTRIC GENERATING PLANT	NACOGDOCHES POWER, LLC	10/14/2015	232	MW	dry low NOx burners, good combustion practices, limited operation	9	PPMVD @ 15% O2	BACT
TX-0768	SHAWNEE ENERGY CENTER	SHAWNEE ENERGY CENTER, LLC	10/9/2015	230	MW	dry low NOx burners and lmiited operation, clean fuel	9	PPMVD @ 15% O2	BACT
TX-0769	VAN ALSTYNE ENERGY CENTER (VAEC)	NAVASOTA NORTH COUNTRY PEAKERS OPERATING COMPANY I	10/27/2015	183	MW	DLN burners and good combustion practices	9	PPMVD @ 15% O2	BACT
TX-0777	UNION VALLEY ENERGY CENTER	NAVASOTA SOUTH PEAKERS OPERATING COMPANY I, LLC.	12/9/2015	183	MW	dry low NOx burners and good combustion practices	9	PPMVD @ 15% O2	BACT
TX-0794	HILL COUNTY GENERATING FACILITY	BRAZOS ELECTRIC COOPERATIVE	4/7/2016	171	MW	Premixing of fuel and air enhances combustion efficiency and minimizes emissions.	9	PPMVD @ 15% O2	BACT
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		catalytic oxidation and carbon monoxide turndown	10	PPMVD @15%O2	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	386	MMBtu/hr	Good Combustion Practices and burning clean fuels (NG)	15	PPMV @ 15% O2	BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	1069	mm btu/hr	good combustion practices and fueled by natural gas	15	PPMVD	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	927	MM BTU/h	Proper Equipment Design, Proper Operation, and Good Combustion Practices.	25	PPMV	BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	540	mm btu/hr	Good Combustion Practices	25	PPMVD	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	0		Minimizing duration of startup/shutdown, using good air pollution control practices and safe operating practices.	0.01	TON/YR	BACT

Table C-1c: RBLC PM Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
FL-0355	FORT MYERS PLANT	FLORIDA POWER & LIGHT (FPL)	9/10/2015	2262.4	MMBtu/hr gas	Use of clean fuels, and annual VE test	2	GR S / 100 SCF GAS	BACT
FL-0355	FORT MYERS PLANT	FLORIDA POWER & LIGHT (FPL)	9/10/2015	2262.4	MMBtu/hr gas	Use of clean fuels	2	GR S / 100 SCF GAS	BACT
FL-0355	FORT MYERS PLANT	FLORIDA POWER & LIGHT (FPL)	9/10/2015	2262.4	MMBtu/hr gas	Use of clean fuels	2	GR S / 100 SCF GAS	BACT
FL-0354	LAUDERDALE PLANT	FLORIDA POWER & LIGHT	8/25/2015	2100	MMBtu/hr (approx)	Clean fuel prevents PM formation	2	GR. S / 100 SCF	BACT
FL-0354	LAUDERDALE PLANT	FLORIDA POWER & LIGHT	8/25/2015	2100	MMBtu/hr (approx)	Clean fuel prevents PM formation	2	GR. S / 100 SCF	BACT
FL-0354	LAUDERDALE PLANT	FLORIDA POWER & LIGHT	8/25/2015	2100	MMBtu/hr (approx)	Clean fuel prevents PM formation	2	GR. S / 100 SCF GAS	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	240	MW Each	good combustion practices, low sulfur fuels	0.0031	LB	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	240	MW Each	good combustion practices and low sulfur fuel content	0.0031	LB	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	240	MW Each	good combustion practices and low sulfur fuel content.	0.0031	LB	BACT
*TN-0187	JOHNSONVILLE COMBUSTION TURBINE	TENNESSEE VALLEY AUTHORITY	8/31/2022	465.8	MMBtu/hr	good combustion design and operating practices and the use of low sulfur fuel	3.65	LB/HR	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	667	MMBTU/H	Pipeline quality natural gas, inlet air conditioning and good combustion practices.	4.5	LB/H	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	667	MMBTU/H	Pipeline quality natural gas, inlet air conditioning and good combustion practices.	4.5	LB/H	BACT
MI-0447	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	1/7/2021	667	MMBTU/H	Pipeline quality natural gas, inlet air conditioning and good combustion practices.	4.5	LB/H	BACT
MI-0447	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	1/7/2021	667	MMBTU/H	Pipeline quality natural gas, inlet air conditioning and good combustion practices.	4.5	LB/H	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	667	MMBTU/H	Pipeline quality natural gas, inlet air conditioning and good combustion practices.	4.5	LB/H	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	667	MMBTU/H	Pipeline quality natural gas, inlet air conditioning and good combustion practices.	4.5	LB/H	BACT
IN-0261	VERMILLION GENERATING STATION	DUKE ENERGY INDIANA, LLC VERMILLION GENERATING STA	2/28/2017	80	MW	GOOD COMBUSTION PRACTICES	5	LB/H	BACT
IN-0261	VERMILLION GENERATING STATION	DUKE ENERGY INDIANA, LLC VERMILLION GENERATING STA	2/28/2017	80	MW	GOOD COMBUSTION PRACTICES	5	LB/H	BACT
IN-0261	VERMILLION GENERATING STATION	DUKE ENERGY INDIANA, LLC VERMILLION GENERATING STA	2/28/2017	80	MW	GOOD COMBUSTION PRACTICES	5	LB/H	BACT
NJ-0086	BAYONNNE ENERGY CENTER	BAYONNNE ENERGY CENTER LLC	8/26/2016	2143980	MMBTU/YR	Use of Natural gas a clean burning fuel	5	LB/H	CASE-BY-CASE
NJ-0086	BAYONNNE ENERGY CENTER	BAYONNNE ENERGY CENTER LLC	8/26/2016	2143980	MMBTU/YR	Use of Natural gas a clean burning fuel	5	LB/H	CASE-BY-CASE
NJ-0086	BAYONNNE ENERGY CENTER	BAYONNNE ENERGY CENTER LLC	8/26/2016	2143980	MMBTU/YR	Use of natural gas a clean burning fuel	5	LB/H	CASE-BY-CASE
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices and the use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices and the use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT

Table C-1c: RBLC PM Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices and the use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices and the use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hR	Good combustion practices and the use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hR	Good combustion practices and the use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices and the use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices and the use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices and the use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices and the use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of low sulfur fuels (pipeline quality natural gas)	6.3	LB/HR	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	927	MM BTU/h	Exclusive Combustion of Fuel Gas and Good Combustion Practices, Including Proper Burner Design.	8	LB/H	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	927	MM BTU/h	Exclusive Combustion of Fuel Gas and Good Combustion Practices, Including Proper Burner Design.	8	LB/H	BACT
TX-0769	VAN ALSTYNE ENERGY CENTER (VAEC)	NAVASOTA NORTH COUNTRY PEAKERS OPERATING COMPANY I	10/27/2015	183	MW	Pipeline Quality Natural Gas	8.6	LB/H	BACT
TX-0769	VAN ALSTYNE ENERGY CENTER (VAEC)	NAVASOTA NORTH COUNTRY PEAKERS OPERATING COMPANY I	10/27/2015	183	MW	Pipeline Quality Natural Gas	8.6	LB/H	BACT
TX-0777	UNION VALLEY ENERGY CENTER	NAVASOTA SOUTH PEAKERS OPERATING COMPANY I, LLC.	12/9/2015	183	MW	pipeline quality natural gas, good combustion practices	8.6	LB/H	BACT
TX-0777	UNION VALLEY ENERGY CENTER	NAVASOTA SOUTH PEAKERS OPERATING COMPANY I, LLC.	12/9/2015	183	MW	pipeline quality natural gas, good combustion practices	8.6	LB/H	BACT
VA-0326	DOSWELL ENERGY CENTER	DOSWELL LIMITED PARTNERSHIP DOSWELL ENERGY CENTER	10/4/2016	1961	MMBTU/HR	Good combustion, operation and maintenance practices and use of pipeline quality natural gas	10	LB	BACT
VA-0326	DOSWELL ENERGY CENTER	DOSWELL LIMITED PARTNERSHIP DOSWELL ENERGY CENTER	10/4/2016	1961	MMBTU/HR	Good combustion, operation and maintenance practices and use of pipeline quality natural gas	12	LB	BACT
VA-0326	DOSWELL ENERGY CENTER	DOSWELL LIMITED PARTNERSHIP DOSWELL ENERGY CENTER	10/4/2016	1961	MMBTU/HR	Good combustion, operation and maintenance practices and use of pipeline quality natural gas	12	LB	BACT
TX-0764	NACOGDOCHES POWER ELECTRIC GENERATING PLANT	NACOGDOCHES POWER, LLC	10/14/2015	232	MW	Pipeline quality natural gas; limited hours; good combustion practices.	12.09	LB/HR	BACT
TX-0764	NACOGDOCHES POWER ELECTRIC GENERATING PLANT	NACOGDOCHES POWER, LLC	10/14/2015	232	MW	Pipeline quality natural gas; limited hours; good combustion practices.	12.09	LB/HR	BACT
TX-0764	NACOGDOCHES POWER ELECTRIC GENERATING PLANT	NACOGDOCHES POWER, LLC	10/14/2015	232	MW	Pipeline quality natural gas; limited hours; good combustion practices.	12.09	LB/HR	BACT
TX-0788	NECHES STATION	APEX TEXAS POWER LLC	3/24/2016	232	MW	good combustion practices, low sulfur fuel	13.4	LB/H	BACT
TX-0788	NECHES STATION	APEX TEXAS POWER LLC	3/24/2016	232	MW	good combustion practices, low sulfur fuel	13.4	LB/H	BACT

Table C-1c: RBLC PM Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
TX-0794	HILL COUNTY GENERATING FACILITY	BRAZOS ELECTRIC COOPERATIVE	4/7/2016	171	MW	Premixing of fuel and air enhances combustion efficiency and minimizes emissions.	14	LB/H	BACT
TX-0794	HILL COUNTY GENERATING FACILITY	BRAZOS ELECTRIC COOPERATIVE	4/7/2016	171	MW	Premixing of fuel and air enhances combustion efficiency and minimizes emissions.	14	LB/H	BACT
WV-0026	WAVERLY FACILITY	PLEASANTS ENERGY, LLC	1/23/2017	1571	mmbtu/hr	Inlet Air Filtration, Use of Natural Gas, Ultra-Low Sulfur Diesel	15	LB/HR	BACT
WV-0028	WAVERLY POWER PLANT	PLEASANTS ENERGY LLC	3/13/2018	167.8	MW	Inlet air filtration.	15.09	LB/HR	BACT
TX-0768	SHAWNEE ENERGY CENTER	SHAWNEE ENERGY CENTER, LLC	10/9/2015	230	MW	Pipeline quality natural gas; limited hours; good combustion practices.	84.1	LB/HR	BACT
TX-0768	SHAWNEE ENERGY CENTER	SHAWNEE ENERGY CENTER, LLC	10/9/2015	230	MW	Pipeline quality natural gas; limited hours; good combustion practices.	84.1	LB/HR	BACT
NY-0106	GREENIDGE STATION	GREENIDGE GENERATION LLC	9/7/2016	107	MW	Baghouse with leak detection system.	0.002	LB/MMBTU	BACT
IL-0121	INVENERGY NELSON EXPANSION LLC	INVENERGY	9/27/2016	190	MW	turbine design and good combustion practices	0.0038	LB/MMBTU	BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	540	mm btu/hr	Good Combustion Practices and Use of low sulfur facility fuel gas	0.0066	LB/MM BTU	BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	540	mm btu/hr	Good Combustion Practices and Use of low sulfur facility fuel gas	0.0066	LB/MM BTU	BACT
IL-0121	INVENERGY NELSON EXPANSION LLC	INVENERGY	9/27/2016	190	MW	turbine design and good combustion practices	0.005	LB/MMBTU	BACT
IL-0121	INVENERGY NELSON EXPANSION LLC	INVENERGY	9/27/2016	190	MW	turbine design and good combustion practices	0.005	LB/MMBTU	BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	228	mw	good combustion practiced and pipeline quality natural gas	0.005	LB/MMBTU	BACT
IN-0264	MONTPELIER GENERATING STATION	AES OHIO GENERATION, LLC	1/6/2017	270.9	MMBTU/H	USE NATURAL GAS AS PRIMARY FUEL; GOOD COMBUSTION PRACTICES	0.0066	LB/MMBTU	BACT
IN-0264	MONTPELIER GENERATING STATION	AES OHIO GENERATION, LLC	1/6/2017	270.9	MMBTU/H	NATURAL GAS PRIMARY FUEL; GOOD COMBUSTION PRACTICES	0.0066	LB/MMBTU	BACT
PA-0305	APPALACHIA/PETROCHEMICALS COMPLEX	SHELL CHEMICAL APPALACHIA	6/18/2015	0	Three 40.6 MW turbines		0.0066	LB/MMBTU	BACT
PA-0305	APPALACHIA/PETROCHEMICALS COMPLEX	SHELL CHEMICAL APPALACHIA	6/18/2015	0	Three 40.6 MW turbines		0.0066	LB/MMBTU	LAER
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	386	MMBtu/hr	Good Combustion Practices and burning clean fuels (NG)	0.007	LB/MMBTU	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	386	MMBtu/hr	Good Combustion Practices and burning clean fuels (NG)	0.007	LB/MMBTU	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	386	MMBtu/hr	Good Combustion Practices and burning clean fuels (NG)	0.007	LB/MMBTU	BACT
AK-0088	LIQUEFACTION PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	7/7/2022	1113	MMBtu/hr	Good combustion practices and burning clean fuel (natural gas)	0.007	LB/MMBTU	BACT
AK-0088	LIQUEFACTION PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	7/7/2022	1113	MMBtu/hr	Good combustion practices and burning clean fuel (natural gas)	0.007	LB/MMBTU	BACT
AK-0088	LIQUEFACTION PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	7/7/2022	1113	MMBtu/hr	Good combustion practices and burning clean fuel (natural gas)	0.007	LB/MMBTU	BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		good combustion practices and the use of gaseous fuel	0.0075	LB/MMBTU	BACT

Table C-1c: RBLC PM Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		good combustion practices and the use of gaseous fuel	0.0075	LB/MMBTU	BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		good combustion practices and the use of gaseous fuel	0.0075	LB/MMBTU	BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	180.2	MW	Pipeline quality natural gas and good combustion practices are used, which meets Tier I BACT. The turbines are proposed to meet 0.0075 lb/MMBtu total particulate matter. It is assumed that emissions of PM2.5 are equal to PM10, which are equal to total PM. No technically feasible post-combustion control technologies are available to reduce particulate matter emissions from gas turbines due to the large amount of excess air inherent to the turbine operation and would create an unacceptable amount of backpressure.	0.0075	LB/MMBTU	BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	1069	mm btu/hr	good combustion practices and fueled by natural gas	0.0076	LB/MM BTU	BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	1069	mm btu/hr	good combustion practices and fueled by natural gas	0.0076	LB/MM BTU	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	229	MW		0.008	LB/MMBTU	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	229	MW		0.008	LB/MMBTU	BACT
NY-0106	GREENIDGE STATION	GREENIDGE GENERATION LLC	9/7/2016	107	MW	Baghouse with leak detection system.	8.25	E-3 LB/MMBTU	BACT
NY-0106	GREENIDGE STATION	GREENIDGE GENERATION LLC	9/7/2016	107	MW	Baghouse with leak detection system.	8.25	E-3 LB/MMBTU	BACT
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	575	mm btu/hr	Good combustion practices.	0.0183	LB/MM BTU	BACT
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	575	mm btu/hr	Good combustion practices.	0.0183	LB/MMBTU	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	0		Minimizing duration of startup/shutdown, using good air pollution control practices and safe operating practices.	0.01	TON/YR	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	0		Minimizing duration of startup/shutdown, using good air pollution control practices and safe operating practices.	0.01	TON/YR	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	0		Minimizing duration of startup/shutdown, using good air pollution control practices and safe operating practices.	0.01	TON/YR	BACT
TX-0819	GAINES COUNTY POWER PLANT	SOUTHWESTERN PUBLIC SERVICE COMPANY	4/28/2017	227.5	MW	Pipeline quality natural gas; limited hours; good combustion practices	8.5	T/YR	BACT
TX-0819	GAINES COUNTY POWER PLANT	SOUTHWESTERN PUBLIC SERVICE COMPANY	4/28/2017	227.5	MW	Pipeline quality natural gas; limited hours; good combustion practices	8.5	T/YR	BACT
TX-0819	GAINES COUNTY POWER PLANT	SOUTHWESTERN PUBLIC SERVICE COMPANY	4/28/2017	227.5	MW	Pipeline quality natural gas; limited hours; good combustion practices	8.5	T/YR	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	920	MW	Use of pipeline quality natural gas and good combustion practices.	11.81	TON/YR	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	920	MW	Use of pipeline quality natural gas and good combustion practices.	11.81	TON/YR	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	920	MW	Use of pipeline quality natural gas and good combustion practices.	11.81	TON/YR	BACT
TX-0826	MUSTANG STATION	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	8/16/2017	162.8	MW	Pipeline quality natural gas and good combustion practices	27	T/YR	BACT

Table C-1c: RBLC PM Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
TX-0826	MUSTANG STATION	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	8/16/2017	162.8	MW	Pipeline quality natural gas and good combustion practices	27	T/YR	BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	333	mm btu/hr	good combustion practices and fueled by natural gas	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	333	mm btu/hr	good combustion practices and fueled by natural gas	0		BACT
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		Good combustion practices and clean natural gas	0		BACT
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		Good combustion practices and clean natural gas	0		BACT
TX-0694	INDECK WHARTON ENERGY CENTER	INDECK WHARTON, L.L.C.	2/2/2015	220	MW		0		BACT
TX-0733	ANTELOPE ELK ENERGY CENTER	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	5/12/2015	202	MW	Pipeline quality natural gas; limited hours; good combustion practices.	0		BACT
TX-0733	ANTELOPE ELK ENERGY CENTER	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	5/12/2015	202	MW	Pipeline quality natural gas; limited hours; good combustion practices.	0		BACT
TX-0733	ANTELOPE ELK ENERGY CENTER	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	5/12/2015	202	MW	Pipeline quality natural gas; limited hours; good combustion practices.	0		BACT
TX-0975	FREESTONE PEAKERS PLANT	FPEC, LLC	6/13/2024	221	MW	Good combustion practices	0		BACT
TX-0975	FREESTONE PEAKERS PLANT	FPEC, LLC	6/13/2024	221	MW	Good combustion practices	0		BACT
TX-0975	FREESTONE PEAKERS PLANT	FPEC, LLC	6/13/2024	221	MW	Good combustion practices	0		BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	180.2	MW	Pipeline quality natural gas and good combustion practices are used, which meets Tier I BACT. The turbines are proposed to meet 0.0075 lb/MMBtu total particulate matter. It is assumed that emissions of PM2.5 are equal to PM10, which are equal to total PM. No technically feasible post-combustion control technologies are available to reduce particulate matter emissions from gas turbines due to the large amount of excess air inherent to the turbine operation and would create an unacceptable amount of backpressure.	0		BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	180.2	MW	Pipeline quality natural gas and good combustion practices are used, which meets Tier I BACT. The turbines are proposed to meet 0.0075 lb/MMBtu total particulate matter. It is assumed that emissions of PM2.5 are equal to PM10, which are equal to total PM. No technically feasible post-combustion control technologies are available to reduce particulate matter emissions from gas turbines due to the large amount of excess air inherent to the turbine operation and would create an unacceptable amount of backpressure.	0		BACT

Table C-1c: RBLC PM Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	0		Lube oil vents are equipped with mist eliminators to minimize emissions. Good design and operating practices are used. Lube oil is assumed to emit as VOC, PM, PM10, and PM2.5. It is assumed that PM is equal to PM10, which is equal to PM2.5. The unloading, storage, and heated recirculation of lube oil in the gas and steam turbine reservoirs will emit less than 0.01 gallon per day of oil per turbine mist eliminator vent, based on oil consumption limits permitted for similar turbines.	0		BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	0		Lube oil vents are equipped with mist eliminators to minimize emissions. Good design and operating practices are used. Lube oil is assumed to emit as VOC, PM, PM10, and PM2.5. It is assumed that PM is equal to PM10, which is equal to PM2.5. The unloading, storage, and heated recirculation of lube oil in the gas and steam turbine reservoirs will emit less than 0.01 gallon per day of oil per turbine mist eliminator vent, based on oil consumption limits permitted for similar turbines.	0		BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	0		Lube oil vents are equipped with mist eliminators to minimize emissions. Good design and operating practices are used. Lube oil is assumed to emit as VOC, PM, PM10, and PM2.5. It is assumed that PM is equal to PM10, which is equal to PM2.5. The unloading, storage, and heated recirculation of lube oil in the gas and steam turbine reservoirs will emit less than 0.01 gallon per day of oil per turbine mist eliminator vent, based on oil consumption limits permitted for similar turbines.	0		BACT

Table C-1d: RBLC SO2 Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
TX-0788	NECHES STATION	APEX TEXAS POWER LLC	3/24/2016	232	MW	good combustion practices, low sulfur fuel	1	GR/100 SCF	BACT
TX-0819	GAINES COUNTY POWER PLANT	SOUTHWESTERN PUBLIC SERVICE COMPANY	4/28/2017	227.5	MW	Pipeline quality natural gas; limited hours; good combustion practices	1.54	GR/100 DSCF	BACT
FL-0355	FORT MYERS PLANT	FLORIDA POWER & LIGHT (FPL)	9/10/2015	2262.4	MMBtu/hr gas	Use of clean fuels	2	GR S / 100 SCF GAS	BACT
FL-0354	LAUDERDALE PLANT	FLORIDA POWER & LIGHT	8/25/2015	2100	MMBtu/hr (approx)	Limitation on S in fuel	2	GR. S / 100 SCF	BACT
IN-0261	VERMILLION GENERATING STATION	DUKE ENERGY INDIANA, LLC VERMILLION GENERATING STA	2/28/2017	80	MW	GOOD COMBUSTION PRACTICES	0.6	LB/H	BACT
NJ-0086	BAYONNNE ENERGY CENTER	BAYONNNE ENERGY CENTER LLC	8/26/2016	2143980	MMBTU/YR	Use of natural gas a low sulfur fuel	0.77	LB/H	OTHER CASE-BY-CASE
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	575	mm btu/hr	Good combustion practices and use of low sulfur fuels	0.0134	LB/MM BTU	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	927	MM BTU/h	Exclusive Combustion of low sulfur fuel - Fuel sulfur content <4 ppm, proper engineering practices.	4	PPMV	BACT
AK-0088	LIQUEFACTION PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	7/7/2022	1113	MMBtu/hr	pipeline quality natural gas and good combustion practices	16	PPMV SULFUR IN FUEL	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	386	MMBtu/hr	Good combustion practices and clean burning fuel low sulfur natural gas	96	PPMV SULFUR IN FUEL	BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	540	mm btu/hr	Good Combustion Practices and Use of low sulfur facility fuel gas	0		BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		Low sulfur content in fuel	0		BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	180.2	MW	Pipeline quality natural gas containing no more than 2 grains sulfur per 100 dry standard cubic foot natural gas (gr S/100 dscf) is used. Tier I BACT for simple cycle natural gas fired turbines is good combustion practices and pipeline quality natural gas with a maximum sulfur content of 2 to 5 gr S/100 dscf on an hourly basis and between 0.5 to 1 gr S/100 dscf on an annual basis. While the hourly value is met, the annual Tier I BACT guideline is exceeded. As justification for the annual Tier I BACT guideline, the limited hours of operation authorized at no more than 2,500 hours per year per turbine results in an effective annual sulfur content of 0.57 gr S/100 dscf when compared to continuous operation of 2 gr S/100 scf at 8,760 hours per year	0		BACT

Table C-1e: RBLC H2SO4 Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	228	mw	low sulfur fuel	6.5	10-4 LB/MMBTU	BACT
TX-0788	NECHES STATION	APEX TEXAS POWER LLC	3/24/2016	232	MW	good combustion practices, low sulfur fuel	1	GR/100 SCF	BACT
FL-0354	LAUDERDALE PLANT	FLORIDA POWER & LIGHT	8/25/2015	2100	MMBtu/hr (approx)	Limitation on S in fuel	2	GR. S / 100 SCF	BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	180.2	MW	Pipeline quality natural gas containing no more than 2 grains sulfur per 100 dry standard cubic foot natural gas (gr S/100 dscf) is used. Tier I BACT for simple cycle natural gas fired turbines is good combustion practices and pipeline quality natural gas with a maximum sulfur content of 2 to 5 gr S/100 dscf on an hourly basis and between 0.5 to 1 gr S/100 dscf on an annual basis. While the hourly value is met, the annual Tier I BACT guideline is exceeded. As justification for the annual Tier I BACT guideline, the limited hours of operation authorized at no more than 2,500 hours per year per turbine results in an effective annual sulfur content of 0.57 gr S/100 dscf when compared to continuous operation of 2 gr S/100 scf at 8,760 hours per year	0		BACT

Table C-1f: RBLC VOC Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	667	MMBTU/H	Good combustion practices.	5	LB/H	BACT
MI-0447	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	1/7/2021	667	MMBTU/H	Good combustion practices	5	LB/H	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	667	MMBTU/H	Good combustion practices.	5	LB/H	BACT
TX-0794	HILL COUNTY GENERATING FACILITY	BRAZOS ELECTRIC COOPERATIVE	4/7/2016	171	MW	Premixing of fuel and air enhances combustion efficiency and minimizes emissions.	5.4	LB/H	BACT
IN-0261	VERMILLION GENERATING STATION	DUKE ENERGY INDIANA, LLC VERMILLION GENERATING STA	2/28/2017	80	MW	GOOD COMBUSTION PRACTICES	17.6	LB/H	BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	540	mm btu/hr	Good Combustion Practices and Use of low sulfur facility fuel gas	0.002	LB/MM BTU	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	386	MMBtu/hr	Good Combustion Practices and burning clean fuels (NG)	0.0022	LB/MMBTU	BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	228	mw	good combustion practices and oxidation catalyst	0.7	PPMVD @ 15% O2	LAER
PA-0305	APPALACHIA/PETROCHEMICALS COMPLEX	SHELL CHEMICAL APPALACHIA	6/18/2015	0	Three 40.6 MW turbines		1	PPMDV @ 15% O2	LAER
*TX-0989	LONE STAR POWER STATION	ENTERGY TEXAS INC	3/17/2025	0		Oxidation catalyst and good combustion practices	1	PPMVD	LAER
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	927	MM BTU/h	Proper Equipment Design, Proper Operation, and Good Combustion Practices.	1.4	PPMV	BACT
TX-0768	SHAWNEE ENERGY CENTER	SHAWNEE ENERGY CENTER, LLC	10/9/2015	230	MW	Pipeline quality natural gas; limited hours; good combustion practices.	1.4	PPMV	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	240	MW Each	CLEAN FUELS AND GOOD COMBUSTION PRACTICES	1.4	PPMVD	BACT
PA-0306	PARTNERS/WESTMORELAND GEN FAC	TENASKA PA PARTNERS LLC	2/12/2016	0		Ox Cat and good combustion practices	1.4	PPMVD @ 15% O2	LAER
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	1069	mm btu/hr	good combustion practices and fueled by natural gas	1.6	PPMVD	BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		Oxidization catalyst, good combustion practices and the use of gaseous fuel	1.7	PPMVD	BACT
AK-0088	LIQUEFACTION PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	7/7/2022	1113	MMBtu/hr	Oxidation catalyst and good combustion practices	2	PPMV @ 15% O2	BACT
TX-0788	NECHES STATION	APEX TEXAS POWER LLC	3/24/2016	232	MW	good combustion practices	2	PPM	BACT
TX-0819	GAINES COUNTY POWER PLANT	SOUTHWESTERN PUBLIC SERVICE COMPANY	4/28/2017	227.5	MW	Pipeline quality natural gas; limited hours; good combustion practices	2	PPMVD	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	920	MW	Good combustion practices	2	PPMVD	BACT
TX-0975	FREESTONE PEAKERS PLANT	FPEC, LLC	6/13/2024	221	MW	Good combustion practices	2	PPMVD	BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	180.2	MW	Each turbine is limited to 2 ppmvd at 15% O2, which meets Tier I BACT. Good combustion practices are used. MSS - Limited to 50 startups and 50 shutdowns per year for each turbine. Startup and shutdown events are each expected to last less than an hour in duration.	2	PPMVD	BACT
TX-0733	ANTELOPE ELK ENERGY CENTER	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	5/12/2015	202	MW	Good combustion practices	2	PPMVD @ 15% O2	BACT

Table C-1f: RBLC VOC Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
TX-0764	NACOGDOCHES POWER ELECTRIC GENERATING PLANT	NACOGDOCHES POWER, LLC	10/14/2015	232	MW	Pipeline quality natural gas; limited hours; good combustion practices.	2	PPMVD @ 15% O2	BACT
CA-1238	PUENTE POWER		10/13/2016	262	MW		2	PPMVD AS METHANE	CASE-BY-CASE
NJ-0086	BAYONNNE ENERGY CENTER	BAYONNNE ENERGY CENTER LLC	8/26/2016	2143980	MMBTU/YR	Add-on VOC control is Oxidation Catalyst, and use of natural gas as fuel for pollution prevention	2	PPMVD@15 %O2	CASE-BY-CASE
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	575	mm btu/hr	Good combustion practices and use of clean fuel.	3	PPMVD	BACT
TX-0833	JACKSON COUNTY GENERATORS	SOUTHERN POWER	1/26/2018	0		Minimizing duration of startup/shutdown, using good air pollution control practices and safe operating practices.	0.06	TON/YR	BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	333	mm btu/hr	good combustion practices and fueled by natural gas	0		BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of pipeline quality natural gas	0		BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of pipeline quality natural gas	0		BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hR	Good combustion practices & use of pipeline quality natural gas	0		BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of pipeline quality natural gas	0		BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of pipeline quality natural gas	0		BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Good combustion practices & use of pipeline quality natural gas	0		BACT
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		Good combustion practices	0		BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	0		Lube oil vents are equipped with mist eliminators to minimize emissions. Good design and operating practices are used. Lube oil is assumed to emit as VOC, PM, PM10, and PM2.5. It is assumed that PM is equal to PM10, which is equal to PM2.5. The unloading, storage, and heated recirculation of lube oil in the gas and steam turbine reservoirs will emit less than 0.01 gallon per day of oil per turbine mist eliminator vent, based on oil consumption limits permitted for similar turbines.	0		BACT

Table C-1g: RBLC GHG Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Facility-wide energy efficiency measures , such as improved combustion measures, and use of pipeline quality natural gas.	50	KG/GJ	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Facility-wide energy efficiency measures , such as improved combustion measures, and use of pipeline quality natural gas.	50	KG/GJ	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	240	MW Each	GOOD COMBUSTION PRACTICES	15420	BTU	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	386	MMBtu/hr	Good combustion practices and clean burning fuel (NG)	117.1	LB/MMBTU	BACT
AK-0088	LIQUEFACTION PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	7/7/2022	1113	MMBtu/hr	Good combustion practices and burning clean fuels (natural gas)	117.1	LB/MMBTU	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hR	Facility-wide energy efficiency measures , such as improved combustion measures, and use of pipeline quality natural gas.	120	LB/MM BTU	BACT
*LA-0327	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER ONE, LLC	5/23/2018	2201	MM BTU/hr	Facility-wide energy efficiency measures , such as improved combustion measures, and use of pipeline quality natural gas.	120	LB/MM BTU	BACT
*TN-0187	JOHNSONVILLE COMBUSTION TURBINE	TENNESSEE VALLEY AUTHORITY	8/31/2022	465.8	MMBtu/hr	Efficient turbine operation and good combustion practices	120	LB/MMBTU	BACT
TX-0826	MUSTANG STATION	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	8/16/2017	162.8	MW	Pipeline quality natural gas and good combustion practices	120	LB/MMBTU	BACT
TX-0975	FREESTONE PEAKERS PLANT	FPEC, LLC	6/13/2024	221	MW	Good combustion practices	800	LB/MWH	BACT
PA-0305	APPALACHIA/PETROCHEMICALS COMPLEX	SHELL CHEMICAL APPALACHIA	6/18/2015	0	Three 40.6 MW turbines		1030	CO2E/MWH	BACT
TX-0819	GAINES COUNTY POWER PLANT	SOUTHWESTERN PUBLIC SERVICE COMPANY	4/28/2017	227.5	MW	Pipeline quality natural gas; limited hours; good combustion practices	1300	LB/MW H	BACT
TX-0735	ANTELOPE ELK ENERGY CENTER	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	5/20/2015	202	MW	Energy efficiency, good design & combustion practices	1304	LB CO2/MWHR	BACT
TX-0788	NECHES STATION	APEX TEXAS POWER LLC	3/24/2016	232	MW	good combustion practiceS	1341	LB/MW H	BACT
FL-0355	FORT MYERS PLANT	FLORIDA POWER & LIGHT (FPL)	9/10/2015	2262.4	MMBtu/hr gas	Use of low-emitting fuel and efficient turbine	1374	LB CO2E / MWH	BACT
TX-0775	CLEAR SPRINGS ENERGY CENTER (CSEC)	NAVASOTA SOUTH PEAKERS OPERATING COMPANY II, LLC.	11/13/2015	183	MW	Low carbon fuel, good combustion, efficient combined cycle design	1461	LB/MW H	BACT
TX-0778	UNION VALLEY ENERGY CENTER	NAVASOTA SOUTH PEAKERS OPERATING COMPANY II, LLC.	12/16/2015	183	MW		1461	LB/MW H	BACT
TX-0824	JACKSON COUNTY GENERATING FACILITY	SOUTHERN POWER	6/30/2017	920	MW	energy efficiency designs, practices, and procedures, CT inlet air cooling, periodic CT burner maintenance and tuning, reduction in heat loss, i.e., insulation of the CT, instrumentation and controls	1316	LB/MW HR	BACT
TX-0771	SHAWNEE ENERGY CENTER	SHAWNEE ENERGY CENTER, LLC	11/10/2015	230	MW		1398	LB/MWH	BACT
TX-0794	HILL COUNTY GENERATING FACILITY	BRAZOS ELECTRIC COOPERATIVE	4/7/2016	171	MW		1434	LB/MWH	BACT
TX-0900	ECTOR COUNTY ENERGY CENTER	ECTOR COUNTY ENERGY CENTER LLC	8/17/2020	0		Best management practices and good combustion practices, clean fuel	1514	LB/MWHR	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	667	MMBTU/H	Low carbon fuel (pipeline quality natural gas), good combustion practices, and energy efficiency measures.	318404	T/YR	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	927	MM BTU/h	Exclusively combust low carbon fuel gas, good combustion practices, good operation and maintenance practices, and insulation	1426146	T/YR	BACT

Table C-1g: RBLC GHG Results for Natural Gas-Fired Simple-Cycle Combustion Turbines

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IL-0121	INVENERGY NELSON EXPANSION LLC	INVENERGY	9/27/2016	190	MW	Turbine-generator design and proper operation	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	333	mm btu/hr	good combustion/operating/maintenance practices and fueled by natural gas; use intake air chiller	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	1069	mm btu/hr	good combustion practices and fueled by natural gas; Use high thermal efficiency turbines	0		BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	540	mm btu/hr	Use Low Carbon Fuel, Energy Efficiency Measures, and Good Combustion Practices	0		BACT
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		Low carbon fuels Energy efficient designs and operation	0		BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	228	mw	max heat rate 7,604 btu/kw-h HHV without duct firing good combustion practice and burning natural gas	0		BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		good combustion practices and the use of gaseous fuel	0		BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	180.2	MW	JCP proposes an output-based standard of 1,400 lb/MW-hr on an annual average, or an input-based standard of 120 lb CO2/MMBtu on an annual average, achieved through energy efficient design and use of low carbon fuels. As stated above, the gas turbines are not subject to 40 CFR 60 Subpart TTTT or 40 CFR 60 Subpart TTTTa, which would require more stringent CO2e emission standards. An RBLC search was conducted to evaluate the applicant-proposed BACT, which showed that previous determinations for natural gas fired simple cycle turbines were between 800 and 1,707 lb/MW-hr, with most in the range of 1,300 and 1,450 lb/MW-hr.	0		BACT
VA-0326	DOSWELL ENERGY CENTER	DOSWELL LIMITED PARTNERSHIP DOSWELL ENERGY CENTER	10/4/2016	1961	MMBTU/HR	Good combustion, maintenance and use of active combustion dynamic monitoring systems.	0		BACT
WV-0028	WAVERLY POWER PLANT	PLEASANTS ENERGY LLC	3/13/2018	167.8	MW	Use of natural gas & use of GE 7FA.004	0		BACT

Table C-2a: RBLC NOx Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	3.8	MMBTU/H	Low NOx burner	0.14	LB/H	BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0.23	LB/HR	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	15	MMBTU/H	Low-NOx gas burner	0.3	LB/H	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.3	LB/HR	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	3.7	MMBTU/H	Good combustion practices.	0.55	LB/H	BACT
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	15.17	MMBTU/H	Low-NOX burners, good combustion practices and the use of natural gas	0.634	LB/H	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	20	MMBtu/hr	Good Combustion & Operation Practices (GCOP) Plan, Low-Nox Burners (LNBs) capable of meeting 0.054 lb/MMBtu.	1.08	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	19.2	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	1.92	LB/HR	BACT
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	15	MMBTU/H	Good combustion practices and the use of natural gas	2.12	LB/H	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	13.31	MMBtu/hr	SCR and ultra low NOx burners, Fired only on natural gas supplied by a public utility.	0.006	LB/MMBTU	LAER
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	10	MMBtu/hr		0.011	LB/MMBTU	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Ultra-low NOx burners.	0.011	LB/MM BTU	BACT
IL-0121	INVENERGY NELSON EXPANSION LLC	INVENERGY	9/27/2016	15	mmbtu/hr		0.033	LB/MMBTU	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	7	MMBTU/H	Good combustion practices and low NOx burners	0.036	LB/MMBTU	BACT
*TX-1000	ANTELOPE ELK ENERGY CENTER	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	8/14/2025	5.5	MMBTU/HR	Low NOx burners and firing pipeline-quality natural gas. RBLC searches for small fuel gas heaters (<10 MMBtu/hr) installed in the last 10 years used low NOx burners achieving emission limits ranging from 0.036 to 0.10 lb/MMBtu.	0.036	LB/MMBTU	BACT
WI-0306	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	14.67	MMBTU/H	Low NOx burners, flue gas recirculation, shall be operated for no more than 500 hours, and shall combust only pipeline quality natural gas.	0.04	LB/MMBTU	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	10	MMBtu/hr	(a) P04 and P05 shall be equipped with low-NOx burners. (b) Only pipeline quality natural gas may be combusted in P04 and P05. (c) The permittee shall operate and maintain P04 and P05 according to the manufacturerâ€™s recommendations. (d) The permittee shall tune within 120 days of the initial operation, and within 24-months of each preceding tune-up. (e) NOx emissions from P04 and P05 may not exceed 0.049 lb/MMBtu each.	0.049	LB/MMBTU	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	10.2	MMBTU/H	Good design and combustion practices, low NOx burners.	0.05	LB/MMBTU	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	10.2	MMBTU/H	Good design and combustion practices, low NOx burners	0.05	LB/MMBTU	BACT

Table C-2a: RBLC NOx Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Low NOz burners	0.097	LB/MMBTU	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of gaseous fuels (fuel gas and/or natural gas). Good combustion practices. Proper equipment (burner) design/operation.	0.098	LB/MM BTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	15	MMBTU/hr each	Low Nox Burners	0.1	LB/MMBTU	BACT
FL-0356	OKEECHOBEE CLEAN ENERGY CENTER	FLORIDA POWER & LIGHT	3/9/2016	10	MMBtu/hr	Must have NOx emission design value less than 0.1 lb/MMBtu	0.1	LB/MMBTU	BACT
FL-0363	DANIA BEACH ENERGY CENTER	FLORIDA POWER AND LIGHT COMPANY	12/4/2017	9.9	MMBtu/hr	Manufacturer certification	0.1	LB/MMBTU	BACT
TX-0694	INDECK WHARTON ENERGY CENTER	INDECK WHARTON, L.L.C.	2/2/2015	3	MMBTU/H		0.1	LB/MMBTU	BACT
TX-0772	PORT OF BEAUMONT PETROLEUM TRANSLOAD TERMINAL (PBPTT)	JEFFERSON RAILPORT TERMINAL I TEXAS LLC	11/6/2015	13.2	MMBTU/H		0.1	LB/MMBTU	BACT
WI-0291	GRAYMONT WESTERN LIME-EDEN	GRAYMONT WESTERN LIME-EDEN	1/28/2019	1.5	mmBTU/hr	Good Combustion Practices	0.1	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Low NOX burners Combustion of clean fuel Good Combustion Practices	0.1	LB/MMBTU	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	20	MMBtu/hr	Low NOx burners and Good Combustion Practices	35	LB/MMSCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	9	MMBtu/hr	low NOx burners, good combustion practices, use of pipeline quality natural gas	50	LB/MMSCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	low NOx burners, good combustion practices and only pipeline quality natural gas shall be combusted	50	LB/MMSCF	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	1.44	MMBtu/hr	Low NOx burners and good combustion practices	50	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	18	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan. Equipped with low-NOx burners.	50	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	4.8	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan. This unit is equipped with low-NOx burners.	50	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	3	MMBtu/hr	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan. This unit is equipped with a low-NOx burner.	70	LB/MMSCF	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	17.5	MMBtu/hr, each	Low-Nox Burner (Designed to maintain 0.08 lb/MMBtu); and a Good Combustion and Operating Practices (GCOP) Plan.	81.6	LB/MMSCF	BACT
MI-0426	DTE GAS COMPANY - MILFORD COMPRESSOR STATION	DTE GAS COMPANY	3/24/2017	3	MMBTU/H	Ultra-low NOx burners and good combustion practices.	20	PPM AT 3% O2	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	17.5	MMBTU/H	LOW NOX BURNER FLUE GAS RECIRCULATION GCP	30	PPMVD	BACT
MI-0420	DTE GAS COMPANY--MILFORD COMPRESSOR STATION	DTE GAS COMPANY	6/3/2016	6	MMBTU/H	Ultra low NOx burners and good combustion practices.	14	PPMVOL	BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion practices	0		BACT

Table C-2a: RBLC NOx Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	16.13	mm btu/hr	ULNB and Good Combustion Practices	0		BACT

Table C-2b: RBLC CO Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	3.8	MMBTU/H	Good combustion controls	0.14	LB/H	BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0.19	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.25	LB/HR	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	3.7	MMBTU/H	Good combustion practices.	0.41	LB/H	BACT
OH-0383	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	7/17/2020	15	MMBTU/H	Good combustion practices and the use of natural gas	0.521	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	15	MMBTU/H	Combustion control	0.83	LB/H	BACT
OH-0383	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	7/17/2020	15.17	MMBTU/H	good combustion practices and the use of natural gas	1.25	LB/H	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	20	MMBtu/hr	Good Combustion & Operation Practices (GCOP) Plan	1.46	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	19.2	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	1.61	LB/HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Combustion of natural gas and good combustion practices	0.0004	LB/MMBTU	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	7	MMBTU/H	Good combustion practices	0.037	LB/MMBTU	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	13.31	MMBtu/hr		0.037	LB/MMBTU	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Good combustion practices.	0.037	LB/MM BTU	BACT
TX-0694	INDECK WHARTON ENERGY CENTER	INDECK WHARTON, L.L.C.	2/2/2015	3	MMBTU/H		0.04	LB/MMBTU	BACT
WI-0306	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	14.67	MMBTU/H	Shall be operated for no more than 500 hours and combust only pipeline quality natural gas.	0.04	LB/MMBTU	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	17.5	MMBTU/H	GCP	0.08	LB/MMBTU	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	10	MMBtu/hr		0.08	LB/MMBTU	BACT
MI-0420	DTE GAS COMPANY--MILFORD COMPRESSOR STATION	DTE GAS COMPANY	6/3/2016	6	MMBTU/H	Good combustion practices and clean burn fuel (pipeline quality natural gas)	0.08	LB/MMBTU	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	10	MMBtu/hr	(a) The permittee shall operate and maintain according to the manufacturerâ€™s recommendations. (b) The permittee shall tune within 120 days of the initial operation, and within 24-months of each preceding tune-up. (c) CO emissions may not exceed 0.08 lb/MMBtu. (d) The permittee may only combust pipeline quality natural gas.	0.08	LB/MMBTU	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of gaseous fuels (fuel gas and/or natural gas). Good combustion practices. Proper equipment (burner) design/operation. Compliance with 40 CFR 63 Subpart DDDDD.	0.082	LB/MM BTU	BACT

Table C-2b: RBLC CO Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	10.2	MMBTU/H	Good design and operation.	0.082	LB/MMBTU	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	10.2	MMBTU/H	Good design and operation	0.082	LB/MMBTU	BACT
WI-0291	GRAYMONT WESTERN LIME-EDEN	GRAYMONT WESTERN LIME-EDEN	1/28/2019	1.5	mmBTU/hr	Good Combustion Practices	0.082	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Combustion of Natural gas and Good Combustion Practice	0.0824	LB/MMBTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	15	MMBTU/hr each	Good combustion practices	0.084	LB/MMBTU	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	17.5	MMBtu/hr, each	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	84	LB/MMSCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	9	MMBtu/hr	good combustion practices	84	LB/MMSCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices	84	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	18	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	84	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	4.8	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	84	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	3	MMBtu/hr	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	84	LB/MMSCF	BACT
MI-0426	DTE GAS COMPANY - MILFORD COMPRESSOR STATION	DTE GAS COMPANY	3/24/2017	3	MMBTU/H	Good combustion practices and clean burn fuel (pipeline quality natural gas).	84	LB/MMSCF	BACT
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	16.8	MMBTU/HR	Good combustion practices	50	PPMVD	BACT
TX-0772	PORT OF BEAUMONT PETROLEUM TRANSLOAD TERMINAL (PBPTT)	JEFFERSON RAILPORT TERMINAL I TEXAS LLC	11/6/2015	13.2	MMBTU/H	Good combustion practice to ensure complete combustion.	50	PPMVD @ 3% O2	BACT
IL-0121	INVENERGY NELSON EXPANSION LLC	INVENERGY	9/27/2016	15	mmbtu/hr		1836	TPY	BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion practices	0		BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	16.13	mm btu/hr	Good Combustion Practices	0		BACT

Table C-2c: RBLC PM Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
SC-0179	CAROLINA PARTICLEBOARD	FLAKEBOARD AMERICA LIMITED	3/18/2015	1.83	MMBTU/H	NATURAL GAS USAGE AND GOOD COMBUSTION PRACTICES.	0.003	LB/H	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.006	LB/HR	BACT
SC-0179	CAROLINA PARTICLEBOARD	FLAKEBOARD AMERICA LIMITED	3/18/2015	1.83	MMBTU/H	USE OF NATURAL GAS AND GOOD COMBUSTION PRACTICES	0.01	LB/H	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.02	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.02	LB/HR	BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0.02	LB/HR	BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0.02	LB/HR	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	3.8	MMBTU/H	Low sulfur fuel	0.03	LB/H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	3.8	MMBTU/H	Low sulfur fuel	0.03	LB/H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	3.8	MMBTU/H	Low sulfur fuel	0.03	LB/H	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	19.2	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.04	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	20	MMBtu/hr	Good Combustion & Operation Practices (GCOP) Plan	0.04	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	20	MMBtu/hr	Good Combustion & Operation Practices (GCOP) Plan	0.07	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	20	MMBtu/hr	Good Combustion & Operation Practices (GCOP) Plan	0.07	LB/HR	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	15	MMBTU/H	Combustion control	0.075	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	15	MMBTU/H	Combustion control	0.075	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	15	MMBTU/H	Combustion control	0.075	LB/H	BACT
IL-0121	INVENERGY NELSON EXPANSION LLC	INVENERGY	9/27/2016	15	mmbtu/hr		0.105	LB/HR	BACT
IL-0121	INVENERGY NELSON EXPANSION LLC	INVENERGY	9/27/2016	15	mmbtu/hr		0.105	LB/HR	BACT
IL-0121	INVENERGY NELSON EXPANSION LLC	INVENERGY	9/27/2016	15	mmbtu/hr		0.105	LB/HR	BACT
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	15	MMBTU/H	Good combustion practices and the use of natural gas	0.112	LB/H	BACT
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	15	MMBTU/H	Good combustion practices and the use of natural gas	0.112	LB/H	BACT
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	15.17	MMBTU/H	Good combustion practices and the use of natural gas	0.113	LB/H	BACT
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	15.17	MMBTU/H	Good combustion practices and the use of natural gas	0.113	LB/H	BACT

Table C-2c: RBLC PM Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	19.2	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.15	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	19.2	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.15	LB/HR	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	10.2	MMBTU/H	Good combustion practices	0.0004	LB/MMBTU	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	10.2	MMBTU/H	Good combustion practices	0.0004	LB/MMBTU	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	10.2	MMBTU/H	Good combustion practices.	0.0005	LB/MMBTU	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	10.2	MMBTU/H	Good combustion practices	0.0005	LB/MMBTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	15	MMBTU/hr each	Good combustion practices	0.0019	LB/MMBTU	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	13.31	MMBtu/hr	Natural gas fired exclusively	0.002	LB/MMBTU	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Good combustion practices.	0.0048	LB/MM BTU	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Good combustion practices.	0.0048	LB/MM BTU	BACT
KS-0029	THE EMPIRE DISTRICT ELECTRIC COMPANY	THE EMPIRE DISTRICT ELECTRIC COMPANY	7/14/2015	18.6	MMBTU/HR		0.005	LB PER MMBTU	BACT
KS-0029	THE EMPIRE DISTRICT ELECTRIC COMPANY	THE EMPIRE DISTRICT ELECTRIC COMPANY	7/14/2015	18.6	MMBTU/HR		0.005	LB PER MMBTU	BACT
KS-0029	THE EMPIRE DISTRICT ELECTRIC COMPANY	THE EMPIRE DISTRICT ELECTRIC COMPANY	7/14/2015	18.6	MMBTU/HR		0.005	LB PER MMBTU	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	3.7	MMBTU/H	Good combustion practices.	0.007	LB/MMBTU	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	13.31	MMBtu/hr		0.007	LB/MMBTU	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	13.31	MMBtu/hr		0.007	LB/MMBTU	BACT
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	16.8	MMBTU/HR	Good combustion practices	0.007	LB/MMBTU	BACT
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	16.8	MMBTU/HR	Good combustion practices	0.007	LB/MMBTU	BACT
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	16.8	MMBTU/HR	Good combustion practices	0.007	LB/MMBTU	BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	16.13	mm btu/hr	Good Combustion Practices and Use of low sulfur facility fuel gas	0.0075	LB/MM BTU	BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	16.13	mm btu/hr	Good Combustion Practices and Use of low sulfur facility fuel gas	0.0075	LB/MM BTU	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of gaseous fuels (fuel gas and/or natural gas). Good combustion practices. Proper equipment (burner) design/operation.	0.0075	LB/MM BTU	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of gaseous fuels (fuel gas and/or natural gas). Good combustion practices. Proper equipment (burner) design/operation.	0.0075	LB/MM BTU	BACT

Table C-2c: RBLC PM Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0420	DTE GAS COMPANY--MILFORD COMPRESSOR STATION	DTE GAS COMPANY	6/3/2016	6	MMBTU/H	Good combustion practices and low sulfur fuel (pipeline quality natural gas).	0.0075	LB/MMBTU	BACT
MI-0420	DTE GAS COMPANY--MILFORD COMPRESSOR STATION	DTE GAS COMPANY	6/3/2016	6	MMBTU/H	Good combustion practices and low sulfur fuel (pipeline quality natural gas).	0.0075	LB/MMBTU	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	3.7	MMBTU/H	Good combustion practices.	0.0075	LB/MMBTU	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	3.7	MMBTU/H	Good combustion practices.	0.0075	LB/MMBTU	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	10.2	MMBTU/H	Good combustion practices	0.0075	LB/MMBTU	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	10.2	MMBTU/H	Good combustion practices	0.0075	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Combustion practice. Natural gas combustion only.	0.0075	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Combustion of Natural gas and Good Combustion Practice	0.0075	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Combustion of Natural gas and Good Combustion Practice	0.0075	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Combustion of natural gas and good combustion practices	0.0075	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Combustion of natural gas and good combustion practices	0.0075	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Combustion of natural gas and good combustion practices	0.0075	LB/MMBTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	15	MMBTU/hr each	Good combustion practices	0.0076	LB/MMBTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	15	MMBTU/hr each	Good combustion practices	0.0076	LB/MMBTU	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	10	MMBtu/hr		0.008	LB/MMBTU	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	10	MMBtu/hr		0.008	LB/MMBTU	BACT
WI-0306	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	14.67	MMBTU/H	Combust only pipeline quality natural gas, can be operated for no more than 500 hours.	0.008	LB/MMBTU	BACT
WI-0306	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	14.67	MMBTU/H	Combust only pipeline quality natural gas, can be operated for no more than 500 hours.	0.008	LB/MMBTU	BACT
WI-0306	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	14.67	MMBTU/H	Combust only pipeline quality natural gas, can be operated for no more than 500 hours.	0.008	LB/MMBTU	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	10	MMBtu/hr	(a) The permittee shall operate and maintain according to the manufacturerâ€™s recommendations. (b) The permittee shall tune within 120 days of the initial operation, and within 24-months of each preceding tune-up. (c) The permittee may only combust pipeline quality natural gas.	0.01	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	10	MMBtu/hr	Good combustion practices	7.6	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	10	MMBtu/hr	Good combustion practices	7.6	LB/MMBTU	BACT

Table C-2c: RBLC PM Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0426	DTE GAS COMPANY - MILFORD COMPRESSOR STATION	DTE GAS COMPANY	3/24/2017	3	MMBTU/H	Good combustion practices and low sulfur fuel (pipeline quality natural gas).	0.52	LB/MMSCF	BACT
MI-0426	DTE GAS COMPANY - MILFORD COMPRESSOR STATION	DTE GAS COMPANY	3/24/2017	3	MMBTU/H	Good combustion practices and low sulfur fuel (pipeline quality natural gas).	0.52	LB/MMSCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	9	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	1.9	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	18	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	1.9	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	4.8	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	1.9	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	3	MMBtu/hr	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	1.9	LB/MMSCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	1.9	LB/MMSCF	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	17.5	MMBtu/hr, each	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	1.9	LB/MMSCF	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	7	MMBTU/H	Low sulfur fuel (natural gas) and good combustion practices (efficient combustion).	1.9	LB/MMSCF	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	17.5	MMBtu/hr, each	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	7.6	LB/MMSCF	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	17.5	MMBtu/hr, each	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	7.6	LB/MMSCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	9	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	7.6	LB/MMSCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	7.6	LB/MMSCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	7.6	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	18	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	7.6	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	18	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	7.6	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	4.8	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	7.6	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	4.8	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	7.6	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	3	MMBtu/hr	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	7.6	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	3	MMBtu/hr	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	7.6	LB/MMSCF	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	7	MMBTU/H	Low sulfur fuel (natural gas) and good combustion practices (efficient combustion)	7.6	LB/MMSCF	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	7	MMBTU/H	Low sulfur fuel (natural gas) and good combustion practices (efficient combustion).	7.6	LB/MMSCF	BACT
SC-0193	MERCEDES BENZ VANS, LLC	MERCEDES BENZ VANS, LLC	4/15/2016	14.27	MMBTU/hr	Annual tune ups per 40 CFR 63.7540(a)(10) are required.	7.6	LB/MMSCF	BACT
SC-0193	MERCEDES BENZ VANS, LLC	MERCEDES BENZ VANS, LLC	4/15/2016	14.27	MMBTU/hr	Annual tune ups per 40 CFR 63.7540(a)(10) are required.	7.6	LB/MMSCF	BACT

Table C-2c: RBLC PM Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
SC-0193	MERCEDES BENZ VANS, LLC	MERCEDES BENZ VANS, LLC	4/15/2016	14.27	MMBTU/hr	Annual tune ups per 40 CFR 63.7540(a)(10) are required.	7.6	LB/MMSCF	BACT
TX-0772	PORT OF BEAUMONT PETROLEUM TRANSLOAD TERMINAL (PBPTT)	JEFFERSON RAILPORT TERMINAL I TEXAS LLC	11/6/2015	13.2	MMBTU/H	Good combustion practice to ensure complete combustion.	0.4	T/YR	BACT
TX-0772	PORT OF BEAUMONT PETROLEUM TRANSLOAD TERMINAL (PBPTT)	JEFFERSON RAILPORT TERMINAL I TEXAS LLC	11/6/2015	13.2	MMBTU/H	Good combustion practice to ensure complete combustion.	4	T/YR	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	9	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion practices	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion practices	0		BACT
TX-0694	INDECK WHARTON ENERGY CENTER	INDECK WHARTON, L.L.C.	2/2/2015	3	MMBTU/H		0		BACT
*WV-0031	MOCKINGBIRD HILL COMPRESSOR STATION	DOMINION ENERGY TRANSMISSION, INC.	6/14/2018	8.72	mmBtu/hr	Limited to natural gas	0		BACT
*WV-0031	MOCKINGBIRD HILL COMPRESSOR STATION	DOMINION ENERGY TRANSMISSION, INC.	6/14/2018	8.72	mmBtu/hr	Limited to natural gas	0		BACT
*WV-0031	MOCKINGBIRD HILL COMPRESSOR STATION	DOMINION ENERGY TRANSMISSION, INC.	6/14/2018	8.72	mmBtu/hr	Limited to natural gas.	0		BACT

Table C-2d: RBLC SO2 Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
FL-0356	OKEECHOBEE CLEAN ENERGY CENTER	FLORIDA POWER & LIGHT	3/9/2016	10	MMBtu/hr	Use of low-sulfur fuel	2	GR. S/100 SCF GAS	BACT
FL-0363	DANIA BEACH ENERGY CENTER	FLORIDA POWER AND LIGHT COMPANY	12/4/2017	9.9	MMBtu/hr	Clean fuel	2	GRAINS S / 100 SCF	BACT
TX-0772	PORT OF BEAUMONT PETROLEUM TRANSLOAD TERMINAL (PBPTT)	JEFFERSON RAILPORT TERMINAL I TEXAS LLC	11/6/2015	13.2	MMBTU/H	Good combustion practice to ensure complete combustion.	5	GR/100 SCF	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	15	MMBTU/H	Pipeline natural gas fuel	0.023	LB/H	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	15	MMBTU/hr each	Good combustion practices	0.0006	LB/MMBTU	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	9	MMBtu/hr	Only pipeline quality natural gas shall be combusted	0.0006	LB/MMBTU	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	Only pipeline quality natural gas fuel shall be combusted	0.0006	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Combustion of Natural gas and Good Combustion Practice	0.0006	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Combustion of natural gas and good combustion practices	0.0006	LB/MMBTU	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Use of natural gas fuel.	0.0014	LB/MM BTU	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	17.5	MMBtu/hr, each	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0.6	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	18	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0.6	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	4.8	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0.6	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	3	MMBtu/hr	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0.6	LB/MMSCF	BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	16.13	mm btu/hr	Good Combustion Practices and Use of low sulfur facility fuel gas	0		BACT

Table C-2e: RBLC H2SO4 Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	3.8	MMBTU/H	Low sulfur fuel	0.34	GR S/100 SCF	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	15	MMBTU/H	Pipeline natural gas fuel	0.0035	LB/H	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	13.31	MMBtu/hr		0.0001	LB/MMBTU	BACT
WI-0306	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	14.67	MMBTU/H	Combust only pipeline quality natural gas.	0		BACT

Table C-2f: RBLC VOC Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
SC-0179	CAROLINA PARTICLEBOARD	FLAKEBOARD AMERICA LIMITED	3/18/2015	1.83	MMBTU/H	NATURAL GAS USAGE AND GOOD COMBUSTION PRACTICES.	0.01	LB/H	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.02	LB/HR	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	3.7	MMBTU/H	Good combustion practices.	0.03	LB/H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	3.8	MMBTU/H	Good combustion controls.	0.03	LB/H	BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0.03	LB/HR	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	15	MMBTU/H	Combustion control	0.075	LB/H	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	20	MMBtu/hr	Good Combustion & Operation Practices (GCOP) Plan	0.11	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	19.2	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.11	LB/HR	BACT
FL-0364	SEMINOLE GENERATING STATION	SEMINOLE ELECTRIC COOPERATIVE, INC.	3/21/2018	9.9	MMBtu/hr		0.005	LB/MMBTU	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	13.31	MMBtu/hr		0.005	LB/MMBTU	LAER
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	16.8	MMBTU/HR	Good combustion practices	0.005	LB/MMBTU	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	10	MMBtu/hr	(a) The permittee shall operate and maintain according to the manufacturerâ€™s recommendations. (b) The permittee shall tune within 120 days of the initial operation, and within 24-months of each preceding tune-up. (c) VOC emissions from may not exceed 0.005 lb/MMBtu. (d) The permittee may only combust pipeline quality natural gas.	0.005	LB/MMBTU	BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	16.13	mm btu/hr	Good Combustion Practices and Use of low sulfur facility fuel gas	0.0054	LB/MM BTU	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of gaseous fuels (fuel gas and/or natural gas). Good combustion practices. Proper equipment (burner) design/operation. Compliance with 40 CFR 63 Subpart DDDDD.	0.0054	LB/MM BTU	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	10.2	MMBTU/H	Good design and operating/combustion practices.	0.0054	LB/MMBTU	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	10.2	MMBTU/H	Good Design and Operating/Combustion Practices	0.0054	LB/MMBTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	15	MMBTU/hr each	Good combustion practices	0.0055	LB/MMBTU	BACT
WI-0292	GREEN BAY PACKAGING INC. â€“MILL DIVISION	GREEN BAY PACKAGING INC. â€“MILL DIVISION	4/1/2019	20	mmBTU/hr	Good Combustion Practices, the Use of Low-NOx Burners	0.0055	LB/MMBTU	BACT
WI-0297	GREEN BAY PACKAGING- MILL DIVISION	GREEN BAY PACKAGING INC.	12/10/2019	8.5	MMBtu/H		0.0055	LB/MMBTU	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	17.5	MMBTU/H	GCP	0.006	LB/MMBTU	BACT

Table C-2f: RBLC VOC Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Good combustion practices.	0.008	LB/MM BTU	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	7	MMBTU/H	Good combustion practices	0.025	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	10	MMBtu/hr	Good combustion practices	5.5	LB/MMBTU	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	9	MMBtu/hr	good combustion practices	5.5	LB/MMSCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices	5.5	LB/MMSCF	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	17.5	MMBtu/hr, each	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	5.5	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	18	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	5.5	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	4.8	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	5.5	LB/MMSCF	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	3	MMBtu/hr	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	5.5	LB/MMSCF	BACT
SC-0193	MERCEDES BENZ VANS, LLC	MERCEDES BENZ VANS, LLC	4/15/2016	14.27	MMBTU/hr	Annual tune ups per 40 CFR 63.7540(a)(10) are required.	5.5	LB/MMSCF	BACT
TX-0772	PORT OF BEAUMONT PETROLEUM TRANSLOAD TERMINAL (PBPTT)	JEFFERSON RAILPORT TERMINAL I TEXAS LLC	11/6/2015	13.2	MMBTU/H	Good combustion practice to ensure complete combustion.	0.3	T/YR	BACT
OK-0164	MIDWEST CITY AIR DEPOT	TINKER AIR FORCE BASE LOGISTICS CENTER	1/8/2015	0	MMBTUH	1. Use pipeline-quality natural gas. 2. Good Combustion Practices w/emission rate limit of 0.005 lb/MMBTU based on AP-42 (7/1998).	7.1	TONS PER YEAR	BACT
WI-0306	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	14.67	MMBTU/H	Shall be operated for no more than 500 hours and combust only pipeline quality natural gas.	0	SEE NOTES	BACT
IL-0127	WINPAK HEAT SEAL CORPORATION	WINPAK HEAT SEAL CORPORATION	10/5/2018	1	mmBtu/hr	Units shall be operated in accordance with good combustion practices.	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion practices	0		BACT

Table C-2g: RBLC GHG Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
WI-0297	GREEN BAY PACKAGING- MILL DIVISION	GREEN BAY PACKAGING INC.	12/10/2019	8.5	MMBtu/H	Use only natural gas.	90	% AVG THERM EFF	BACT
WI-0306	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	14.67	MMBTU/H	Combust only pipeline quality natural gas.	118	CO2/MMBTU	BACT
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	15	MMBTU/H	Good combustion practices and the use of natural gas	1764	LB/H	BACT
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	15.17	MMBTU/H	Good combustion practices and the use of natural gas	1784	LB/H	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	20	MMBtu/hr	Good Combustion Practices	117	LB/MMBTU	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	1.44	MMBtu/hr	Good Combustion Practices	117	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good operating practices	117	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Good operating practices	117	LB/MMBTU	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	10	MMBtu/hr		117.1	LB/MMBTU	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	9	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	117.1	LB/MMBTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	15	MMBTU/hr each	Good combustion practices	121	LB/MMBTU	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	3	MMBtu/hr	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan and implement various design and operational efficiency requirements.	30	TONS/YR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Design Requirements, Good Combustion & Operation Practices (GCOP) Plan	1579	TONS/YR	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	3.7	MMBTU/H	Good combustion practices.	1934	T/YR	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Good combustion practices.	2326	T/YR	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	2625	TONS/YR	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	7	MMBTU/H	Energy efficiency	3590	T/YR	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	10.2	MMBTU/H	Good Combustion and Maintenance Practices, Natural Gas Only	5254	T/YR	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	10.2	MMBTU/H	Good combustion and maintenance practices, natural gas only.	5254	T/YR	BACT
MI-0420	DTE GAS COMPANY--MILFORD COMPRESSOR STATION	DTE GAS COMPANY	6/3/2016	6	MMBTU/H	Use of pipeline quality natural gas and energy efficiency measures.	6155	T/YR	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	3.8	MMBTU/H	Natural gas fuel	6310	T/YR	BACT
TX-0772	PORT OF BEAUMONT PETROLEUM TRANSLOAD TERMINAL (PBPTT)	JEFFERSON RAILPORT TERMINAL I TEXAS LLC	11/6/2015	13.2	MMBTU/H	Good combustion practice to ensure complete combustion.	6850	T/YR	BACT
MI-0426	DTE GAS COMPANY - MILFORD COMPRESSOR STATION	DTE GAS COMPANY	3/24/2017	3	MMBTU/H	Use of pipeline quality natural gas and energy efficiency measures.	7324	T/YR	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	15	MMBTU/H	Natural gas, low-emitting fuel	7695	T/YR	BACT

Table C-2g: RBLC GHG Results for Natural Gas Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
KS-0029	THE EMPIRE DISTRICT ELECTRIC COMPANY	THE EMPIRE DISTRICT ELECTRIC COMPANY	7/14/2015	18.6	MMBTU/HR		9521.5	TONS PER YEAR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	19.2	MMBtu/hr (total)	Design Requirements, Good Combustion & Operation Practices (GCOP) Plan	10104	TONS/YR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	20	MMBtu/hr	Design Requirements, Good Combustion & Operation Practices (GCOP) Plan	10258	TONS/YR	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	18	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan and implement various design and operational efficiency requirements.	12675	TONS/YR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	17.5	MMBtu/hr, each	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	18142	TON/YR	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	17.5	MMBTU/H		34189	T/YR	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	4.8	MMBtu/hr, each	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan and implement various design and operational efficiency requirements.	37581	TONS/YR	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	13.31	MMBtu/hr		44107	TON	BACT
OK-0164	MIDWEST CITY AIR DEPOT	TINKER AIR FORCE BASE LOGISTICS CENTER	1/8/2015	0	MMBTUH	1. Use pipeline-quality natural gas. 2. Good Combustion Practices. 3. Tune-ups for applicable boilers/heaters per 40CFR63, Subpart DDDDD.	153716	TONS PER YEAR	BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion/operating/maintenance practices and fueled by natural gas	0		BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	16.13	mm btu/hr	Use Low Carbon Fuel, Energy Efficiency Measures, and Good Combustion Practices	0		BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0		BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of low carbon intensity gaseous fuels. Good combustion and operating practices. Efficiency improvement measures.	0		BACT
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	16.8	MMBTU/HR	Good combustion practices	0		BACT
WI-0292	GREEN BAY PACKAGING INC. â€“MILL DIVISION	GREEN BAY PACKAGING INC. â€“MILL DIVISION	4/1/2019	20	mmBTU/hr	Good Combustion Practices, the Use of Low-NOx Burners	0		BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	10	MMBtu/hr	(a) Equipped with low-NOx burners. (b) Only pipeline quality natural gas may be combusted. (c) The permittee shall operate and maintain according to the manufacturerâ€™s recommendations. (d) The permittee shall tune within 120 days of the initial operation of, and within 24-months of each preceding tune-up.	0		BACT
*WV-0031	MOCKINGBIRD HILL COMPRESSOR STATION	DOMINION ENERGY TRANSMISSION, INC.	6/14/2018	8.72	mmBtu/hr	Limited to natural gas; and tune-up the boiler once every five years.	0		BACT

Table C-3a: RBLC NOx Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	3000	kWe, each	GOOD COMBUSTION PRACTICES AND LOW SULFUR FUELS	4.56	G	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	1500	kWe	good combustion practices and low sulfur fuel	4.56	G	BACT
TX-0728	PEONY CHEMICAL MANUFACTURING FACILITY	BASF	4/1/2015	1500	hp	Minimized hours of operations Tier II engine	0.0218	G/HP HR	LAER
LA-0346	GULF COAST METHANOL COMPLEX	IGP METHANOL LLC	1/4/2018	13410	hp (each)	Comply with standards of 40 CFR 60 Subpart JJJJ	2	G/BHP-HR	BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	3000	KW	Generator equipped with selective catalytic reduction. Compliance demonstrated with vendor emission certification and adherence to vendor-specified maintenance recommendations.	2.11	G/BHP-H	LAER
AL-0328	PLANT BARRY	ALABAMA POWER COMPANY	11/9/2020	0			3	GR/BHP-HR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	186.6	gph	Good combustion practices, limit operation to 500 hours per year.	3.3	G/HP-HR	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	1000	kW	Good Combustion Practices and meeting NSPS Subpart IIII requirements.	3.81	G/HP-HR	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	2000	kW	Good Combustion Practices and meeting NSPS Subpart IIII requirements.	3.81	G/HP-HR	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	3.9	G/BHP-HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	3.9	G/BHP-HR	BACT
TX-0955	INEOS OLIGOMERS CHOCOLATE BAYOU	INEOS OLIGOMERS USA LLC	3/14/2023	0		TIER III ENGINE, OPERATIONS LIMITED TO 100 HRS/YR	3.9	G/HP HR	LAER
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	3600	HP EACH	GOOD COMBUSTION PRACTICES	4.42	G/HP-H EACH	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	3600	HP		4.42	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	4.77	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	4.77	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	4.77	G/HP-HR	BACT
WV-0027	INWOOD	KNAUF INSULATION INC.	9/15/2017	900	bhp	Engine Design	4.77	G/HP-HR	BACT
PA-0310	CPV FAIRVIEW ENERGY CENTER	CPV FAIRVIEW, LLC	9/2/2016	0			4.8	G/BHP-HR	LAER
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	H/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	4.8	G/HP H	BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	H/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	4.8	G/HP-H	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	1490	HP	Operation limited to 500 hours/year and operate and maintain according to the manufacturerâ€™s recommendations.	4.8	G/HP-H	BACT

Table C-3a: RBLC NOx Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	4.8	G/HP-HR	BACT
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	4.8	G/HP-HR	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	3000	Horsepower	certified engine	4.8	G/HP-HR	BACT
*IN-0365	MAPLE CREEK ENERGY LLC		6/19/2023	2012	hp		4.8	G/HP-HR	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	2937	hp	Compliance with 40 CFR 60 Subpart IIII, good combustion practices, and use of ultra-low sulfur diesel fuel.	4.8	G/HP-HR	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	1140	hp	Proper equipment design and good combustion practices. Use ultra-low sulfur diesel as fuel. Low emission combustion. Compliance with 40 CFR 60 Subpart IIII.	4.8	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, and NSPS compliance	4.8	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, and NSPS compliance	4.8	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, and NSPS compliance	4.8	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, and NSPS compliance	4.8	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, and NSPS compliance	4.8	G/HP-HR	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	1490	HP	Good combustion practices and combusting clean fuels	4.8	G/HP-HR	BACT
AK-0082	POINT THOMSON PRODUCTION FACILITY	EXXON MOBIL CORPORATION	1/23/2015	2695	hp		4.8	GRAMS/HP-H	BACT
PA-0311	MOXIE FREEDOM GENERATION PLANT	MOXIE FREEDOM LLC	9/1/2015	0			4.93	G/HP-HR	LAER
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	1100	KW		5.3	G/HP-H	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	0			5.45	GM/HP-HR	LAER
*AR-0192	GREEN BAY PACKAGING	GREEN BAY PACKAGING	10/26/2023	0		ULSD, Good Design and Operating Practices	6.39	G/BHP-HR	BACT
AR-0161	SUN BIO MATERIAL COMPANY	SUN BIO MATERIAL COMPANY	9/23/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.4	G/KW-H	BACT
AR-0163	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	6/9/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	4.86	G/KW-HR	BACT
WI-0284	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT		4/24/2018	0		The Use of Ultra-Low Sulfur Fuel and Good Combustion Practices	5.36	G/KWH	BACT
WI-0286	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT	SIO INTERNATIONAL	4/24/2018	0		Good Combustion Practices, The Use of an Engine Turbocharger and Aftercooler.	5.36	G/KWH	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	2923	hp	Good combustion practices including compliance with 40 CFR 60 Subpart IIII.	5.36	G/KW-H	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	5364	HP	Good Combustion and Operating Practices	5.6	G/KW-H	BACT

Table C-3a: RBLC NOx Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
AR-0177	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	11/21/2022	3634	Horsepower		5.6	G/KW-HR	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Good Combustion Practices/Maintenance	6.4	G/KW	N/A
IN-0317	RIVERVIEW ENERGY CORPORATION	RIVERVIEW ENERGY CORPORATION	6/11/2019	2800	HP	Tier II diesel engine	6.4	G/KWH	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	22.68	MMBTU/H	Good combustion practices and meeting NSPS IIII requirements.	6.4	G/KW-H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	6.4	G/KW-H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Good combustion practices and meeting NSPS IIII requirements.	6.4	G/KW-H	BACT
MI-0434	FLAT ROCK ASSEMBLY PLANT	FORD MOTOR COMPANY	3/22/2018	3633	BHP	Good combustion practices.	6.4	G/KW-H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	2	MW	State of the art combustion design.	6.4	G/KW-H	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	1500	HP	Good combustion practices and will be NSPS compliant.	6.4	G/KW-H	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	6000	HP	Good combustion practices and will be NSPS compliant.	6.4	G/KW-H	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	22.68	MMBTU/H	Good Combustion Practices and meeting NSPS Subpart IIII requirements	6.4	G/KW-H	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	6.4	G/KW-H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Good Combustion Practices and meeting NSPS Subpart IIII requirements	6.4	G/KW-H	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	4474.2	KW	Good combustion practices and will be NSPS compliant.	6.4	G/KW-H	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	762	bhp		6.4	G/KW-H	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	5051	HP	Certified to meet Tier 2 standards and good combustion practices	6.4	G/KW-H	BACT
FL-0367	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	7/27/2018	14.82	MMBtu/hour	Operate and maintain the engine according to the manufacturer's written instructions	6.4	G/KW-HOUR	BACT
FL-0371	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	6/7/2021	14.82	MMBtu/hour		6.4	G/KW-HOUR	BACT
IL-0130	JACKSON ENERGY CENTER	JACKSON GENERATION, LLC	12/31/2018	1500	kW		6.4	G/KW-HR	LAER
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	3985	hp		6.4	G/KW-HR	BACT
LA-0309	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	6/4/2015	2922	hp (each)	Complying with 40 CFR 60 Subpart IIII	6.4	G/KW-HR	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	1250	kW		6.4	GRAMS	BACT
MI-0418	WARREN TECHNICAL CENTER	GENERAL MOTORS TECHNICAL CENTER - WARREN	1/14/2015	2710	KW	No add-on controls, but injection timing retardation (ITR) is good design. Engines are tuned for low-NOx operation versus low CO operation.	7.13	G/KW-H	BACT

Table C-3a: RBLC NOx Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0418	WARREN TECHNICAL CENTER	GENERAL MOTORS TECHNICAL CENTER - WARREN	1/14/2015	3490	KW	No add-on controls, but injection timing retardation (ITR) is good design. Engines are tuned for low-NOx operation versus low CO operation.	8	G/KW-H	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	1500	kWe	Good Combustion Practices	8	G/KW-HR	BACT
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	4290	kw	Compliance with 40 CFR 60 Subpart IIII and operating engines per manufacturers' instructions and written procedures designed to maximize combustion efficiency and minimize fuel usage.	8.46	G/KW-HR	BACT
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	3131	HP	Tier IV engine Tier IV NSPS standards certified by engine manufacturer.	3.45	LB/H	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Certified engines, limited operating hours	4.4	LB/H	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Certified Engines, Limited Operating Hours	4.4	LB/H	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	5000	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	5.5	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	760	horsepower	Compliance with 40 CFR 60 Subpart IIII	5.97	LB/HR	BACT
LA-0292	HOLBROOK COMPRESSOR STATION	CAMERON INTERSTATE PIPELINE LLC	1/22/2016	1341	HP	Good equipment design, proper combustion techniques, use of low sulfur fuel, and compliance with 40 CFR 60 Subpart IIII	14.16	LB/HR	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	1341	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	14.96	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2012	horsepower	Compliance with 40 CFR 60 Subpart IIII	15.79	LB/HR	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	1529	HP	State-of-the-art combustion design	16.07	LB/H	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	1529	HP	State-of-the-art combustion design	16.1	LB/H	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	1860	HP	Good combustion practices (ULSD) and compliance with 40 CFR Part 60, Subpart IIII	19.68	LB/H	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Certified engines, limited operating hours.	21.2	LB/H	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Certified engines, limited operating hours	21.2	LB/H	BACT
OH-0366	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	8/25/2015	2346	HP	State-of-the-art combustion design	21.6	LB/H	BACT
MI-0421	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	8/26/2016	500	H/YR	Certified engines, limited operating hours.	22.6	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	2206	HP	Certified to the meet the emissions standards in 40 CFR 89.112 and 89.113 pursuant to 40 CFR 60.4205(b) and 60.4202(a)(2). Good combustion practices per the manufacturerâ€™s operating manual.	23.21	LB/H	BACT
OH-0375	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	11/7/2017	2206	HP	Good combustion design	24.71	LB/H	BACT

Table C-3a: RBLC NOx Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	2947	HP	State-of-the-art combustion design	27.18	LB/H	BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	2584	HP	Compliance with NESHAP 40 CFR 63 Subpart ZZZZ and NSPS 40 CFR 60 Subpart IIII, and good combustion practices (use of ultra-low sulfur diesel fuel).	27.34	LB/H	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	2682	HP	Comply with NSPS 40 CFR 60 Subpart IIII	28.2	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	3353	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	37.41	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	18.41	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	18.41	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	18.41	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	18.41	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	18.41	LB/HR	BACT
*LA-0312	ST. JAMES METHANOL PLANT	SOUTH LOUISIANA METHANOL LP	6/30/2017	1474	horsepower	Compliance with NSPS Subpart IIII	19.23	LB/HR	BACT
*WV-0033	MAIDSVILLE	MOUNTAIN STATE CLEAN ENERGY, LLC	1/5/2022	2100	hp	Combustion Control (retarded timing and/or lean burn)	24.6	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	2923	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	28.48	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	2923	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	28.48	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	3634	horsepower	Compliance with 40 CFR 60 Subpart IIII	38.24	LB/HR	BACT
AL-0318	TALLADEGA SAWMILL	GEORGIA PACIFIC WOOD PRODUCTS, LLC	12/18/2017	0			0		N/A
IL-0129	CPV THREE RIVERS ENERGY CENTER	CPV THREE RIVERS, LLC	7/30/2018	0			0		LAER
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2922	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0		BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2937	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0		BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2937	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0		BACT
LA-0305	LAKE CHARLES METHANOL FACILITY	LAKE CHARLES METHANOL, LLC	6/30/2016	4023	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	0		good combustion practices, Use ultra low sulfur diesel, and comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	3353	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0317	METHANEX - GEISMAR METHANOL PLANT	METHANEX USA, LLC	12/22/2016	0		complying with 40 CFR 60 Subpart IIII and 40 CFR 63 Subpart ZZZZ	0		BACT

Table C-3a: RBLC NOx Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
LA-0318	FLOPAM FACILITY	FLOPAM, INC.	1/7/2016	0		Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0350	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	3/28/2018	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0382	BIG LAKE FUELS METHANOL PLANT	BIG LAKE FUELS LLC	4/25/2019	0		Comply with standards of 40 CFR 60 Subpart IIII	0		BACT
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
TX-0876	PORT ARTHUR ETHANE CRACKER UNIT	MOTIVA ENTERPRISE LLC	2/6/2020	0		Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101, limited to 100 hours per year of non-emergency operation	0		BACT
TX-0888	ORANGE POLYETHYLENE PLANT	CHEVRON PHILLIPS CHEMICAL COMPANY LP	4/23/2020	0		well-designed and properly maintained engines and each limited to 100 hours per year of non-emergency use.	0		BACT
TX-0904	MOTIVA POLYETHYLENE MANUFACTURING COMPLEX		9/9/2020	0		100 HOURS OPERATIONS, Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101	0		BACT
TX-0905	DIAMOND GREEN DIESEL PORT ARTHUR FACILITY	DIAMOND GREEN DIESEL	9/16/2020	0		limited to 100 hours per year of non-emergency operation	0		BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		limited to 100 hours per year of non-emergency operation. EPA Tier 2 (40 CFR Â§ 1039.101) exhaust emission standards	0		BACT
*TX-0976	CORPUS CHRISTI LIQUEFACTION STAGE 3	CORPUS CHRISTI LIQUEFACTION, LLC	5/30/2025	0		EPA Tier 2, limited to 100 hours per year of non-emergency operation. GHGs from the emergency engines will be limited through engine design and certification in accordance with standards, limited operational hours, and proper operation and maintenance	0		BACT

Table C-3b: RBLC CO Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	3000	kWe, each	Good combustion practices, low sulfur fuel, and limited operating hours	2.6	G	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	1500	kWe	Good combustion practices, low sulfur fuel	2.6	G	BACT
*PA-0313	FIRST QUALITY TISSUE LOCK HAVEN PLT	FIRST QUALITY TISSUE, LLC	7/27/2017	2500	bhp		3.5	G	
TX-0728	PEONY CHEMICAL MANUFACTURING FACILITY	BASF	4/1/2015	1500	hp	Minimized hours of operations Tier II engine	0.0126	G/HP HR	CASE-BY-CASE
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	1100	KW		0.15	G/HP-H	BACT
PA-0311	MOXIE FREEDOM GENERATION PLANT	MOXIE FREEDOM LLC	9/1/2015	0			0.26	G/HP-HR	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	0			0.6	GM/HP-HR	BACT
*AR-0192	GREEN BAY PACKAGING	GREEN BAY PACKAGING	10/26/2023	0		ULSD, Good Design and Operating Practices	0.7	G/BHP-HR	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.9	G/BHP-HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.9	G/BHP-HR	BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	H/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	2.6	G/HP H	BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	3000	KW	Compliance demonstrated with vendor emission certification and adherence to vendor-specified maintenance recommendations.	2.6	G/BHP-H	BACT
AL-0328	PLANT BARRY	ALABAMA POWER COMPANY	11/9/2020	0			2.6	G/BHP-HR	BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	H/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	2.6	G/HP-H	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	1490	HP	Operation limited to 500 hours/year, and operate and maintain generator according to the manufacturerâ€™s recommendations.	2.6	G/HP-H	BACT
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	2.6	G/HP-HR	BACT
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	2.6	G/HP-HR	BACT
*IN-0365	MAPLE CREEK ENERGY LLC		6/19/2023	2012	hp		2.6	G/HP-HR	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	2937	hp	Compliance with 40 CFR 60 Subpart IIII, good combustion practices, and the use of ultra-low sulfur diesel fuel.	2.6	G/HP-HR	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	1140	hp	Proper equipment design and good combustion practices. Use ultra-low sulfur diesel as fuel. Low emission combustion. Compliance with 40 CFR 60 Subpart IIII.	2.6	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	2.6	G/HP-HR	BACT

Table C-3b: RBLC CO Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	2.6	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	2.6	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	2.6	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	2.6	G/HP-HR	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	1490	HP	Good combustion practices which include operational and design elements.	2.6	G/HP-HR	BACT
KY-0109	FRITZ WINTER NORTH AMERICA, LP	FRITZ WINTER NORTH AMERICA, LP	10/24/2016	53.6	gal/hr	The permittee shall prepare and maintain for EU72, EU73, and EU74, within 90 days of startup, a good combustion and operation practices plan (GCOP) that defines, measures and verifies the use of operational and design practices determined as BACT for minimizing CO, VOC, PM, PM10, and PM2.5 emissions. Any revisions requested by the Division shall be made and the plan shall be maintained on site. The permittee shall operate according to the provisions of this plan at all times, including periods of startup, shutdown, and malfunction. The plan shall be incorporated into the plant standard operating procedures (SOP) and shall be made available for the Divisionâ€™s inspection. The plan shall include, but not be limited to: i. A list of combustion optimization practices and a means of verifying the practices have occurred. ii. A list of combustion and operation practices to be used to lower energy consumption and a means of verifying the practices have occurred. iii. A list of the design choices determined to be BACT and verification that designs were implemented in the final construction.	2.6	G/HP-HR (EU72 &EU73)	BACT
AK-0082	POINT THOMSON PRODUCTION FACILITY	EXXON MOBIL CORPORATION	1/23/2015	2695	hp		2.6	GRAMS/HP-H	BACT
PA-0310	CPV FAIRVIEW ENERGY CENTER	CPV FAIRVIEW, LLC	9/2/2016	0			2.61	G/BHP-HR	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	3600	HP EACH	GOOD COMBUSTION PRACTICES	2.61	G/HP-H EACH	BACT
TX-0915	UNIT 5	NRG CEDAR BAYOU LLC	3/17/2021	0		LIMITED 500 HR/YR OPERATION	2.61	G/HPHR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	3600	HP		2.61	G/HP-HR	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	3000	Horsepower	oxidation catalyst and certified engine	2.61	G/HP-HR	BACT
IN-0382	DUKE ENERGY INDIANA, INC.-CAYUGA GENERATING STATION	DUKE ENERGY INDIANA, INC.	2/14/2025	2000	HP	good combustion practices	2.61	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	2.61	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	2.61	G/HP-HR	BACT

Table C-3b: RBLC CO Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	2.61	G/HP-HR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	186.6	gph	Oxidation Catalyst, Good Combustion Practices, and 500 hour limit per year.	3.3	G/HP-HR	BACT
LA-0346	GULF COAST METHANOL COMPLEX	IGP METHANOL LLC	1/4/2018	13410	hp (each)	Comply with standards of 40 CFR 60 Subpart JJJJ	4	G/BHP-HR	BACT
TX-0872	CONDENSATE SPLITTER FACILITY	MAGELLAN PROCESSING, L.P.	10/31/2019	0		Limiting duration and frequency of generator use to 100 hr/yr. Good combustion practices will be used to reduce VOC including maintaining proper air-to-fuel ratio.	0.6	G/KW HR	BACT
WI-0284	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT		4/24/2018	0		Good Combustion Practices	0.6	G/KWH	BACT
WI-0286	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT	SIO INTERNATIONAL	4/24/2018	0		Good Combustion Practices	0.6	G/KWH	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	2923	hp	Good combustion practices including compliance with 40 CFR 60 Subpart IIII.	0.6	G/KW-H	BACT
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	4290	kw	Compliance with 40 CFR 60 Subpart IIII and operating engines per manufacturers' instructions and written procedures designed to maximize combustion efficiency and minimize fuel usage.	1.21	G/KW-HR	BACT
FL-0356	OKEECHOBEE CLEAN ENERGY CENTER	FLORIDA POWER & LIGHT	3/9/2016	0		Use of clean engine	3.5	G / KW-HR	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Good Combustion Practices/Maintenance	3.5	G/KW	N/A
IN-0317	RIVERVIEW ENERGY CORPORATION	RIVERVIEW ENERGY CORPORATION	6/11/2019	2800	HP	Tier II diesel engine	3.5	G/KWH	BACT
AR-0161	SUN BIO MATERIAL COMPANY	SUN BIO MATERIAL COMPANY	9/23/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	3.5	G/KW-H	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	5364	HP	Good Combustion and Operating Practices.	3.5	G/KW-H	BACT
MI-0421	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	8/26/2016	500	H/YR	Good design and combustion practices.	3.5	G/KW-H	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	22.68	MMBTU/H	Good combustion practices and meeting NSPS Subpart IIII requirements.	3.5	G/KW-H	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Good design and combustion practices.	3.5	G/KW-H	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Good design and combustion practices.	3.5	G/KW-H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	3.5	G/KW-H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Good combustion practices and meeting NSPS IIII requirements.	3.5	G/KW-H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	2	MW	State of the art combustion design.	3.5	G/KW-H	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	1500	HP	Good combustion practices and will be NSPS compliant.	3.5	G/KW-H	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	6000	HP	Good combustion practices and will be NSPS compliant.	3.5	G/KW-H	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	22.68	MMBTU/H	Good Combustion Practices and meeting NSPS Subpart IIII requirements	3.5	G/KW-H	BACT

Table C-3b: RBLC CO Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0447	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	1/7/2021	4474.2	KW	Good combustion practices and will be NSPS compliant.	3.5	G/KW-H	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Good design and combustion practices	3.5	G/KW-H	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Good Design and Combustion Practices	3.5	G/KW-H	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Good combustion practices and meeting NSPS IIII requirements.	3.5	G/KW-H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Good Combustion Practices and meeting NSPS Subpart IIII requirements	3.5	G/KW-H	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	4474.2	KW	Good combustion practices and will be NSPS compliant.	3.5	G/KW-H	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	762	bhp	Good combustion practices	3.5	G/KW-H	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	5051	HP	Certified to meet Tier 2 standards and good combustion practices	3.5	G/KW-H	BACT
FL-0367	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	7/27/2018	14.82	MMBtu/hour	Operate and maintain the engine according to the manufacturer's written instructions	3.5	G/KW-HOUR	BACT
FL-0371	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	6/7/2021	14.82	MMBtu/hour		3.5	G/KW-HOUR	BACT
AR-0163	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	6/9/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	3.5	G/KW-HR	BACT
AR-0177	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	11/21/2022	3634	Horsepower		3.5	G/KW-HR	BACT
IL-0130	JACKSON ENERGY CENTER	JACKSON GENERATION, LLC	12/31/2018	1500	kW		3.5	G/KW-HR	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	3985	hp		3.5	G/KW-HR	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	1250	kW		3.5	GRAMS	BACT
FL-0363	DANIA BEACH ENERGY CENTER	FLORIDA POWER AND LIGHT COMPANY	12/4/2017	0		Certified engine	3.5	GRAMS PER KWH	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	1500	kWe	Good Combustion Practices	4.38	G/KW-HR	BACT
*LA-0312	ST. JAMES METHANOL PLANT	SOUTH LOUISIANA METHANOL LP	6/30/2017	1474	horsepower	Compliance with NSPS Subpart IIII	0.51	LB/HR	BACT
*WV-0033	MAIDSVILLE	MOUNTAIN STATE CLEAN ENERGY, LLC	1/5/2022	2100	hp	Good Combustion Practices w/ OxCat. Applicant did not justify why an oxcat is infeasible for an emergency engine	1.94	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	2923	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	2.9	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	2923	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	2.9	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	760	horsepower	Compliance with 40 CFR 60 Subpart IIII	4.37	LB/HR	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	1341	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	7.7	LB/H	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	1529	HP	State-of-the-art combustion design	8.8	LB/H	BACT

Table C-3b: RBLC CO Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	1529	HP	State-of-the-art combustion design	8.8	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2012	horsepower	Compliance with 40 CFR 60 Subpart IIII	11.57	LB/HR	BACT
OH-0375	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	11/7/2017	2206	HP	Good combustion design	12.64	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	2206	HP	Certified to the meet the emissions standards in 40 CFR 89.112 and 89.113 pursuant to 40 CFR 60.4205(b) and 60.4202(a)(2). Good combustion practices per the manufacturerâ€™s operating manual.	12.69	LB/H	BACT
OH-0366	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	8/25/2015	2346	HP	State-of-the-art combustion design	13.5	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII.	13.5	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	13.5	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	13.5	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	13.5	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	13.5	LB/HR	BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	2584	HP	Compliance with NESHAP 40 CFR 63 Subpart ZZZZ and NSPS 40 CFR 60 Subpart IIII, and good combustion practices (use of ultra-low sulfur diesel fuel).	14.81	LB/H	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	2682	HP	Comply with NSPS 40 CFR 60 Subpart IIII	15.4	LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	2947	HP	State-of-the-art combustion design	16.96	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	3353	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	19.25	LB/H	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	3634	horsepower	Compliance with 40 CFR 60 Subpart IIII	20.91	LB/HR	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	5000	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	28.8	LB/H	BACT
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	18.7	MMBTU/HR	GOOD COMBUSTION PRACTICES, LIMITED TO 100 HR/YR	0.006	LB/HP HR	BACT
AL-0318	TALLADEGA SAWMILL	GEORGIA PACIFIC WOOD PRODUCTS, LLC	12/18/2017	0			0		N/A
IL-0129	CPV THREE RIVERS ENERGY CENTER	CPV THREE RIVERS, LLC	7/30/2018	0			0		BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2922	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0		BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2937	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0		BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2937	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0		BACT

Table C-3b: RBLC CO Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
LA-0305	LAKE CHARLES METHANOL FACILITY	LAKE CHARLES METHANOL, LLC	6/30/2016	4023	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	0		good combustion practices, Use ultra low sulfur diesel, and comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0309	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	6/4/2015	2922	hp (each)	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	3353	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0317	METHANEX - GEISMAR METHANOL PLANT	METHANEX USA, LLC	12/22/2016	0		complying with 40 CFR 60 Subpart IIII and 40 CFR 63 Subpart ZZZZ	0		BACT
LA-0318	FLOPAM FACILITY	FLOPAM, INC.	1/7/2016	0		Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0350	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	3/28/2018	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0382	BIG LAKE FUELS METHANOL PLANT	BIG LAKE FUELS LLC	4/25/2019	0		Comply with standards of 40 CFR 60 Subpart IIII	0		BACT
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
OH-0383	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	7/17/2020	3131	HP	Tier IV engine Good combustion practices	0		BACT
TX-0876	PORT ARTHUR ETHANE CRACKER UNIT	MOTIVA ENTERPRISE LLC	2/6/2020	0		Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101, limited to 100 hours per year of non-emergency operation	0		BACT
TX-0888	ORANGE POLYETHYLENE PLANT	CHEVRON PHILLIPS CHEMICAL COMPANY LP	4/23/2020	0		well-designed and properly maintained engines and each limited to 100 hours per year of non-emergency use.	0		BACT
TX-0904	MOTIVA POLYETHYLENE MANUFACTURING COMPLEX		9/9/2020	0		100 HOURS OPERATIONS, Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101	0		BACT
TX-0905	DIAMOND GREEN DIESEL PORT ARTHUR FACILITY	DIAMOND GREEN DIESEL	9/16/2020	0		limited to 100 hours per year of non-emergency operation	0		BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		limited to 100 hours per year of non-emergency operation. EPA Tier 2 (40 CFR Â§ 1039.101) exhaust emission standards	0		BACT
*TX-0976	CORPUS CHRISTI LIQUEFACTION STAGE 3	CORPUS CHRISTI LIQUEFACTION, LLC	5/30/2025	0		EPA Tier 2, limited to 100 hours per year of non-emergency operation. GHGs from the emergency engines will be limited through engine design and certification in accordance with standards, limited operational hours, and proper operation and maintenance	0		BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	1500	kWe	Good combustion practices, low sulfur fuel	0.05	G	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	1500	kWe	Good combustion practices, low sulfur fuel	0.07	G	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	1500	kWe	Good combustion practices, low sulfur fuel	0.07	G	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	3000	kWe, each	Good combustion Practices, low sulfur fuel	0.15	G	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	3000	kWe, each	Good combustion practices, low sulfur fuel	0.17	G	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	3000	kWe, each	Good combustion practices, low sulfur fuel	0.17	G	BACT
*PA-0313	FIRST QUALITY TISSUE LOCK HAVEN PLT	FIRST QUALITY TISSUE, LLC	7/27/2017	2500	bhp		0.2	G	
FL-0356	OKEECHOBEE CLEAN ENERGY CENTER	FLORIDA POWER & LIGHT	3/9/2016	0		Use of clean fuel	0.2	G / KW-HR	BACT
TX-0915	UNIT 5	NRG CEDAR BAYOU LLC	3/17/2021	0		LIMITED 500 HR/YR OPERATION	0.022	G/HPHR	BACT
TX-0915	UNIT 5	NRG CEDAR BAYOU LLC	3/17/2021	0		LIMITED 500 HR/YR OPERATION	0.022	G/HPHR	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	0			0.025	GM/HP-HR	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	0			0.025	GM/HP-HR	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	0			0.025	GM/HP-HR	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	1100	KW	Good combustion practices and ultra low sulfur diesel	0.04	G/HP-H	BACT
PA-0311	MOXIE FREEDOM GENERATION PLANT	MOXIE FREEDOM LLC	9/1/2015	0			0.04	G/HP-HR	BACT
PA-0311	MOXIE FREEDOM GENERATION PLANT	MOXIE FREEDOM LLC	9/1/2015	0			0.04	G/HP-HR	BACT
PA-0311	MOXIE FREEDOM GENERATION PLANT	MOXIE FREEDOM LLC	9/1/2015	0			0.04	G/HP-HR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	186.6	gph	Good combustion practices, ULSD, and limit operation to 500 hours per year.	0.045	G/HP-HR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	186.6	gph	Good combustion practices, ULSD, and limit operation to 500 hours per year.	0.045	G/HP-HR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	186.6	gph	Good combustion practices, ULSD, and limit operation to 500 hours per year.	0.045	G/HP-HR	BACT
WI-0294	CARDINAL FG COMPANY	CARDINAL FG COMPANY	8/26/2019	0			0.05	G/BHP-H	BACT
WI-0294	CARDINAL FG COMPANY	CARDINAL FG COMPANY	8/26/2019	0			0.05	G/BHP-H	BACT
WI-0294	CARDINAL FG COMPANY	CARDINAL FG COMPANY	8/26/2019	0			0.05	G/B-HP-H	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.1	G/BHP-HR	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.1	G/BHP-HR	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.1	G/BHP-HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.1	G/BHP-HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0			0.1	G/BHP-HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.1	G/BHP-HR	BACT
KY-0109	FRITZ WINTER NORTH AMERICA, LP	FRITZ WINTER NORTH AMERICA, LP	10/24/2016	53.6	gal/hr	The permittee shall prepare and maintain for EU72, EU73, and EU74, within 90 days of startup, a good combustion and operation practices plan (GCOP) that defines, measures and verifies the use of operational and design practices determined as BACT for minimizing CO, VOC, PM, PM10, and PM2.5 emissions. Any revisions requested by the Division shall be made and the plan shall be maintained on site. The permittee shall operate according to the provisions of this plan at all times, including periods of startup, shutdown, and malfunction. The plan shall be incorporated into the plant standard operating procedures (SOP) and shall be made available for the Divisionâ€™s inspection. The plan shall include, but not be limited to: i. A list of combustion optimization practices and a means of verifying the practices have occurred. ii. A list of combustion and operation practices to be used to lower energy consumption and a means of verifying the practices have occurred. iii. A list of the design choices determined to be BACT and verification that designs were implemented in the final construction.	0.149	G/HP-HR (EU72 &EU73)	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
KY-0109	FRITZ WINTER NORTH AMERICA, LP	FRITZ WINTER NORTH AMERICA, LP	10/24/2016	53.6	gal/hr	The permittee shall prepare and maintain for EU72, EU73, and EU74, within 90 days of startup, a good combustion and operation practices plan (GCOP) that defines, measures and verifies the use of operational and design practices determined as BACT for minimizing CO, VOC, PM, PM10, and PM2.5 emissions. Any revisions requested by the Division shall be made and the plan shall be maintained on site. The permittee shall operate according to the provisions of this plan at all times, including periods of startup, shutdown, and malfunction. The plan shall be incorporated into the plant standard operating procedures (SOP) and shall be made available for the Divisionâ€™s inspection. The plan shall include, but not be limited to: i. A list of combustion optimization practices and a means of verifying the practices have occurred. ii. A list of combustion and operation practices to be used to lower energy consumption and a means of verifying the practices have occurred. iii. A list of the design choices determined to be BACT and verification that designs were implemented in the final construction.	0.149	G/HP-HR (EU72 &EU73)	BACT
KY-0109	FRITZ WINTER NORTH AMERICA, LP	FRITZ WINTER NORTH AMERICA, LP	10/24/2016	53.6	gal/hr	The permittee shall prepare and maintain for EU72, EU73, and EU74, within 90 days of startup, a good combustion and operation practices plan (GCOP) that defines, measures and verifies the use of operational and design practices determined as BACT for minimizing CO, VOC, PM, PM10, and PM2.5 emissions. Any revisions requested by the Division shall be made and the plan shall be maintained on site. The permittee shall operate according to the provisions of this plan at all times, including periods of startup, shutdown, and malfunction. The plan shall be incorporated into the plant standard operating procedures (SOP) and shall be made available for the Divisionâ€™s inspection. The plan shall include, but not be limited to: i. A list of combustion optimization practices and a means of verifying the practices have occurred. ii. A list of combustion and operation practices to be used to lower energy consumption and a means of verifying the practices have occurred. iii. A list of the design choices determined to be BACT and verification that designs were implemented in the final construction.	0.149	G/HP-HR (EU72 &EU73)	BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	3000	KW	Compliance demonstrated with vendor emission certification and adherence to vendor-specified maintenance recommendations.	0.15	G/BHP-H	BACT
*IN-0365	MAPLE CREEK ENERGY LLC		6/19/2023	2012	hp		0.15	G PER HP-HR	BACT
*IN-0365	MAPLE CREEK ENERGY LLC		6/19/2023	2012	hp		0.15	G PER HP-HR	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
AL-0328	PLANT BARRY	ALABAMA POWER COMPANY	11/9/2020	0			0.15	G/BHP-HR	BACT
*AR-0192	GREEN BAY PACKAGING	GREEN BAY PACKAGING	10/26/2023	0		ULSD, Good Design and Operating Practices	0.15	G/BHP-HR	BACT
*AR-0192	GREEN BAY PACKAGING	GREEN BAY PACKAGING	10/26/2023	0		ULSD, Good Design and Operating Practices	0.15	G/BHP-HR	BACT
*AR-0192	GREEN BAY PACKAGING	GREEN BAY PACKAGING	10/26/2023	0		ULSD, Good Design and Operating Practices	0.15	G/BHP-HR	BACT
PA-0310	CPV FAIRVIEW ENERGY CENTER	CPV FAIRVIEW, LLC	9/2/2016	0			0.15	G/BHP-HR	BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	H/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.15	G/HP H	BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	H/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.15	G/HP H	BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	H/YR	Good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.15	G/HP H	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	3000	Horsepower	certified engine	0.15	G/HP-H	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	3000	Horsepower	certified engine	0.15	G/HP-H	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	3000	Horsepower	certified engine	0.15	G/HP-H	BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	H/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.15	G/HP-H	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	1490	HP	Limited to operate 500 hours/year, sulfur content of the diesel fuel oil fired may not exceed 15 ppm, and operate and maintain according to the manufacturerâ€™s recommendations.	0.15	G/HP-H	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	1490	HP	Limited to operate 500 hours/year, sulfur content of the diesel fuel oil fired may not exceed 15 ppm, and operate and maintain according to the manufacturerâ€™s recommendations.	0.15	G/HP-H	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	1490	HP	Limited to operate 500 hours/year, sulfur content of the diesel fuel oil fired may not exceed 15 ppm, and operate and maintain according to the manufacturerâ€™s recommendations.	0.15	G/HP-H	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	3600	HP EACH	GOOD COMBUSTION PRACTICES	0.15	G/HP-H EACH	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	3600	HP EACH	GOOD COMBUSTION PRACTICES	0.15	G/HP-H EACH	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	3600	HP EACH	GOOD COMBUSTION PRACTICES	0.15	G/HP-H EACH	BACT
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	0.15	G/HP-HR	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	0.15	G/HP-HR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	3600	HP		0.15	G/HP-HR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	3600	HP		0.15	G/HP-HR	BACT
*IN-0365	MAPLE CREEK ENERGY LLC		6/19/2023	2012	hp		0.15	G/HP-HR	BACT
IN-0382	DUKE ENERGY INDIANA, INC.-CAYUGA GENERATING STATION	DUKE ENERGY INDIANA, INC.	2/14/2025	2000	HP	good combustion practices	0.15	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0.15	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0.15	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0.15	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0.15	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0.15	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0.15	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0.15	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0.15	G/HP-HR	BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0.15	G/HP-HR	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2922	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan.	0.15	G/HP-HR	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2922	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0.15	G/HP-HR	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2922	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0.15	G/HP-HR	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2937	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0.15	G/HP-HR	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2937	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0.15	G/HP-HR	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2937	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0.15	G/HP-HR	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2937	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0.15	G/HP-HR	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2937	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0.15	G/HP-HR	BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	2937	HP	The permittee must develop a Good Combustion and Operating Practices (GCOP) Plan	0.15	G/HP-HR	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	2937	hp	Compliance with 40 CFR 60 Subpart IIII, good combustion practices, and use of ultra-low sulfur diesel fuel.	0.15	G/HP-HR	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	2937	hp	Compliance with 40 CFR 60 Subpart IIII (Tier 2 non-road engines) standards, good combustion practices, and the use of ultra-low sulfur diesel fuel.	0.15	G/HP-HR	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	1140	hp	Proper equipment design and good combustion practices. Use ultra-low sulfur diesel as fuel. Compliance with 40 CFR 60 Subpart IIII.	0.15	G/HP-HR	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	1140	hp	Proper equipment design and good combustion practices. Use ultra-low sulfur diesel as fuel. Compliance with 40 CFR 60 Subpart IIII.	0.15	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.15	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.15	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.15	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.15	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.15	G/HP-HR	BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	H/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.15	G/HP-HR	BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	H/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.15	G/HP-HR	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	1490	HP	Good combustion practices and clean fuels. The sulfur content of the diesel fuel oil fired may not exceed 15 ppm.	0.15	G/HP-HR	BACT
AK-0082	POINT THOMSON PRODUCTION FACILITY	EXXON MOBIL CORPORATION	1/23/2015	2695	hp		0.15	GRAMS/HP-H	BACT
AK-0082	POINT THOMSON PRODUCTION FACILITY	EXXON MOBIL CORPORATION	1/23/2015	2695	hp		0.15	GRAMS/HP-H	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.16	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.16	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.16	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.16	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.16	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.16	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.16	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.16	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.16	G/HP-HR	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, ULSD, and NSPS compliance	0.16	G/HP-HR	BACT
WV-0027	INWOOD	KNAUF INSULATION INC.	9/15/2017	900	bhp	ULSD	0.2	G/HP-HR	BACT
AR-0161	SUN BIO MATERIAL COMPANY	SUN BIO MATERIAL COMPANY	9/23/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.02	G/KW-H	BACT
AR-0161	SUN BIO MATERIAL COMPANY	SUN BIO MATERIAL COMPANY	9/23/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.02	G/KW-H	BACT
AR-0161	SUN BIO MATERIAL COMPANY	SUN BIO MATERIAL COMPANY	9/23/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.02	G/KW-HR	BACT
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	4290	kw	Compliance with 40 CFR 60 Subpart IIII and operating engines per manufacturers' instructions and written procedures designed to maximize combustion efficiency and minimize fuel usage.	0.067	G/KW-HR	BACT
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	4290	kw	Compliance with 40 CFR 60 Subpart IIII and operating engines per manufacturers' instructions and written procedures designed to maximize combustion efficiency and minimize fuel usage.	0.067	G/KW-HR	BACT
WI-0284	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT		4/24/2018	0		The Use of Ultra-Low Sulfur Fuel and Good Combustion Practices	0.17	G/KWH	BACT
WI-0284	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT		4/24/2018	0		The Use of Ultra-Low Sulfur Fuel and Good Combustion Practices	0.17	G/KWH	BACT
WI-0284	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT		4/24/2018	0		The Use of Ultra-Low Sulfur Fuel and Good Combustion Practices	0.17	G/KWH	BACT
WI-0286	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT	SIO INTERNATIONAL	4/24/2018	0		Good Combustion Practices and The Use of Ultra-low Sulfur Fuel	0.17	G/KWH	BACT
WI-0286	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT	SIO INTERNATIONAL	4/24/2018	0		Good Combustion Practices and The Use of Ultra-low Sulfur Fuel	0.17	G/KWH	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	2923	hp	Good combustion practices including compliance with 40 CFR 60 Subpart IIII.	0.17	G/KW-H	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	2923	hp	Good combustion practices including compliance with 40 CFR 60 Subpart IIII.	0.17	G/KW-H	BACT
IN-0317	RIVERVIEW ENERGY CORPORATION	RIVERVIEW ENERGY CORPORATION	6/11/2019	2800	HP	Tier II diesel engine	0.2	G/KWH	BACT
IN-0317	RIVERVIEW ENERGY CORPORATION	RIVERVIEW ENERGY CORPORATION	6/11/2019	2800	HP	Tier II diesel engine	0.2	G/KWH	BACT
IN-0317	RIVERVIEW ENERGY CORPORATION	RIVERVIEW ENERGY CORPORATION	6/11/2019	2800	HP	Tier II diesel engine	0.2	G/KWH	BACT
KS-0040	JOHNS MANVILLE AT MCPHERSON	JOHNS MANVILLE	12/3/2019	0		One diesel engine and fire pump subject to NSPS Subpart IIII - Combustion Control and Limited Operating Hours.	0.2	G/KWH	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	4474.2	KW	Good combustion practices, burn ultra-low diesel fuel, and will be NSPS compliant.	0.2	G/KWH	CASE-BY-CASE
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	5364	HP	Good combustion and operating practices.	0.2	G/KW-H	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	5364	HP	Good combustion and operating practices.	0.2	G/KW-H	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	22.68	MMBTU/H	Good combustion practices and meeting NSPS Subpart IIII requirements.	0.2	G/KW-H	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.2	G/KW-H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Diesel particulate filter, good combustion practices and meeting NSPS IIII requirements.	0.2	G/KW-H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	2	MW	State of the art combustion design	0.2	G/KW-H	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	22.68	MMBTU/H	Good Combustion Practices and meeting NSPS Subpart IIII requirements	0.2	G/KW-H	BACT
MI-0447	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	1/7/2021	4474.2	KW	Good combustion practices, burn ultra-low diesel fuel, and will be NSPS compliant.	0.2	G/KW-H	CASE-BY-CASE
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.2	G/KW-H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Diesel particulate filter, Good Combustion Practices and meeting NSPS Subpart IIII requirements	0.2	G/KW-H	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	762	bhp	Good combustion practices, use of ULSD	0.2	G/KW-H	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	5051	HP	Certified to meet Tier 2 standards and good combustion practices	0.2	G/KW-H	BACT
FL-0367	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	7/27/2018	14.82	MMBtu/hour	Operate and maintain the engine according to the manufacturer's written instructions	0.2	G/KW-HOUR	BACT
FL-0371	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	6/7/2021	14.82	MMBtu/hour		0.2	G/KW-HOUR	BACT
AR-0163	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	6/9/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.2	G/KW-HR	BACT
AR-0163	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	6/9/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.2	G/KW-HR	BACT
AR-0163	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	6/9/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.2	G/KW-HR	BACT
AR-0177	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	11/21/2022	3634	Horsepower		0.2	G/KW-HR	BACT
AR-0177	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	11/21/2022	3634	Horsepower		0.2	G/KW-HR	BACT
AR-0177	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	11/21/2022	3634	Horsepower		0.2	G/KW-HR	BACT
IL-0130	JACKSON ENERGY CENTER	JACKSON GENERATION, LLC	12/31/2018	1500	kW		0.2	G/KW-HR	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	3985	hp		0.2	G/KW-HR	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	3985	hp		0.2	G/KW-HR	BACT
*IL-0137	MARQUIS CARBON CAPTURE LLC	MARQUIS CARBON CAPTURE LLC	6/23/2025	0		Use of ultra-low sulfur diesel (ULSD) as fuel; good combustion practices; compliance with emission limits of the NSPS, 40 CFR 60 Subpart IIII; 500 hours/year operation	0.2	G/KW-HR	BACT
*IL-0137	MARQUIS CARBON CAPTURE LLC	MARQUIS CARBON CAPTURE LLC	6/23/2025	0		Use of ultra-low sulfur diesel (ULSD) as fuel; good combustion practices; compliance with emission limits of the NSPS, 40 CFR 60 Subpart IIII; 500 hours/year operation	0.2	G/KW-HR	BACT
LA-0309	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	6/4/2015	2922	hp (each)	Complying with 40 CFR 60 Subpart IIII	0.2	G/KW-HR	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
LA-0309	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	6/4/2015	2922	hp (each)	Complying with 40 CFR 60 Subpart IIII	0.2	G/KW-HR	BACT
KS-0040	JOHNS MANVILLE AT MCPHERSON	JOHNS MANVILLE	12/3/2019	0		Emergency Diesel Engine and Fire Pump Subject to NSPS Subpart IIII - Combustion Control and Limited Operating Hours.	0.2	GR/KWH	BACT
KS-0040	JOHNS MANVILLE AT MCPHERSON	JOHNS MANVILLE	12/3/2019	0		One diesel fuel emergency engine and one fire pump subject to NSPS Subpart IIII - Combustion Control and Limited Operating Hours.	0.2	GR/KWH	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	1250	kW		0.2	GRAMS	BACT
FL-0363	DANIA BEACH ENERGY CENTER	FLORIDA POWER AND LIGHT COMPANY	12/4/2017	0		Clean fuel	0.2	GRAMS PER KWH	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	1500	kWe	Clean Fuel and Good Combustion Practices	0.25	G/KW-HR	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	1500	kWe	Clean Fuel and Good Combustion Practices	0.25	G/KW-HR	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	1500	kWe	Clean Fuel and Good Combustion Practices	0.25	G/KW-HR	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Ultra Low Sulfur Diesel/Fuel (15 ppm max)	0.4	G/KR	N/A
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Ultra Low Sulfur Diesel/Fuel (15 ppm max)	0.4	G/KW	N/A
WI-0286	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT	SIO INTERNATIONAL	4/24/2018	0		Good Combustion Practices and The Use of Ultra-low Sulfur Fuel	17	G/KWH	BACT
SC-0193	MERCEDES BENZ VANS, LLC	MERCEDES BENZ VANS, LLC	4/15/2016	1500	hp	Meet the standards of 40 CFR 60, Subpart IIII	100	HR/YR	BACT
SC-0193	MERCEDES BENZ VANS, LLC	MERCEDES BENZ VANS, LLC	4/15/2016	1500	hp	Must meet the standards of 40 CFR 60, Subpart IIII	100	HR/YR	BACT
SC-0193	MERCEDES BENZ VANS, LLC	MERCEDES BENZ VANS, LLC	4/15/2016	1500	hp	Meet emission standards of 40 CFR 60, Subpart IIII	100	HRS/YR	BACT
VA-0333	NORFOLK NAVAL SHIPYARD	US NAVY NORFOLK NAVAL SHIPYARD	12/9/2020	2220	HP		1.1	LB	BACT
VA-0333	NORFOLK NAVAL SHIPYARD	US NAVY NORFOLK NAVAL SHIPYARD	12/9/2020	2220	HP		1.1	LB	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	1100	KW	Good combustion practices and ultra low sulfur diesel.	7.55	LB/1000 GAL	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	1100	KW	Good combustion practices and ultra low sulfur diesel	7.85	LB/1000 GAL	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	5051	HP	Certified to meet Tier 2 standards and good combustion practices	0.07	LB/H	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	5051	HP	Certified to meet Tier 2 standards and good combustion practices	0.07	LB/H	BACT
*LA-0312	ST. JAMES METHANOL PLANT	SOUTH LOUISIANA METHANOL LP	6/30/2017	1474	horsepower	Compliance with NSPS Subpart IIII	0.08	LB/HR	BACT
*LA-0312	ST. JAMES METHANOL PLANT	SOUTH LOUISIANA METHANOL LP	6/30/2017	1474	horsepower	Compliance with NSPS Subpart IIII	0.08	LB/HR	BACT
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	3131	HP	Tier IV engine Good combustion practices	0.15	LB/H	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	3131	HP	Tier IV engine Good combustion practices	0.15	LB/H	BACT
TX-0728	PEONY CHEMICAL MANUFACTURING FACILITY	BASF	4/1/2015	1500	hp	Minimized hours of operations Tier II engine	0.15	LB/H	CASE-BY-CASE
TX-0728	PEONY CHEMICAL MANUFACTURING FACILITY	BASF	4/1/2015	1500	hp	Minimized hours of operations Tier II engine	0.15	LB/H	CASE-BY-CASE
TX-0728	PEONY CHEMICAL MANUFACTURING FACILITY	BASF	4/1/2015	1500	hp	Minimized hours of operations Tier II engine	0.15	LB/H	CASE-BY-CASE
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	5000	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	0.2	LB/H	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	5000	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	0.2	LB/H	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Certified engines, good design, operation and combustion practices. Operational restrictions/limited use.	0.22	LB/H	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Certified engines. Good design, operation and combustion practices. Operational restrictions/limited use.	0.22	LB/H	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Certified engines. Good design, operation and combustion practices. Operational restrictions/limited use.	0.22	LB/H	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Certified Engines, Good Design, Operation, and Combustion Practices, Operational Restrictions/Limited Use	0.22	LB/H	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Certified Engines, Good Design, Operation, and Combustion Practices, Operational Restrictions/Limited Use	0.22	LB/H	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Certified Engines, Good Design, Operation, and Combustion Practices, Operational Restrictions/Limited Use	0.22	LB/H	BACT
*WV-0033	MAIDSVILLE	MOUNTAIN STATE CLEAN ENERGY, LLC	1/5/2022	2100	hp	Clean Fuels and Good Combustion Practices.	0.23	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	760	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.25	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	760	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.25	LB/HR	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	762	bhp	Good combustion practices; use of ULSD.	0.3	LB/H	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	762	bhp	Good combustion practices, use of ULSD	0.3	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	1341	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	0.44	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	1341	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	0.44	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	1341	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	0.44	LB/H	BACT
LA-0292	HOLBROOK COMPRESSOR STATION	CAMERON INTERSTATE PIPELINE LLC	1/22/2016	1341	HP	Use of a certified engine, low sulfur diesel, and limiting non-emergency use to no more than 100 hours per year	0.44	LB/HR	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	1529	HP	Ultra low sulfur diesel fuel	0.5	LB/H	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	1529	HP	Ultra low sulfur diesel fuel	0.5	LB/H	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	1529	HP	Ultra low sulfur diesel fuel	0.5	LB/H	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	1529	HP	Ultra low sulfur diesel fuel	0.5	LB/H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.52	LB/H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.52	LB/H	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.52	LB/H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Diesel particulate filter, Good Combustion Practices and meeting NSPS Subpart IIII requirements	0.52	LB/H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.54	LB/H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.54	LB/H	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.54	LB/H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Diesel particulate filter, Good Combustion Practices and meeting NSPS Subpart IIII requirements	0.54	LB/H	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	1860	HP	Good combustion practices (ULSD) and compliance with 40 CFR Part 60, Subpart IIII	0.62	LB/H	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	1860	HP	Good combustion practices (ULSD) and compliance with 40 CFR Part 60, Subpart IIII	0.62	LB/H	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	1860	HP	Good combustion practices (ULSD) and compliance with 40 CFR Part 60, Subpart IIII	0.62	LB/H	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Certified engines, good design, operation and combustion practices. Operational restrictions/limited use.	0.66	LB/H	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Certified engines, good design, operation and combustion practices. Operational restrictions/limited use.	0.66	LB/H	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Certified engines, good design, operation and combustion practices. Operational restrictions/limited use.	0.66	LB/H	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Certified Engines, Good Design, Operation, and Combustion Practices, Operational Restrictions/Limited Use	0.66	LB/H	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Certified Engines, Good Design, Operation, and Combustion Practices, Operational Restrictions/Limited Use	0.66	LB/H	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Certified Engines, Good Design, Operation, and Combustion Practices, Operational Restrictions/Limited Use	0.66	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2012	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.66	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2012	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.66	LB/HR	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	1500	HP	Good combustion practices, burn ultra-low sulfur diesel fuel and be NSPS compliant.	0.69	LB/H	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	1500	HP	Ultra low-sulfur diesel fuel.	0.69	LB/H	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	2206	HP	Certified to the meet the emissions standards in 40 CFR 89.112 and 89.113 pursuant to 40 CFR 60.4205(b) and 60.4202(a)(2). Good combustion practices per the manufacturerâ€™s operating manual.	0.73	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	2206	HP	Certified to the meet the emissions standards in 40 CFR 89.112 and 89.113 pursuant to 40 CFR 60.4205(b) and 60.4202(a)(2). Good combustion practices per the manufacturerâ€™s operating manual.	0.73	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	2206	HP	Certified to the meet the emissions standards in 40 CFR 89.112 and 89.113 pursuant to 40 CFR 60.4205(b) and 60.4202(a)(2). Good combustion practices per the manufacturerâ€™s operating manual.	0.73	LB/H	BACT
OH-0375	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	11/7/2017	2206	HP	Good combustion design	0.73	LB/H	BACT
OH-0375	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	11/7/2017	2206	HP	Good combustion design	0.73	LB/H	BACT
OH-0375	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	11/7/2017	2206	HP	Good combustion design	0.73	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.76	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.76	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.76	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.76	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.76	LB/HR	BACT
OH-0366	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	8/25/2015	2346	HP	State-of-the-art combustion design	0.77	LB/H	BACT
OH-0366	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	8/25/2015	2346	HP	State-of-the-art combustion design	0.77	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.77	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.77	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.77	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.77	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.77	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	2923	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	0.84	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	2923	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	0.84	LB/HR	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	2923	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	0.84	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	2923	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	0.84	LB/HR	BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	2584	HP	Compliance with NESHAP 40 CFR 63 Subpart ZZZZ and NSPS 40 CFR 60 Subpart IIII, and good combustion practices (use of ultra-low sulfur diesel fuel).	0.86	LB/H	BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	2584	HP	Compliance with NESHAP 40 CFR 63 Subpart ZZZZ and NSPS 40 CFR 60 Subpart IIII, and good combustion practices (use of ultra-low sulfur diesel fuel).	0.86	LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	2947	HP	State-of-the-art combustion design	0.97	LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	2947	HP	State-of-the-art combustion design	0.97	LB/H	BACT
MI-0447	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	1/7/2021	4474.2	KW	Good combustion practices, burn ultra-low diesel fuel and be NSPS compliant.	1	LB/H	BACT
MI-0447	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	1/7/2021	4474.2	KW	ultra-low sulfur diesel fuel	1	LB/H	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	4474.2	KW	Good combustion practices, burn ultra-low diesel fuel, and be NSPS compliant.	1	LB/H	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	4474.2	KW	Ultra-low sulfur diesel fuel	1	LB/H	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	2682	HP	Comply with NSPS 40 CFR 60 Subpart IIII	1.01	LB/H	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	2682	HP	Comply with NSPS 40 CFR 60 Subpart IIII	1.01	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	3353	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	1.1	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	3353	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	1.1	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	3353	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	1.1	LB/H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	2	MW	State of the art combustion design	1.18	LB/H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	2	MW	State of the art combustion design.	1.18	LB/H	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	3634	horsepower	Compliance with 40 CFR 60 Subpart IIII	1.19	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	3634	horsepower	Compliance with 40 CFR 60 Subpart IIII	1.19	LB/HR	BACT
MI-0421	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	8/26/2016	500	H/YR	Certified engines, good design, operation and combustion practices. Operational restrictions/limited use.	1.41	LB/H	BACT
MI-0421	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	8/26/2016	500	H/YR	Certified engines, good design, operation and combustion practices. Operational restrictions/limited use.	1.41	LB/H	BACT
MI-0421	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	8/26/2016	500	H/YR	Certified engines, good design, operation and combustion practices. Operational restrictions/limited use.	1.41	LB/H	BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	22.68	MMBTU/H	Good combustion practices.	1.58	LB/H	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	22.68	MMBTU/H	Good combustion practices.	1.58	LB/H	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	22.68	MMBTU/H	Good combustion practices	1.58	LB/H	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	22.68	MMBTU/H	Good combustion practices	1.58	LB/H	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	6000	HP	Good combustion practices, burn ultra low sulfur diesel fuel, and be NSPS compliant.	2.7	LB/H	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	6000	HP	Ultra low sulfur diesel fuel.	2.7	LB/H	BACT
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	18.7	MMBTU/HR	GOOD COMBUSTION PRACTICES, LIMITED TO 100 HR/YR	0.0003	LB/HP HR	BACT
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	18.7	MMBTU/HR	GOOD COMBUSTION PRACTICES, LIMITED TO 100 HR/YR	0.0003	LB/HP HR	BACT
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	18.7	MMBTU/HR	GOOD COMBUSTION PRACTICES, LIMITED TO 100 HR/YR	0.003	LB/HP HR	BACT
AL-0318	TALLADEGA SAWMILL	GEORGIA PACIFIC WOOD PRODUCTS, LLC	12/18/2017	0			0		N/A
AL-0318	TALLADEGA SAWMILL	GEORGIA PACIFIC WOOD PRODUCTS, LLC	12/18/2017	0			0		N/A
AL-0318	TALLADEGA SAWMILL	GEORGIA PACIFIC WOOD PRODUCTS, LLC	12/18/2017	0			0		N/A
IL-0129	CPV THREE RIVERS ENERGY CENTER	CPV THREE RIVERS, LLC	7/30/2018	0			0		BACT
LA-0305	LAKE CHARLES METHANOL FACILITY	LAKE CHARLES METHANOL, LLC	6/30/2016	4023	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0305	LAKE CHARLES METHANOL FACILITY	LAKE CHARLES METHANOL, LLC	6/30/2016	4023	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	0		good combustion practices, Use ultra low sulfur diesel, and comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	0		good combustion practices, Use ultra low sulfur diesel, and comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	3353	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	3353	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0317	METHANEX - GEISMAR METHANOL PLANT	METHANEX USA, LLC	12/22/2016	0		complying with 40 CFR 60 Subpart IIII and 40 CFR 63 Subpart ZZZZ	0		BACT
LA-0317	METHANEX - GEISMAR METHANOL PLANT	METHANEX USA, LLC	12/22/2016	0		complying with 40 CFR 60 Subpart IIII and 40 CFR 63 Subpart ZZZZ	0		BACT
LA-0318	FLOPAM FACILITY	FLOPAM, INC.	1/7/2016	0		Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0346	GULF COAST METHANOL COMPLEX	IGP METHANOL LLC	1/4/2018	13410	hp (each)		0		BACT
LA-0346	GULF COAST METHANOL COMPLEX	IGP METHANOL LLC	1/4/2018	13410	hp (each)		0		BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
LA-0350	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	3/28/2018	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0350	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	3/28/2018	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0382	BIG LAKE FUELS METHANOL PLANT	BIG LAKE FUELS LLC	4/25/2019	0		Comply with standards of 40 CFR 60 Subpart IIII	0		BACT
LA-0382	BIG LAKE FUELS METHANOL PLANT	BIG LAKE FUELS LLC	4/25/2019	0		Comply with standards of 40 CFR 60 Subpart IIII	0		BACT
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
TX-0876	PORT ARTHUR ETHANE CRACKER UNIT	MOTIVA ENTERPRISE LLC	2/6/2020	0		Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101, limited to 100 hours per year of non-emergency operation	0		BACT
TX-0876	PORT ARTHUR ETHANE CRACKER UNIT	MOTIVA ENTERPRISE LLC	2/6/2020	0		Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101, limited to 100 hours per year of non-emergency operation	0		BACT
TX-0888	ORANGE POLYETHYLENE PLANT	CHEVRON PHILLIPS CHEMICAL COMPANY LP	4/23/2020	0		well-designed and properly maintained engines and each limited to 100 hours per year of non-emergency use.	0		BACT
TX-0888	ORANGE POLYETHYLENE PLANT	CHEVRON PHILLIPS CHEMICAL COMPANY LP	4/23/2020	0		well-designed and properly maintained engines and each limited to 100 hours per year of non-emergency use.	0		BACT
TX-0888	ORANGE POLYETHYLENE PLANT	CHEVRON PHILLIPS CHEMICAL COMPANY LP	4/23/2020	0		well-designed and properly maintained engines and each limited to 100 hours per year of non-emergency use.	0		BACT
TX-0904	MOTIVA POLYETHYLENE MANUFACTURING COMPLEX		9/9/2020	0		100 HOURS OPERATIONS, Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101	0		BACT
TX-0904	MOTIVA POLYETHYLENE MANUFACTURING COMPLEX		9/9/2020	0		100 HOURS OPERATIONS, Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101	0		BACT
TX-0904	MOTIVA POLYETHYLENE MANUFACTURING COMPLEX		9/9/2020	0		100 HOURS OPERATIONS, Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101	0		BACT
TX-0905	DIAMOND GREEN DIESEL PORT ARTHUR FACILITY	DIAMOND GREEN DIESEL	9/16/2020	0		limited to 100 hours per year of non-emergency operation	0		BACT
TX-0905	DIAMOND GREEN DIESEL PORT ARTHUR FACILITY	DIAMOND GREEN DIESEL	9/16/2020	0		limited to 100 hours per year of non-emergency operation	0		BACT
TX-0905	DIAMOND GREEN DIESEL PORT ARTHUR FACILITY	DIAMOND GREEN DIESEL	9/16/2020	0		limited to 100 hours per year of non-emergency operation	0		BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		limited to 100 hours per year of non-emergency operation. EPA Tier 2 (40 CFR Â§ 1039.101) exhaust emission standards	0		BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		limited to 100 hours per year of non-emergency operation. EPA Tier 2 (40 CFR Â§ 1039.101) exhaust emission standards	0		BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		limited to 100 hours per year of non-emergency operation. EPA Tier 2 (40 CFR Â§ 1039.101) exhaust emission standards	0		BACT
*TX-0976	CORPUS CHRISTI LIQUEFACTION STAGE 3	CORPUS CHRISTI LIQUEFACTION, LLC	5/30/2025	0		EPA Tier 2, limited to 100 hours per year of non-emergency operation. GHGs from the emergency engines will be limited through engine design and certification in accordance with standards, limited operational hours, and proper operation and maintenance	0		BACT

Table C-3c: RBLC PM Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*TX-0976	CORPUS CHRISTI LIQUEFACTION STAGE 3	CORPUS CHRISTI LIQUEFACTION, LLC	5/30/2025	0		EPA Tier 2, limited to 100 hours per year of non-emergency operation. GHGs from the emergency engines will be limited through engine design and certification in accordance with standards, limited operational hours, and proper operation and maintenance	0		BACT
*TX-0976	CORPUS CHRISTI LIQUEFACTION STAGE 3	CORPUS CHRISTI LIQUEFACTION, LLC	5/30/2025	0		EPA Tier 2, limited to 100 hours per year of non-emergency operation. GHGs from the emergency engines will be limited through engine design and certification in accordance with standards, limited operational hours, and proper operation and maintenance	0		BACT

Table C-3d: RBLC SO2 Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
FL-0356	OKEECHOBEE CLEAN ENERGY CENTER	FLORIDA POWER & LIGHT	3/9/2016	0		Use of ULSD	0.0015	% S IN ULSD	BACT
AR-0163	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	6/9/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.0015	% SULFUR FUEL	BACT
AL-0328	PLANT BARRY	ALABAMA POWER COMPANY	11/9/2020	0			15	PPM	BACT
IN-0317	RIVERVIEW ENERGY CORPORATION	RIVERVIEW ENERGY CORPORATION	6/11/2019	2800	HP	ULSD	15	PPM	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	22.68	MMBTU/H	Good combustion practices and meeting NSPS Subpart IIII requirements.	15	PPM	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	15	PPM	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	15	PPM	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	22.68	MMBTU/H	Good Combustion Practices and meeting NSPS Subpart IIII requirements	15	PPM	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	15	PPM	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Good Combustion Practices and meeting NSPS Subpart IIII requirements	15	PPM	BACT
AR-0177	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	11/21/2022	3634	Horsepower	Fuel Specification	15	PPM FUEL	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	15	FUEL CONTENT	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	15	LESS S IN FUE	BACT
*AR-0192	GREEN BAY PACKAGING	GREEN BAY PACKAGING	10/26/2023	0		ULSD, Good Design and Operating Practices	15	LESS S IN FUE	BACT
FL-0363	DANIA BEACH ENERGY CENTER	FLORIDA POWER AND LIGHT COMPANY	12/4/2017	0		Clean fuel	15	PPM S IN FUEL	BACT
FL-0367	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	7/27/2018	14.82	MMBtu/hour	Clean fuel	15	PPM S IN FUEL	BACT
FL-0371	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	6/7/2021	14.82	MMBtu/hour		15	PPM S IN FUEL	BACT
TX-0876	PORT ARTHUR ETHANE CRACKER UNIT	MOTIVA ENTERPRISE LLC	2/6/2020	0		Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101, limited to 100 hours per year of non-emergency operation	15	PPMW	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	186.6	gph	Good combustion practices, ULSD, and limit operation to 500 hours per year.	15	SULFUR IN FUEL	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	5364	HP	Ultra-low sulfur diesel fuel with sulfur content of 15 ppmv.	0	LB/HP-H	BACT
AR-0161	SUN BIO MATERIAL COMPANY	SUN BIO MATERIAL COMPANY	9/23/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	0.007	G/KW-HR	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	1529	HP	Ultra low sulfur diesel fuel	0.016	LB/H	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	1529	HP	Ultra low sulfur diesel fuel	0.016	LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	2947	HP	Ultra low sulfur diesel fuel	0.03	LB/H	BACT

Table C-3d: RBLC SO2 Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
TX-0728	PEONY CHEMICAL MANUFACTURING FACILITY	BASF	4/1/2015	1500	hp	Low sulfur fuel 15 ppmw	0.61	LB/H	CASE-BY-CASE
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	4290	kw	Compliance with 40 CFR 60 Subpart IIII and operating engines per manufacturers' instructions and written procedures designed to maximize combustion efficiency and minimize fuel usage.	0.0015	LB/MM BTU	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	2923	hp	Use of ultra-low sulfur diesel as fuel.	0.0078	LB/MM BTU	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	2206	HP	ultra-low sulfur diesel (ULSD) fuel with a sulfur content of less than 15 ppm (0.0015 percent by weight)	0.0015	LB/MMBTU	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	1860	HP	ultra-low sulfur diesel (ULSD) fuel with a sulfur content of less than 15 ppm (0.0015 percent by weight)	0.0015	LB/MMBTU	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Ultra Low Sulfur Diesel/Fuel (15 ppm max)	0.0015	LB/MMBTU	N/A
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	H/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.0015	LB/MMBTU	BACT
*IN-0365	MAPLE CREEK ENERGY LLC		6/19/2023	2012	hp		0.0016	LB PER MMBTU	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	1250	kW	Use of ultra-low sulfur diesel, with a sulfur content < 15 ppm sulfur.	0		BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	3000	Horsepower	ultra-low sulfur diesel fuel (0.0015%S)	0		BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0		BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0		BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0		BACT
LA-0305	LAKE CHARLES METHANOL FACILITY	LAKE CHARLES METHANOL, LLC	6/30/2016	4023	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0309	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	6/4/2015	2922	hp (each)		0		BACT
LA-0350	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	3/28/2018	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
TX-0888	ORANGE POLYETHYLENE PLANT	CHEVRON PHILLIPS CHEMICAL COMPANY LP	4/23/2020	0		ultra-low sulfur diesel (15 ppmw sulfur content).	0		BACT
TX-0904	MOTIVA POLYETHYLENE MANUFACTURING COMPLEX		9/9/2020	0		100 HOURS OPERATIONS, Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101	0		BACT
TX-0911	FORMOSA POINT COMFORT PLANT	FORMOSA PLASTICS CORPORATION, TEXAS	12/15/2020	0		ULTRA LOW SULFUR FUEL	0		BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		limited to 100 hours per year of non-emergency operation. EPA Tier 2 (40 CFR Â§ 1039.101) exhaust emission standards	0		BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	H/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0		BACT
WI-0284	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT		4/24/2018	0		The Use of Ultra-Low Sulfur Fuel and Good Combustion Practices	0		BACT

Table C-3d: RBLC SO2 Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
WI-0286	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT	SIO INTERNATIONAL	4/24/2018	0		Good Combustion Practices and The Use of Ultra-low Sulfur Fuel	0		BACT

Table C-3e: RBLC H2SO4 Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	2	MW	Good combustion practices, low sulfur fuel.	15	PPM	BACT
PA-0311	MOXIE FREEDOM GENERATION PLANT	MOXIE FREEDOM LLC	9/1/2015	0			0.0006	G/HP-HR	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	0			0.0007	GM/HP-HR	BACT
OH-0375	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	11/7/2017	2206	HP	Low sulfur fuel	0.0016	LB/H	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Ultra Low Sulfur Diesel/Fuel (15 ppm max)	0.0001	LB/MMBTU	N/A
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	H/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.0001	LB/MMBTU	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	1529	HP	Ultra low sulfur diesel fuel	3.3	X10-4 LB/H	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	1529	HP	Ultra low sulfur diesel fuel	3.3	X10-4 LB/H	BACT
OH-0366	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	8/25/2015	2346	HP	Low sulfur fuel	5.1	X10-4 LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	2947	HP	Ultra low sulfur diesel fuel	6.4	X10-4 LB/H	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	1860	HP	ultra-low sulfur diesel (ULSD) fuel with a sulfur content of less than 15 ppm (0.0015 percent by weight)	7.3	X10-4 LB/MMBTU	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	2206	HP	ultra-low sulfur diesel (ULSD) fuel with a sulfur content of less than 15 ppm (0.0015 percent by weight)	3.4	X10-3 LB/H	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	1250	kW	Use of ultra-low sulfur diesel, with a sulfur content < 15 ppm sulfur.	0		BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	H/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0		BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	1490	HP	Operation limited to 500 hours/year, sulfur content of the diesel fuel oil fired may not exceed 15 ppm, and permittee shall operate and maintain generator according to the manufacturerâ€™s recommendations.	0		BACT

Table C-3f: RBLC VOC Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	3000	kWe, each	Good combustion practices, low sulfur fuel	0.24	G	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	1500	kWe	Good combustion practices, low sulfur fuel	0.24	G	BACT
*PA-0313	FIRST QUALITY TISSUE LOCK HAVEN PLT	FIRST QUALITY TISSUE, LLC	7/27/2017	2500	bhp		3.5	G	
PA-0311	MOXIE FREEDOM GENERATION PLANT	MOXIE FREEDOM LLC	9/1/2015	0			0.02	G/HP-HR	LAER
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	3000	KW	Compliance demonstrated with vendor emission certification and adherence to vendor-specified maintenance recommendations.	0.11	G/BHP-H	LAER
*AR-0192	GREEN BAY PACKAGING	GREEN BAY PACKAGING	10/26/2023	0		ULSD, Good Design and Operating Practices	0.14	G/BHP-HR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	186.6	gph	Oxidation Catalyst, Good combustion practices, and limit operation to 500 hours per year.	0.18	G/HP-HR	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	0			0.22	GM/HP-HR	LAER
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	1490	HP	Operation limited to 500 hours/year and operate and maintain generator according to the manufacturerâ€™s recommendations	0.32	G/HP-H	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	3000	Horsepower	certified engine	0.32	G/HP-HR	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	1490	HP	Good combustion practices include operational and design elements.	0.32	G/HP-HR	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	3600	HP EACH	GOOD COMBUSTION PRACTICES	0.35	G/HP-H EACH	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	3600	HP		0.35	G/HP-HR	BACT
TX-0915	UNIT 5	NRG CEDAR BAYOU LLC	3/17/2021	0		LIMITED 500 HR/YR OPERATION	0.5	G/HPHR	BACT
LA-0346	GULF COAST METHANOL COMPLEX	IGP METHANOL LLC	1/4/2018	13410	hp (each)	Comply with standards of 40 CFR 60 Subpart JJJJ	1	G/BHP-HR	BACT
OK-0175	WILDHORSE TERMINAL	WILDHORSE TERMINAL LLC	6/29/2017	0		Good combustion practices. Certified to meet EPA Tier 3 engine standards. Shall be limited to operate at no more than 500 hr/yr.	3	GM/HP-HR	BACT
OK-0181	WILDHORSE TERMINAL	KEYERA ENERGY INC	9/11/2019	0		Good combustion practices. Certified to meet EPA Tier 3 engine standards. Each engine shall be limited to operate not more than 500 hours per year.	3	GM/HP-HR	BACT

Table C-3f: RBLC VOC Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
KY-0109	FRITZ WINTER NORTH AMERICA, LP	FRITZ WINTER NORTH AMERICA, LP	10/24/2016	53.6	gal/hr	The permittee shall prepare and maintain for EU72, EU73, and EU74, within 90 days of startup, a good combustion and operation practices plan (GCOP) that defines, measures and verifies the use of operational and design practices determined as BACT for minimizing CO, VOC, PM, PM10, and PM2.5 emissions. Any revisions requested by the Division shall be made and the plan shall be maintained on site. The permittee shall operate according to the provisions of this plan at all times, including periods of startup, shutdown, and malfunction. The plan shall be incorporated into the plant standard operating procedures (SOP) and shall be made available for the Divisionâ€™s inspection. The plan shall include, but not be limited to: i. A list of combustion optimization practices and a means of verifying the practices have occurred. ii. A list of combustion and operation practices to be used to lower energy consumption and a means of verifying the practices have occurred. iii. A list of the design choices determined to be BACT and verification that designs were implemented in the final construction.	4.77	G/HP-HR (EU72 &EU73)	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	2937	hp	Compliance with 40 CFR 60 Subpart IIII standards, good combustion practices, and the use of ultra-low sulfur diesel fuel.	4.8	G/HP-HR	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	1140	hp	Proper equipment design and good combustion practices. Use ultra-low sulfur diesel as fuel. Compliance with 40 CFR 60 Subpart IIII.	4.8	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	4.8	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	4.8	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	4.8	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	4.8	G/HP-HR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	4.8	G/HP-HR	BACT
TX-0872	CONDENSATE SPLITTER FACILITY	MAGELLAN PROCESSING, L.P.	10/31/2019	0		Limiting duration and frequency of generator use to 100 hr/yr. Good combustion practices will be used to reduce VOC including maintaining proper air-to-fuel ratio.	0.12	G/KW HR	BACT
LA-0324	COMMONWEALTH LNG FACILITY	COMMONWEALTH LNG, LLC	3/28/2023	4290	kw	Compliance with 40 CFR 60 Subpart IIII and operating engines per manufacturers' instructions and written procedures designed to maximize combustion efficiency and minimize fuel usage.	0.322	G/KW-HR	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	5051	HP	Certified to meet Tier 2 standards and good combustion practices	0.4	G/KW-H	BACT
WI-0284	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT		4/24/2018	0		Good Combustion Practices	0.56	G/KWH	BACT

Table C-3f: RBLC VOC Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
WI-0286	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT	SIO INTERNATIONAL	4/24/2018	0		Good Combustion Practices	0.56	G/KWH	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	2923	hp	Good combustion practices including compliance with 40 CFR 60 Subpart IIII.	0.56	G/KW-H	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	5364	HP	Good combustion and operating practices.	0.79	G/KW-H	BACT
AR-0177	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	11/21/2022	3634	Horsepower		0.8	G/KW-HR	BACT
AR-0163	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	6/9/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1.55	G/KW-HR	BACT
AR-0161	SUN BIO MATERIAL COMPANY	SUN BIO MATERIAL COMPANY	9/23/2019	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1.9	G/KW-HR	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Good Combustion Practices/Maintenance	6.4	G/KW	N/A
IN-0317	RIVERVIEW ENERGY CORPORATION	RIVERVIEW ENERGY CORPORATION	6/11/2019	2800	HP	Tier II diesel engine	6.4	G/KWH	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	762	bhp	Good combustion practices	6.4	G/KW-H	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	3985	hp		6.4	G/KW-HR	BACT
IN-0382	DUKE ENERGY INDIANA, INC.- CAYUGA GENERATING STATION	DUKE ENERGY INDIANA, INC.	2/14/2025	2000	HP		6.4	G/KW-HR	BACT
SC-0193	MERCEDES BENZ VANS, LLC	MERCEDES BENZ VANS, LLC	4/15/2016	1500	hp	Must meet the standards of 40 CFR 60, Subpart IIII	100	HR/YR	BACT
AK-0082	POINT THOMSON PRODUCTION FACILITY	EXXON MOBIL CORPORATION	1/23/2015	2695	hp		0.0007	LB/HP-H	BACT
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	18.7	MMBTU/HR	GOOD COMBUSTION PRACTICES, LIMITED TO 100 HR/YR	0.001	LB/HP HR	BACT
*LA-0312	ST. JAMES METHANOL PLANT	SOUTH LOUISIANA METHANOL LP	6/30/2017	1474	horsepower	Compliance with NSPS Subpart IIII	0.04	LB/HR	BACT
*WV-0033	MAIDSVILLE	MOUNTAIN STATE CLEAN ENERGY, LLC	1/5/2022	2100	hp	Good Combustion Practices w/ OxCat. Applicant did not justify why an oxcat is infeasible for an emergency engine	0.46	LB/HR	BACT
VA-0327	PERDUE GRAIN AND OILSEED, LLC	PERDUE AGRIBUSINESS, LLC	7/12/2017	0			0.49	LB/HR	BACT
TX-0728	PEONY CHEMICAL MANUFACTURING FACILITY	BASF	4/1/2015	1500	hp	Minimized hours of operations Tier II engine	0.7	LB/H	CASE-BY-CASE
LA-0292	HOLBROOK COMPRESSOR STATION	CAMERON INTERSTATE PIPELINE LLC	1/22/2016	1341	HP	Good combustion practices consistent with the manufacturer's recommendations to maximize fuel efficiency and minimize emissions	0.83	LB/HR	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Good combustion practices.	0.86	LB/H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Good combustion practices	0.86	LB/H	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	1100	KW		0.86	LB/H	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Good combustion practices	0.86	LB/H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Good combustion practices.	0.86	LB/H	BACT

Table C-3f: RBLC VOC Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	1.33	LB/HR	BACT
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	1.33	LB/HR	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	5000	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	1.6	LB/H	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	22.68	MMBTU/H	Good combustion practices.	1.87	LB/H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	2	MW	State of the art combustion design.	1.89	LB/H	BACT
LA-0390	DERIDDER SAWMILL	CSP DERIDDER, LLC	5/10/2022	750	horsepower	Good combustion practices and maintenance and compliance with applicable 40 CFR 60 Subpart JJJJ limitation for VOC.	1.98	LB/HR	BACT
LA-0390	DERIDDER SAWMILL	CSP DERIDDER, LLC	5/10/2022	750	horsepower	Good combustion practices and maintenance and compliance with applicable 40 CFR 60 Subpart JJJJ limitation for VOC	1.98	LB/HR	BACT
LA-0390	DERIDDER SAWMILL	CSP DERIDDER, LLC	5/10/2022	750	horsepower	Good Combustion practices and maintenance and compliance with applicable 40 CFR 60 Subpart JJJJ limitations for VOC	1.98	LB/HR	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	1529	HP	State-of-the-art combustion design	2	LB/H	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	1529	HP	State-of-the-art combustion design	2	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	760	horsepower	Compliance with 40 CFR 60 Subpart IIII	2.03	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	2923	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	2.06	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	2923	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	2.06	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	3634	horsepower	Compliance with 40 CFR 60 Subpart IIII	2.29	LB/HR	BACT
OH-0366	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	8/25/2015	2346	HP		3.1	LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	2947	HP	State-of-the-art combustion design	3.84	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2012	horsepower	Compliance with 40 CFR 60 Subpart IIII	5.38	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	6.27	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	6.27	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	6.27	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	6.27	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	Compliance with 40 CFR 60 Subpart IIII	6.27	LB/HR	BACT

Table C-3f: RBLC VOC Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	1341	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	14.96	LB/H	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	1860	HP	Good combustion practices (ULSD) and compliance with 40 CFR Part 60, Subpart IIII	19.68	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	2206	HP	Certified to the meet the emissions standards in 40 CFR 89.112 and 89.113 pursuant to 40 CFR 60.4205(b) and 60.4202(a)(2). Good combustion practices per the manufacturerâ€™s operating manual.	23.21	LB/H	BACT
OH-0375	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	11/7/2017	2206	HP	Good combustion design	24.71	LB/H	BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	2584	HP	Good combustion practices	27.34	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	3353	HP	certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII, shall employ good combustion practices per the manufacturerâ€™s operating manual	37.41	LB/H	BACT
AL-0318	TALLADEGA SAWMILL	GEORGIA PACIFIC WOOD PRODUCTS, LLC	12/18/2017	0			0		N/A
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0		BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0		BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	0		good combustion practices, Use ultra low sulfur diesel, and comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0309	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	6/4/2015	2922	hp (each)	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	3353	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0318	FLOPAM FACILITY	FLOPAM, INC.	1/7/2016	0		Complying with 40 CFR 60 Subpart IIII	0		LAER
LA-0350	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	3/28/2018	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0382	BIG LAKE FUELS METHANOL PLANT	BIG LAKE FUELS LLC	4/25/2019	0		Comply with standards of 40 CFR 60 Subpart IIII	0		BACT
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
TX-0876	PORT ARTHUR ETHANE CRACKER UNIT	MOTIVA ENTERPRISE LLC	2/6/2020	0		Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101, limited to 100 hours per year of non-emergency operation	0		BACT
TX-0888	ORANGE POLYETHYLENE PLANT	CHEVRON PHILLIPS CHEMICAL COMPANY LP	4/23/2020	0		well-designed and properly maintained engines and each limited to 100 hours per year of non-emergency use.	0		BACT
TX-0904	MOTIVA POLYETHYLENE MANUFACTURING COMPLEX		9/9/2020	0		100 HOURS OPERATIONS, Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101	0		BACT
TX-0905	DIAMOND GREEN DIESEL PORT ARTHUR FACILITY	DIAMOND GREEN DIESEL	9/16/2020	0		limited to 100 hours per year of non-emergency operation	0		BACT

Table C-3f: RBLC VOC Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		limited to 100 hours per year of non-emergency operation. EPA Tier 2 (40 CFR Â§ 1039.101) exhaust emission standards	0		BACT
TX-0955	INEOS OLIGOMERS CHOCOLATE BAYOU	INEOS OLIGOMERS USA LLC	3/14/2023	0		TIER III	0		LAER
*TX-0976	CORPUS CHRISTI LIQUEFACTION STAGE 3	CORPUS CHRISTI LIQUEFACTION, LLC	5/30/2025	0		EPA Tier 2, limited to 100 hours per year of non-emergency operation. GHGs from the emergency engines will be limited through engine design and certification in accordance with standards, limited operational hours, and proper operation and maintenance	0		BACT

Table C-3g: RBLC GHG Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	1140	hp	Proper equipment design and good combustion practices. Use ultra-low sulfur diesel as fuel. Compliance with 40 CFR 60 Subpart IIII.	1.16	G/HP-HR	BACT
TX-0766	GOLDEN PASS LNG EXPORT TERMINAL	GOLDEN PASS PRODUCTS, LLC	9/11/2015	750	hp	Equipment specifications & work practices - Good combustion practices and limited operational hours	40	HR/YR	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	1490	HP	Selection of the most efficient reciprocating internal combustion engine and minimizing the hours of operation limits greenhouse gas emissions. EPA Tier 2 certified reciprocating internal combustion engine and limiting its hours of operation to 500 hours per each 12 consecutive calendar months.	500	HOURS	BACT
VA-0333	NORFOLK NAVAL SHIPYARD	US NAVY NORFOLK NAVAL SHIPYARD	12/9/2020	2220	HP		2.543	LB	BACT
OH-0379	PETMIN USA INCORPORATED	PETMIN USA INCORPORATED	2/6/2019	3131	HP	Tier IV engine Good combustion practices	3632	LB/H	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	2937	hp	Compliance with 40 CFR 60 Subpart IIII, good combustion practices, and the use of ultra-low sulfur diesel fuel.	74.21	KG/MM BTU	BACT
AR-0177	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	11/21/2022	3634	Horsepower		163	LB/MMBTU	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	186.6	gph	Good combustion practices and limit operation to 500 hours per year	163.6	LB/MMBTU	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	3000	Horsepower	Good engineering design and manufacturerâ€™s recommended operating and maintenance procedures.	163.6	LB/MMBTU	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	2682	HP	Equipment design and maintenance requirements	163.6	LB/MMBTU	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Good Combustion Practices/Maintenance	163.6	LB/MMBTU	N/A
AR-0161	SUN BIO MATERIAL COMPANY	SUN BIO MATERIAL COMPANY	9/23/2019	0		Good Combustion Practices	164	LB/MMBTU	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good combustion practices	164	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Combustion Practices	164	LB/MMBTU	BACT
PA-0311	MOXIE FREEDOM GENERATION PLANT	MOXIE FREEDOM LLC	9/1/2015	0			44	TPY	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Good combustion and design practices.	70	T/YR	BACT
LA-0292	HOLBROOK COMPRESSOR STATION	CAMERON INTERSTATE PIPELINE LLC	1/22/2016	1341	HP		77	TPY	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	1341	HP	good operating practices (proper maintenance and operation)	80	T/YR	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	0			81	TONS	BACT
*LA-0312	ST. JAMES METHANOL PLANT	SOUTH LOUISIANA METHANOL LP	6/30/2017	1474	horsepower	Compliance with NSPS Subpart IIII	84	TPY	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	1860	HP	Efficient design and proper maintenance and operation	109.2	T/YR	BACT
OH-0375	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	LONG RIDGE ENERGY GENERATION LLC - HANNIBAL POWER	11/7/2017	2206	HP	Efficient design	116.8	T/YR	BACT

Table C-3g: RBLC GHG Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	2206	HP	good operating practices (proper maintenance and operation)	120	T/YR	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	3985	hp		160	TONS/YEAR	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	2	MW	Energy efficient design.	161	T/YR	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	2923	hp	Good combustion practices including compliance with 40 CFR 60 Subpart IIII.	180	T/YR	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	3353	HP	good operating practices (proper maintenance and operation)	200	T/YR	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	762	bhp	Good combustion practices, use of current energy efficiency measures.	205	T/YR	BACT
MI-0425	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	5/9/2017	500	H/YR	Good combustion and design practices.	209	T/YR	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Good Combustion and Design Practices	209	T/YR	BACT
*IL-0137	MARQUIS CARBON CAPTURE LLC	MARQUIS CARBON CAPTURE LLC	6/23/2025	0		Use of ultra-low sulfur diesel (ULSD) as fuel; good combustion practices; compliance with emission limits of the NSPS, 40 CFR 60 Subpart IIII; 500 hours/year operation	222.5	TONS/YEAR	BACT
MI-0421	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	8/26/2016	500	H/YR	Good combustion and design practices.	223	T/YR	BACT
IL-0130	JACKSON ENERGY CENTER	JACKSON GENERATION, LLC	12/31/2018	1500	kW		225	TONS/YEAR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	231	TON/YR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	772	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	231	TON/YR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, and good combustion practices	314	TON/YR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	314	TON/YR	BACT
MN-0095	BLUE LAKE	XCEL ENERGY	8/7/2024	1049	horsepower	Emergency classification, hourly restriction, good combustion practices, and NSPS compliance	314	TON/YR	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Good combustion practices.	383	T/YR	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	1341	HP	Good combustion practices.	383	T/YR	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Good combustion practices	383	T/YR	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	1341	HP	Good combustion practices	383	T/YR	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	1000	kW	Good Combustion Practices	389	TONS	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	1500	HP	Good combustion practices and energy efficiency measures.	406	T/YR	BACT
MI-0442	THOMAS TOWNSHIP ENERGY, LLC	THOMAS TOWNSHIP ENERGY, LLC	8/21/2019	1100	KW		444	T/YR	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	1529	HP	Efficient design	445	T/YR	BACT

Table C-3g: RBLC GHG Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	1529	HP	state of the art combustion design	445	T/YR	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	1250	kW		508	TONS/YEAR	BACT
MI-0447	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	1/7/2021	4474.2	KW	low carbon fuel (pipeline quality natural gas), good combustion practices, and energy efficiency measures.	590	T/YR	BACT
MI-0448	GRAYLING PARTICLEBOARD	ARAUCO NORTH AMERICA	12/18/2020	500	h/yr	Good Combustion and Design Practices	590	T/YR	BACT
MI-0454	LBWL-ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/20/2022	4474.2	KW	low carbon fuel (pipeline quality natural gas), good combustion practices, and energy efficiency measures.	590	T/YR	BACT
*IN-0365	MAPLE CREEK ENERGY LLC		6/19/2023	2012	hp		625	TONS PER YEAR	BACT
OH-0366	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	CLEAN ENERGY FUTURE - LORDSTOWN, LLC	8/25/2015	2346	HP	Efficient design	683	T/YR	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	2000	kW	Good Combustion Practices	778	TONS	BACT
IN-0317	RIVERVIEW ENERGY CORPORATION	RIVERVIEW ENERGY CORPORATION	6/11/2019	2800	HP	Tier II diesel engine	811	TONS	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	2947	HP	Efficient design	858	T/YR	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	22.68	MMBTU/H	Good combustion practices	928	T/YR	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	22.68	MMBTU/H	Good combustion practices	928	T/YR	BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	H/YR	use of S15 ULSD and high efficiency design and operation	981	T/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	3600	HP		1044	TON/YR	BACT
*AR-0192	GREEN BAY PACKAGING	GREEN BAY PACKAGING	10/26/2023	0		ULSD, Good Design and Operating Practices	1200	TPY	BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	H/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	1203	T/YR	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	5000	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	1289	T/YR	BACT
LA-0331	CALCASIEU PASS LNG PROJECT	VENTURE GLOBAL CALCASIEU PASS, LLC	9/21/2018	5364	HP	Good Combustion of Practices and Good Operation and Maintenance Practices	1481	T/YR	BACT
MI-0441	LBWL--ERICKSON STATION	LANSING BOARD OF WATER AND LIGHT	12/21/2018	6000	HP	Good combustion practices and energy efficiency measures.	1590	T/YR	BACT
AK-0082	POINT THOMSON PRODUCTION FACILITY	EXXON MOBIL CORPORATION	1/23/2015	2695	hp		2332	TONS/YEAR	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	1500	kWe	Good Combustion Practices	2781	TPY	BACT
IL-0129	CPV THREE RIVERS ENERGY CENTER	CPV THREE RIVERS, LLC	7/30/2018	0			0		BACT
IN-0382	DUKE ENERGY INDIANA, INC.- CAYUGA GENERATING STATION	DUKE ENERGY INDIANA, INC.	2/14/2025	2000	HP	good combustion practices	0		BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	3000	kWe, each		0		BACT

Table C-3g: RBLC GHG Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	1500	kWe	GHG emissions of the plant emergency generator shall be controlled through the use of a good engineering design fuel efficient design, and limited operating hours.	0		BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0		BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0		BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	2922	HP	This EP is required to have a Good Combustion and Operating Practices (GCOP) Plan.	0		BACT
LA-0305	LAKE CHARLES METHANOL FACILITY	LAKE CHARLES METHANOL, LLC	6/30/2016	4023	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	0		good combustion/operating/maintenance practices	0		BACT
LA-0309	BENTELER STEEL TUBE FACILITY	BENTELER STEEL / TUBE MANUFACTURING CORPORATION	6/4/2015	2922	hp (each)		0		BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	2584	HP	Good combustion practices	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	3353	hp	good combustion practices	0		BACT
LA-0317	METHANEX - GEISMAR METHANOL PLANT	METHANEX USA, LLC	12/22/2016	0		complying with 40 CFR 60 Subpart IIII and 40 CFR 63 Subpart ZZZZ	0		BACT
LA-0383	LAKE CHARLES LNG EXPORT TERMINAL	LAKE CHARLES LNG EXPORT COMPANY, LLC	9/3/2020	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	BACT is determined to be the implementation of the following energy efficiency measures: Improved Combustion Measures, Insulation, and Minimization of Air Infiltration	0		BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	BACT is determined to be the implementation of the following energy efficiency measures: Improved Combustion Measures, Insulation, and Minimization of Air Infiltration	0		BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2346	horsepower	BACT is determined to be the implementation of the following energy efficiency measures: Improved Combustion Measures, Insulation, and Minimization of Air Infiltration	0		BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	2012	horsepower	BACT is determined to be the implementation of the following energy efficiency measures: Improved Combustion Measures, Insulation, and Minimization of Air Infiltration	0		BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	760	horsepower	BACT is determined to be the implementation of the following energy efficiency measures: Improved Combustion Measures, Insulation, and Minimization of Air Infiltration	0		BACT
TX-0872	CONDENSATE SPLITTER FACILITY	MAGELLAN PROCESSING, L.P.	10/31/2019	0		Limiting duration and frequency of generator use to 100 hr/yr. Good combustion practices will be used to reduce VOC including maintaining proper air-to-fuel ratio.	0		BACT
TX-0876	PORT ARTHUR ETHANE CRACKER UNIT	MOTIVA ENTERPRISE LLC	2/6/2020	0		Tier 4 exhaust emission standards specified in 40 CFR Â§ 1039.101, limited to 100 hours per year of non-emergency operation	0		BACT
TX-0888	ORANGE POLYETHYLENE PLANT	CHEVRON PHILLIPS CHEMICAL COMPANY LP	4/23/2020	0		well-designed and properly maintained engines and each limited to 100 hours per year of non-emergency use.	0		BACT
TX-0905	DIAMOND GREEN DIESEL PORT ARTHUR FACILITY	DIAMOND GREEN DIESEL	9/16/2020	0		limited to 100 hours per year of non-emergency operation	0		BACT

Table C-3g: RBLC GHG Results for Emergency Diesel Generators

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
TX-0915	UNIT 5	NRG CEDAR BAYOU LLC	3/17/2021	0		LIMITED 500 HR/YR OPERATION	0		BACT
TX-0933	NACERO PENWELL FACILITY	NACERO TX 1 LLC	11/17/2021	0		limited to 100 hours per year of non-emergency operation. EPA Tier 2 (40 CFR Â§ 1039.101) exhaust emission standards	0		BACT
TX-0939	ORANGE COUNTY ADVANCED POWER STATION	ENTERGY TEXAS, INC.	3/13/2023	18.7	MMBTU/HR	GOOD COMBUSTION PRACTICES, LIMITED TO 100 HR/YR	0		BACT
*TX-0976	CORPUS CHRISTI LIQUEFACTION STAGE 3	CORPUS CHRISTI LIQUEFACTION, LLC	5/30/2025	0		EPA Tier 2, limited to 100 hours per year of non-emergency operation. GHGs from the emergency engines will be limited through engine design and certification in accordance with standards, limited operational hours, and proper operation and maintenance	0		BACT
WI-0284	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT		4/24/2018	0		The Use of Ultra-Low Sulfur Fuel and Good Combustion Practices	0		BACT
WI-0286	SIO INTERNATIONAL WISCONSIN, INC. -ENERGY PLANT	SIO INTERNATIONAL	4/24/2018	0		Good Combustion Practices and The Use of Ultra-low Sulfur Fuel	0		BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	1490	HP	Certified to at least meet EPAâ€™s criteria for Tier 2 reciprocating internal combustion engines and the 40 CFR 60, Subpart IIII emission limitations, operation limited to 500 hours/year, and operate and maintain generator according to the manufacturerâ€™s recommendations.	0		BACT

Table C-4a: RBLC NOx Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
NE-0064	NORFOLK CRUSH, LLC	NORFOLK CRUSH, LLC	11/21/2022	510	hp		2.38	G/HP-HR	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	282	bhp	Good combustion practices, low sulfur fuel	2.85	G	BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	460	hp	Compliance demonstrated with vendor emission certification and adherence to vendor-specified maintenance recommendations.	2.6	G/BHP-H	LAER
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	1.66	MMBTU/H	Good combustion practices and meeting NSPS Subpart IIII requirements.	3	G/BHP-H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	3	G/BHP-H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	3	G/BHP-H	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	1.66	MMBTU/H	Good Combustion Practices and meeting NSPS Subpart IIII requirements	3	G/BHP-H	BACT
MI-0434	FLAT ROCK ASSEMBLY PLANT	FORD MOTOR COMPANY	3/22/2018	250	BHP	Good combustion practices.	3	G/B-HP-H	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	3	G/B-HP-H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Good Combustion Practices and meeting NSPS Subpart IIII requirements	3	G/B-HP-H	BACT
PA-0310	CPV FAIRVIEW ENERGY CENTER	CPV FAIRVIEW, LLC	9/2/2016	0			3	G/BHP-HR	LAER
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	500	H/YR	Good combustion practices.	3	G/HP-H	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	282	HP	Operation limited to 500 hours/year and shall be operated and maintained according to the manufacturerâ€™s recommendations.	3	G/HP-H	BACT
KS-0030	MID-KANSAS ELECTRIC COMPANY, LLC - RUBART STATION	MID-KANSAS ELECTRIC COMPANY, LLC - RUBART STATION	3/31/2016	197	HP		3	G/HP-HR	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	355	hp	Compliance with 40 CFR 60 Subpart IIII, good combustion practices, and use of ultra-low sulfur diesel fuel.	3	G/HP-HR	BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	HR/YR	Good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	3	G/HP-HR	BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	HR/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	3	G/HP-HR	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	282	HP	Good combustion practices and combusting clean fuels: (a) P06 may not operate more than 500 hours per each 12 rolling calendar months. (b) P06 shall be operated and maintained according to the manufacturerâ€™s recommendations. (c) NOx emissions from P06 may not exceed 3.0 g/hp-hr.	3	G/HP-HR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	19.4	gph	Good combustion practices, limit operation to 500 hours per year per engine	3.6	G/HP-HR	BACT
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	7.8	G/HP-HR	BACT

Table C-4a: RBLC NOx Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
AR-0173	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	1/31/2022	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	14.06	G/BHP-HR	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	14.06	G/BHP-HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	14.06	G/BHP-HR	BACT
TX-0846	MOTOR VEHICLE ASSEMBLY PLANT	TOYOTA MOTORS	9/23/2018	214	kW	Meets EPA Tier 4 requirements	0.4	G/KW	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	282	hp	Good combustion practices including compliance with 40 CFR 60 Subpart IIII.	3.31	G/KW-H	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	252	hp	Good Combustion Practices	3.7	G/KW-HR	BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	375	HP	Good combustion practices and NSPS IIII	4	G/KW-H	BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	300	HP	Good combustion practices and NSPS Subpart IIII	4	G/KW-H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	399	BHP	State of the art combustion design.	4	G/KW-H	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	500	bhp		4	G/KW-H	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	275	HP	Certified to meet the standards in Table 4 of 40 CFR Part 60, Subpart IIII and good combustion practices	4	G/KW-H	BACT
FL-0371	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	6/7/2021	2.46	MMBtu/hour		4	G/KW-HOUR	BACT
FL-0367	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	7/27/2018	8700	gal/year	Operate and maintain the engine according to the manufacturer's written instructions	4	G/KW-HR	BACT
IL-0130	JACKSON ENERGY CENTER	JACKSON GENERATION, LLC	12/31/2018	420	horsepower		4	G/KW-HR	LAER
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	369	hp		4	G/KW-HR	BACT
LA-0339	SHINTECH PLAQUEMINE PLANT 3	SHINTECH LOUISIANA LLC	1/19/2021	0		Compliance with 40 CFR 60 Subpart IIII	4	G/KW-HR	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	320	horsepower		4	GRAMS	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	460	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	0.3	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.98	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.98	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.98	LB/HR	BACT
*LA-0370	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER LLC	4/27/2020	1.1	MM BTU/hr	The use of low sulfur fuels and compliance with 40 CFR 60 Subpart IIII	1.15	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	237	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	1.49	LB/HR	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	250	HP	Comply with NSPS 40 CFR 60 Subpart IIII	1.6	LB/H	BACT

Table C-4a: RBLC NOx Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
NJ-0084	PSEG FOSSIL LLC SEWAREN GENERATING STATION	PSEG FOSSIL LLC	3/10/2016	100	H/YR	use of ULSD a clean burning fuel, and limited hours of operation	1.7	LB/H	LAER
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	311	HP	State-of-the-art combustion design	1.79	LB/H	BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	282	HP	Compliance with NESHAP 40 CFR 63 Subpart ZZZZ and NSPS 40 CFR 60 Subpart IIII, and good combustion practices (use of ultra-low sulfur diesel fuel).	1.87	LB/H	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	300	HP	State-of-the-art combustion design	1.97	LB/H	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	300	HP	State-of-the-art combustion design	1.97	LB/H	BACT
NJ-0085	MIDDLESEX ENERGY CENTER, LLC	STONEGATE POWER, LLC	7/19/2016	100	H/YR	Use of Ultra Low Sulfur Diesel (ULSD) Oil a clean burning fuel and limited hours of operation	2.05	LB/H	LAER
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	320	HP	Good combustion practices (ULSD) and compliance with 40 CFR Part 60, Subpart IIII	2.12	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	402	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII and employ good combustion practices per the manufacturerâ€™s operating manual	2.64	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	410	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII. Good combustion practices per the manufacturerâ€™s operating manual	2.7	LB/H	BACT
WI-0302	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	290	HP	Only use diesel fuel oil with a sulfur content of no greater than 0.0015% by weight	3.64	LB/H	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	422	horsepower	Compliance with 40 CFR 60 Subpart IIII	3.96	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	422	horsepower	Compliance with 40 CFR 60 Subpart IIII	3.96	LB/HR	BACT
AK-0086	KENAI NITROGEN OPERATIONS	AGRIUM U.S. INC.	3/26/2021	2.7	MMBtu/hr	Good Combustion Practices and Limited Use	4.41	LB/MMBTU	BACT
SC-0182	FIBER INDUSTRIES LLC	FIBER INDUSTRIES LLC	10/31/2017	0		Use of Ultra Low Sulfur Diesel Fuel (15 ppm), good combustion, operation, and maintenance practices; compliance with NESHAP Subpart ZZZZ	200	OPERATING HR/YR	BACT
IL-0129	CPV THREE RIVERS ENERGY CENTER	CPV THREE RIVERS, LLC	7/30/2018	0			0		LAER
LA-0314	INDORAMA LAKE CHARLES FACILITY	INDORAMA VENTURES OLEFINS, LLC	8/3/2016	425	hp	complying with 40 CFR 63 subpart ZZZZ	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	460	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0345	DIRECT REDUCED IRON FACILITY	NUCOR STEEL LOUISIANA LLC	6/13/2019	0		Comply with requirements of 40 CFR 60 Subpart IIII	0		BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	0		Comply with 40 CFR 60 Subpart IIII and Good Combustion Practices	0		BACT
*LA-0381	EUEG-5 UNIT - GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2019	0		Comply with standards of 40 CFR 60 Subpart IIII	0		BACT
LA-0384	DIRECT REDUCED IRON FACILITY	NUCOR STEEL LOUISIANA, LLC	6/13/2019	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0386	LASALLE BIOENERGY LLC	DRAX BIOMASS INC.	5/5/2021	0		Comply with 40 CFR 60 Subpart IIII	0		BACT

Table C-4a: RBLC NOx Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
LA-0397	WESTLAKE ETHYLENE PLANT	INDORAMA VENTURES OLEFINS, LLC	4/29/2022	0		Compliance with applicable requirements of 40 CFR 60 Subpart IIII	0		BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Good Combustion Practices/Maintenance	0		N/A
WI-0263	WISCONSIN POWER & LIGHT - NEENAH GENERATING STATION	ALLIANT ENERGY	2/15/2016	1.27	mmBtu/hr	Good combustion practices, use diesel fuel, and operate <500 hr/yr	0		BACT

Table C-4b: RBLC CO Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	460	hp	Compliance demonstrated with vendor emission certification and adherence to vendor-specified maintenance recommendations.	0.53	G/BHP-H	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	282	bhp	Good Combustion practices, low sulfur fuel	2.6	G	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	1.66	MMBTU/H	Good combustion practices and meeting NSPS Subpart IIII requirements.	2.6	G/BHP-H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	2.6	G/BHP-H	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	1.66	MMBTU/H	Good Combustion Practices and meeting NSPS Subpart IIII requirements	2.6	G/BHP-H	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Good combustion practices and meeting NSPS Subpart IIII requirements	2.6	G/B-HP-H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Good Combustion Practices and meeting NSPS Subpart IIII requirements	2.6	G/B-HP-H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	2.6	G/BPH-H	BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	HR/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	2.6	G/HP HR	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Good Combustion Practices/Maintenance	2.6	G/HP-H	N/A
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	HR/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	2.6	G/HP-H	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	282	HP	Operation limited to 500 hours/year and shall be operated and maintained according to the manufacturerâ€™s recommendations.	2.6	G/HP-H	BACT
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	2.6	G/HP-HR	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	355	hp	Compliance with 40 CFR 60 Subpart IIII, good combustion practices, and the use of ultra-low sulfur diesel fuel.	2.6	G/HP-HR	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	282	HP	Good combustion practices: (a) P06 may not operate more than 500 hours per each 12 rolling calendar months. (b) P06 shall be operated and maintained according to the manufacturerâ€™s recommendations. (c) Carbon monoxide emission from P06 may not exceed 2.6 g/hp-hr.	2.6	G/HP-HR	BACT
PA-0310	CPV FAIRVIEW ENERGY CENTER	CPV FAIRVIEW, LLC	9/2/2016	0			2.61	G/BHP-HR	BACT
AR-0173	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	1/31/2022	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	3.03	G/BHP-HR	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	3.03	G/BHP-HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	3.03	G/BHP-HR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	19.4	gph	Good combustion practices, limit operation to 500 hours per year per engine	3.3	G/HP-HR	BACT

Table C-4b: RBLC CO Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	500	H/YR	Good combustion practices.	3.7	G/HP-H	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	282	hp	Good combustion practices including compliance with 40 CFR 60 Subpart IIII.	0.6	G/KW-H	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	252	hp	Good Combustion Practices	3.3	G/KW-HR	BACT
FL-0363	DANIA BEACH ENERGY CENTER	FLORIDA POWER AND LIGHT COMPANY	12/4/2017	0		Certified engine	3.5	G / KWH	BACT
FL-0356	OKEECHOBEE CLEAN ENERGY CENTER	FLORIDA POWER & LIGHT	3/9/2016	0		Use of clean engine technology	3.5	G / KW-HR	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	399	BHP	State of the art combustion design.	3.5	G/KW-H	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	500	bhp	Good combustion practices	3.5	G/KW-H	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	275	HP	Certified to meet the standards in Table 4 of 40 CFR Part 60, Subpart IIII and good combustion practices	3.5	G/KW-H	BACT
FL-0367	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	7/27/2018	8700	gal/year	Operate and maintain the engine according to the manufacturer's written instructions	3.5	G/KW-HOUR	BACT
FL-0371	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	6/7/2021	2.46	MMBtu/hour		3.5	G/KW-HOUR	BACT
IL-0130	JACKSON ENERGY CENTER	JACKSON GENERATION, LLC	12/31/2018	420	horsepower		3.5	G/KW-HR	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	369	hp		3.5	G/KW-HR	BACT
LA-0339	SHINTECH PLAQUEMINE PLANT 3	SHINTECH LOUISIANA LLC	1/19/2021	0		Compliance with 40 CFR 60 Subpart IIII	3.5	G/KW-HR	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	320	horsepower		3.5	GRAMS	BACT
TX-0846	MOTOR VEHICLE ASSEMBLY PLANT	TOYOTA MOTORS	9/23/2018	214	kW	Meets EPA Tier 4 requirements	3.58	G/KW	BACT
WI-0302	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	290	HP	Good combustion practices	0.33	LB/H	BACT
*LA-0370	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER LLC	4/27/2020	1.1	MM BTU/hr	The use of low sulfur fuels and compliance with 40 CFR 60 Subpart IIII	0.4	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	237	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	0.5	LB/HR	BACT
NJ-0084	PSEG FOSSIL LLC SEWAREN GENERATING STATION	PSEG FOSSIL LLC	3/10/2016	100	H/YR	use of ULSD a clean burning fuel, and limited hours of operation	1.1	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	1.15	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	1.15	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	1.15	LB/HR	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	250	HP	Comply with NSPS 40 CFR 60 Subpart IIII	1.4	LB/H	BACT
*LA-0306	TOPCHEM POLLOCK, LLC	TOPCHEM POLLOCK, LLC	12/20/2016	225	horsepower	Meet NSPS Subpart IIII Limitations and Good Combustion Practices	1.55	LB/H	BACT

Table C-4b: RBLC CO Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*LA-0306	TOPCHEM POLLOCK, LLC	TOPCHEM POLLOCK, LLC	12/20/2016	225	horsepower	Meet NSPS Subpart IIII Limitations and Good Combustion Practices	1.55	LB/H	BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	282	HP	Compliance with NESHAP 40 CFR 63 Subpart ZZZZ and NSPS 40 CFR 60 Subpart IIII, and good combustion practices (use of ultra-low sulfur diesel fuel).	1.62	LB/H	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	300	HP	State-of-the-art combustion design	1.73	LB/H	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	300	HP	state of the art combustion design	1.73	LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	311	HP	State-of-the-art combustion design	1.79	LB/H	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	320	HP	Good combustion practices (ULSD) and compliance with 40 CFR Part 60, Subpart IIII	1.83	LB/H	BACT
NJ-0085	MIDDLESEX ENERGY CENTER, LLC	STONEGATE POWER, LLC	7/19/2016	100	H/YR	Use of Ultra Low Sulfur Diesel (ULSD) Oil a clean burning fuel and limited hours of operation	1.87	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	402	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII and employ good combustion practices per the manufacturerâ€™s operating manual	2.31	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	410	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII. Good combustion practices per the manufacturerâ€™s operating manual.	2.36	LB/H	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	460	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	2.6	LB/H	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	422	horsepower	Compliance with 40 CFR 60 Subpart IIII	3.44	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	422	horsepower	Compliance with 40 CFR 60 Subpart IIII	3.44	LB/HR	BACT
AK-0086	KENAI NITROGEN OPERATIONS	AGRIUM U.S. INC.	3/26/2021	2.7	MMBtu/hr	Good Combustion Practices and Limited Use	0.95	LB/MMBTU	BACT
SC-0182	FIBER INDUSTRIES LLC	FIBER INDUSTRIES LLC	10/31/2017	0		Use of Ultra Low Sulfur Diesel Fuel (15 ppm), good combustion, operation, and maintenance practices; compliance with NESHAP Subpart ZZZZ	200	OPERATING HR/YR	BACT
IL-0129	CPV THREE RIVERS ENERGY CENTER	CPV THREE RIVERS, LLC	7/30/2018	0			0		BACT
IN-0382	DUKE ENERGY INDIANA, INC.-CAYUGA GENERATING STATION	DUKE ENERGY INDIANA, INC.	2/14/2025	300	hp	good combustion practices	0		BACT
LA-0314	INDORAMA LAKE CHARLES FACILITY	INDORAMA VENTURES OLEFINS, LLC	8/3/2016	425	hp	complying with 40 CFR 63 subpart ZZZZ	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	460	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0345	DIRECT REDUCED IRON FACILITY	NUCOR STEEL LOUISIANA LLC	6/13/2019	0		Comply with requirements of 40 CFR 60 Subpart IIII	0		BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	0		Comply with 40 CFR 60 Subpart IIII and Good Combustion Practices	0		BACT
*LA-0381	EUEG-5 UNIT - GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2019	0		Comply with standards of 40 CFR 60 Subpart IIII	0		BACT
LA-0384	DIRECT REDUCED IRON FACILITY	NUCOR STEEL LOUISIANA, LLC	6/13/2019	0		Comply with 40 CFR 60 Subpart IIII	0		BACT

Table C-4b: RBLC CO Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
LA-0386	LASALLE BIOENERGY LLC	DRAX BIOMASS INC.	5/5/2021	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0397	WESTLAKE ETHYLENE PLANT	INDORAMA VENTURES OLEFINS, LLC	4/29/2022	0		Compliance with applicable requirements of 40 CFR 60 Subpart IIII	0		BACT
WI-0263	WISCONSIN POWER & LIGHT - NEENAH GENERATING STATION	ALLIANT ENERGY	2/15/2016	1.27	mmBtu/hr	Good combustion practices, use diesel fuel, and operate <500 hr/yr	0		BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	375	HP	Compliance with 40 CFR 60 Subpart IIII	3.5		BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	300	HP	Compliance with 40 CFR 60 Subpart IIII	3.5		BACT

Table C-4c: RBLC PM Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	282	bhp	Good combustion practice, low sulfur fuel	0.15	G	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	282	bhp	Good combustion practice, low sulfur fuel	0.17	G	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	282	bhp	Good combustion practices, low sulfur fuel	0.17	G	BACT
FL-0363	DANIA BEACH ENERGY CENTER	FLORIDA POWER AND LIGHT COMPANY	12/4/2017	0		Certified engine	0.2	G / KWH	BACT
FL-0356	OKEECHOBEE CLEAN ENERGY CENTER	FLORIDA POWER & LIGHT	3/9/2016	0		Use of clean fuel	0.2	G / KW-HR	BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	460	hp	Compliance demonstrated with vendor emission certification and adherence to vendor-specified maintenance recommendations.	0.087	G/BHP-H	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	1.66	MMBTU/H	Good combustion practices and meeting NSPS Subpart IIII requirements.	0.15	G/BHP-H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.15	G/BHP-H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.15	G/BHP-H	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	1.66	MMBTU/H	Good Combustion Practices and meeting NSPS Subpart IIII requirements	0.15	G/BHP-H	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.15	G/B-HP-H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Diesel particulate filter, Good Combustion Practices and meeting NSPS Subpart IIII requirements	0.15	G/B-HP-H	BACT
MI-0453	GENERAL MOTORS LLC ORION ASSEMBLY	GENERAL MOTORS LLC	9/27/2022	500	h/yr	Hours of Operation Restriction, Good Combustion Practices, Compliance with NSPS	0.15	G/B-HP-H	BACT
MI-0453	GENERAL MOTORS LLC ORION ASSEMBLY	GENERAL MOTORS LLC	9/27/2022	500	h/yr	Hours of Operation Restriction, Good Combustion Practices, Compliance with NSPS	0.15	G/B-HP-H	BACT
PA-0310	CPV FAIRVIEW ENERGY CENTER	CPV FAIRVIEW, LLC	9/2/2016	0			0.15	G/BHP-HR	BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	HR/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.15	G/HP HR	BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	HR/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.15	G/HP HR	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	282	HP	Operation limited to 500 hours/year, sulfur content of diesel fuel oil fired may not exceed 15 ppm, and shall be operated and maintained according to the manufacturerâ€™s recommendations.	0.15	G/HP-H	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	282	HP	Operation limited to 500 hours/year, sulfur content of diesel fuel oil fired may not exceed 15 ppm, and operate and maintain according to the manufacturerâ€™s recommendations.	0.15	G/HP-H	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	282	HP	Operation limited to 500 hours/year, sulfur content of diesel fuel oil fired may not exceed 15 ppm, and shall be operated and maintained according to the manufacturerâ€™s recommendations.	0.15	G/HP-H	BACT
IN-0382	DUKE ENERGY INDIANA, INC.-CAYUGA GENERATING STATION	DUKE ENERGY INDIANA, INC.	2/14/2025	300	hp	good combustion practices	0.15	G/HP-HR	BACT

Table C-4c: RBLC PM Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
KS-0030	MID-KANSAS ELECTRIC COMPANY, LLC - RUBART STATION	MID-KANSAS ELECTRIC COMPANY, LLC - RUBART STATION	3/31/2016	197	HP		0.15	G/HP-HR	BACT
KS-0030	MID-KANSAS ELECTRIC COMPANY, LLC - RUBART STATION	MID-KANSAS ELECTRIC COMPANY, LLC - RUBART STATION	3/31/2016	197	HP		0.15	G/HP-HR	BACT
KS-0030	MID-KANSAS ELECTRIC COMPANY, LLC - RUBART STATION	MID-KANSAS ELECTRIC COMPANY, LLC - RUBART STATION	3/31/2016	197	HP		0.15	G/HP-HR	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	355	hp	Compliance with 40 CFR 60 Subpart IIII standards, good combustion practices, and the use of ultra-low sulfur diesel fuel.	0.15	G/HP-HR	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	355	hp	Compliance with 40 CFR 60 Subpart IIII, good combustion practices, and the use of ultra-low sulfur diesel fuel.	0.15	G/HP-HR	BACT
NE-0064	NORFOLK CRUSH, LLC	NORFOLK CRUSH, LLC	11/21/2022	510	hp		0.15	G/HP-HR	BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	HR/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.15	G/HP-HR	BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	HR/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.15	G/HP-HR	BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	HR/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.15	G/HP-HR	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	282	HP	Good combustion practices and clean fuels: (a) P06 may not operate more than 500 hours per each 12 rolling calendar months. (b) The sulfur content of the diesel fuel oil fired may not exceed 15 ppm. (c) P06 shall be operated and maintained according to the manufacturerâ€™s recommendations.	0.15	G/HP-HR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	19.4	gph	Good combustion practices, ULSD, and limit operation to 500 hours per year per engine	0.19	G/HP-HR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	19.4	gph	Good combustion practices, ULSD, and limit operation to 500 hours per year per engine	0.19	G/HP-HR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	19.4	gph	Good combustion practices, ULSD, and limit operation to 500 hours per year per engine	0.19	G/HP-HR	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	500	H/YR	Good combustion practices.	0.22	G/HP-H	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Ultra Low Sulfur Diesel/Fuel (15 ppm max)	0.3	G/HP-H	N/A
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Ultra Low Sulfur Diesel/Fuel (15 ppm max)	0.3	G/HP-H	N/A
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	0.4	G/HP-HR	BACT
AR-0173	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	1/31/2022	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1	G/BHP-HR	BACT
AR-0173	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	1/31/2022	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1	G/BHP-HR	BACT
AR-0173	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	1/31/2022	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1	G/BHP-HR	BACT

Table C-4c: RBLC PM Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1	G/BHP-HR	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1	G/BHP-HR	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1	G/BHP-HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1	G/BHP-HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1	G/BHP-HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1	G/BHP-HR	BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	HR/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	15	G/HP/HR	BACT
TX-0846	MOTOR VEHICLE ASSEMBLY PLANT	TOYOTA MOTORS	9/23/2018	214	kW	Meets EPA Tier 4 requirements	0.02	G/KW	BACT
TX-0846	MOTOR VEHICLE ASSEMBLY PLANT	TOYOTA MOTORS	9/23/2018	214	kW	Meets EPA Tier 4 requirements	0.02	G/KW	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	282	hp	Good combustion practices including compliance with 40 CFR 60 Subpart IIII.	0.1	G/KW-H	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	282	hp	Use of ultra-low sulfur diesel as fuel.	0.1	G/KW-H	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	252	hp	Clean Fuel and Good Combustion Practices	0.19	G/KW-HR	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	252	hp	Clean Fuel and Good Combustion Practices	0.19	G/KW-HR	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	252	hp	Clean Fuel and Good Combustion Practices	0.19	G/KW-HR	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	399	BHP	State of the art combustion design	0.2	G/KW-H	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	500	bhp	Good combustion practices, use of ULSD	0.2	G/KW-H	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	275	HP	Certified to meet the standards in Table 4 of 40 CFR Part 60, Subpart IIII and good combustion practices	0.2	G/KW-H	BACT
FL-0367	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	7/27/2018	8700	gal/year	Operate and maintain the engine according to the manufacturer's written instructions	0.2	G/KW-HOUR	BACT
FL-0371	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	6/7/2021	2.46	MMBtu/hour		0.2	G/KW-HOUR	BACT
IL-0130	JACKSON ENERGY CENTER	JACKSON GENERATION, LLC	12/31/2018	420	horsepower		0.2	G/KW-HR	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	369	hp		0.2	G/KW-HR	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	369	hp		0.2	G/KW-HR	BACT
LA-0339	SHINTECH PLAQUEMINE PLANT 3	SHINTECH LOUISIANA LLC	1/19/2021	0		Compliance with 40 CFR 60 Subpart IIII	0.2	G/KW-HR	BACT
LA-0339	SHINTECH PLAQUEMINE PLANT 3	SHINTECH LOUISIANA LLC	1/19/2021	0		Compliance with 40 CFR 60 Subpart IIII	0.2	G/KW-HR	BACT

Table C-4c: RBLC PM Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	320	horsepower		0.2	GRAMS	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	460	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	0.02	LB/H	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	460	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	0.02	LB/H	BACT
*LA-0370	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER LLC	4/27/2020	1.1	MM BTU/hr	The use of low sulfur fuels and compliance with 40 CFR 60 Subpart IIII	0.04	LB/HR	BACT
*LA-0370	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER LLC	4/27/2020	1.1	MM BTU/hr	The use of low sulfur fuels and compliance with 40 CFR 60 Subpart IIII	0.04	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	237	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	0.06	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	237	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	0.06	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.07	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.07	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.07	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.07	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.07	LB/HR	BACT
*LA-0306	TOPCHEM POLLOCK, LLC	TOPCHEM POLLOCK, LLC	12/20/2016	225	horsepower	Meet NSPS Subpart IIII Limitations and Good Combustion Practices	0.09	LB/H	BACT
*LA-0306	TOPCHEM POLLOCK, LLC	TOPCHEM POLLOCK, LLC	12/20/2016	225	horsepower	Meet NSPS Subpart IIII Limitations and Good Combustion Practices	0.09	LB/H	BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	282	HP	Compliance with NESHAP 40 CFR 63 Subpart ZZZZ and NSPS 40 CFR 60 Subpart IIII, and good combustion practices (use of ultra-low sulfur diesel fuel).	0.09	LB/H	BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	282	HP	Compliance with NESHAP 40 CFR 63 Subpart ZZZZ and NSPS 40 CFR 60 Subpart IIII, and good combustion practices (use of ultra-low sulfur diesel fuel).	0.09	LB/H	BACT
NJ-0084	PSEG FOSSIL LLC SEWAREN GENERATING STATION	PSEG FOSSIL LLC	3/10/2016	100	H/YR	use of ULSD a clean burning fuel, and limited hours of operation	0.1	LB/H	BACT
NJ-0084	PSEG FOSSIL LLC SEWAREN GENERATING STATION	PSEG FOSSIL LLC	3/10/2016	100	H/YR	use of ULSD a clean burning fuel, and limited hours of operation	0.1	LB/H	BACT
NJ-0084	PSEG FOSSIL LLC SEWAREN GENERATING STATION	PSEG FOSSIL LLC	3/10/2016	100	H/YR	use of ULSD a clean burning fuel, and limited hours of operation	0.1	LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	311	HP	State-of-the-art combustion design	0.1	LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	311	HP	State-of-the-art combustion design	0.1	LB/H	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	300	HP	Ultra low sulfur diesel fuel	0.1	LB/H	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	300	HP	Ultra low sulfur diesel fuel	0.1	LB/H	BACT

Table C-4c: RBLC PM Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	300	HP	Ultra low sulfur diesel fuel	0.1	LB/H	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	300	HP	Ultra low sulfur diesel fuel	0.1	LB/H	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	250	HP	Comply with NSPS 40 CFR 60 Subpart IIII	0.1	LB/H	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	250	HP	Comply with NSPS 40 CFR 60 Subpart IIII	0.1	LB/H	BACT
NJ-0085	MIDDLESEX ENERGY CENTER, LLC	STONEGATE POWER, LLC	7/19/2016	100	H/YR	Use of Ultra Low Sulfur Diesel (ULSD) Oil a clean burning fuel and limited hours of operation	0.108	LB/H	BACT
NJ-0085	MIDDLESEX ENERGY CENTER, LLC	STONEGATE POWER, LLC	7/19/2016	100	H/YR	Use of Ultra Low Sulfur Diesel (ULSD) Oil a clean burning fuel and limited hours of operation	0.108	LB/H	BACT
NJ-0085	MIDDLESEX ENERGY CENTER, LLC	STONEGATE POWER, LLC	7/19/2016	100	H/YR	Use of ULSD a clean burning fuel and limited hours of operation	0.108	LB/H	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	320	HP	Good combustion practices (ULSD) and compliance with 40 CFR Part 60, Subpart IIII	0.11	LB/H	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	320	HP	Good combustion practices (ULSD) and compliance with 40 CFR Part 60, Subpart IIII	0.11	LB/H	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	320	HP	Good combustion practices (ULSD) and compliance with 40 CFR Part 60, Subpart IIII	0.11	LB/H	BACT
WI-0302	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	290	HP	Good combustion practices, use diesel fuel oil with sulfur content of no greater than 0.0015% by weight.	0.11	LB/H	BACT
WI-0302	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	290	HP	Good combustion practices, use diesel fuel oil with sulfur content of no greater than 0.0015% by weight	0.11	LB/H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	399	BHP	State of the art combustion design.	0.13	LB/H	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	399	BHP	State of the art combustion design.	0.13	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	410	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII. Good combustion practices per the manufacturerâ€™s operating manual.	0.13	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	410	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII. Good combustion practices per the manufacturerâ€™s operating manual.	0.13	LB/H	RACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	410	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII. Good combustion practices per the manufacturerâ€™s operating manual.	0.13	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	402	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII and employ good combustion practices per the manufacturerâ€™s operating manual	0.13	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	402	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII and employ good combustion practices per the manufacturerâ€™s operating manual	0.13	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	402	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII and employ good combustion practices per the manufacturerâ€™s operating manual	0.13	LB/H	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	422	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.2	LB/HR	BACT

Table C-4c: RBLC PM Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	422	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.2	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	422	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.2	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	422	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.2	LB/HR	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	1.66	MMBTU/H	Good combustion practices	0.57	LB/H	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	1.66	MMBTU/H	Good combustion practices	0.57	LB/H	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	1.66	MMBTU/H	Good combustion practices	0.57	LB/H	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	1.66	MMBTU/H	Good combustion practices	0.57	LB/H	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	275	HP	Certified to meet the standards in Table 4 of 40 CFR Part 60, Subpart IIII and good combustion practices	0.6	LB/H	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	275	HP	Certified to meet the standards in Table 4 of 40 CFR Part 60, Subpart IIII and good combustion practices	0.6	LB/H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.66	LB/H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.66	LB/H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.66	LB/H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.66	LB/H	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.66	LB/H	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Diesel particulate filter, good combustion practices and meeting NSPS Subpart IIII requirements.	0.66	LB/H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Diesel particulate filter, Good Combustion Practices and meeting NSPS Subpart IIII requirements	0.66	LB/H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Diesel particulate filter, Good Combustion Practices and meeting NSPS Subpart IIII requirements	0.66	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.76	LB/HR	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	500	bhp	Good combustion practices, use of ULSD	1.1	LB/H	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	500	bhp	Good combustion practices, use of ULSD	1.1	LB/H	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	500	H/YR	Good combustion practices.	0.09	LB/MMBTU	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	500	H/YR	Good combustion practices.	0.09	LB/MMBTU	BACT
AK-0086	KENAI NITROGEN OPERATIONS	AGRIUM U.S. INC.	3/26/2021	2.7	MMBtu/hr	Good Combustion Practices and Limited Use	0.31	LB/MMBTU	BACT
AK-0086	KENAI NITROGEN OPERATIONS	AGRIUM U.S. INC.	3/26/2021	2.7	MMBtu/hr	Good Combustion Practices and Limited Use	0.31	LB/MMBTU	BACT

Table C-4c: RBLC PM Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
AK-0086	KENAI NITROGEN OPERATIONS	AGRIUM U.S. INC.	3/26/2021	2.7	MMBtu/hr	Good Combustion Practices and Limited Use	0.31	LB/MMBTU	BACT
SC-0182	FIBER INDUSTRIES LLC	FIBER INDUSTRIES LLC	10/31/2017	0		Use of Ultra Low Sulfur Diesel Fuel (15 ppm), good combustion, operation, and maintenance practices; compliance with NESHAP Subpart ZZZZ	200	OPERATING HR/YR	BACT
SC-0182	FIBER INDUSTRIES LLC	FIBER INDUSTRIES LLC	10/31/2017	0		Use of Ultra Low Sulfur Diesel Fuel (15 ppm), good combustion, operation, and maintenance practices; compliance with NESHAP Subpart ZZZZ	200	OPERATING HR/YR	BACT
SC-0182	FIBER INDUSTRIES LLC	FIBER INDUSTRIES LLC	10/31/2017	0		Use of Ultra Low Sulfur Diesel Fuel (15 ppm), good combustion, operation, and maintenance practices; compliance with NESHAP Subpart ZZZZ	200	OPERATING HR/YR	BACT
IL-0129	CPV THREE RIVERS ENERGY CENTER	CPV THREE RIVERS, LLC	7/30/2018	0			0		BACT
LA-0314	INDORAMA LAKE CHARLES FACILITY	INDORAMA VENTURES OLEFINS, LLC	8/3/2016	425	hp	complying with 40 CFR 63 subpart ZZZZ	0		BACT
LA-0314	INDORAMA LAKE CHARLES FACILITY	INDORAMA VENTURES OLEFINS, LLC	8/3/2016	425	hp	complying with 40 CFR 63 subpart ZZZZ	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	460	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	460	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0345	DIRECT REDUCED IRON FACILITY	NUCOR STEEL LOUISIANA LLC	6/13/2019	0		Comply with requirements of 40 CFR 60 Subpart IIII	0		BACT
LA-0345	DIRECT REDUCED IRON FACILITY	NUCOR STEEL LOUISIANA LLC	6/13/2019	0		Comply with requirements of 40 CFR 60 Subpart IIII	0		BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	0		Comply with 40 CFR 60 Subpart IIII and Good Combustion Practices	0		BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	0		Comply with 40 CFR 60 Subpart IIII and Good Combustion Practices	0		BACT
*LA-0381	EUEG-5 UNIT - GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2019	0		Comply with standards of 40 CFR 60 Subpart IIII, limit operation to <= 100 hrs/yr	0		BACT
*LA-0381	EUEG-5 UNIT - GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2019	0		Comply with standards of 40 CFR 60 Subpart IIII Limit operation to <= 100 hrs/yr	0		BACT
LA-0384	DIRECT REDUCED IRON FACILITY	NUCOR STEEL LOUISIANA, LLC	6/13/2019	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0384	DIRECT REDUCED IRON FACILITY	NUCOR STEEL LOUISIANA, LLC	6/13/2019	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0386	LASALLE BIOENERGY LLC	DRAX BIOMASS INC.	5/5/2021	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0386	LASALLE BIOENERGY LLC	DRAX BIOMASS INC.	5/5/2021	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0397	WESTLAKE ETHYLENE PLANT	INDORAMA VENTURES OLEFINS, LLC	4/29/2022	0		Compliance with applicable requirements of 40 CFR 60 Subpart IIII	0		BACT
LA-0397	WESTLAKE ETHYLENE PLANT	INDORAMA VENTURES OLEFINS, LLC	4/29/2022	0		Compliance with applicable requirements of 40 CFR 60 Subpart IIII	0		BACT
WI-0263	WISCONSIN POWER & LIGHT - NEENAH GENERATING STATION	ALLIANT ENERGY	2/15/2016	1.27	mmBtu/hr	Good combustion practices, use diesel fuel with sulfur content < 15 ppm, and operate <500 hr/yr	0		BACT
WI-0263	WISCONSIN POWER & LIGHT - NEENAH GENERATING STATION	ALLIANT ENERGY	2/15/2016	1.27	mmBtu/hr	Good combustion practices, use diesel fuel with sulfur content < 15 ppm, and operate <500 hr/yr	0		BACT

Table C-4c: RBLC PM Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
WI-0263	WISCONSIN POWER & LIGHT - NEENAH GENERATING STATION	ALLIANT ENERGY	2/15/2016	1.27	mmBtu/hr	Good combustion practices, use diesel fuel with sulfur content < 15 ppm, and operate <500 hr/yr	0		BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	375	HP	Compliance with 40 CFR 60 Subpart IIII.	0.2		BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	375	HP	Compliance with 40 CFR 60 Subpart IIII	0.2		BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	300	HP	Compliance with 40 CFR 60 Subpart IIII	0.2		BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	300	HP	Compliance with 40 CFR 60 Subpart III	0.2		BACT

Table C-4d: RBLC SO2 Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	300	HP	Ultra low sulfur diesel fuel	3.2	X10-3 LB/H	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	300	HP	Ultra low sulfur diesel fuel	3.2	X10-3 LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	311	HP	Ultra low sulfur diesel fuel	0.004	LB/H	BACT
WI-0302	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	290	HP	Only use diesel fuel oil with a sulfur content of no greater than 0.0015% by weight.	0.36	LB/H	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	282	hp		0.0078	LB/MM BTU	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	410	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII. Good combustion practices per the manufacturerâ€™s operating manual.	0.0015	LB/MMBTU	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	320	HP	ultra-low sulfur diesel (ULSD) fuel with a sulfur content of less than 15 ppm (0.0015 percent by weight)	0.0015	LB/MMBTU	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Ultra Low Sulfur Diesel/Fuel (15 ppm max)	0.0015	LB/MMBTU	N/A
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	HR/YR	: good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.0015	LB/MMBTU	BACT
FL-0356	OKEECHOBEE CLEAN ENERGY CENTER	FLORIDA POWER & LIGHT	3/9/2016	0		Use of ULSD	0.0015	% S IN ULSD	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	1.66	MMBTU/H	Good combustion practices and meeting NSPS Subpart IIII requirements.	15	PPM	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	15	PPM	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	15	PPM	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	1.66	MMBTU/H	Good Combustion Practices and meeting NSPS Subpart IIII requirements	15	PPM	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Good combustion practices and meeting NSPS Subpart IIII requirements.	15	PPM	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Good Combustion Practices and meeting NSPS Subpart IIII requirements	15	PPM	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	15	PPM MAX FUEL CONTENT	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	15	PPM OR LESS S IN FUE	BACT
FL-0363	DANIA BEACH ENERGY CENTER	FLORIDA POWER AND LIGHT COMPANY	12/4/2017	0		Clean fuel	15	PPM S IN FUEL	BACT
FL-0367	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	7/27/2018	8700	gal/year	Clean fuel	15	PPM S IN FUEL	BACT
FL-0371	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	6/7/2021	2.46	MMBtu/hour		15	PPM S IN FUEL	BACT
AR-0173	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	1/31/2022	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	15	PPM SULFUR IN FUEL	BACT

Table C-4d: RBLC SO2 Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	19.4	gph	Good combustion practices, ULSD, and limit operation to 500 hours per year.	15	PPMW SULFUR IN FUEL	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	320	horsepower	Use of ultra-low sulfur diesel, with a sulfur content < 15 ppm sulfur.	0		BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	0		Comply with 40 CFR 60 Subpart IIII and Good Combustion Practices	0		BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	HR/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0		BACT
WI-0263	WISCONSIN POWER & LIGHT - NEENAH GENERATING STATION	ALLIANT ENERGY	2/15/2016	1.27	mmBtu/hr	Use diesel fuel with sulfur content < 15 ppm	0		BACT

Table C-4e: RBLC H2SO4 Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	460	hp	Ultra low sulfur diesel with maximum sulfur content 0.0015 percent.	0.0001	LB/MMBTU	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Ultra Low Sulfur Diesel/Fuel (15 ppm max)	0.0001	LB/MMBTU	N/A
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	HR/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.0001	LB/MMBTU	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	410	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII. Good combustion practices per the manufacturerâ€™s operating manual.	0.0015	LB/MMBTU	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	399	BHP	Good combustion practices, low sulfur fuel.	15	PPM	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	320	HP	ultra-low sulfur diesel (ULSD) fuel with a sulfur content of less than 15 ppm (0.0015 percent by weight)	7.3	X10-4 LB/MMBTU	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	300	HP	Ultra low sulfur diesel fuel	6.5	X10-5 LB/H	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	300	HP	Ultra low sulfur diesel fuel	6.5	X10-5 LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	311	HP	Ultra low sulfur diesel fuel	6.7	X10-5 LB/H	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	320	horsepower	Use of ultra-low sulfur diesel, with a sulfur content < 15 ppm sulfur.	0		BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	HR/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0		BACT
WI-0263	WISCONSIN POWER & LIGHT - NEENAH GENERATING STATION	ALLIANT ENERGY	2/15/2016	1.27	mmBtu/hr	Use diesel fuel with sulfur content < 15 ppm	0		BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	282	HP	Operation limited to 500 hours/year, operate and maintain according to the manufacturerâ€™s recommendations, and sulfur content of the diesel fuel oil fired may not exceed 15 ppm.	0		BACT
WI-0302	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	290	HP	Only use diesel fuel oil with a sulfur content of no greater than 0.015% by weight	0		BACT

Table C-4f: RBLC VOC Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	282	bhp	Good combustion practices, low sulfur fuel	0.15	G	BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	460	hp	Compliance demonstrated with vendor emission certification and adherence to vendor-specified maintenance recommendations.	0.1	G/BHP-H	LAER
MI-0443	MACK AVENUE ASSEMBLY PLANT	FCA US LLC	4/26/2019	500	h/yr		0.1	G/B-HP-H	LAER
MI-0443	MACK AVENUE ASSEMBLY PLANT	FCA US LLC	4/26/2019	500	h/yr		0.1	G/B-HP-H	LAER
MI-0443	MACK AVENUE ASSEMBLY PLANT	FCA US LLC	4/26/2019	500	h/yr		0.1	G/B-HP-H	LAER
*MI-0446	MACK AVENUE ASSEMBLY PLANT	FCA US LLC	10/30/2020	500	h/yr		0.1	G/B-HP-H	LAER
*MI-0446	MACK AVENUE ASSEMBLY PLANT	FCA US LLC	10/30/2020	500	h/yr		0.1	G/B-HP-H	LAER
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	HR/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0.11	G/HP-HR	BACT
MI-0453	GENERAL MOTORS LLC ORION ASSEMBLY	GENERAL MOTORS LLC	9/27/2022	500	h/yr	Hours of Operation Restriction, Good Combustion Practices, Compliance with NSPS	0.19	G/B-HP-H	LAER
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	19.4	gph	Good combustion practices, ULSD, and limit operation to 500 hours per year.	0.19	G/HP-HR	BACT
NE-0064	NORFOLK CRUSH, LLC	NORFOLK CRUSH, LLC	11/21/2022	510	hp		0.62	G/HP-HR	BACT
OK-0177	CUSHING SOUTH TANK FARM	TALLGRASS TERMLS LLC	1/4/2018	0		Good combustion practices, certified to meet EPA Tier III engine standards. Limited to operate no more than 500 hr/yr.	1	GM/HP-HR	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	282	HP	Operation limited to 500 hours/year and operate and maintain according to the manufacturerâ€™s recommendations.	1.1	G/HP-H	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	282	HP	Good combustion practices: (a) P06 may not operate more than 500 hours per each 12 rolling calendar months. (b) P06 shall be operated and maintained according to the manufacturerâ€™s recommendations. (c) VOC emission from P06 may not exceed 1.1 g/hp-hr.	1.1	G/HP-HR	BACT
AR-0173	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	1/31/2022	0		Good Operating Practices, limited hours of operation, Compliance with NSPS Subpart IIII	1.12	G/BHP-HR	BACT
KS-0030	MID-KANSAS ELECTRIC COMPANY, LLC - RUBART STATION	MID-KANSAS ELECTRIC COMPANY, LLC - RUBART STATION	3/31/2016	197	HP		1.14	G/HP-HR	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Good Combustion Practices/Maintenance	3	G/HP-H	N/A
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	355	hp	Compliance with 40 CFR 60 Subpart IIII, good combustion practices, and the use of ultra-low sulfur diesel fuel.	3	G/HP-HR	BACT
OK-0175	WILDHORSE TERMINAL	WILDHORSE TERMINAL LLC	6/29/2017	0		Good combustion practices, certified to meet EPA Tier 3 engine standards. Gen-1, FP-1, and FP-2 shall be limited to operate no more than 500 hr/yr.	3	GM/HP-HR	BACT
OK-0181	WILDHORSE TERMINAL	KEYERA ENERGY INC	9/11/2019	0		Good Combustion Practices. Certified to meet EPA Tier 3 engine standards. Gen-1 and FP-1 shall be limited to operate not more than 500 hours per year. SP-1 shall be limited to operate not more than 876 hours per year.	3	GM/HP-HR	BACT

Table C-4f: RBLC VOC Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	282	hp	Good combustion practices including compliance with 40 CFR 60 Subpart IIII.	0.11	G/KW-H	BACT
TX-0846	MOTOR VEHICLE ASSEMBLY PLANT	TOYOTA MOTORS	9/23/2018	214	kW	Meets EPA Tier 4 requirements	0.19	G/KW	BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	375	HP	Good combustion practices and NSPS Subpart IIII	4	G/KW-H	BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	300	HP	Good combustion practices and NSPS Subpart IIII	4	G/KW-H	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	500	bhp	Good combustion practices	4	G/KW-H	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	369	hp		4	G/KW-HR	BACT
IN-0382	DUKE ENERGY INDIANA, INC.- CAYUGA GENERATING STATION	DUKE ENERGY INDIANA, INC.	2/14/2025	300	hp	good combustion practices	4	G/KW-HR	BACT
LA-0339	SHINTECH PLAQUEMINE PLANT 3	SHINTECH LOUISIANA LLC	1/19/2021	0		Compliance with 40 CFR 60 Subpart IIII	4	G/KW-HR	BACT
WI-0292	GREEN BAY PACKAGING INC. â€“MILL DIVISION	GREEN BAY PACKAGING INC. â€“MILL DIVISION	4/1/2019	0		Hours of Operation	200	HOURS	BACT
SC-0182	FIBER INDUSTRIES LLC	FIBER INDUSTRIES LLC	10/31/2017	0		Use of Ultra Low Sulfur Diesel Fuel (15 ppm), good combustion, operation, and maintenance practices; compliance with NESHAP Subpart ZZZZ	200	OPERATING HR/YR	BACT
*AL-0336	TWO RIVERS LUMBER- ALEXANDER CITY	TWO RIVERS LUMBER COMPANY, LLC	7/2/2024	321	bHp		500	HOURS/YEAR	BACT
LA-0366	HOLDEN WOOD PRODUCTS MILL	WEYERHAEUSER NR COMPANY	2/3/2021	0		Good Combustion Practices and Compliance with NSPS 40 CFR 60 Subpart IIII	804.6	HP	BACT
NJ-0084	PSEG FOSSIL LLC SEWAREN GENERATING STATION	PSEG FOSSIL LLC	3/10/2016	100	H/YR	use of ULSD a clean burning fuel, and limited hours of operation	0.1	LB/H	LAER
NJ-0085	MIDDLESEX ENERGY CENTER, LLC	STONEGATE POWER, LLC	7/19/2016	100	H/YR	Use of Ultra Low Sulfur Diesel (ULSD) Oil a clean burning fuel and limited hours of operation	0.117	LB/H	LAER
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	399	BHP	State of the art combustion design.	0.13	LB/H	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	460	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	0.14	LB/H	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	300	HP	State-of-the-art combustion design	0.24	LB/H	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	300	HP	State-of-the-art combustion design	0.24	LB/H	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	311	HP	State-of-the-art combustion design	0.25	LB/H	BACT
WI-0302	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	2/28/2020	290	HP	Good combustion practices	0.26	LB/H	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.33	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.33	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	200	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.33	LB/HR	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	500	H/YR	Good combustion practices	0.47	LB/H	BACT

Table C-4f: RBLC VOC Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	237	horsepower	Compliance with the requirements of 40 CFR 60 Subpart IIII	0.61	LB/HR	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	1.66	MMBTU/H	Good combustion practices	0.64	LB/H	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	275	HP	Certified to meet the standards in Table 4 of 40 CFR Part 60, Subpart IIII and good combustion practices	0.7	LB/H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Good combustion practices.	0.75	LB/H	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Good combustion practices	0.75	LB/H	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Good combustion practices.	0.75	LB/H	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Good Combustion Practices	0.75	LB/H	BACT
*AL-0337	KRONOSPAN, LLC	KRONOSPAN, LLC	9/21/2022	0		Catalyst	0.79	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	422	horsepower	Compliance with 40 CFR 60 Subpart IIII	1.47	LB/HR	BACT
*LA-0401	KOCH METHANOL (KME) FACILITY	KOCH METHANOL ST. JAMES, LLC	12/20/2023	422	horsepower	Compliance with 40 CFR 60 Subpart IIII	1.47	LB/HR	BACT
LA-0390	DERIDDER SAWMILL	CSP DERIDDER, LLC	5/10/2022	500	horsepower	Good combustion practices and maintenance and compliance with applicable 40 CFR 60 Subpart JJJJ limitation for VOC.	1.85	LB/HR	BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	282	HP	Good combustion practices	1.87	LB/H	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	320	HP	Good combustion practices (ULSD) and compliance with 40 CFR Part 60, Subpart IIII	2.12	LB/H	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	402	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII and employ good combustion practices per the manufacturerâ€™s operating manual	2.64	LB/H	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	410	HP	Certified to the meet the emissions standards in Table 4 of 40 CFR Part 60, Subpart IIII. Good combustion practices per the manufacturerâ€™s operating manual.	2.7	LB/H	BACT
AK-0086	KENAI NITROGEN OPERATIONS	AGRIUM U.S. INC.	3/26/2021	2.7	MMBtu/hr	Good Combustion Practices and Limited Use	0.36	LB/MMBTU	BACT
*LA-0387	TAYLOR SAWMILL	HUNT FOREST PRODUCTS, LLC	4/12/2022	274	horsepower	Compliance with 40 CFR 60 Subpart IIII	0.02	TPY	BACT
*AL-0346	BUCKS SAWMILL	NEW SOUTH LUMBER, LLC	5/30/2024	305	BHP		0		BACT
LA-0314	INDORAMA LAKE CHARLES FACILITY	INDORAMA VENTURES OLEFINS, LLC	8/3/2016	425	hp	complying with 40 CFR 63 subpart ZZZZ	0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	460	hp	Complying with 40 CFR 60 Subpart IIII	0		BACT
LA-0349	DRIFTWOOD LNG FACILITY	DRIFTWOOD LNG LLC	7/10/2018	0		Comply with 40 CFR 60 Subpart IIII and Good Combustion Practices	0		BACT
LA-0386	LASALLE BIOENERGY LLC	DRAX BIOMASS INC.	5/5/2021	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0397	WESTLAKE ETHYLENE PLANT	INDORAMA VENTURES OLEFINS, LLC	4/29/2022	0		Compliance with applicable requirements of 40 CFR 60 Subpart IIII	0		BACT

Table C-4f: RBLC VOC Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	HR/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	0		BACT
WI-0263	WISCONSIN POWER & LIGHT - NEENAH GENERATING STATION	ALLIANT ENERGY	2/15/2016	1.27	mmBtu/hr	Good combustion practices, use diesel fuel, and operate <500 hr/yr	0		BACT

Table C-4g: RBLC GHG Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
WI-0292	GREEN BAY PACKAGING INC. â€“MILL DIVISION	GREEN BAY PACKAGING INC. â€“MILL DIVISION	4/1/2019	0		Hours of Operation	200	HOURS	BACT
SC-0182	FIBER INDUSTRIES LLC	FIBER INDUSTRIES LLC	10/31/2017	0		Use of Ultra Low Sulfur Diesel Fuel (15 ppm), good combustion, operation, and maintenance practices; compliance with NESHAP Subpart ZZZZ	200	OPERATING HR/YR	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	282	HP	Selection of the most efficient reciprocating internal combustion engine and minimizing the hours of operation limits greenhouse gas emissions. - EPA Tier 3 certified reciprocating internal combustion engine and - limiting its hours of operation to 500 hours per each 12 consecutive calendar months. - Part of ensuring efficient engine operation is properly maintaining the engine and the combustion of clean fuels. In this case, clean fuel is determined to be ultra-low sulfur diesel fuel (less than 15 ppm sulfur content).	500	HOURS	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	355	hp	Compliance with 40 CFR 60 Subpart IIII, good combustion practices, and the use of ultra-low sulfur diesel fuel.	74.21	KG/MM BTU	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	19.4	gph	Good combustion practices and limit operation to 500 hours per year per engine	163.6	LB/MMBTU	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	250	HP	Equipment design and maintenance requirements	163.6	LB/MMBTU	BACT
AK-0086	KENAI NITROGEN OPERATIONS	AGRIUM U.S. INC.	3/26/2021	2.7	MMBtu/hr	Good Combustion Practices and Limited Use	164	LB/MMBTU	BACT
AR-0173	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	1/31/2022	0		Good Operating Practices	164	LB/MMBTU	BACT
AR-0180	HYBAR LLC	HYBAR LLC	4/28/2023	0		Good combustion practices	164	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Good Combustion Practices	164	LB/MMBTU	BACT
*LA-0370	WASHINGTON PARISH ENERGY CENTER	WASHINGTON PARISH ENERGY CENTER LLC	4/27/2020	1.1	MM BTU/hr	Good combustion practices in order to comply with 40 CFR 60 Subpart IIII	9	TPY	BACT
*LA-0306	TOPCHEM POLLOCK, LLC	TOPCHEM POLLOCK, LLC	12/20/2016	225	horsepower	Good Combustion Practices	13	T/YR	BACT
*LA-0306	TOPCHEM POLLOCK, LLC	TOPCHEM POLLOCK, LLC	12/20/2016	225	horsepower	Good Combustion Practices	13	T/YR	BACT
TX-0824	JACKSON COUNTY GENERATING FACILITY	SOUTHERN POWER	6/30/2017	160	HP	Good operating and maintenance practices, efficient design, and low annual capacity	13	T/YR	BACT
MI-0423	INDECK NILES, LLC	INDECK NILES, LLC	1/4/2017	1.66	MMBTU/H	Good combustion practices	13.58	T/YR	BACT
*MI-0445	INDECK NILES, LLC	INDECK NILES, LLC	11/26/2019	1.66	MMBTU/H	Good combustion practices	13.58	T/YR	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	282	hp	Good combustion practices including compliance with 40 CFR 60 Subpart IIII.	15	T/YR	BACT
OH-0377	HARRISON POWER	HARRISON POWER	4/19/2018	320	HP	Efficient design and proper maintenance and operation	18.67	T/YR	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	402	HP	good operating practices (proper maintenance and operation)	23	T/YR	BACT

Table C-4g: RBLC GHG Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	369	hp		25	TONS/YEAR	BACT
MI-0459	LANSING BOARD OF WATER AND LIGHT--DELTA ENERGY PARK	LANSING BOARD OF WATER AND LIGHT	6/27/2024	500	bhp	Good combustion practices, use of current energy efficiency measures.	27	T/YR	BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	375	HP	Good Combustion Practices	28	T/YR	BACT
LA-0328	PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/2/2018	300	HP	Good Combustion Practices	28	T/YR	BACT
OH-0374	GUERNSEY POWER STATION LLC	GUERNSEY POWER STATION LLC	10/23/2017	410	HP	good operating practices (proper maintenance and operation)	29	T/YR	BACT
MI-0424	HOLLAND BOARD OF PUBLIC WORKS - EAST 5TH STREET	HOLLAND BOARD OF PUBLIC WORKS	12/5/2016	500	H/YR	Good combustion practices.	55.6	T/YR	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Good combustion practices.	85.6	T/YR	BACT
MI-0433	MEC NORTH, LLC AND MEC SOUTH LLC	MARSHALL ENERGY CENTER LLC	6/29/2018	300	HP	Good combustion practices.	85.6	T/YR	BACT
MI-0451	MEC NORTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Good combustion practices	85.6	T/YR	BACT
MI-0452	MEC SOUTH, LLC	MARSHALL ENERGY CENTER, LLC	6/23/2022	300	HP	Good combustion practices.	85.6	T/YR	BACT
MI-0435	BELLE RIVER COMBINED CYCLE POWER PLANT	DTE ELECTRIC COMPANY	7/16/2018	399	BHP	Energy efficient design	86	T/YR	BACT
OH-0370	TRUMBULL ENERGY CENTER	TRUMBULL ENERGY CENTER	9/7/2017	300	HP	Efficient design	87	T/YR	BACT
OH-0372	OREGON ENERGY CENTER	OREGON ENERGY CENTER	9/27/2017	300	HP	State-of-the-art combustion design	87	T/YR	BACT
OH-0367	SOUTH FIELD ENERGY LLC	SOUTH FIELD ENERGY LLC	9/23/2016	311	HP	Efficient design	90	T/YR	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	320	horsepower		92	TONS/YEAR	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Good Combustion Practices/Maintenance	104	T/YR	N/A
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	500	HR/YR	good combustion practices, high efficiency design, and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	106	T/YR	BACT
NY-0103	CRICKET VALLEY ENERGY CENTER	CRICKET VALLEY ENERGY CENTER LLC	2/3/2016	460	hp	Good combustion practice and efficient engine design.	115	TPY	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	460	HP	good combustion control and operating practices and engines designed to meet the stands of 40 CFR Part 60, Subpart IIII	123	T/YR	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	252	hp	Good Combustion Practices	216	TPY (COMBINED)	BACT
IL-0130	JACKSON ENERGY CENTER	JACKSON GENERATION, LLC	12/31/2018	420	horsepower		241	TONS/YEAR	BACT
MI-0453	GENERAL MOTORS LLC ORION ASSEMBLY	GENERAL MOTORS LLC	9/27/2022	500	h/yr	Hours of Operation Restriction, Good Combustion Practices, Compliance with NSPS	656.2	T/YR	BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	500	HR/YR	good combustion practices and the use of ultra low sulfur diesel (S15 ULSD) fuel oil with a maximum sulfur content of 15 ppmw.	1040	T/YR	BACT

Table C-4g: RBLC GHG Results for Emergency Diesel Fire Pumps

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*AL-0346	BUCKS SAWMILL	NEW SOUTH LUMBER, LLC	5/30/2024	305	BHP		0		BACT
IL-0129	CPV THREE RIVERS ENERGY CENTER	CPV THREE RIVERS, LLC	7/30/2018	0			0		BACT
IN-0382	DUKE ENERGY INDIANA, INC.- CAYUGA GENERATING STATION	DUKE ENERGY INDIANA, INC.	2/14/2025	300	hp	good combustion practices	0		BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	282	bhp		0		BACT
LA-0313	ST. CHARLES POWER STATION	ENTERGY LOUISIANA, LLC	8/31/2016	282	HP	Good combustion practices	0		BACT
LA-0314	INDORAMA LAKE CHARLES FACILITY	INDORAMA VENTURES OLEFINS, LLC	8/3/2016	425	hp		0		BACT
LA-0316	CAMERON LNG FACILITY	CAMERON LNG LLC	2/17/2017	460	hp	good combustion practices	0		BACT
LA-0339	SHINTECH PLAQUEMINE PLANT 3	SHINTECH LOUISIANA LLC	1/19/2021	0		Energy efficiency measures to minimize the amount of fuel used	0		BACT
*LA-0381	EUEG-5 UNIT - GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2019	0		Good engine design Good work practices	0		BACT
LA-0384	DIRECT REDUCED IRON FACILITY	NUCOR STEEL LOUISIANA, LLC	6/13/2019	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
LA-0386	LASALLE BIOENERGY LLC	DRAX BIOMASS INC.	5/5/2021	0		Comply with 40 CFR 60 Subpart IIII	0		BACT
TX-0846	MOTOR VEHICLE ASSEMBLY PLANT	TOYOTA MOTORS	9/23/2018	214	kW	Meets EPA Tier 4 requirements . Fuels with a low carbon density, regular equipment maintenance, the use of efficient equipment and operation limited to less than 100 hr/yr	0		BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	282	HP	Be certified by manufacturer to EPAâ€™s criteria for Tier 3 reciprocating internal combustion engines and to the 40 CFR 60, Subpart IIII emission limitations, operation limited to 500 hours/year, and operate and maintain according to the manufacturerâ€™s recommendations.	0		BACT

Table C-5: RBLC PM Results for Paved Roads

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IL-0129	CPV THREE RIVERS ENERGY CENTER	CPV THREE RIVERS, LLC	7/30/2018	0		Paving is required for roads used by trucks transporting bulk materials.	10	% OPACITY	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	219000	vehicles/yr	Use of Hard pavement	0.04	LB/HR	BACT
LA-0379	SHINTECH PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/4/2021	0		Paving plant road as much as practicable.	0.08	LB/HR	BACT
LA-0379	SHINTECH PLAQUEMINES PLANT 1	SHINTECH LOUISIANA, LLC	5/4/2021	0		Paving plant road as much as practicable.	0.08	LB/HR	BACT
*LA-0403	SHINTECH PLAQUEMINE PLANT 4	SHINTCH LOUISIANA, LLC	12/16/2024	219000	vehicles/yr	Use of Hard pavement	0.15	LB/HR	BACT
AR-0172	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	9/1/2021	0		Water Sprays, sweeping,	0.5	LB/HR	BACT
AR-0172	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	9/1/2021	0		Water Sprays, sweeping,	3.9	LB/HR	BACT
AR-0172	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	9/1/2021	0		Water Sprays, sweeping,	15.2	LB/HR	BACT
SC-0181	RESOLUTE FP US INC. - CATAWBA LUMBER MILL	RESOLUTE FP US INC.	11/3/2017	0		Good housekeeping practices.	0.01	LB/VMT	BACT
SC-0181	RESOLUTE FP US INC. - CATAWBA LUMBER MILL	RESOLUTE FP US INC.	11/3/2017	0		Good housekeeping practices.	0.03	LB/VMT	BACT
SC-0181	RESOLUTE FP US INC. - CATAWBA LUMBER MILL	RESOLUTE FP US INC.	11/3/2017	0		Good housekeeping practices.	0.13	LB/VMT	BACT
IN-0317	RIVERVIEW ENERGY CORPORATION	RIVERVIEW ENERGY CORPORATION	6/11/2019	0		Fugitive dust control plan	1	MIN	BACT
IN-0317	RIVERVIEW ENERGY CORPORATION	RIVERVIEW ENERGY CORPORATION	6/11/2019	0		Fugitive dust control plan	1	MIN	BACT
IN-0317	RIVERVIEW ENERGY CORPORATION	RIVERVIEW ENERGY CORPORATION	6/11/2019	0		Fugitive dust control plan	1	MIN	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	0		Paving of all roadways and Fugitive Dust Control Plan, including mitigative measures (sweeping, water sprays, prompt cleanups)	10	PERCENT	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	0		Paving of all roadways and Fugitive Dust Control Plan, including mitigative measures (sweeping, water sprays, prompt cleanups)	10	PERCENT	BACT
IL-0130	JACKSON ENERGY CENTER	JACKSON GENERATION, LLC	12/31/2018	0			10	PERCENT OPACITY	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	0			10	PERCENT OPACITY	BACT
OH-0380	AMG VANADIUM LLC	AMG VANADIUM LLC	8/7/2019	31689	MI/YR	Pave all in-plant haul roads and parking areas. Implement best management practices including posting and limiting vehicle speeds to 15 miles per hour in production areas. Utilize a vacuum sweeper as needed based on the daily inspections	0.01	T/YR	BACT
*IL-0137	MARQUIS CARBON CAPTURE LLC	MARQUIS CARBON CAPTURE LLC	6/23/2025	0		10% opacity fugitive dust; all roads and parking areas shall be paved; dust control measures (weekly sweeping, prompt cleanup of spills)	0.01	TON/YEAR	BACT

Table C-5: RBLC PM Results for Paved Roads

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*IL-0137	MARQUIS CARBON CAPTURE LLC	MARQUIS CARBON CAPTURE LLC	6/23/2025	0		10% opacity fugitive dust; all roads and parking areas shall be paved; dust control measures (weekly sweeping, prompt cleanup of spills)	0.01	TON/YEAR	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	0		Pave all roadways and parking areas and implement best management practices, including limiting vehicle speeds and water spraying or sweeping as needed based on daily inspections.	0.04	T/YR	BACT
OH-0380	AMG VANADIUM LLC	AMG VANADIUM LLC	8/7/2019	31689	MI/YR	Pave all in-plant haul roads and parking areas. Implement best management practices including posting and limiting vehicle speeds to 15 miles per hour in production areas. Utilize a vacuum sweeper as needed based on the daily inspections	0.06	T/YR	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	182865	MI/YR	i.Pave all in-plant haul roads and parking areas; ii.Implement best management practices including posting and limiting vehicle speeds to 20 miles per hour and water spraying or sweeping as needed based on the daily inspections conducted	0.09	T/YR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Development and Implementation of Fugitive Dust Control Plan	0.1	TPY	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	0		water flushing and sweeping	0.15	T/YR	BACT
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	0		Pave all roadways and parking areas and implement best management practices, including limiting vehicle speeds and water spraying or sweeping as needed based on daily inspections.	0.16	T/YR	BACT
AR-0173	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	1/31/2022	0		Development and Implementation of Fugitive Dust Control Plan	0.2	TPY	BACT
OH-0391	VALENCIA PROJECT LLC	VALENCIA PROJECT LLC	10/27/2023	0		Best available control measures â€“ pave all roadways, speed reduction, good housekeeping practices, and/or watering	0.23	T/YR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Development and Implementation of Fugitive Dust Control Plan	0.3	TPY	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	182865	MI/YR	i.Pave all in-plant haul roads and parking areas; ii.Implement best management practices including posting and limiting vehicle speeds to 20 miles per hour and water spraying or sweeping as needed based on the daily inspections conducted	0.38	T/YR	BACT
AR-0173	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	1/31/2022	0		Development and Implementation of Fugitive Dust Control Plan	0.6	TPY	BACT
OH-0376	IRONUNITS LLC - TOLEDO HBI	IRONUNITS LLC - TOLEDO HBI	2/9/2018	0		water flushing and sweeping	0.63	T/YR	BACT
OH-0392	NUCOR STEEL MARION, INC.	NUCOR STEEL MARION, INC.	2/27/2024	0		Best available control measures â€“ speed reduction, good housekeeping practices, watering, resurfacing, and/or chemical stabilization	0.63	T/YR	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	686399	MI/YR	Paved: sweeping, vacuuming, washing with water, and posted speed limits to comply with the applicable requirements. Unpaved: use of dust suppressant as necessary to comply with the applicable requirements.	0.75	T/YR	BACT

Table C-5: RBLC PM Results for Paved Roads

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0387	INTEL OHIO SITE	INTEL OHIO SITE	9/20/2022	0		Pave all roadways and parking areas and implement best management practices, including limiting vehicle speeds and water spraying or sweeping as needed based on daily inspections.	0.8	T/YR	BACT
OH-0391	VALENCIA PROJECT LLC	VALENCIA PROJECT LLC	10/27/2023	0		Best available control measures â€“ pave all roadways, speed reduction, good housekeeping practices, and/or watering	0.92	T/YR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	0		Development and Implementation of Fugitive Dust Control Plan	1.3	TPY	BACT
OH-0378	PTTGCA PETROCHEMICAL COMPLEX	PTTGCA PETROCHEMICAL COMPLEX	12/21/2018	182865	MI/YR	i.Pave all in-plant haul roads and parking areas; ii.Implement best management practices including posting and limiting vehicle speeds to 20 miles per hour and water spraying or sweeping as needed based on the daily inspections conducted	1.88	T/YR	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	70000	MI/YR	i.Paving of all plant roads that will be used for raw material and product transport; ii.Covering, at all times, of open-bodied vehicles when transporting materials likely to become airborne; and iii.Compliance with the opacity limits. Specifically, additional mitigation measures potentially including road sweeping or wet suppression will be implemented on an as-needed basis determined through visual observation of emissions associated with truck movements on the plant site.	2.6	T/YR	BACT
AR-0173	BIG RIVER STEEL LLC	BIG RIVER STEEL LLC	1/31/2022	0		Development and Implementation of Fugitive Dust Control Plan	2.8	TPY	BACT
*IA-0117	SHELL ROCK SOY PROCESSING	SHELL ROCK SOY PROCESSING	3/17/2021	0		sweeping	2.97	TONS PER YEAR	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	686399	MI/YR	Paved: sweeping, vacuuming, washing with water, and posted speed limits to comply with the applicable requirements. Unpaved: use of dust suppressant as necessary to comply with the applicable requirements.	3.55	T/YR	BACT
OH-0392	NUCOR STEEL MARION, INC.	NUCOR STEEL MARION, INC.	2/27/2024	0		Best available control measures â€“ speed reduction, good housekeeping practices, watering, resurfacing, and/or chemical stabilization	4.4	T/YR	BACT
OH-0391	VALENCIA PROJECT LLC	VALENCIA PROJECT LLC	10/27/2023	0		Best available control measures â€“ pave all roadways, speed reduction, good housekeeping practices, and/or watering	4.6	T/YR	BACT
OH-0368	PALLAS NITROGEN LLC	PALLAS NITROGEN LLC	4/19/2017	70000	MI/YR	i.Paving of all plant roads that will be used for raw material and product transport; ii.Covering, at all times, of open-bodied vehicles when transporting materials likely to become airborne; and iii.Compliance with the opacity limits. Specifically, additional mitigation measures potentially including road sweeping or wet suppression will be implemented on an as-needed basis determined through visual observation of emissions associated with truck movements on the plant site.	13.2	T/YR	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	686399	MI/YR	Paved: sweeping, vacuuming, washing with water, and posted speed limits to comply with the applicable requirements. Unpaved: use of dust suppressant as necessary to comply with the applicable requirements.	16.74	T/YR	BACT

Table C-5: RBLC PM Results for Paved Roads

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
OH-0392	NUCOR STEEL MARION, INC.	NUCOR STEEL MARION, INC.	2/27/2024	0		Best available control measures â€“ speed reduction, good housekeeping practices, watering, resurfacing, and/or chemical stabilization	18.15	T/YR	BACT
WI-0310	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	7/8/2021	0		All roads and parking lots within the property boundary must be paved, 5 mph speed limits posted for all vehicles and develop, maintain, and implement a fugitive dust control plan.	520	TRUCK TRIPS/YR	BACT
WI-0310	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	7/8/2021	0		All roads and parking lots within the property boundary must be paved, 5 mph speed limits posted for all vehicles; and develop, maintain, and implement a fugitive dust control plan.	520	TRUCK TRIPS/YR	BACT
WI-0310	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	7/8/2021	0		All roads and parking lots within the property boundary must be paved, 5 mph speed limits posted for all vehicles; and develop, maintain, and implement a fugitive dust control plan.	520	TRUCK TRIPS/YR	BACT
IL-0132	NUCOR STEEL KANKAKEE, INC.	NUCOR STEEL KANKAKEE, INC.	1/25/2021	0		Roadways shall be paved; speed limit posting of 15 miles/hour; best management practices to reduce fugitive emissions in accordance with written operating program that provides for cleaning or treatment of roadways	0		BACT
IL-0132	NUCOR STEEL KANKAKEE, INC.	NUCOR STEEL KANKAKEE, INC.	1/25/2021	0		Roadways shall be paved; speed limit posting of 15 miles/hour; best management practices to reduce fugitive emissions in accordance with written operating program that provides for cleaning or treatment of roadways	0		BACT
IL-0132	NUCOR STEEL KANKAKEE, INC.	NUCOR STEEL KANKAKEE, INC.	1/25/2021	0		Roadways shall be paved; speed limit posting of 15 miles/hour; best management practices to reduce fugitive emissions in accordance with written operating program that provides for cleaning or treatment of roadways	0		BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	0			0		BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	0			0		BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	0		comply with fugitive dust control plan	0		BACT
*IN-0366	CARGILL, INC. - SOYBEAN PROCESSING DIVISION		7/5/2023	0			0		BACT
IN-0382	DUKE ENERGY INDIANA, INC.-CAYUGA GENERATING STATION	DUKE ENERGY INDIANA, INC.	2/14/2025	0		Paved haul roads and dust mitigation	0		BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	0			0		BACT
KY-0109	FRITZ WINTER NORTH AMERICA, LP	FRITZ WINTER NORTH AMERICA, LP	10/24/2016	0.43	Miles (length)	The permittee shall vacuum sweep the pavement at least weekly, except during recent rain events, or as needed in the event of a spill.	0		BACT
KY-0109	FRITZ WINTER NORTH AMERICA, LP	FRITZ WINTER NORTH AMERICA, LP	10/24/2016	0.43	Miles (length)	The permittee shall vacuum sweep the pavement at least weekly, except during recent rain events, or as needed in the event of a spill.	0		BACT
KY-0109	FRITZ WINTER NORTH AMERICA, LP	FRITZ WINTER NORTH AMERICA, LP	10/24/2016	0.43	Miles (length)	The permittee shall vacuum sweep the pavement at least weekly, except during recent rain events, or as needed in the event of a spill.	0		BACT
KY-0110	NUCOR STEEL BRANDENBURG	NUCOR	7/23/2020	374840	VMT/yr	surface improvements (pavement), sweeping (good work practice) and watering	0		BACT

Table C-5: RBLC PM Results for Paved Roads

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	0		Sweeping & Watering	0		BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	0		Sweeping & Watering	0		BACT
KY-0115	NUCOR STEEL GALLATIN, LLC	NUCOR STEEL GALLATIN, LLC	4/19/2021	0		Sweeping & Watering	0		BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	426.5	VMT/day	Sweeping (minimum of monthly), speed limit, paving, and reasonable precautions.	0		BACT
LA-0382	BIG LAKE FUELS METHANOL PLANT	BIG LAKE FUELS LLC	4/25/2019	0		Proper maintenance	0		BACT
LA-0382	BIG LAKE FUELS METHANOL PLANT	BIG LAKE FUELS LLC	4/25/2019	0		Proper maintenance	0		BACT
MO-0089	OWENS CORNING INSULATION SYSTEMS, LLC	OWENS CORNING	5/12/2016	0		vacuum sweep, wash, etc	0		BACT
SC-0193	MERCEDES BENZ VANS, LLC	MERCEDES BENZ VANS, LLC	4/15/2016	10.66	VMT/hr	Proper Maintenance of all roads. Fugitive dust minimization.	0		BACT
SC-0205	CORPORATION - BLYTHEWOOD PLANT	SCOUT MOTORS INC A DELAWARE CORPORATION	10/31/2023	0		Paving and maintaining all roads	0		BACT
SC-0205	CORPORATION - BLYTHEWOOD PLANT	SCOUT MOTORS INC A DELAWARE CORPORATION	10/31/2023	0		Paving and maintaining all roads	0		BACT
SC-0205	CORPORATION - BLYTHEWOOD PLANT	SCOUT MOTORS INC A DELAWARE CORPORATION	10/31/2023	0		Paving and maintaining all roads	0		BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	0		(a) All roads within the property boundary shall be paved; (b) The permittee shall post a 5 miles per hour speed limit for all vehicle traffic; and (c) The permittee shall develop, maintain, and implement a fugitive dust control plan no later than the date facility roads are in use after commencing construction.	0		BACT

Table C-6: RBLC GHG Results for Natural Gas Piping Fugitives

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Enclosed pressure type breaker and leak detector	19	T/YR	N/A
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	0		Audio, Visual, and Olfactory (AVO) inspection walk-throughs once per day	65.7	TONS	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	0		Proper piping design and installation.	104	T/YR	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	0		Proper design and good work practices.	375	T/YR	BACT
TX-0824	JACKSON COUNTY GENERATING FACILITY	SOUTHERN POWER	6/30/2017	0		weekly checks for leaks using audio, visual, and olfactory (AVO) sensing for natural gas leaks	693.3	T/YR	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Enclosed pressure type breaker and leak detection	1032	T/YR	N/A
MD-0045	MATTAWOMAN ENERGY CENTER	MATTAWOMAN ENERGY, LLC	11/13/2015	0		IMPLEMENTATION OF AN AUDIO, VISUAL AND OLFACTORY (AVO) PROGRAM ON FILE AT POWER PLANT SITE FOR REVIEW UPON REQUEST BY THE AGENCY. LEAKS DETECTED SHALL BE REPAIRED WITHIN FIVE DAYS OF DISCOVERY.	0		BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Audible/visual/olfactory (AVO) monitoring and leak repair	0		N/A
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	0.5	%	Enclosed-pressure design with low-pressure detection system (with alarm).	0		BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	0		Best management practices to prevent, detect and repair leaks of natural gas from the piping components.	0		BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	0		Best management practices to prevent, detect and repair leaks of natural gas from the piping components.	0		BACT
WI-0314	WISCONSIN PUBLIC SERVICE CORPORATION- WESTON PLANT	WISCONSIN PUBLIC SERVICE CORPORATION- WESTON PLANT	3/10/2022	0		Establish and implement a leak detection and repair (LDAR) program	0		BACT
TX-0986	JACK COUNTY GENERATION FACILITY	JACK COUNTY POWER LLC	12/27/2024	0		For natural gas fugitive piping components, emissions are represented	0		BACT

Table C-7a: RBLC SF_6 Results for Circuit Breakers

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0382	DUKE ENERGY INDIANA, INC.- CAYUGA GENERATING STATION	DUKE ENERGY INDIANA, INC.	2/14/2025	0		fully enclosed pressurized SF6 circuit breakers outfitted with density alarms for leak detection	0.5	%	BACT
FL-0363	DANIA BEACH ENERGY CENTER	FLORIDA POWER AND LIGHT COMPANY	12/4/2017	0		Certified leak rate < 0.5% per year	0.5	% LEAK PER YEAR	BACT
FL-0367	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	7/27/2018	0		Certified leak rate < 0.5% per year	0.5	% LEAK PER YEAR	BACT
FL-0371	SHADY HILLS COMBINED CYCLE FACILITY	SHADY HILLS ENERGY CENTER, LLC	6/7/2021	0		Certified leak rate < 0.5% per year	0.5	% LEAK PER YEAR	BACT
IL-0129	CPV THREE RIVERS ENERGY CENTER	CPV THREE RIVERS, LLC	7/30/2018	0			0.5	% LEAK RATE	BACT
WI-0299	WPL- RIVERSIDE ENERGY CENTER	WPL- RIVERSIDE ENERGY CENTER	8/20/2020	0			0.5	RATE, BY WGHT	BACT
FL-0354	LAUDERDALE PLANT	FLORIDA POWER & LIGHT	8/25/2015	0		Limitation on leaks	0.5	% PER YEAR	BACT
FL-0355	FORT MYERS PLANT	FLORIDA POWER & LIGHT (FPL)	9/10/2015	0		Limitation on leak of SF6 from circuit breakers	0.5	PERCENT	BACT
IL-0130	JACKSON ENERGY CENTER	JACKSON GENERATION, LLC	12/31/2018	0			0.5	PERCENT LEAK RATE	BACT
IL-0133	LINCOLN LAND ENERGY CENTER	LINCOLN LAND ENERGY CENTER (A/K/A EMBERCLEAR)	7/29/2022	0			0.5	PERCENT LEAK RATE	BACT
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	0		low pressure alarms and low pressure lockout system	6	LB/12MO	BACT
IN-0384	SYCAMORE RIVERSIDE ENERGY LLC	INVENERGY THERMAL DEVELOPMENT LLC	3/31/2025	0		totally enclosed and pressurized circuit breakers with a design leak rate of no more than 0.5% and a density alarm for leak detection	73.6	LB	BACT
PA-0310	CPV FAIRVIEW ENERGY CENTER	CPV FAIRVIEW, LLC	9/2/2016	0		State-of-the-art sealed enclosed-pressure circuit breakers with leak detection	1500	PPM	BACT
*IN-0400	AES INDIANA, LLC- EAGLE VALLEY GENERATING STATION	AES INDIANA, LLC	10/7/2025	0		fully enclosed SF6 circuit breakers outfitted with low pressure alarms for leak detection	192.4	TONS CO2E	BACT
FL-0356	OKEECHOBEE CLEAN ENERGY CENTER	FLORIDA POWER & LIGHT	3/9/2016	0		Leak prevention. Must have manufacturer-guaranteed leak rate no more than 0.5% per year. Must be equipped with leakage detection systems and alarms.	0		BACT
IN-0294	ST. JOSEPH ENERGY CENTER, LLC	ST. JOSEPH ENERGY CENTER, LLC	8/8/2018	0			0		CASE-BY-CASE
MD-0045	MATTAWOMAN ENERGY CENTER	MATTAWOMAN ENERGY, LLC	11/13/2015	0			0		CASE-BY-CASE

Table C-7b: RBLC GHG Results for Circuit Breakers

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	0			0.5	% BY WEIGHT/YEAR	BACT
*IL-0137	MARQUIS CARBON CAPTURE LLC	MARQUIS CARBON CAPTURE LLC	6/23/2025	0		Low-leak design (leak rate of no more than 0.5% per year); insulating (or dielectric) material with a GWP of no greater than 292; LDAR	0.02	TON/YEAR	BACT
KS-0029	THE EMPIRE DISTRICT ELECTRIC COMPANY	THE EMPIRE DISTRICT ELECTRIC COMPANY	7/14/2015	0		Installation of modern, totally enclosed SF6 circuit breakers with density (leak detection) alarms and a guaranteed loss rate of < 0.5 % by weight per year.	6.9	TONS PER YEAR	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Enclosed pressure type breaker and leak detector	19	T/YR	N/A
PA-0309	LACKAWANNA ENERGY CTR/JESSUP	LACKAWANNA ENERGY CENTER, LLC	12/23/2015	0			79.8	TONS	BACT
LA-0391	MAGNOLIA POWER GENERATING STATION UNIT 1	MAGNOLIA POWER LLC	6/3/2022	0		Enclosed pressure design with a low pressure detection system with an alarm to limit SF6 leak rate to 0.5 % per year.	85	T/YR	BACT
VA-0325	GREENSVILLE POWER STATION	VIRGINIA ELECTRIC AND POWER COMPANY	6/17/2016	0		Enclosed pressure type breaker and leak detection	1032	T/YR	N/A
MD-0045	MATTAWOMAN ENERGY CENTER	MATTAWOMAN ENERGY, LLC	11/13/2015	0		GHG BACT FOR THE CIRCUIT BREAKERS SHALL BE INSTALLATION OF STATE-OF-THE-ART CIRCUIT BREAKERS THAT ARE DESIGNED TO MEET ANSI C37.013 OR EQUIVALENT TO DETECT AND MINIMIZE SF6 LEAKS	0		BACT
VA-0328	C4GT, LLC	NOVI ENERGY	4/26/2018	0.5	%	Enclosed-pressure design with low-pressure detection system (with alarm).	0		BACT
VA-0332	CHICKAHOMINY POWER LLC	CHICKAHOMINY POWER LLC	6/24/2019	0.5	%	Enclosed-pressure design with low-pressure detection system (with alarm).	0		BACT

Table C-8: RBLC VOC Results for Diesel Storage Tanks

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
AK-0088	LIQUEFACTION PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	7/7/2022	0		Submerged Fill	0.01	TPY COMBINED	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4788	gal/yr	Good design practices and use of submerged fill pipe.	0.03	T/YR	BACT
AK-0085	GAS TREATMENT PLANT	ALASKA GASLINE DEVELOPMENT CORPORATION	8/13/2020	0		Submerged Fill	0.59	TPY	BACT
AK-0084	DONLIN GOLD PROJECT	DONLIN GOLD LLC.	6/30/2017	0		Submerged Fill	1.7	TPY	BACT
IN-0273	ST. JOSEPH ENERGY CENTER	ST. JOSEPH ENERGY CENTER	6/22/2017	650	GALLONS	THE USE OF GOOD DESIGN AND OPERATING PRACTICES. EACH TANK SHALL UTILIZE A FIXED ROOF.	0		BACT
IN-0273	ST. JOSEPH ENERGY CENTER	ST. JOSEPH ENERGY CENTER	6/22/2017	5000	GALLONS	THE USE OF GOOD DESIGN AND OPERATING PRACTICES. EACH TANK SHALL UTILIZE A FIXED ROOF.	0		BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	0		(a) Use of fixed roof tanks; (b) Performing submerged-filling or bottom loading of each fixed roof storage tank; and (c) These storage tanks may only store diesel fuel.	0		BACT
*TX-0997	PORT ARTHUR REFINERY	MOTIVA ENTERPRISES LLC	9/22/2025	0		Tanks 2127 and 1821 are insulated tanks with the liquid maintained at temperatures of 450Â°F and 250Â°F, respectively. Temperature monitoring is implemented but not required since the vapor pressure of the stored material during routine operation will not exceed a relevant vapor pressure cutoff for BACT purposes.	0		BACT

Table C-9a: RBLC NOx Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.1	pound per MMBtu	The pilot and purge gas fuels shall be natural gas	168	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas.	240	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	336	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	336	HR/YR	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	1.2	MMBTU/H	Use of natural gas, good combustion practices and design	0.12	LB/H	BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0.23	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.3	LB/HR	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	9.5	mmbtu/hr	Use of natural gas, good combustion practices and design	0.95	LB/H	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.1	MMBTU/H		125	LB/H	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Ultra-low NOx burners.	0.011	LB/MM BTU	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	10	MMBtu/hr		0.011	LB/MMBTU	BACT
PA-0310	CPV FAIRVIEW ENERGY CENTER	CPV FAIRVIEW, LLC	9/2/2016	3.2	MMBtu/hr		0.035	LB/MMBTU	LAER
*TX-1000	ANTELOPE ELK ENERGY CENTER	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	8/14/2025	5.5	MMBTU/HR	Low NOx burners and firing pipeline-quality natural gas. RBLC searches for small fuel gas heaters (<10 MMBtu/hr) installed in the last 10 years used low NOx burners achieving emission limits ranging from 0.036 to 0.10 lb/MMBtu.	0.036	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Low-NOx burners, only use pipeline quality natural gas, maintain and operate according to manufacturerâ€™s recommendations.	0.049	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Low-NOx burners, only combust pipeline quality natural gas, and operate and maintain the heater according to the manufacturerâ€™s recommendations.	0.049	LB/MMBTU	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	10	MMBtu/hr	(a) P04 and P05 shall be equipped with low-NOx burners. (b) Only pipeline quality natural gas may be combusted in P04 and P05. (c) The permittee shall operate and maintain P04 and P05 according to the manufacturerâ€™s recommendations. (d) The permittee shall tune within 120 days of the initial operation, and within 24-months of each preceding tune-up. (e) NOx emissions from P04 and P05 may not exceed 0.049 lb/MMBtu each.	0.049	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		0.068	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		0.068	LB/MMBTU	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	2.1	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	0.068	LB/MMBTU	BACT

Table C-9a: RBLC NOx Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	4	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	0.068	LB/MMBTU	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	1.4	MMBtu/hr	Good combustion practices and pilot and purge gas fuels shall be natural gas.	0.068	LB/MMBTU	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	4.44	MMBTU/H	LOW NOX BURNER	0.07	LB/MMBTU	CASE-BY-CASE
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	1.37	MMBTU/H	LOW NOX BURNER	0.07	LB/MMBTU	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	8.3	MMBTU/H	LOW NOX BURNER	0.08	LB/MMBTU	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of gaseous fuels (fuel gas and/or natural gas). Good combustion practices. Proper equipment (burner) design/operation.	0.098	LB/MM BTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	3	MMBTU/hr each	Low Nox Burners	0.1	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Combustion of clean fuel Good Combustion Practices	0.1	LB/MMBTU	BACT
FL-0356	OKEECHOBEE CLEAN ENERGY CENTER	FLORIDA POWER & LIGHT	3/9/2016	10	MMBtu/hr	Must have NOx emission design value less than 0.1 lb/MMBtu	0.1	LB/MMBTU	BACT
FL-0363	DANIA BEACH ENERGY CENTER	FLORIDA POWER AND LIGHT COMPANY	12/4/2017	9.9	MMBtu/hr	Manufacturer certification	0.1	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare minimization plan and root cause analysis, smokeless, steam-assist flare design, work practices in accordance with 40 CFR 63.11(b), GCP, nitrogen purge gas	0.1	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist flare design, Work Practices in accordance with 40 CFR 63.11(b), GCP, use of nitrogen as purge gas	0.1	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	0.4	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist flare design, Work Practices in accordance with 40 CFR 60.18(b), GCP, use of nitrogen as purge gas	0.1	LB/MMBTU	BACT
TX-0694	INDECK WHARTON ENERGY CENTER	INDECK WHARTON, L.L.C.	2/2/2015	3	MMBTU/H		0.1	LB/MMBTU	BACT
WI-0291	GRAYMONT WESTERN LIME-EDEN	GRAYMONT WESTERN LIME-EDEN	1/28/2019	1.5	mmBTU/hr	Good Combustion Practices	0.1	LB/MMBTU	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	low NOx burners, good combustion practices and only pipeline quality natural gas shall be combusted	50	LB/MMSCF	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	1.44	MMBtu/hr	Low NOx burners and good combustion practices	50	LB/MMSCF	BACT
MI-0420	DTE GAS COMPANY--MILFORD COMPRESSOR STATION	DTE GAS COMPANY	6/3/2016	6	MMBTU/H	Ultra low NOx burners and good combustion practices.	14	PPMVOL	BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion practices	0		BACT

Table C-9b: RBLC CO Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.1	pound per MMBtu	The pilot and purge gas fuels shall be natural gas	168	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	240	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	shall be natural gas	336	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	336	HR/YR	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	1.2	MMBTU/H	Use of natural gas, good combustion practices and design	0.02	LB/H	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	9.5	mmbtu/hr	Use of natural gas, good combustion practices and design	0.19	LB/H	BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0.19	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.25	LB/HR	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Combustion of natural gas and good combustion practices	0.0004	LB/MMBTU	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	1.37	MMBTU/H	GCP	0.03	LB/MMBTU	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Good combustion practices.	0.037	LB/MM BTU	BACT
TX-0694	INDECK WHARTON ENERGY CENTER	INDECK WHARTON, L.L.C.	2/2/2015	3	MMBTU/H		0.04	LB/MMBTU	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	10	MMBtu/hr		0.08	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist flare design, Work Practices in accordance with 40 CFR 63.11(b), GCP, use of nitrogen as purge gas	0.08	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist flare design, Work Practices in accordance with 40 CFR 63.11(b), GCP, use of nitrogen as purge gas	0.08	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	0.4	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist flare design, Work Practices in accordance with 40 CFR 60.18(b), GCP, use of nitrogen as purge gas	0.08	LB/MMBTU	BACT
MI-0420	DTE GAS COMPANY--MILFORD COMPRESSOR STATION	DTE GAS COMPANY	6/3/2016	6	MMBTU/H	Good combustion practices and clean burn fuel (pipeline quality natural gas)	0.08	LB/MMBTU	BACT
PA-0310	CPV FAIRVIEW ENERGY CENTER	CPV FAIRVIEW, LLC	9/2/2016	3.2	MMBtu/hr		0.08	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Only combust pipeline quality natural gas and operate and maintain the process according to manufacturerâ€™s recommendations.	0.08	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Only combust pipeline quality natural gas and operate and maintain heater according to the manufacturerâ€™s recommendations.	0.08	LB/MMBTU	BACT

Table C-9b: RBLC CO Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	10	MMBtu/hr	(a) The permittee shall operate and maintain according to the manufacturerâ€™s recommendations. (b) The permittee shall tune within 120 days of the initial operation, and within 24-months of each preceding tune-up. (c) CO emissions may not exceed 0.08 lb/MMBtu. (d) The permittee may only combust pipeline quality natural gas.	0.08	LB/MMBTU	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of gaseous fuels (fuel gas and/or natural gas). Good combustion practices. Proper equipment (burner) design/operation. Compliance with 40 CFR 63 Subpart DDDDD.	0.082	LB/MM BTU	BACT
WI-0291	GRAYMONT WESTERN LIME-EDEN	GRAYMONT WESTERN LIME-EDEN	1/28/2019	1.5	mmBTU/hr	Good Combustion Practices	0.082	LB/MMBTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	3	MMBTU/hr each	Good Combustion Practices	0.084	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.1	MMBTU/H		0.37	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		0.37	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		0.37	LB/MMBTU	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	2.1	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	0.37	LB/MMBTU	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices	84	LB/MMSCF	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	4.44	MMBTU/H	GCP	0		BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	8.3	MMBTU/H	GCP	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion practices	0		BACT
*TX-0996	SECONDARY ALUMINUM PRODUCTION FACILITY	AUDUBON METALS TEXAS LLC	5/27/2025	0	MMBTU/HR	Add-on control devices have not historically been required for VOC and CO control for sources of this type at Secondary Aluminum Production Facilities. Good operating practices will be used to minimize emissions of VOC and CO.	0		BACT

Table C-9c: RBLC PM Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.1	pound per MMBtu	The pilot and purge gas fuels shall be natural gas	168	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels used shall be natural gas	240	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels used shall be natural gas.	240	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	336	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	336	HR/YR	BACT
SC-0179	CAROLINA PARTICLEBOARD	FLAKEBOARD AMERICA LIMITED	3/18/2015	1.83	MMBTU/H	NATURAL GAS USAGE AND GOOD COMBUSTION PRACTICES.	0.003	LB/H	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	1.2	MMBTU/H	Use of natural gas, good combustion practices and design	0.004	LB/H	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	1.2	MMBTU/H	Use of natural gas, good combustion practices and design	0.004	LB/H	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	1.2	MMBTU/H	Use of natural gas, good combustion practices and design	0.004	LB/H	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.006	LB/HR	BACT
SC-0179	CAROLINA PARTICLEBOARD	FLAKEBOARD AMERICA LIMITED	3/18/2015	1.83	MMBTU/H	USE OF NATURAL GAS AND GOOD COMBUSTION PRACTICES	0.01	LB/H	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.02	LB/HR	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.02	LB/HR	BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0.02	LB/HR	BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0.02	LB/HR	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	9.5	mmbtu/hr	Use of natural gas, good combustion practices and design	0.03	LB/H	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	9.5	mmbtu/hr	Use of natural gas, good combustion practices and design	0.03	LB/H	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	9.5	mmbtu/hr	Use of natural gas, good combustion practices and design	0.03	LB/H	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	3	MMBTU/hr each	Good Combustion Practices	0.0019	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.1	MMBTU/H		0.0019	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		0.0019	LB/MMBTU	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Good combustion practices.	0.0048	LB/MM BTU	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Good combustion practices.	0.0048	LB/MM BTU	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of gaseous fuels (fuel gas and/or natural gas). Good combustion practices. Proper equipment (burner) design/operation.	0.0075	LB/MM BTU	BACT

Table C-9c: RBLC PM Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of gaseous fuels (fuel gas and/or natural gas). Good combustion practices. Proper equipment (burner) design/operation.	0.0075	LB/MM BTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Combustion of natural gas and good combustion practices	0.0075	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Combustion of natural gas and good combustion practices	0.0075	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Combustion of natural gas and good combustion practices	0.0075	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist, smokeless flare design, Work Practices in accordance with 40 CFR 63.11(b), GCP, use of nitrogen as purge gas	0.0075	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist, smokeless flare design, Work Practices in accordance with 40 CFR 63.11(b), GCP, use of nitrogen as purge gas	0.0075	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist, smokeless flare design, Work Practices in accordance with 40 CFR 63.11(b), GCP, use of nitrogen as purge gas	0.0075	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist, smokeless flare design, Work Practices in accordance with 40 CFR 63.11(b), GCP, use of nitrogen as purge gas	0.0075	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.1	MMBTU/H		0.0075	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.1	MMBTU/H		0.0075	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		0.0075	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		0.0075	LB/MMBTU	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels used shall be natural gas, Flares shall be designed for and operated with no visible emissions, except for periods not to exceed 5 minutes during any two consecutive hours, Flares shall be operated with a flame present at all times, Flares shall be continuously monitored to assure the presence of a pilot flame with a thermocouple, infrared monitor, or other approved device	0.0075	LB/MMBTU	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels used shall be natural gas, Flares shall be designed for and operated with no visible emissions, except for periods not to exceed 5 minutes during any two consecutive hours, Flares shall be operated with a flame present at all times, Flares shall be continuously monitored to assure the presence of a pilot flame with a thermocouple, infrared monitor, or other approved device	0.0075	LB/MMBTU	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	2.1	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	0.0075	LB/MMBTU	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	2.1	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	0.0075	LB/MMBTU	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.1	pound per MMBtu	The pilot and purge gas fuels shall be natural gas	0.0075	LB/MMBTU	BACT

Table C-9c: RBLC PM Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
MI-0420	DTE GAS COMPANY--MILFORD COMPRESSOR STATION	DTE GAS COMPANY	6/3/2016	6	MMBTU/H	Good combustion practices and low sulfur fuel (pipeline quality natural gas).	0.0075	LB/MMBTU	BACT
MI-0420	DTE GAS COMPANY--MILFORD COMPRESSOR STATION	DTE GAS COMPANY	6/3/2016	6	MMBTU/H	Good combustion practices and low sulfur fuel (pipeline quality natural gas).	0.0075	LB/MMBTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	3	MMBTU/hr each	Good Combustion Practices	0.0076	LB/MMBTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	3	MMBTU/hr each	Good Combustion Practices	0.0076	LB/MMBTU	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	10	MMBtu/hr		0.008	LB/MMBTU	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	10	MMBtu/hr		0.008	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Operate and maintain heaters according to manufacturer's recommendations. Only combust pipeline quality natural gas.	0.01	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Operate and maintain according to manufacturerâ€™s recommendations. Only combust pipeline quality natural gas.	0.01	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Operate and maintain heater according to manufacturerâ€™s recommendations. Only combust pipeline quality natural gas.	0.01	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Operate and maintain heater according to the manufacturerâ€™s recommendations and only combust pipeline quality natural gas.	0.01	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Operate and maintain heater according to the manufacturerâ€™s recommendations and only combust pipeline quality natural gas.	0.01	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Operate and maintain heater according to the manufacturerâ€™s recommendations and only combust pipeline quality natural gas.	0.01	LB/MMBTU	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	10	MMBtu/hr	(a) The permittee shall operate and maintain according to the manufacturerâ€™s recommendations. (b) The permittee shall tune within 120 days of the initial operation, and within 24-months of each preceding tune-up. (c) The permittee may only combust pipeline quality natural gas.	0.01	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	10	MMBtu/hr	Good combustion practices	7.6	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	10	MMBtu/hr	Good combustion practices	7.6	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	10	MMBtu/hr	Good combustion practices	7.6	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	10	MMBtu/hr	Good combustion practices	7.6	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	0.5	MMBtu/hr	Good combustion practices	7.6	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	0.5	MMBtu/hr	Good combustion practices	7.6	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	0.5	MMBtu/hr	Good combustion practices	7.6	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	0.5	MMBtu/hr	Good combustion practices	7.6	LB/MMBTU	BACT

Table C-9c: RBLC PM Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		1.9	LB/MMCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	1.9	LB/MMSCF	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		7.6	LB/MMCF	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		7.6	LB/MMCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	7.6	LB/MMSCF	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	7.6	LB/MMSCF	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	0.4	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist, smokeless flare design, Work Practices in accordance with 40 CFR 60.18(b), GCP, use of nitrogen as purge gas	0.01	TONS/YEAR	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	0.4	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist, smokeless flare design, Work Practices in accordance with 40 CFR 60.18(b), GCP, use of nitrogen as purge gas	0.01	TONS/YEAR	BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion practices	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion practices	0		BACT
TX-0694	INDECK WHARTON ENERGY CENTER	INDECK WHARTON, L.L.C.	2/2/2015	3	MMBTU/H		0		BACT
*WV-0031	MOCKINGBIRD HILL COMPRESSOR STATION	DOMINION ENERGY TRANSMISSION, INC.	6/14/2018	8.72	mmBtu/hr	Limited to natural gas	0		BACT
*WV-0031	MOCKINGBIRD HILL COMPRESSOR STATION	DOMINION ENERGY TRANSMISSION, INC.	6/14/2018	8.72	mmBtu/hr	Limited to natural gas	0		BACT
*WV-0031	MOCKINGBIRD HILL COMPRESSOR STATION	DOMINION ENERGY TRANSMISSION, INC.	6/14/2018	8.72	mmBtu/hr	Limited to natural gas.	0		BACT

Table C-9d: RBLC SO2 Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
FL-0356	OKEECHOBEE CLEAN ENERGY CENTER	FLORIDA POWER & LIGHT	3/9/2016	10	MMBtu/hr	Use of low-sulfur fuel	2	GR. S/100 SCF GAS	BACT
FL-0363	DANIA BEACH ENERGY CENTER	FLORIDA POWER AND LIGHT COMPANY	12/4/2017	9.9	MMBtu/hr	Clean fuel	2	GRAINS S / 100 SCF	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	1.2	MMBTU/H	Use of natural gas, good combustion practices and design	0.001	LB/H	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	9.5	mmbtu/hr	Use of natural gas, good combustion practices and design	0.01	LB/H	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	3	MMBTU/hr each	Good Combustion Practices	0.0006	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Combustion of natural gas and good combustion practices	0.0006	LB/MMBTU	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	Only pipeline quality natural gas fuel shall be combusted	0.0006	LB/MMBTU	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Use of natural gas fuel.	0.0014	LB/MM BTU	BACT

Table C-9e: RBLC H2SO4 Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Only combust pipeline quality natural gas and maintain heater according to manufacturerâ€™s recommendations.	0		BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Only combust pipeline quality natural gas and operate and maintain heater according to manufacturerâ€™s recommendations.	0		BACT

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.1	pound per MMBtu	The pilot and purge gas fuels shall be natural gas	168	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	240	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	336	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	336	HR/YR	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	1.2	MMBTU/H	Use of natural gas, good combustion practices and design	0.01	LB/H	BACT
SC-0179	CAROLINA PARTICLEBOARD	FLAKEBOARD AMERICA LIMITED	3/18/2015	1.83	MMBTU/H	NATURAL GAS USAGE AND GOOD COMBUSTION PRACTICES.	0.01	LB/H	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Good Combustion & Operation Practices (GCOP) Plan	0.02	LB/HR	BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0.03	LB/HR	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	9.5	mmbtu/hr	Use of natural gas, good combustion practices and design	0.05	LB/H	BACT
FL-0364	SEMINOLE GENERATING STATION	SEMINOLE ELECTRIC COOPERATIVE, INC.	3/21/2018	9.9	MMBtu/hr		0.005	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Only combust pipeline quality natural gas, operate and maintain process according to manufacturerâ€™s recommendations.	0.005	LB/MMBTU	BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Only combust pipeline quality natural gas and operate and maintain according to manufacturerâ€™s recommendations.	0.005	LB/MMBTU	BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	10	MMBtu/hr	(a) The permittee shall operate and maintain according to the manufacturerâ€™s recommendations. (b) The permittee shall tune within 120 days of the initial operation, and within 24-months of each preceding tune-up. (c) VOC emissions from may not exceed 0.005 lb/MMBtu. (d) The permittee may only combust pipeline quality natural gas.	0.005	LB/MMBTU	BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of gaseous fuels (fuel gas and/or natural gas). Good combustion practices. Proper equipment (burner) design/operation. Compliance with 40 CFR 63 Subpart DDDDD.	0.0054	LB/MM BTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist flare design, Work Practices in accordance with 40 CFR 63.11(b), GCP, use of nitrogen as purge gas	0.0054	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist flare design, Work Practices in accordance with 40 CFR 63.11(b), GCP, use of nitrogen as purge gas	0.0054	LB/MMBTU	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	0.4	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist flare design, Work Practices in accordance with 40 CFR 60.18(b), GCP, use of nitrogen as purge gas	0.0054	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.1	MMBTU/H		0.0054	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		0.0054	LB/MMBTU	BACT
IN-0263	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	3/23/2017	1.12	MMBTU/H		0.0054	LB/MMBTU	BACT

Table C-9f: RBLC VOC Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	2.1	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	0.0054	LB/MMBTU	BACT
WI-0297	GREEN BAY PACKAGING- MILL DIVISION	GREEN BAY PACKAGING INC.	12/10/2019	8.5	MMBtu/H		0.0055	LB/MMBTU	BACT
WI-0297	GREEN BAY PACKAGING- MILL DIVISION	GREEN BAY PACKAGING INC.	12/10/2019	8.5	MMBtu/H	Use only natural gas.	0.0055	LB/MMBTU	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	4.44	MMBTU/H		0.006	LB/MMBTU	CASE-BY-CASE
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	1.37	MMBTU/H	GCP	0.006	LB/MMBTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	3	MMBTU/hr each	Good Combustion Practices	0.0076	LB/MMBTU	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Good combustion practices.	0.008	LB/MM BTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	10	MMBtu/hr	Good combustion practices	5.5	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	10	MMBtu/hr	Good combustion practices	5.5	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	0.5	MMBtu/hr	Good combustion practices	5.5	LB/MMBTU	BACT
MN-0096	USG CLOQUET	USG CORPORATION	4/3/2024	0.5	MMBtu/hr	Good combustion practices	5.5	LB/MMBTU	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices	5.5	LB/MMSCF	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	8.3	MMBTU/H	GCP	0		BACT
IL-0127	WINPAK HEAT SEAL CORPORATION	WINPAK HEAT SEAL CORPORATION	10/5/2018	1	mmBtu/hr	Units shall be operated in accordance with good combustion practices.	0		BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion practices	0		BACT

Table C-9g: RBLC GHG Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
WI-0297	GREEN BAY PACKAGING- MILL DIVISION	GREEN BAY PACKAGING INC.	12/10/2019	8.5	MMBtu/H	Use only natural gas.	90	% AVG THERM EFF	BACT
WI-0297	GREEN BAY PACKAGING- MILL DIVISION	GREEN BAY PACKAGING INC.	12/10/2019	8.5	MMBtu/H	Only fire natural gas.	90	% AVG THERM EFF	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	1.4	MMBtu/hr	Good combustion practices and pilot and purge gas fuels used shall be natural gas	168	HOURS VENTING	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	240	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	336	HR/YR	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.12	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	336	HR/YR	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	1.2	MMBTU/H	Use of natural gas and energy efficient design	140.22	LB/H	BACT
OH-0381	NORTHSTAR BLUESCOPE STEEL, LLC	NORTHSTAR BLUESCOPE STEEL, LLC	9/27/2019	9.5	mmbtu/hr	Use of natural gas and energy efficient design	1110.1	LB/H	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	2.1	MMBtu/hr	The pilot and purge gas fuels shall be natural gas	116.89	LB/MMBTU	BACT
*AR-0191	HYBAR	HYBAR	10/2/2025	9	mmbtu/hr	Good operating practices	117	LB/MMBTU	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	1.44	MMBtu/hr	Good Combustion Practices	117	LB/MMBTU	BACT
IN-0371	WABASH VALLEY RESOURCES, LLC		1/11/2024	4	MMBtu/hr	The pilot and purge gas fuels shall be natural gas.	117	LB/MMBTU	BACT
AL-0329	COLBERT COMBUSTION TURBINE PLANT	TENNESSEE VALLEY AUTHORITY	9/21/2021	10	MMBtu/hr		117.1	LB/MMBTU	BACT
AR-0171	NUCOR STEEL ARKANSAS	NUCOR CORPORATION	2/14/2019	3	MMBTU/hr each	Good Combustion Practices	121	LB/MMBTU	BACT
IN-0324	MIDWEST FERTILIZER COMPANY LLC	MIDWEST FERTILIZER COMPANY LLC	5/6/2022	1.1	pound per MMBtu	The pilot and purge gas fuels shall be natural gas	563	LB/MMBTU	BACT
KY-0116	NOVELIS CORPORATION - GUTHRIE	NOVELIS CORPORATION	7/25/2022	3	MMBtu/hr (total)	Design Requirements, Good Combustion & Operation Practices (GCOP) Plan	1579	TONS/YR	BACT
*LA-0412	ENTERGY LOUISIANA, LLC - FRANKLIN FARMS POWER STATION	ENTERGY LOUSIANA, LLC	12/23/2025	4.54	mm BTU/h	Good combustion practices.	2326	T/YR	BACT
IN-0359	NUCOR STEEL	NUCOR STEEL	3/30/2023	5.12	MMBtu/hr	good combustion practices and only pipeline quality natural gas shall be combusted	2625	TONS/YR	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	0.4	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist flare design, Work Practices in accordance with 40 CFR 60.18(b), GCP, use of nitrogen as purge gas	3305	TONS/YEAR	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	8.3	MMBTU/H		4256	T/YR	BACT
MI-0420	DTE GAS COMPANY--MILFORD COMPRESSOR STATION	DTE GAS COMPANY	6/3/2016	6	MMBTU/H	Use of pipeline quality natural gas and energy efficiency measures.	6155	T/YR	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist flare design, Work Practices in accordance with 40 CFR 63.11(b), GCP, use of nitrogen as purge gas	16093	TONS/YEAR	BACT
IL-0134	CRONUS CHEMICALS	CRONUS CHEMICALS, LLC	12/21/2023	2.22	mmBtu/hr	Flare Minimization Plan and Root Cause Analysis, Steam-assist flare design, Work Practices in accordance with 40 CFR 63.11(b), GCP, use of nitrogen as purge gas	18603	TONS/YEAR	BACT

Table C-9g: RBLC GHG Results for Building Heaters

RBLC ID	Facility Name	Company Name	Permit Date	Throughput	Units	Controls	Emission Limit	Units	Type
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	4.44	MMBTU/H		36251	T/YR	BACT
AL-0307	ALLOYS PLANT	CONSTELLIUM	10/9/2015	1.37	MMBTU/H		36251	T/YR	BACT
LA-0307	MAGNOLIA LNG FACILITY	MAGNOLIA LNG, LLC	3/21/2016	7.37	mm btu/hr	good combustion/operating/maintenance practices and fueled by natural gas	0		BACT
*LA-0394	GEISMAR PLANT	SHELL CHEMICAL LP	12/12/2023	2	MM BTU/hr	Use of good thermal oxidizer design and implementation of good combustion practices	0		BACT
*LA-0406	MCA GEIMAR SITE	MITSUBISHI CHEMICAL AMERICA, INC.	7/24/2024	8	mm btu/hr	Use of low carbon intensity gaseous fuels. Good combustion and operating practices. Efficiency improvement measures.	0		BACT
*TX-1000	ANTELOPE ELK ENERGY CENTER	GOLDEN SPREAD ELECTRIC COOPERATIVE, INC.	8/14/2025	5.5	MMBTU/HR	GOOD COMBUSTION PRACTICES	0		BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Low-NOx burners, only combust pipeline quality natural gas, and operate and maintain heater according to manufacturerâ€™s recommendations.	0		BACT
WI-0300	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/1/2020	10	MMBTU/H	Low-NOx burners, only combust pipeline quality natural gas, and operate and maintain according to manufacturerâ€™s recommendations.	0		BACT
*WI-0326	NEMADJI TRAIL ENERGY CENTER	NEMADJI TRAIL ENERGY CENTER	9/19/2023	10	MMBtu/hr	(a) Equipped with low-NOx burners. (b) Only pipeline quality natural gas may be combusted. (c) The permittee shall operate and maintain according to the manufacturerâ€™s recommendations. (d) The permittee shall tune within 120 days of the initial operation of, and within 24-months of each preceding tune-up.	0		BACT
*WV-0031	MOCKINGBIRD HILL COMPRESSOR STATION	DOMINION ENERGY TRANSMISSION, INC.	6/14/2018	8.72	mmBtu/hr	Limited to natural gas; and tune-up the boiler once every five years.	0		BACT

Appendix D – Economic Tables

Table D-1
SCR System Capital Cost Analysis - Simple Cycle Turbines (Each)

Item	Value	Basis
Direct Costs		
Purchased Equipment Cost		
Equipment cost + auxiliaries[b] **[A]**	**$14,702,650**	A
Instrumentation[a]	$1,470,265	0.10 x A
Sales taxes[a]	$441,080	0.03 x A
Freight[a]	$735,133	0.05 x A
Total Purchased Equipment Cost (PEC) [B]	**$17,349,127**	1.18 x A
Direct Installation Costs		
Total Direct Installation Cost[a]	$5,204,738	0.30 x B
Site Preparation (SP)	$0	As required
Buildings (Bldg.)	$0	As required
Total Direct Cost (DC)	**$22,553,865**	1.30B + SP + Bldg.
Indirect Costs (Installation)		
Engineering[a]	$1,734,913	0.10 x B
Construction and field expenses[a]	$867,456	0.05 x B
Contractor fees[a]	$1,734,913	0.10 x B
Start-up[a]	$346,983	0.02 x B
Performance test[c]	$10,000	
Contingencies[a]	$867,456	0.05 x B
Other	$0	As required
Total Indirect Cost (IC)	**$5,561,721**	0.32B + Perf. Test + Other
Total Capital Investment (TCI) = DC + IC	**$28,115,586**	1.62B + Performance test + Other + SP + Bldg.

[a]U.S. Environmental Protection Agency Office of Air Quality Planning and Standards, Air Pollution Control Cost Manual, Sixth Edition (2002)

[b]Cost obtained from vendor

[c]Cost obtained from vendor

[d]U.S. Energy Information Administration Electric Power Monthly (June 2025)

Table D-2
SCR System Annual Cost Analysis - Simple Cycle Turbines (Each)

Item	Value	Basis
Direct Annual Costs (DC)		
Electricity		
Press. Drop (in W.C.)	3.0	Pressure drop - catalyst bed
Power output of Turbine (kW)	239,000	ISO Rating
Power Loss Due to Pressure Drop (%)	0.30%	0.1% for every 1" pressure drop
Power Loss Due to Pressure Drop (kW)	717.00	
Unit cost ($/kWh)[d]	$0.089	Estimated market value
Cost of Power Loss ($/yr)	$224,349	Based on operational hours/yr
Operating Labor		
Catalyst labor req.	$6,563	1/2 hr/shift @ $30/hr
Ammonia delivery requirement (SCR)	$720	24 hr/yr (3 deliveries per year) @ $30/hr
Ammonia recordkeeping and reporting (SCR)	$1,200	40 hours per year @ $30/hr
Catalyst cleaning	$1,200	40 hours per year @ $30/hr
Supervisor	$984	15% Operating labor
Total Cost ($/yr)	$10,667	
Maintenance[a]		
Total Cost ($/yr)	$421,734	0.015 x B
Ammonia		
Requirement (tons/yr)	106.4	29% aqueous ammonia @ $375/ton
Unit Cost ($/ton)	$375	Estimate
Total Cost ($/yr)	$39,886	
Process Air		
Requirement (scf/lb NH_3)	350	
Requirement (mscf/yr)	130,295	
Unit Cost ($/mscf)	$0.20	$0.20 per 1000 scf
Total Cost ($/yr)	$26,059	
Catalyst		
Catalyst Cost ($)[b]	$2,475,000	
Catalyst Disposal Cost ($)	$38	Disposal of catalyst modules
Sales Tax ($)[a]	$74,250	0.03 x Catalyst Cost
Catalyst Life (yrs)[b]	5	n
Interest Rate (%)[a]	7.0%	i
CRF[a]	0.244	Amortization of catalyst for 5 yrs
Total Cost ($/yr)	$621,748	(Volume) * (Unit Cost) * (CRF)
Indirect Annual Costs (IC)		
Overhead[a]	$0	OAQPS SCR Assumption
Administrative charges[a]	$0	OAQPS SCR Assumption
Annual Contingency[a]	$0	OAQPS SCR Assumption
Property taxes[a]	$0	OAQPS SCR Assumption
Insurance[a]	$0	OAQPS SCR Assumption
Capital Recovery[a]	$2,653,912	CRF x TCI (20 yr life, 7.0% interest)
Total Indirect Costs ($/yr)	$2,653,912	
Total Annualized Costs (TAC) ($)	**$3,998,355**	
Total NOx Controlled (ton/yr) - Natural Gas	44.7	50% reduction (2.5 ppm)
COST EFFECTIVENESS ($/ton)[a]	**$89,389**	

[a]U.S. Environmental Protection Agency Office of Air Quality Planning and Standards, Air Pollution Control Cost Manual, Sixth Edition (2002)

[b]Cost obtained from vendor

[c]Cost obtained from vendor

[d]U.S. Energy Information Administration Electric Power Monthly (December 2025)

Table D-3
Oxidation Catalyst Capital Cost Analysis - Simple Cycle Turbines (Each)

Item	Value	Basis
Direct Costs		
Purchased Equipment Cost		
Equipment cost + auxiliaries[b] **[A]**	**$5,297,350**	A
Instrumentation[a]	$529,735	0.10 x (A)
Sales taxes[a]	$0	Pollution Control Equipment Exempt
Freight[a]	$264,868	0.05 x (A)
Total Purchased Equipment Cost (PEC) [B]	**$6,091,953**	B = 1.15 x (A)
Direct Installation Costs		
Total Direct Installation Cost[a]	$1,827,586	0.30 x B
Site Preparation (SP)	$0	As required
Buildings (Bldg.)	$0	As required (5-18% PEC)
Total Direct Cost (DC)	**$7,919,538**	1.3B + SP + Bldg.
Indirect Costs (Installation)		
Engineering[a]	$609,195	0.10 x B
Construction and field expenses[a]	$304,598	0.05 x B
Contractor fees[a]	$609,195	0.10 x B
Start-up[a]	$121,839	0.02 x B
Performance test[c]	$7,500	Stack Test Vendor Quote
Contingencies[a]	$304,598	0.05 x B
Other	$0	As required
Total Indirect Cost (IC)	**$1,956,925**	0.32B + Other + Perf. Test
Total Capital Investment (TCI) = DC + IC	**$9,876,463**	1.62B + Performance test + Other + SP + Bldg.

[a]U.S. Environmental Protection Agency Office of Air Quality Planning and Standards, Air Pollution Control Cost Manual, Sixth Edition (2002)

[b]Cost obtained from vendor

[c]Cost obtained from vendor

[d]U.S. Energy Information Administration Electric Power Monthly (June 2025)

Table D-4
Oxidation Catalyst Annual Cost Analysis - Simple Cycle Turbines (Each)

Item	Value	Basis
Direct Annual Costs (DC)		
Steam		
Press. Drop (in W.C.)	3.0	Pressure drop - catalyst bed
Power output of turbine (kW)	239,000	ISO Rating
Output Loss Due to Pressure Drop (%)	0.30%	0.1% for every 1" pressure drop
Output Loss Due to Pressure Drop (kW)	717.00	
Unit cost ($/kWh)[d]	$0.089	Current Purchase Price
Cost of Heat Rate Loss ($/yr)	$224,349	Based on operation hours/yr
Operating Labor		Assumed $30/hr
Catalyst labor req.	$6,563	216 hr/yr (1/2 hr/shift. 1095 shifts/yr)
Supervisor	$984	15% Operating labor
Total Cost ($/yr)	$7,547	
Maintenance[a]		
Catalyst replacement labor	$3,200	107 hr/yr(8 worker, 40 hr, every 3 years)
Material	$3,200	100% of maintenance labor
Total Cost ($/yr)	$6,400	
Catalyst		
Catalyst Cost ($)[b]	$920,000	Catalyst modules
Catalyst Disposal Cost ($)	$1,500	Disposal of catalyst modules
Sales Tax ($)[a]	$0	Assume exempt from taxes
Catalyst Life (yrs)[b]	3	n
Interest Rate (%)[a]	7%	I
CRF[a]	0.381	Amortization of catalyst over 3 yrs
Total Cost ($/yr)	$351,139	(Volume)(Unit Cost)(CRF)
Indirect Annual Costs (IC)		
Overhead[a]	$0	OAQPS SCR Assumption
Administrative charges[a]	$0	OAQPS SCR Assumption
Annual Contingency[a]	$0	OAQPS SCR Assumption
Property taxes[a]	$0	OAQPS SCR Assumption
Insurance[a]	$0	OAQPS SCR Assumption
Capital Recovery[a]	$932,268	CRF x TCI (20 yr life, 7.0% interest)
Total Indirect Costs ($/yr)	$932,268	
Total Annualized Costs (TAC) ($)	**$1,521,704**	
Total CO Controlled (ton/yr)	28.3	78% removal
Total VOC Controlled (ton/yr)	0.5	30% removal
COST EFFECTIVENESS ($/ton)	**$52,771**	

[a]U.S. Environmental Protection Agency Office of Air Quality Planning and Standards, Air Pollution Control Cost Manual, Sixth Edition (2002)
[b]Cost obtained from vendor
[c]Cost obtained from vendor
[d]U.S. Energy Information Administration Electric Power Monthly (December 2025)

Table D-5
SCR System Capital Cost Analysis - Hot Water Boilers (Each)

Item	Value	Basis
Direct Costs		
Purchased Equipment Cost		
Equipment cost + auxiliaries **[A]**	**$42,000**	A
Instrumentation	$4,200	0.10 x A
Sales taxes	$1,260	0.03 x A
Freight	$2,100	0.05 x A
Total Purchased Equipment Cost (PEC) [B]	**$49,560**	1.18 x A
Direct Installation Costs		
Foundations and supports	$3,965	0.08 x B
Handling and erection	$6,938	0.14 x B
Electrical	$1,982	0.04 x B
Piping	$991	0.02 x B
Insulation for ductwork	$496	0.01 x B
Painting	$496	0.01 x B
Total Direct Installation Cost	$14,868	0.30 x B
Site Preparation (SP)	$0	As required
Buildings (Bldg.)	$0	As required
Total Direct Cost (DC)	**$64,428**	1.30B + SP + Bldg.
Indirect Costs (Installation)		
Engineering	$4,956	0.10 x B
Construction and field expenses	$2,478	0.05 x B
Contractor fees	$4,956	0.10 x B
Start-up	$991	0.02 x B
Performance test	$7,500	Stack Test Vendor Quote
Contingencies	$2,478	0.05 x B
Other	$0	As required
Total Indirect Cost (IC)	**$23,359**	0.32B + Perf. Test + Other
Total Capital Investment (TCI) = DC + IC	**$87,787**	1.62B + Performance test + Other + SP + Bldg.

Table D-6
SCR System Annual Cost Analysis - Hot Water Boilers (Each)

Item	Value	Basis
Direct Annual Costs (DC)		
Operating Labor		
Catalyst labor req.	$16,425	1/2 hr/shift @ $30/hr
Ammonia delivery requirement (SCR)	$720	24 hr/yr (3 deliveries per year) @ $30/hr
Ammonia recordkeeping and reporting (SCR)	$1,200	40 hours per year @ $30/hr
Catalyst cleaning	$1,200	40 hours per year @ $30/hr
Supervisor	$2,464	15% Operating labor
Total Cost ($/yr)	$22,009	
Maintenance		
Total Cost ($/yr)	$1,317	0.015 x B
Ammonia		
Requirement (tons/yr)	2.7	29% aqueous ammonia @ $375/ton
Unit Cost ($/ton)	$375	Estimate
Total Cost ($/yr)	$1,019	
Process Air		
Requirement (scf/lb NH_3)	350	
Requirement (mscf/yr)	8,334	
Unit Cost ($/mscf)	$0.20	$0.20 per 1000 scf
Total Cost ($/yr)	$1,667	
Catalyst		
Catalyst Cost ($)	$37,000	
Catalyst Disposal Cost ($)	$38	Disposal of catalyst modules
Sales Tax ($)	$1,110	0.03 x Catalyst Cost
Catalyst Life (yrs)	5	n
Interest Rate (%)	7.0%	i
CRF	0.244	Amortization of catalyst for 5 yrs
Total Cost ($/yr)	$9,304	(Volume) * (Unit Cost) * (CRF)
Indirect Annual Costs (IC)		
Overhead	$0	OAQPS SCR Assumption
Administrative charges	$0	OAQPS SCR Assumption
Annual Contingency	$0	OAQPS SCR Assumption
Property taxes	$0	OAQPS SCR Assumption
Insurance	$0	OAQPS SCR Assumption
Capital Recovery	$8,286	CRF x TCI (20 yr life, 7.0% interest)
Total Indirect Costs ($/yr)	$8,286	
Total Annualized Costs (TAC) ($)	**$43,602**	
Total NOx Controlled (ton/yr)	1.1	90% reduction
COST EFFECTIVENESS ($/ton)	**$38,141**	

Table D-7
Oxidation Catalyst Capital Cost Analysis - Hot Water Boilers (Each)

Item	Value	Basis
Direct Costs		
Purchased Equipment Cost		
Equipment cost + auxiliaries **[A]**	**$8,400**	A
Instrumentation	$840	0.10 x A
Sales taxes	$252	0.03 x A
Freight	$420	0.05 x A
Total Purchased Equipment Cost (PEC) [B]	**$9,912**	B = 1.18 x A
Direct Installation Costs		
Foundations and supports	$792.96	0.08 x B
Handling and erection	$1,388	0.14 x B
Electrical	$396	0.04 x B
Piping	$198	0.02 x B
Insulation for ductwork	$99	0.01 x B
Painting	$99	0.01 x B
Total Direct Installation Cost	$2,974	0.30 x B
Site Preparation (SP)	$0	As required
Buildings (Bldg.)	$0	As required
Total Direct Cost (DC)	**$12,886**	1.3B + SP + Bldg.
Indirect Costs (Installation)		
Engineering	$991	0.10 x B
Construction and field expenses	$496	0.05 x B
Contractor fees	$991	0.10 x B
Start-up	$198	0.02 x B
Performance test	$7,500	Stack Test Vendor Quote
Contingencies	$496	0.05 x B
Other	$0	As required
Total Indirect Cost (IC)	**$10,672**	0.32B + Other + Perf. Test
Total Capital Investment (TCI) = DC + IC	**$23,557**	1.62B + Performance test + Other + SP + Bldg.

Table D-8
Oxidation Catalyst Annual Cost Analysis - Hot Water Boilers (Each)

Item	Value	Basis
Direct Annual Costs (DC)		
Operating Labor		Assumed $30/hr
Catalyst labor req.	$16,425	216 hr/yr (1/2 hr/shift. 1095 shifts/yr)
Supervisor	$2,464	15% Operating labor
Total Cost ($/yr)	$18,889	
Maintenance		
Catalyst replacement labor	$3,200	107 hr/yr(8 worker, 40 hr, every 3 years)
Material	$3,200	100% of maintenance labor
Total Cost ($/yr)	$6,400	
Catalyst		
Catalyst Cost ($)	$35,000	Catalyst modules
Catalyst Disposal Cost ($)	$1,500	Disposal of catalyst modules
Sales Tax ($)	$0	Assume exempt from taxes
Catalyst Life (yrs)	3	n
Interest Rate (%)	7%	I
CRF	0.381	Amortization of catalyst over 3 yrs
Total Cost ($/yr)	$13,908	(Volume)(Unit Cost)(CRF)
Indirect Annual Costs (IC)		
Overhead	$0	OAQPS SCR Assumption
Administrative charges	$0	OAQPS SCR Assumption
Annual Contingency	$0	OAQPS SCR Assumption
Property taxes	$0	OAQPS SCR Assumption
Insurance	$0	OAQPS SCR Assumption
Capital Recovery	$2,224	CRF x TCI (20 yr life, 7.0% interest)
Total Indirect Costs ($/yr)	$2,224	
Total Annualized Costs (TAC) ($)	**$41,421**	
Total CO Controlled (ton/yr)	1.1	50% reduction
Total VOC Controlled (ton/yr)	0.07	50% reduction
COST EFFECTIVENESS ($/ton)	**$36,372**	

Table D-9
Water Truck Cost Analysis

Item	Value	Basis
Direct Costs		
Purchased Equipment Cost		
Equipment cost + auxiliaries	**$100,000**	Conservative estimate of cost of new water truck
Direct Annual Costs (DC)		
Operating Labor		Assumed $30/hr
Driver labor	$1,800	1 hr per water truck cycle
Total Cost ($/yr)	$1,800	
Water Costs		
Water Cost ($/cubic foot)	$0.0302	Water rate in Cedar Rapids, Iowa
Gallons of water per truck tank	2,000	Typical water truck size for this type of facility
Expected number of water trucks cycles per year	60	
Total Cost ($/yr)	$484	
Indirect Annual Costs (IC)		
Capital Recovery	$9,439	CRF x TCI (20 yr life, 7.0% interest)
Total Indirect Costs ($/yr)	$9,439	
Total Annualized Costs (TAC) ($)	**$11,724**	
Total PM Controlled (ton/yr)	0.035	69% reduction
COST EFFECTIVENESS ($/ton)	**$339,818**	

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